STANDARD DRAWINGS FOR ACCESSORIES AND EQUIPMENT OF VESSELS

FWEL CONTRACT No : 1-14-16601/19

CONTRACT NO. : POGC-760-81-156

PROJECT : SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)

COMPANY : NATIONAL IRANIAN OIL COMPANY

SITE : ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN

REV.	DATE	DESCRIPTION	ORIG.	PRPD	CHKD	APP'D	CONTRACTOR's APP'D
8	08/12/04	AFC (Revised as marked)	LG	S.R.Kim	S.S.Park	H.S.Kim	S.C.Kim
7	31/08/04	Approved for Construction	LG	S.R.Kim	S.S.Park	H.S.Kim	S.C.Kim
6	15/07/04	Issued to Company	LG	S.R.Kim	S.S.Park	H.S.Kim	S.C.Kim
5	21/12/03	As per COMPANY's comments	LG/OIEC	S.R.Kim	H.C.Shin	K.S.Lee	
4	27/11/03	Issued for Feed Package	FWEL	S. Jacobs	S. Sim		
3	14/11/03	Issued for Feed Package	FWEL	S. Jacobs	S. Sim		S. Sim
2	12/11/03	Issued for Feed Package	FWEL	S. Jacobs	S. Sim		S. Sim
1	17.07.03	Issued for Company Review	FWEL	S. Jacobs	S. Sim		S. Sim
0	14.12.00	Issued for Basic Design Package	TECHNIP	V. Zarna	V. Zarna		F. Durot

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TABULATION OF REVISED PAGES

PAGE	REV.4	REV.5	REV.6	REV.7	REV.8
1	X	X			
2	X	X			
3					
4	X	X	X	X	X
5	X	X			
6	X	X	X	X	X
7	X	X			
8	X	X	X		X
9	X		X		X

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CONTENTS

– GENERAL STATEMENT
– ADDENDUM
– TEP GENERAL SPECIFICATION N° SP TCS 202
– ATTACHMENT :
APPENDIX 1 - APPLICABLE STANDARDS.

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GENERAL STATEMENT

The attached TOTAL EXPLORATION PRODUCTION (TEP) General Specification n° SP TCS 202 Rev. 1 is confirmed as the Project Specification, except as amended herein, and renumbered/reissued as specification n° RP 6340 S 999 0400 005 Rev. 8 for SOUTH PARS PHASES 9 & 10 Project.

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ADDENDUM

The following items refer to the attached TEP General Specification n° SP TCS 202 Rev. 1.

The clauses set out below shall modify or replace the existing clauses in the original specification as noted.

1. THE FOLLOWING STANDARDS ARE DELETED AND REPLACED

- PS TCS 204 by 6340S-L-C022
- PS TCS 205 by 6340S 000 STC 000 400 301 & 304
- PS TCS 208 by 6340S 000 STC 000 400 301 & 304
- PS TCS 209 by 6340S-L-C021
- PS TCS 218 by 6340S-L-C008
- PS TCS 219 by 6340S-L-C008
- PS TCS 221 by 6340S 000 STC 000 400 402
- PS TCS 223 by 6340S-L-E021
- PS TCS 224 by 6340S 000 STC 000 400 505 and by 6340S-L-E020
- PS TCS 220 by 6340S-L-C027 and by 6340S-L-C028

2. FOLLOWING STANDARDS ARE DELETED

- PS TCS 210 (Name of connections not used)
- PS TCS 225 (Davit not supplied by Vessel Manufacturer)
- PS TCS 202 (Vessel dimensions are indicated in Process Specifications)
- PS TCS 203 (Vessel dimensions are indicated in Process Specifications)
- PS-TCS-304 (Davit not supplied by Vessel Manufacturer)
- PS-TCS-301 (Storage spheres tolerances)
- PS-TCS-302 (Fire proofing supports)
- PS-TCS-303 (Insulation supports)
- PS-TCS-305 (Platform and stairway)

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3. FOLLOWING STANDARDS ARE ADDED

– 6340 S 000 STC 000 400 905
– 6340 S 000 STC 000 400 401
– 6340 S L C011
– 6340 S L C013 (1/3 - 3/3)
– 6340 S L C014
– 6340 S L C015
– 6340 S L C016
– 6340 S L C019
– 6340 S L C020
– *6340 S L C021*
– *6340 S L C022*
– 6340 S L C023
– DELETED
– *6340 S L C027*
– *6340 S L C028*
– 6340 S L C040
– 6340 S L C041
– 6340 S L C043
– 6340 S L C046
– 6340 S L E020
– 6340 S L E021
– 6340 S L C015

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4. FOLLOWING STANDARDS ARE AMENDED

– PS.TCS.201: Out of plumbness shall be limited to 0.1% of length, not to exceed 19mm for vessels up to 30m long and 25mm for vessels exceeding 30m in length.

– PS.TCS.215: Item numbers are as per PS.TCS.214

– 6340S-L-C027 & C028: Following comments should be added as note;
Distance between center line of manhole and lower head or vessel interior wall shall not exceed 800mm. This shall be added on 6340S-L-C027..
Rungs in dished area of vertical vessels below manways are required for vessel diameters in excess of 2000mm. This shall be added on 6340S-L-C-028.

– PS.TCS.222: For nozzles in excess of 14” NB, removable baffles are required.

– 6340S 000 STC 000400505: Detail titled “PT INOX” - STAINLESS STEEL. “PT NORMAL” shall be read CARBON STEEL”

– L-C027: Drawing number shall read 6340S-L-C027
– L-C028: Drawing number shall read 6340S-L-C028
– L-C041: Drawing number shall read 6340S-L-C041
– L-C043: Drawing number shall read 6340S-L-C043

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APPENDIX I

FOLLOWING STANDARDS ARE APPLICABLE :

– PS-TCS-201 Rev. 0 : Pressure Vessel Tolerances
– PS-TCS-206 Rev. 0 : Beds for Horizontal Drums and Tanks
– PS-TCS-207 Rev. 0 : Saddles Welded on Vessels
– PS-TCS-211 Rev. 0 : Minimum thickness of wall nozzles welded on vessels
– PS-TCS-212 Rev. 0 : Nozzle diameter < 2” on vessel
– PS-TCS-213 Rev. 0 : Projection and drilling of nozzles
– PS-TCS-214 Rev. 0 : Vertical manhole with Davit (welding neck flange)
– PS-TCS-215 Rev. 0 : Vertical manhole with Davis (slip - on flange)
– PS-TCS-216 Rev. 0 : Horizontal Manhole with Davit (WN Flange)
– PS-TCS-217 Rev. 0 : Horizontal Manhole with Davit (SO Flange)

– PS-TCS-222 Rev. 0 : Deflector for steam inlet
– PS-TCS-229 Rev. 0 : Fireproofing supports
– PS-TCS-230 Rev. 0 : Insulation supports for hot service vertical apparatus
– PS-TCS-231 Rev. 0 : Insulation supports for hot service horizontal apparatus
– PS-TCS-232 Rev. 0 : Clad alloy vessel welding procedure
– PS-TCS-233 Rev. 0 : Nozzles lining
– PS-TCS-234 Rev. 0 : Lined ring joint face

– 6340S 000 STC 000 400 301 Rev. 0 : Skirts –Access holes-Vents
– *DELETED*
– 6340S 000 STC 000 400 303.1 Rev. 0 : Support saddles for horizontal vessels 3800 $\leq \varnothing <$ 5000
– 6340S 000 STC 000 400 304 *Rev. 1* : Anchor chairs

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– 6340S 000 STC 000 400 401 Rev. 0 : Flanges for removable internal piping
– 6340S 000 STC 000 400 402 Rev. 0 : Internal arrangement of vessels
– 6340S 000 STC 000 400 505 *Rev. 1* : Earth connection
– 6340S 000 STC 000 400 905 Rev. 0 : Nozzle Reinforcements Details
– 6340S-L-C008 Rev. 0: Manhole Cover Hinge
– 6340S-L-C011 Rev. 0: Pipe Davit
– 6340S-L-C013-1/3 Rev. 0: Platform for Vertical Vessel
– 6340S-L-C013-2/3 Rev. 0: Top Platform for Vertical Vessel
– 6340S-L-C013-3/3 Rev. 0: Platform for Horizontal Vessel
– 6340S-L-C014 Rev. 0: Ladder
– 6340S-L-C016 Rev. 0: Template
– 6340S-L-C019 Rev. 0: Support Lugs for Vertical Vessel (Light Duty)
– 6340S-L-C020 Rev. 0: Support Lugs for Vertical Vessel (Heavy Duty)
– 6340S-L-C021 Rev. 0: Support Legs for Vertical Vessel
– 6340S-L-C022 Rev. 0: Lifting Lugs
– 6340S-L-C023 Rev. 0: Foundation Bolts (Metric)
– DELETED
– 6340S-L-C027 Rev. 0: Internal Ladder Rung
– 6340S-L-C028 Rev. 0: Internal Ladder for Horizontal Vessel
– 6340S-L-C040 Rev. 0: Retaining Plate
– 6340S-L-C041 Rev. 0: Demister & Support
– 6340S-L-C043 Rev. 0: Preferential Reboiler Baffle
– 6340S-L-C046 Rev. 0: Bolt Hole Orientation for Flange
– 6340S-L-E020 Rev. 0: Earth Connection
– 6340S-L-E021 *Rev. 1*: Manufacturer Nameplate
– 6340S-L-C015 Rev. 0: Typical Skirt & Base Details for Vertical Vessels (for reference)

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TOTAL EXPLORATION PRODUCTION

GENERAL SPECIFICATION

PIPING - VESSELS - WELDING

SP - TCS - 202

STANDARD DRAWINGS FOR ACCESSORIES AND EQUIPMENT OF VESSELS

1	July 1993	New revision
0	February 1981	First issue
REV. N°	DATE	SUBJECT OF THE REVISION

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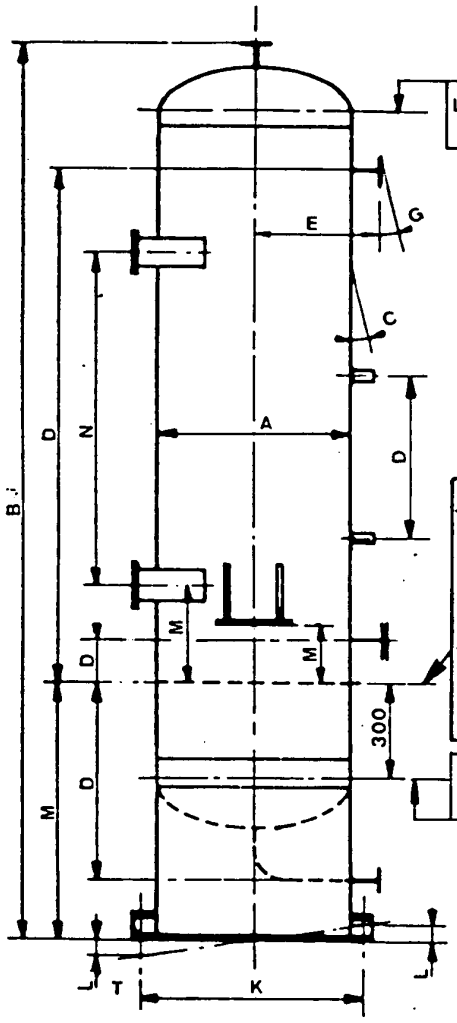
Plan	TITRE	Page	Revisions											
			0	1	2	3	3	4	5	5	6	7	8	9
PS-TCS-201	Pressure vessel tolerances	1 2 3	X X X											
PS-TCS-202	Vessels dished ends with large knucle radius	1	X											
PS-TCS-203	Vessels ellipticals dished ends	1	X											
PS-TCS-204	Standard lifting lug for apparatus	1	X											
PS-TCS-205	Lug and base ring for verticals apparatus	1	X											
PS-TCS-206	Beds for horizontals drums and tanks	1	X											
PS-TCS-207	Saddles welded on vessels	1 2	X X											
PS-TCS-208	Column skirt standard equipment	1	X											
PS-TCS-209	Feet for small apparatus	1 2	X X											
PS-TCS-210	Name of connections	1	X											
PS-TCS-211	Minimum thickness of wall nozzles welded on vessels	1 2	X X											
PS-TCS-212	Nozzle diameter ≤ 2" on vessel	1	X											
PS-TCS-213	Projection and drilling of nozzles	1	X											
PS-TCS-214	Vertical manhole with davit (welding neck flange)	1 2	X X											

Plan	TITRE	Page	Revisions										
			0	1	2	3	3	4	5	5	6	7	8	9
PS-TCS-215	Vertical manhole with davit (slip-on flange)	1	X											
		2	X											
PS-TCS-216	Horizontal manhole with davit (welding neck flange)	1	X											
PS-TCS-217	Horizontal manhole with davit (slip-on flange)	1	X											
PS-TCS-218	Hinged manhole (welding neck flange)	1	X											
		2	X											
PS-TCS-219	Hinged manhole (slip-on flange)	1	X											
		2	X											
PS-TCS-220	Hand grip and ladder rung for manhole	1	X											
PS-TCS-221	Vortex breakers on fluid outlet	1	X											
		2	X											
PS-TCS-222	Deflector on steam inlet	1	X											
PS-TCS-223	Names plates for vessels	1	X											
PS-TCS-224	Grounding plate	1	X											
		2	X											
PS-TCS-225	Monorail davit	1	X											
		2	X											
		3	X											
		4	X											
		5	X											

Plan	TITRE	Page	Revisions										
			0	1	2	3	3	4	5	5	6	7	8	9
PS-TCS-229	Fire proofing supports	1	X											
PS-TCS-230	Insulation supports for hot service vertical apparatus	1	X											
PS-TCS-231	Insulation supports for hot service horizontal apparatus	1	X											
PS-TCS-232	Clad alloy vessel welding procedure	1	X											
		2	X											
		3	X											
		4	X											
		5	X											
PS-TCS-233	Nozzles lining	1	X											
		2	X											
PS-TCS-234	Lined ring joint face	1	X											
PS-TCS-301	Storage spheres tolerances	1	X											
		2	X											
PS-TCS-302	Fire proofing supports	1	X											
PS-TCS-303	Insolation supports	1	X											
PS-TCS-304	Monorail davit and slide	1	X											
		2	X											
		3	X											
		4	X											
		5	X											
PS-TCS-305	Platform and stairway	1	X											
		2	X											

TOTAL TEP/DD	TOLERANCE DES APPAREILS A PRESSION PRESSURE VESSEL TOLERANCES	PS.TCS. 201 Page 1 de 3 Date : février 1981

DIMENSIONS EXTERIEURES
OUTSIDE DIMENSIONS

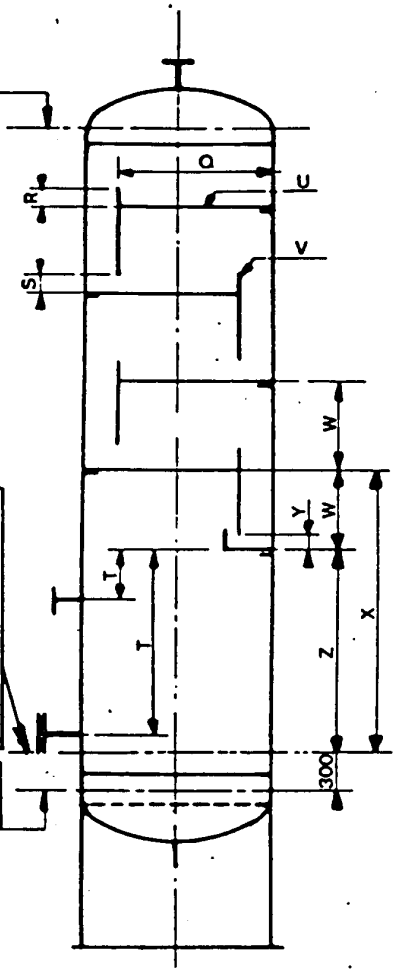


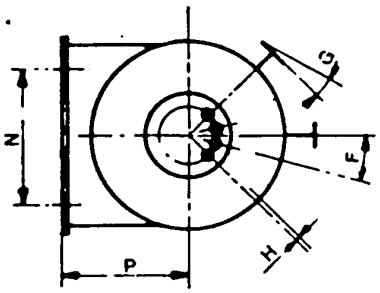
PLAN DE REFERENCE:
POINTAGE INTERIEUR ET EXTERIEUR TOUS LES 80 SUR LA VIROLE POUR CONTROLE

REFERENCE PLANE:
INSIDE AND OUTSIDE HAMMER-PUNCHING EACH EVERY 80 ON SHELL FOR CONTROL

LIGNE DE TANGENCE
TANGENT LINE

DIMENSIONS INTERIEURES
INSIDE DIMENSIONS





DIMENSIONS	U	V
ø EXT. COLONNE VESSEL OUT. DIA. mm	TOLÉRANCES mm	
ø ≤ 1200	3	3
1200 < ø ≤ 2100	5	3
2100 < ø ≤ 3900	6,5	5
3900 < ø ≤ 4800	8	6,5
ø > 4800	9,5	8

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOLERANCES SUR LES DIMENSIONS EXTERIEURES
TOLERANCES ON OUTSIDE DIMENSIONS

A – OVALISATION PERMISE PAR LE CODE.
– OUT OF ROUNDNESS AS PERMITTED BY CODE.

B – TOLERANCE SUR LONGUEUR : ± 1,3 mm/m (MAX : 13 mm)
– TOLERANCE ON LENGTH : ± 1,3 mm/m (MAX : 13 mm)

C – RECTITUDE DES GENERATRICES : UNE PARTIE QUELCONQUE AVANT UNE LONGUEUR – DE 6,5 mm D'UNE GENERATRICE DU CYLINDRE DOIT ETRE COMPRISE ENTRE DEUX DROITES PARALLELES A L'AXE DISTANTES DE 6,5 mm.
– RECTITUDE OF THE GENERATRIX : WHATEVER PART OF A CYLINDER GENERATRIX WITH 6,5 mm IN LENGTH SHALL RANGE WITHIN 2 X 6,5 mm SPACED LINES BOTH PARALLEL TO THE CENTERLINE.

D – TOLERANCE SUR LES ELEVATIONS :
POUR LES TROUS D'HOMME : ± 13 mm
POUR LES TUBULURES : ± 6,5 mm ENTRE LES AXES DES CONNEXIONS DES PRISES D'INSTRUMENTS : ± 3 mm
TOLERANCE ON ELEVATIONS :
FOR MANHOLES : ± 13 mm FOR NOZZLES : ± 6,5 mm
CLEARANCE BETWEEN INSTRIMENT BRANCH CONNECTIONS : ± 3 mm.

E – TOLERANCE SUR LE DEPASSEMENT DES FACES DE BRIDES :
POUR LES TROUS D'HOMME : ± 6,5 mm
POUR LES TUBULURES : ± 5 mm
– FLANGE FACES PROJECTING FROM SHELL :
FOR MANHOLES : ± 6,5 mm
FOR NOZZLES : ± 5 mm

F – TOLERANCE SUR LA DEVIATION CIRCONFERENTIELLE ENTRE L'AXE REEL ET L'AXE THEORIQUE DES TUBULURES, TROUS D'HOMME, SUPPORT : ± 3 mm MESURES SUR LA VIROLE.
– CIRCUMFERENTIAL DEVIATION BETWEEN THE THEORETICAL AXIS AND CENTERLINE OF NOZZLES, MANHOLES AND SUPPORTS SHALL NOT EXCEED ± 3 mm MEASURED ON SHELL.

G – ANGLE DU PLAN DE LA FACE D'UNE BRIDE OU D'UN SUPPORT PAR RAPPORT AU PLAN THEORIQUE : ± 1/2°
– ANGLE OF FLANGE FACE IN RATIO TO THE THEORICAL PLANE : ± 1/2°

H – TOLERANCE SUR L'ORIENTATION DU PERCAGE DES TUBULURES : ± 1,5 mm MESURES SUR LE CERCLE DE PERCAGE.
– ORIENTATION OF NOZZLE BOLT HOLES : ± 1,5 mm MEASURED ON DRILLING CIRCLE.

K – TOLERANCE SUR PERCAGE DE L'EMBASE :
SUR DIAMETRE MESURES A 90° : ± 5 mm
ENTRE TROUS : ± 5 mm
– BASE RING DRILLING TOLERANCE :
MEASURED ON PERPENDICULAR DIAMETERS : ± 5 mm
MEASUPED ON DRILLING CIRCLE BETWEEN HOLES : ± 5 mm

L – TOLERANCE SUR L'EQUERRAGE DE L'EMABSE :
– TOLERANCE ON BASE RING PERPENDICULARITY :
$K \leqslant 1,2$ mm : ± 3 mm
$1,2 \text{ mm} < K \leqslant 1,2$ m : ± 5 mm
$K > 1,2$ m : ± 6,5 mm

M – TOLERANCE SUR POSITION DU PLAN DE REFERENCE :
– TOELRANCE ON REFERENCE PLANE : + 6,5 mm
– 0

N – TOLERANCE SUR DISTANCE DES TROUS DE SUPPORTS POUR RESERVOIRS HORIZONTAUX : ± 3 mm.
– CLEARANCE BETWEEN SUPPORT BOLT HOLES ON HORIZONTAL VESSELS : ± 3 mm.

P – DISTANCE ENTRE PLAN DE POSE DU SUPPORT ET L'AXE D'UN RESERVOIR HORIZONTAL : ± 6,5 mm
– CLEARANCE BETWEEN SUPPORT SETTING LEVEL AND HORIZONTAL VESSEL CENTERLINE : ± 6,5 mm

Rev.N°	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOLERANCES SUR LES DIMENSIONS INTERIEURES
TOLERANCES ON INSIDE DIMENSIONS

Q – DISTANCE DE LA PAROI INTERIEURE DE L'APPAREIL AU DEVERSOIR : ± 6,5 mm
– CLEARANCE FROM VESSEL INSIDE WALL TO WEIR : ± 6,5 mm

R – HAUTEUR DU DEVERSOIR FIXE AU DESSUS DU SUPPORT DE PLATEAU : ± 3 mm
– HEIGHT OF FIXED WEIR ABOVE TRAY SUPPORTING RING : ± 3 mm

S – DISTANCE DU FOND DE LA PLAQUE DE DESCENTE D'ECOULEMENT AU SUPPORT DE PLATEAU INFERIEUR : ± 3 mm
– CLEARANCE FROM BOTTOM OF DOWN-FLOW PLATE TO TRAY SUPPORTING RING : ± 3 mm.

T – DISTANCE DU SUPPORT DE PLATEAU DE SOUTIRAGE A L'AXE DU TROU D'HOMME OU A L'AXE DE LA TUBULURE QUAND LE PLATEAU EST COTE PAR RAPPORT A CEUX-CI : ± 3 mm.
– CLEARANCE FROM DRAW-OFF PAN SUPPORT TO MANHOLE OR NOZZLE CENTERLINE WHEN LOCATED IN RATIO TO ONE ANOTHER : ± 3 mm.

U – DEVIATION MAXIMUM DU SUPPORT DE PLATEAU PAR RAPPORT A UN PLAN PARALLELE AU PLAN DE REFERENCE : VOIR TABLEAU.
– MAXIMUM DEVIATION OF TRAY SUPPORTING FROM A PLANE PARALLEL TO REFERENCE PLANE : SEE TABLE.

V – DEVIATION MAXIMUM DU SOMMET DE LA PLAQUE DE DEVERSOIR AVEC UN PLAN PARALLELE AU PLAN DE REFERENCE : VOIR TABLEAU.
– MAXIMUM DEVIATION OF TOP OF WEIR PLATE FROM A PLANE PARALLEL TO REFERENCE PLANE : SEE TABLE.

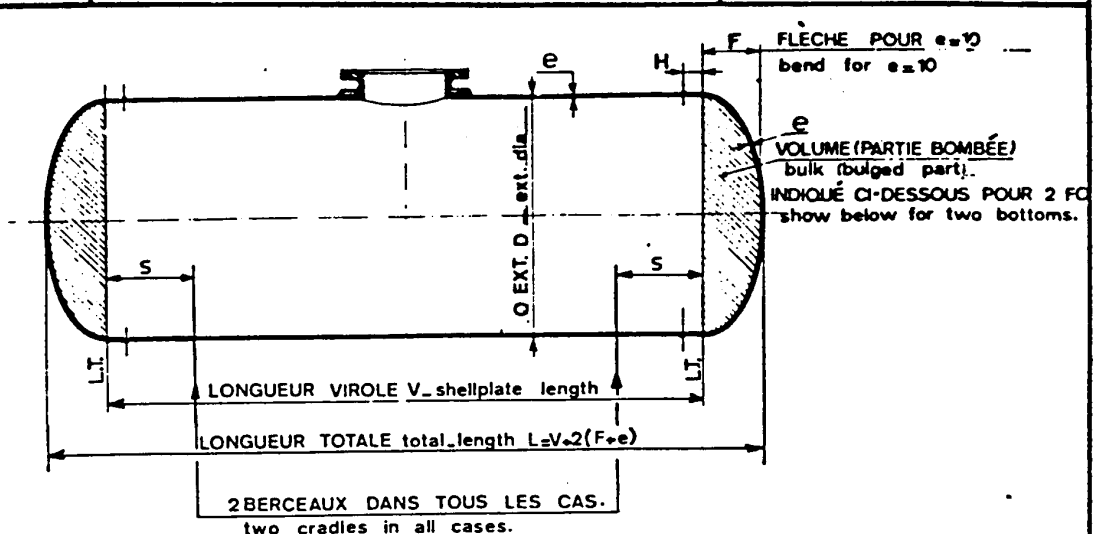
W – DISTANCE ENTRE LES SUPPORTS DE PLATEAUX ADJACENTS : ± 3 mm.
– CLEARANCE BETWEEN ADJOINING TRAY SUPPORTS : ± 3 mm.

X – DISTANCE D'UN SUPPORT DE PLATEAU AU PLAN DE REFERENCE : ± 6,5 mm.
– CLEARANCE FROM ONE TRAY SUPPORTING TO THE REFERENCE PLANE : ± 6,5 mm.

Y – DISTANCE DU BAS DE LA PLAQUE DE DESCENTE DE L'ECOULEMENT AU SUPPORT DU DERNIER DEVERSOIR : ± 3 mm.
– CLEARANCE FROM BOTTOM OF DOWN-FLOW PLATE TO LOWER DOWNCOMER SUPPORT : ± 3 mm

Z – DISTANCE DES SUPPORTS INTERIEURS AU PLAN DE REFERENCE : ± 6,5 mm.
– CLEARANCE FROM ANY INTERNAL SUPPORT TO REFERENCE PLANE : ± 6,5 mm.

TOTAL · TEP/DD	RESERVOIRS HORIZONTAUX FONDS A GRAND RAYON DE CARRE VESSELS DISHED ENDS WITH LARGE KNUCLE RADIUS	PS.TCS. 202 Page 1 de 1 Date: février 1981



CAPACITÉ NOM. EN LITRES nom. capacity in liters	Ø EXT. D ext. dia. D	LONG. VIROLE V shell-plate lenght V	VOL. 2 FONDS P^A e=10 EN LITRES vol. for bottoms for e=10	VOL. VIR. P^A V=100 shell-plate volume for V=100	S	POUR e=10 FLÈCHE F bend	POIDS ENS. AVEC H=50 total weight with H=50	POIDS VIR. POUR V=100
250	500	1200	306	18	240	126	198	12
300	550	↓	41	22	240	140	223	135
400	600	1400	54	26	280	153	280	145
500	650	↓	688	31	280	166	308	16
600	700		866	36	300	179	354	17
700	750	1500	107	42	300	192	382	18
800	800	↓	131	48	300	205	418	195
↑	850	1600	1574	54	300	219	475	21
1000	900	1500	1876	61	300	232	490	22
↓	950	1300	222	70	300	245	483	23
1500	1000	1800	252	75	300	258	632	245
2000	1100	1900	347	92	350	284	745	27
2500	1150	2100	397	100	350	297	848	28
3000	1200	2400	453	110	350	310	988	29
3500	1250	2600	512	120	350	324	1095	305
4000	1300	2800	578	130	400	337	1215	32
4500	1400	2600	724	150	400	363	1265	34

CAPACITÉ NOM. EN LITRES nom. capacity in liters	Ø EXT. D ext. dia. D	LONG. VIROLE V shell-plate lenght V	VOL. 2 FONDS P^A e=10 EN LITRES vol. for bottoms for e=10	VOL. VIR. P^A V=100 shell plate volume for V=100	S	POUR e=10 FLÈCHE F bend	POIDS ENS. AVEC H=50 total weight with H=50	shellplate weight for V=1m
5000	1500	↑ 2600	894	170	400	389	1385	37
6000	1600	↓	1088	195	400	416	1502	39
7000	1700	2800	1306	220	400	442	1705	42
8000	1800	↓	1554,4	250	400	468	1835	44
10000	1900	3000	1832	280	400	495	2055	47
12000	2000	3400	2139	305	400	521	2400	49
↑	2100	3800	2478	340	450	546	2760	52
15000	2150	3600	2658	355	450	559	2735	53
↓	2200	↑ 4200	2854	375	450	572	3140	54
18000	2300	↓	3266	410	450	599	3330	56
20000	2400	5000	3716	445	500	625	3990	59
25000	2500	5400	4202	485	500	652	4450	61
30000	2600	6800	4726	520	500	677	5550	64
40000	2700	8000	5300	560	500	703	6600	66
50000	2800	9000	5918	605	600	730	7570	69
60000	2900	9600	6576	650	600	756	8350	71
70000	3000	10400	7332	695	600	782	9230	74
80000								

ÉPAISSEURS NORMALISÉES ET HAUTEURS DE BORDS CORRESPONDANTS.
standard thicknesses and heights of corresponding edges.

ÉPAISSEUR thickness e	8	9	10	11	12	14	16	18	20	22	25	28	32	35	40
HAUTEUR height H	50			55		60	65	70		75	80	90	100	110	120

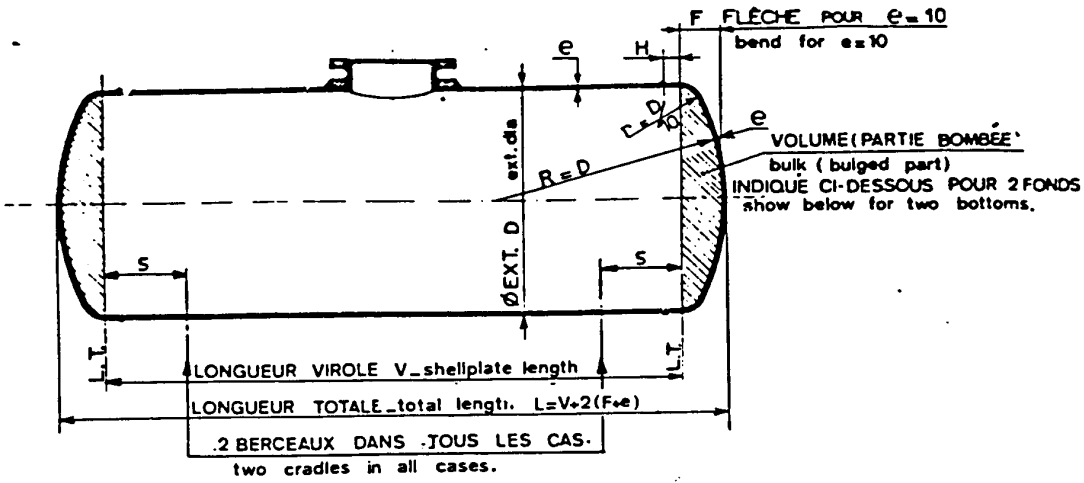
(LES FLÈCHES SONT DÉTERMINÉES A L'AIDE DE LA FORMULE : $F = \frac{D - 2e}{3,8}$
the bend are settled with formula :
POUR CAPACITÉS SUPÉRIEURES PRENDRE S=0,2 D ENVIRON _ for higher capacities make use of S=0,2D nearly

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL
TEP/DD

RESERVOIRS HORIZONTAUX FONDS ELLIPTIQUES
VESSELS ELLIPTICAL DISHED ENDS

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Date: février 1981



CAPACITÉ NOM. EN LITRES nom. capacity in liters	Ø EXT. D ext. dia. D	LONG. VIROLE V shell-plate lenght V	VOL. FONDS P^A e=10 EN LITRES vol. for bottoms for e=10	VOL. VIR. P^A V=300 LITRES shellpla-te vol. for V=300	S	POUR for e=10 FLÈCHE F bend	POIDS ENSEMB. AVEC H=50 tot. weight with H=50	POIDS VIR. POUR V=300
250	500	1200	22	18	240	92	198	12
300	550		30	22	240	102	223	13,5
400	600	1400	39	26	280	111	280	14,5
500	650		50	31	280	121	308	16
600	700	1500	63	36	300	131	352	17
700	750		78	42	300	140	382	18,3
800	800		95	48	300	150	415	19,5
1000	850	1600	114	54	300	160	470	21
	900	1500	136	61	300	169	480	22
	950	1300	165	70	300	179	470	23
1200	1000	1400	187	75	300	189	530	24,5
1500	1050	1600	217	83	300	199	620	25,5
2000	1100	1900	250	92	350	208	740	27
2300	1150	2100	287	100	350	218	840	28
2500	1200		326	110	350	228	880	29,3
2800	1250		369	120	350	237	930	30,5
3000	1300	2400	418	130	350	247	975	32
3500					350		1070	
4000		2800			400		1200	
4500	1400	2800	521	150	400	266	1315	34
5000	1500	2600			400		1340	
5500		2800	643	170	400	286	1415	37
6000		3200			400		1560	
6500	1600	2800	782	195	400	305	1550	39
7000	1700	2600	940	220	400	324	1595	42
7500	1800		1120	250	400	344	1720	44
	1900	2200	1320	280	400	363	1655	47
		2400			400		1745	

CAPACITÉ NOM. EN LITRES nom. capacity in liters	Ø EXT. D ext. dia. D	LONG. VIROLE V shell-plate lenght V	VOL. FONDS P^A e=10 EN LITRES vol. for bottoms for e=10	VOL. VIR. P^A V=300 LITRES shellpla-te vol. for V=300	S	POUR for e=10 FLÈCHE F bend	POIDS ENSEMB. AVEC H=50 tot. weight with H=50	POIDS VIR. shellplate V=300
8000	1900	2600			400		1835	
10000		3200			400		2120	
12000		4000	1320	280	400	363	2500	
15000		5000			400		2960	
17000	2000	4600	1542	305	400	383	2950	49
18000	2100	4000	1785	340	450	402	2820	52
20000	2150	4400	1916	355	450	412	3120	53
25000	2200	4200	2060	375	450	421	3100	54
30000	2300	4000	2350	410	450	441	3200	56
40000	2400	3800	2670	445	500	460	3270	59
45000	2500	3600			500		3320	
50000		4600	3030	485	500	480	3940	61
		5600			500		4550	
55000		7600			500		5770	
60000	2600	7200	3355	520	600	499	5780	64
	2700	6600	3815	560	600	518	5700	66
70000	2800	6800	4270	605	600	538	6080	69
80000	2900	6400	4735	650	600	557	6070	71
90000	2500	9800			600		7090	
100000	2700	8200			600		6740	
	3000	6400	5240	695	600	576	6340	74
	3100	6800	5760	745	700	596	6900	76
	3200	7000	6320	795	700	615	7370	78
	3000	8000			700		7500	
	3300	7600	6940	845	700	634	8150	81
		8800			700		9100	
	3400	9400	7580	900	700	654	9900	83
		10400			700		10800	

ÉPAISSEURS NORMALISÉES ET HAUTEURS DE BORDS CORRESPONDANTS.
standard thicknesses and heights of corresponding edges.

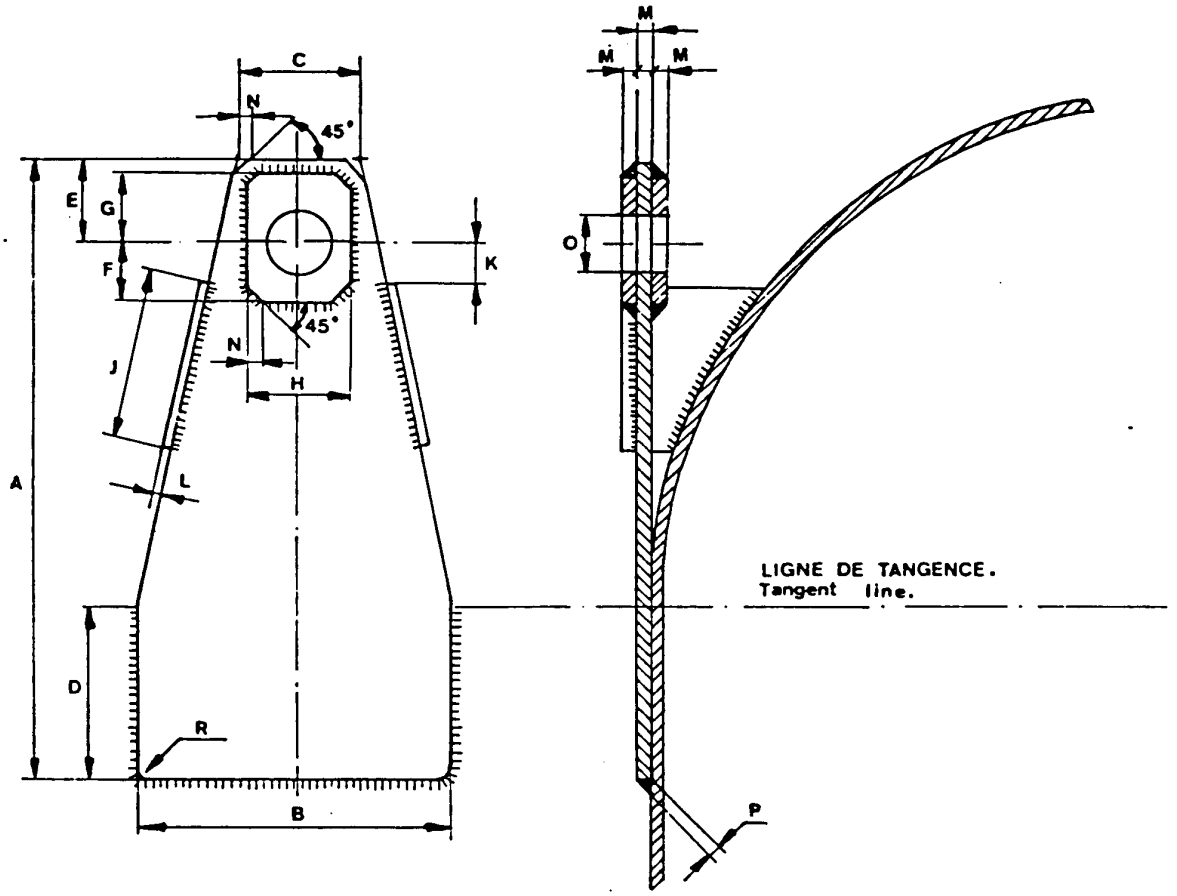
ÉPAISSEUR thickness e	6	7	8	9	10	11	12	14	16	18	20	22	25	28	32	35	40
HAUTEUR height H	40		50			55		60	65	70		75	80	90	100	110	120

LES FLÈCHES SONT DÉTERMINÉES A L'AIDE DE LA FORMULE : the bend are settled with formula : $F = R - \sqrt{(R-r)^2 - (a-r)^2}$ où $a = \frac{D-2e}{2}$

POUR CAPACITÉS SUPÉRIEURES PRENDRE S=0,2D ENVIRON _ for highter capacities make use of S=0,2D nearly.

Rev. No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	OREILLES DE LEVAGE POUR APPAREILS STANDARD LIFTING LUG FOR APPARATUS	PS.TCS.204 Page 1 de 1 Date: Février 1981



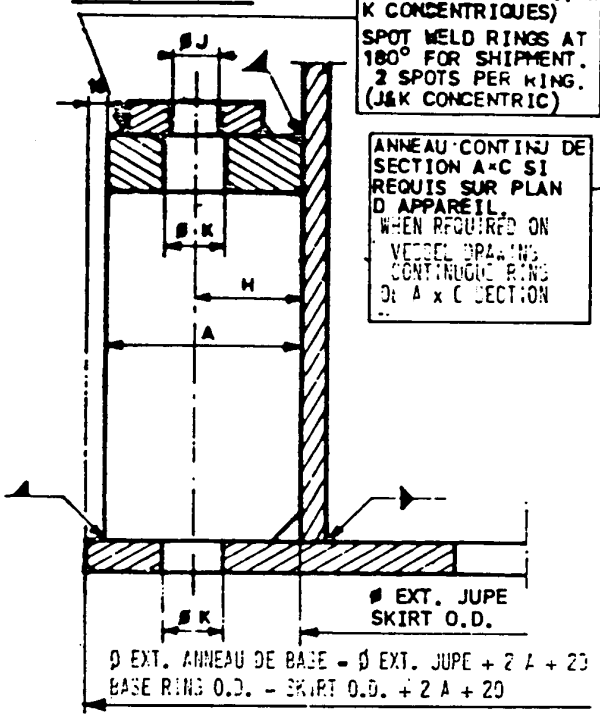
POIDS MAXI DE L'APPAREIL EN TONNES (POUR 2 OREILLES). Max. weight of vessel in tons (for two lugs).	DIMENSIONS EN MILLIMETRES.															
	A	B	C	D	E	F	G	H	J	K	L	M	N	O	P	R
20 ≤ 50	910	450	175	260	120	85	105	150	250	60	10	20	20	75	12	35
> 50 ≤ 100	1210	600	240	350	160	130	145	210	350	105	10	25	25	100	15	40
> 100 ≤ 150	1425	750	300	430	200	160	185	270	450	135	12	30	25	130	15	45

NOTA : MATIERE EQUIVALENTE A CELLE DE L'APPAREIL.
Note : Material equivalent to vessel wall.

Rev.No.	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	OREILLES D'ANCRAGE ET ANNEAU DE BASE POUR APPAREILS VERTICAUX LUGS AND BASE RING FOR VERTICAL APPARATUS	PS.TCS. 205 Page 1 de 1 Date: Février 1981

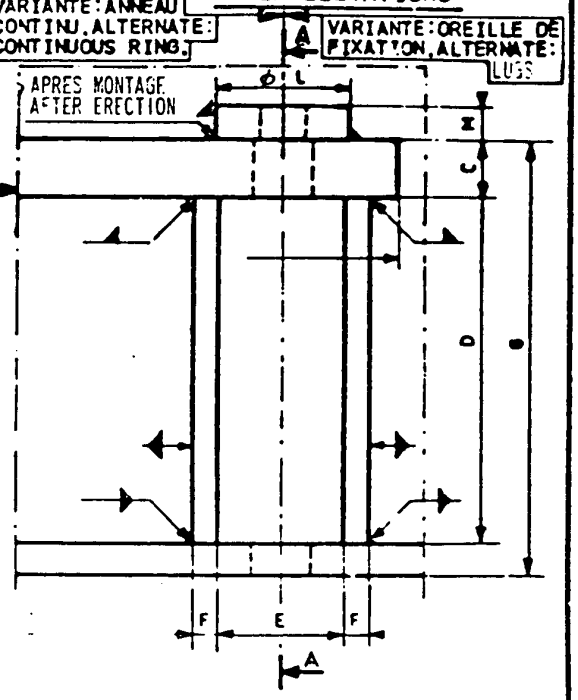
COUPE A
SECTION A



2 POINTS DE SOUDURE A 180° PAR RONDELLE POUR EXPEDITION. (J & K CONCENTRIQUES)
SPOT WELD RINGS AT 180° FOR SHIPMENT. 2 SPOTS PER RING. (J&K CONCENTRIC)

ANNEAU CONTINU DE SECTION A×C SI REQUIS SUR PLAN D APPAREIL.
WHEN REQUIRED ON VESSEL DRAWING CONTINUOUS RING OF A x C SECTION

DEMI-ELEVATIONS
HALF ELEVATIONS



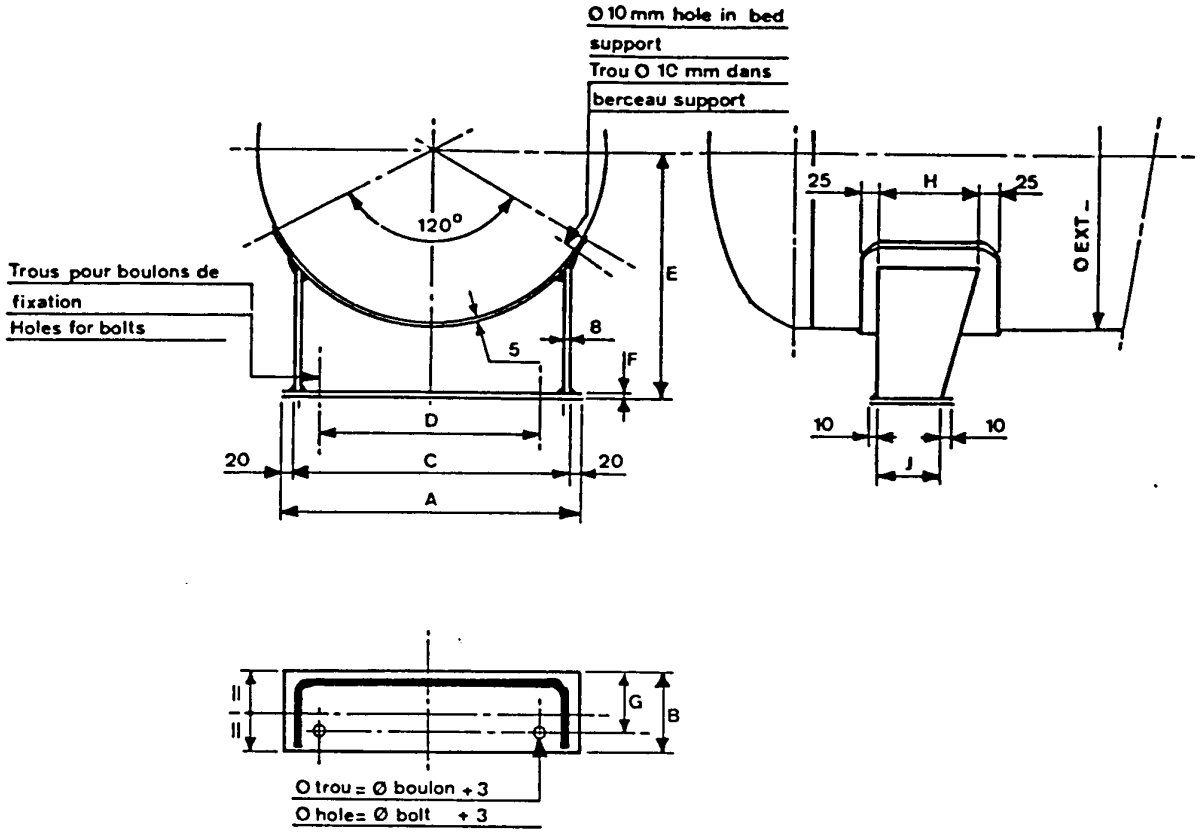
NOTA : MATIERE : E.24.1 OU E.26.1
DANS LE CAS DE RESERVOIR A FOND PLAT , L ANNEAU DE BASE SERA REMPLACÉ PAR LA TOLE DE FOND DU RESERVOIR .

NOTE : MATERIAL : E.24.1 OU E.26.1
IN CASE OF FLAT BOTTOM VESSEL , THE BASE RING IS REPLACED BY THE VESSEL BOTTOM PLATE .

Ø BOULON BOLT DIA	A	B	C	D	E	F	G	H	J	K	L	M	ANNEAU DE BASE: PLAT BASE RING DIMEN'S	EP. CORDON SOUD WELD SEAM THK.	POIDS OREILLE WEIGHT PER LUG KG	POIDS 2 PLATS A×F×D WEIGHT 2 FLATS A×F×D KG
27	100	120	20	148	70	10	180	55	30	41	55	12	12 x 200	10	4,1	2,3
30	100	120	20	148	70	10	180	55	33	44	60	12	12 x 200	10	4,1	2,3
33	100	120	25	155	70	10	200	55	36	47	65	20	20 x 200	10	4,5	2,4
36	100	120	25	155	70	10	200	55	39	50	65	20	20 x 200	10	4,5	2,4
39	120	140	40	185	80	16	250	65	42	53	75	25	25 x 250	10	12	5,6
42	120	140	40	185	80	16	250	65	45	56	80	25	25 x 250	10	12	5,6
45	120	140	40	185	80	16	250	65	48	59	85	25	25 x 250	12	12	5,6
48	120	140	40	185	80	16	250	65	51	62	85	25	25 x 250	12	12	5,6
52	140	170	40	285	100	20	350	75	55	66	100	25	25 x 300	12	21,8	12,6
56	140	170	50	318	100	20	400	75	59	70	105	32	32 x 300	12	22,6	14
64	140	170	50	318	100	20	400	75	67	78	185	32	32 x 300	12	22,6	14
72	150	200	50	418	120	25	500	80	75	86	113	32	32 x 300	12	35,5	24,6

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	BERCEAUX POUR CAPACITES HORIZONTALES Ø 350 A 550 MM BEDS FOR HORIZONTAL DRUMS AND TANKS	PS.TCS. 206 Page 1 de 1 Date: février 1981



DIMENSIONS EN MILLIMETRE / SIZE IN MILLIMETER											Masse du support en kg	Support weight in kg
Ø EXT	A	B	C	D	E	F	G	H	J	Ø BOLT Ø BOULON		
350	290	130	250	150	380	13	95	160	110	22	23	
400	330		290	190	405						25	
450	370		330	230	430						27	
500	390	170	350	250	455		125	200	150		30	
550	420		380	280	485						32	

NOTA _POUR LES BALLONS ET RESERVOIRS D'UN Ø INTERMEDIAIRE, PRENDRE LES DIMENSIONS DE BERCEAU DU Ø IMMEDIATEMENT SUPERIEUR.

_CES BERCEAUX SONT A SOUDER SUR L'APPAREIL. (SOUDURE CONTINUE)

NOTE _FOR DRUMS & TANKS OF AN INTERMEDIATE DIAMETER TAKE BED SIZE OF IMMEDIATE UPPER DIAMETER.

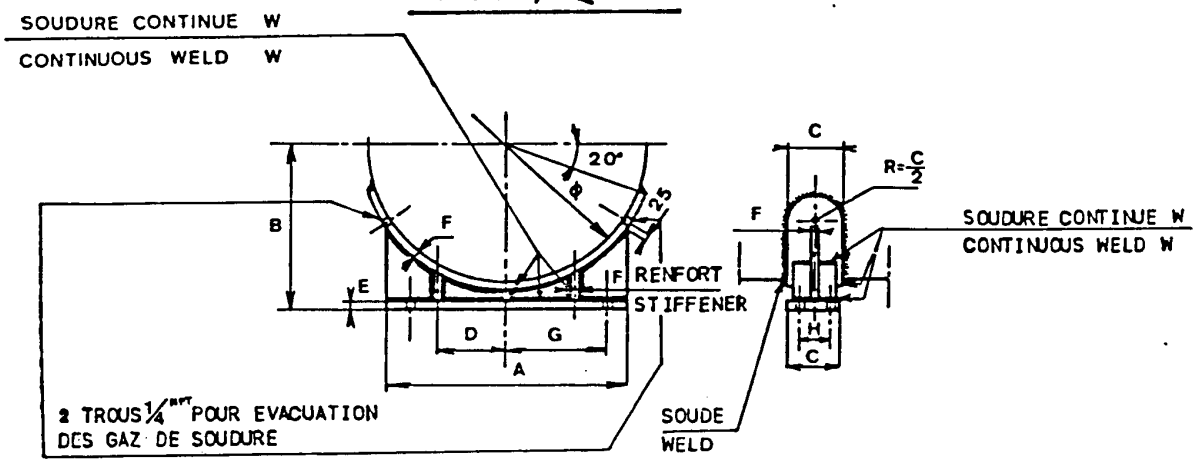
_THESE BEDS ARE TO WELD ON VESSEL. (UNCEASING WELDING)

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

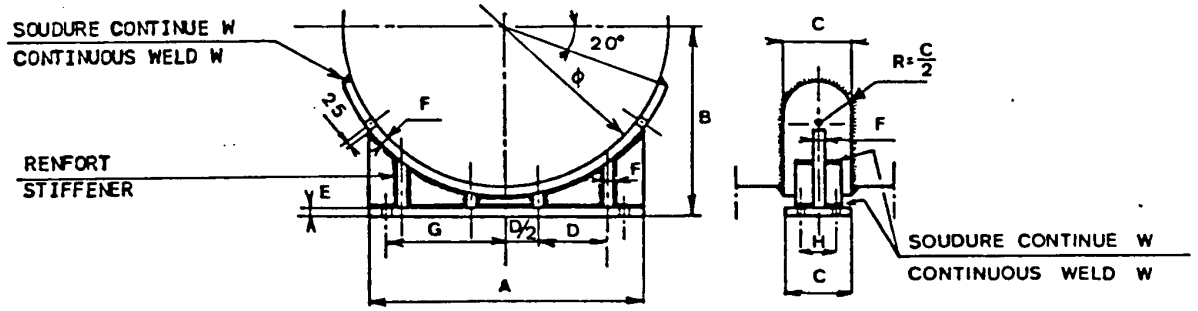
TOTAL TEP/DD	BERCEAUX POUR RESERVOIRS HORIZONTAUX SADDLES WELDED ON VESSELS	PS.TCS. 207 Page 1 de 2 Date : Frévrier 1981

RESERVOIRS HORIZONTAUX — SOUDES SUR BERCEAUX
HORIZONTAL VESSELS — WELD ON SADDLES

TYPE : 1 Ø ≤ 1200



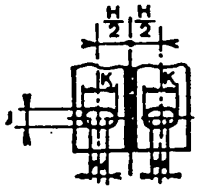
TYPE : 2 Ø > 1200



POUR SPECIFICATION SOUDURE VOIR TYPE 1

FOR WELDING SPECIFICATION SEE TYPE 1

DETAIL DES TROUS OVALISES
OVALIZED HOLES DETAIL



DIMENSIONS DES SOUDURES : — W = 6 POUR Ø ≤ 1700
— W = 10 POUR Ø > 1800

WELD SIZES : — W = 6 FOR Ø ≤ 1700
— W = 10 FOR Ø > 1800

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	BERCEAUX POUR RESERVOIRS HORIZONTAUX SADDLES WELDED ON VESSELS	PS.TCS. 207 Page 2 de 2 Date: février 1981

Ø	A	B	C	D	E	F	G	H	Dimension Trous Ovalisés Slotted Holes Sizes J	K	Ø Boulons Bolts	Dimensions des Renforts Stiffeners Sizes	Masse KG
600	540	650	180	180	12	10	230	95	26	40	20	80×10	43
750	670	725	↓	205	↓	↓	270	↓	↓	↓	↓	↓	52
900	800	800		280			340						62
1000	890	850		295			370						74
1200	1080	950	220	350	16	12	440	140	28	45	22	100×12	117
1400	1240	1050	↓	305	↓	↓	540	↓	↓	↓	↓	↓	136
1500	1320	1100		325			575						144
1700	1500	1200		365			650						165
1800	1580	1250		390			690						174
2000	1760	1350	↓	440	↓	↓	770	↓				↓	194
2100	1850	1400	250	475	20	16	820	150				110×16	289
2300	2020	1500	↓	525	↓	↓	900	↓	↓	↓	↓	↓	316
2400	2110	1550		550			940		33	50	27		332
2600	2280	1650	↓	600			1020	↓	↓	↓	↓	↓	360
2700	2370	1700	300	620			1080	190				130×16	420
2900	2540	1800	↓	670			1140	↓	↓	↓	↓	↓	452
3000	2630	1850	↓	690			1170	↓				↓	474
3200	2800	1950	350	720			1240	210				150×16	559
3400	2980	2050	↓	780			1330	↓	↓	↓	↓	↓	604
3500	3080	2100		795			1380						617
3700	3230	2200	↓	840	↓	↓	1440	↓	↓	↓	↓	↓	658
3900	3420	2300	400	880	↓	20	1500	230	↓	↓	↓	170×20	897
4000	3500	2400	↓	875	25	↓	1530	↓	36	55	30	↓	1004
4200	3680	2500	↓	900	↓	↓	1600	↓	↓	↓	↓	↓	1060
4400	3850	2600	440	945			1670	250				190×20	1190
4500	3940	2650	↓	980			1710	↓	↓	↓	↓	↓	1222
4700	4100	2750	↓	990			1750	↓	↓	↓	↓	↓	1271
4900	4280	2850	↓	990	↓	↓	1800	↓	↓	↓	↓	↓	1336
5000	4380	2900	↓	1000	↓	↓	1840	↓	↓	↓	↓	↓	1344

PAR APPAREIL IL FAUT

- 1 BERCEAU FIXE AVEC 4 TROUS DE ø = ø DES BOULONS + 6mm
- 1 BERCEAU MOBILE AVEC 4 TROUS OVALISES (VOIR DIMENSIONS FOLIO I)

FOR EACH VESSEL

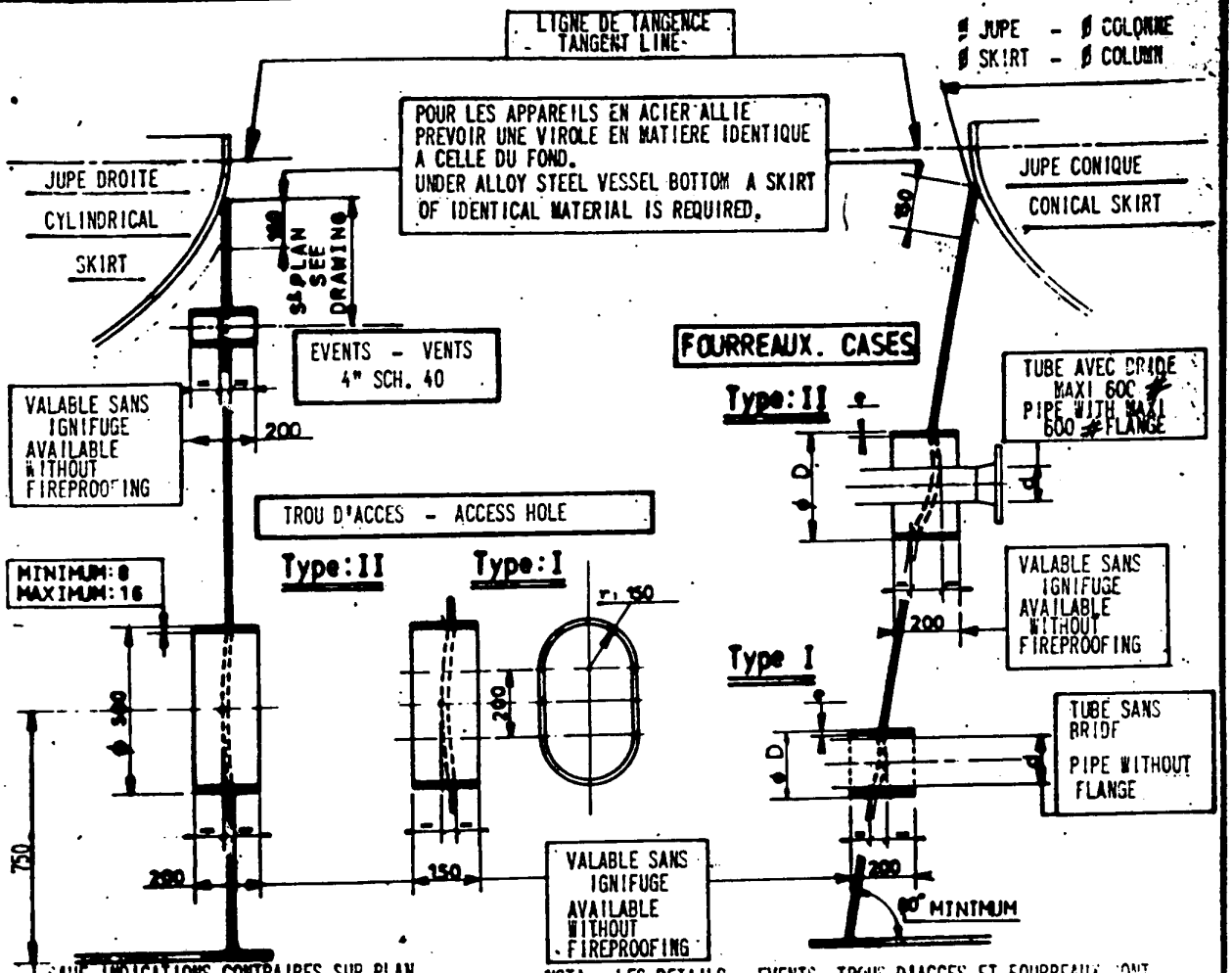
- 1 FIXED SADDLE WITH 4 HOLES OF DIA = BOLTS DIAMETER + 6mm
- 1 FREE SADDLE WITH 4 SLOTTED HOLES (FOLIO I)

-CES BERCEAUX SONT A SOUDER SUR L'APPAREIL.
-POUR LES BALLONS ET RESERVOIRS D'UN ø INTERMEDIAIRE PRENDRE LES DIMENSIONS DE BERCEAU DU ø IMMEDIATEMENT SUPERIEUR.

-ALL SADDLES ARE TO BE WELDED ON VESSEL
-FOR VESSELS OF AN INTERMEDIATE ø SELECT SADDLE DIMENSIONS OF THE NEXT HIGHER DIAMETER.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	EQUIPEMENT STANDARD DES JUPES DE COLONNES COLUMN SKIRT STANDARD EQUIPMENT	PS.TCS. 208 Page 1 de 1 Date: Février 1981



SAUF INDICATIONS CONTRAIRES SUR PLAN
UNLESS NOTED OTHERWISE ON DRAWING

NOTA : LES DETAILS : EVENTS, TROUS D'ACCES ET FOURREAUX SONT VALABLES POUR CHAQUE TYPE DE JUPE.
NOTE : DETAILS : VENTS, ACCESS HOLE, AND CASES ARE AVAILABLE FOR BOTH SKIRT TYPE.

EVENTS. VENTS.		
DIAMETRE EXTÉRIEUR JUPE. SKIRT OUTSIDE DIAMETER.	QUANTITÉ. QUANTITY.	
500 ≤ Ø ≤ 950	2	ÉQUIDISTANTS EQUALLY SPACER
1000 ≤ Ø ≤ 3000	3	
3100 ≤ Ø ≤ 4000	4	
4200 ≤ Ø ≤ 5800	6	
6100 ≤ Ø	8	

TROUS D'ACCÈS. ACCESS HOLE.		
DIAMÈTRE EXTÉRIEUR JUPE. SKIRT OUTSIDE DIAMETER.	TYPE	QUANTITÉ. QUANTITY.
500 ≤ Ø ≤ 950	I	1
1000 ≤ Ø	II	2 EQUI.

FOURREAUX Type:1 CASES Type:1		
d	D	e
3/4"	TUBE 1½	SCH.40
1"	TUBE 1½	SCH.40
1½"	TUBE 2½	SCH.40
2"	TUBE 3"	SCH.40
3"	TUBE 4"	SCH.40
4"	TUBE 6"	SCH.40
6"	TUBE 8"	SCH.40
8"	TUBE 10"	SCH.40
10"	TUBE 12"	SCH.40
12"	425 mm	13mm
14"	460mm	13mm
16"	500mm	13mm
18"	560mm	13mm

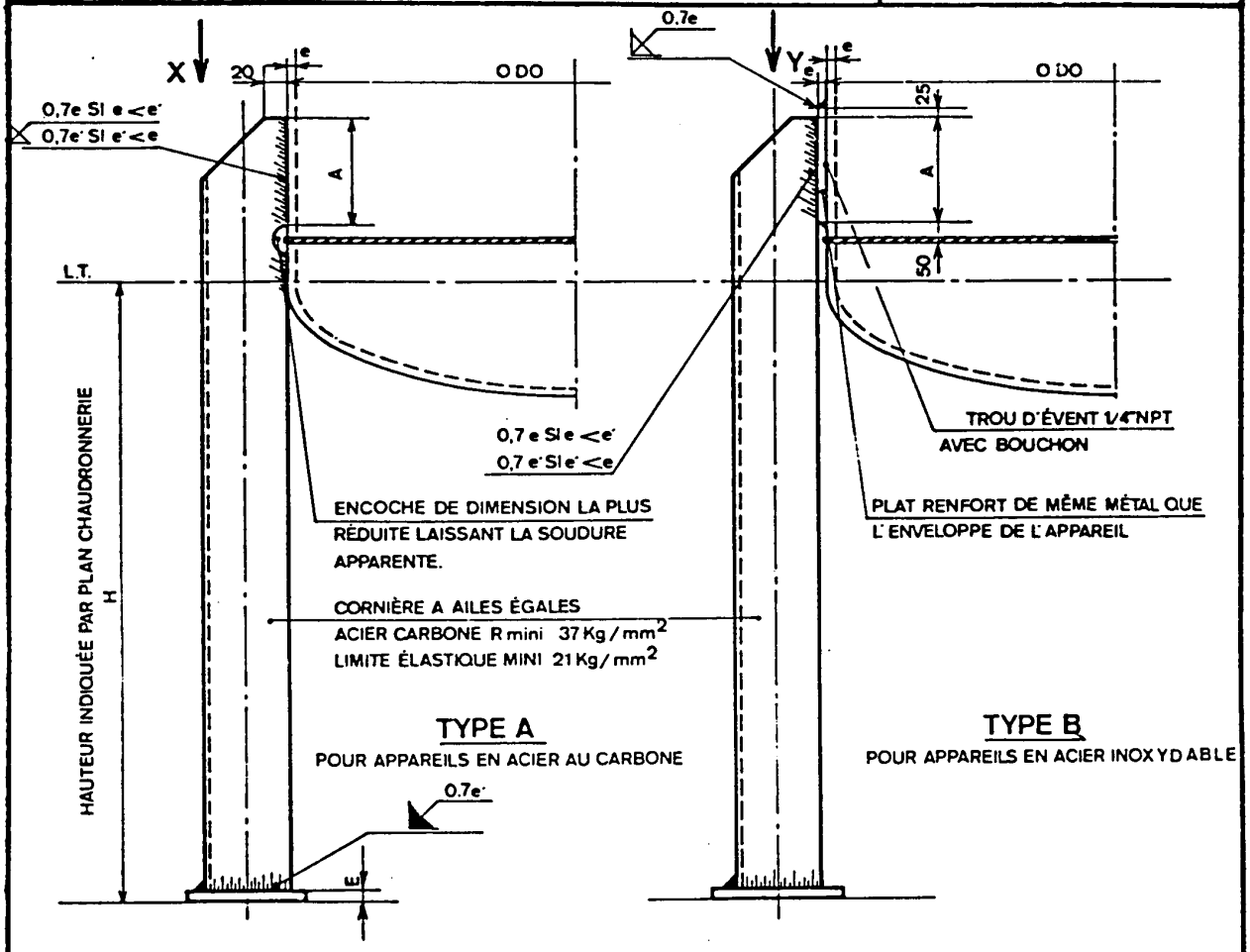
FOURREAUX Type:11 CASES Type:11		
d	D	e
3/4"	TUBE 6"	SCH.40
1"	TUBE 6"	SCH.40
1½	TUBE 8"	SCH.40
2"	TUBE 8"	SCH 40
3"	TUBE 10"	SCH.40
4"	TUBE 12"	SCH.40
6"	400 mm	MIN¹ 8mm
8"	460 mm	
10"	550 mm	
12"	610 mm	MAXI 16mm
14"	650 mm	
16"	750 mm	
18"	800 mm	

MATIERE - JUPE - SKIRT - : ASTM A 283 GRADE C.
MATERIAL - EVENTS ET FOURREAUX - VENTS AND CASES : ASTM A 106 OU A 53

} SAUF INDICATIONS CONTRAIRES.
UNLESS NOTED OTHERWISE .

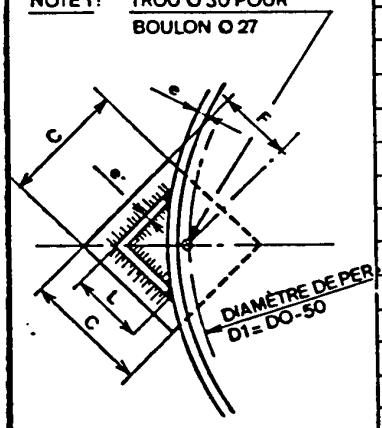
DATE		OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	PIEDS POUR PETITS APPAREILS LEGS FOR SMALL APPARATUS	PS.TCS. 209 Page 1 de 2 Date: février 1981



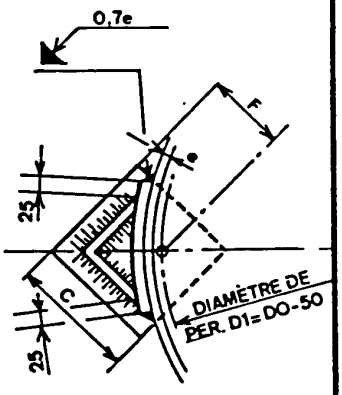
VUE SUIVANT X

NOTE 1 : TROU O 30 POUR BOULON O 27



PIEDS L	A	C(1)	E(1)	F
50×50×5	100	100	10	68
60×60×6	150	110	12	73
70×70×7	150	130	14	83
80×80×8	200	150	16	93
90×90×9	200	170	18	103
100×100×10	250	200	20	118
100×100×14	250	200	22	118
120×120×12	250	230	24	133
120×120×14	300	250	26	143
150×150×15	300	300	28	168
150×150×18	350	300	30	168

VUE SUIVANT Y



NOTE 1 : LORSQUE LE COUPLE DE RENVERSEMENT SOUS L'EFFET DU VENT (APPAREIL VIDE) OU SOUS CHARGE SISMIQUE (APPAREIL PLEIN) NÉCÉSSITE UN ANCRAGE PLUS IMPORTANT, L'ANCRAGE DÉFINI PAR CE STANDARD DEVRA ÊTRE ÉTUDIÉ EN CONSÉQUENCE.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

CHARGES UTILES PAR PIED (Kg) (VOIR NOTE 1)															
HAUTEUR (mm)	1000	1500	2000	2400	2500	2800	3000	3100	3500	4000	4500	4700	5000	5500	6000
L 50×50×5	3800	1650	950	(3)	–	–	–	–	–	–	–	–	–	–	–
L 60×60×6	6000	4500	2000	1350	(3)	–	–	–	–	–	–	–	–	–	–
L 70×70×7	8900	7100	3600	2500	2300	1850	(3)	–	–	–	–	–	–	–	–
L 80×80×8	12200	10000	8000	4200	3900	3100	2700	2500	(3)	–	–	–	–	–	–
L 90×90×9	16000	13500	11300	6800	6400	5100	4400	4100	3300	(3)	–	–	–	–	–
L 100×100×10	20000	17500	15000	13100	12500	7900	6700	6400	4900	3800	(3)	–	–	–	–
L 100×100×14	27000	24000	20500	17900	17000	10800	9100	8700	6700	5400	(3)	–	–	–	–
L 120×120×12	30000	27000	24000	21500	21000	19200	18000	12300	10300	7900	6100	5500	–	–	–
L 120×120×14	34500	31000	27500	24800	24000	22000	20500	14200	11900	9100	7000	6300	–	–	–
L 150×150×15	48000	45000	42000	38400	38000	35200	34000	33000	30000	20000	15000	13700	12300	10200	8500
L 150×150×18	57000	53400	49700	45500	45000	41600	40300	39000	35600	23600	17800	16200	14500	12000	10000

NOTES : 1 – LE TABLEAU INDIQUE LES CHARGES UTILES PAR PIED EN FONCTION DES PROFILS ET DES HAUTEURS H. PAR CHARGE UTILE IL EST ENTENDU LA SOMME DES CHARGES SUIVANTES :
- POIDS PROPRE DU RÉCIPIENT.
- POIDS DU CONTENU (AVEC DENSITE $\geq$ 1).
- COMPRESSION DÛE AU VENT OU CHARGES SISMIQUES SUR LE PIED LE PLUS SOLLICITÉ.
- EFFET DES FORCES ET MOMENTS ÉVENTUELS DÛS AUX TUYAUTERIES.

2 – LES APPAREILS REPOSERONT EN GÉNÉRAL SUR 3 PIEDS, CEPENDANT UN NOMBRE SUPÉRIEUR POURRA ÊTRE ADOPTÉ SI NÉCÉSSAIRE.

3 – DANS LE TABLEAU, LES CHARGES NON MENTIONNÉES CORRESPONDENT A DES HAUTEURS INTERDITES AU PROFIL CONSIDÉRÉ.

4 – LE CONSTRUCTEUR EST HABILITÉ A SÉLECTIONNER DES PROFILS SUPÉRIEURS S'IL JUGE NÉCÉSSAIRE POUR ASSUMER SA RESPONSABILITÉ.

REVISION	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL

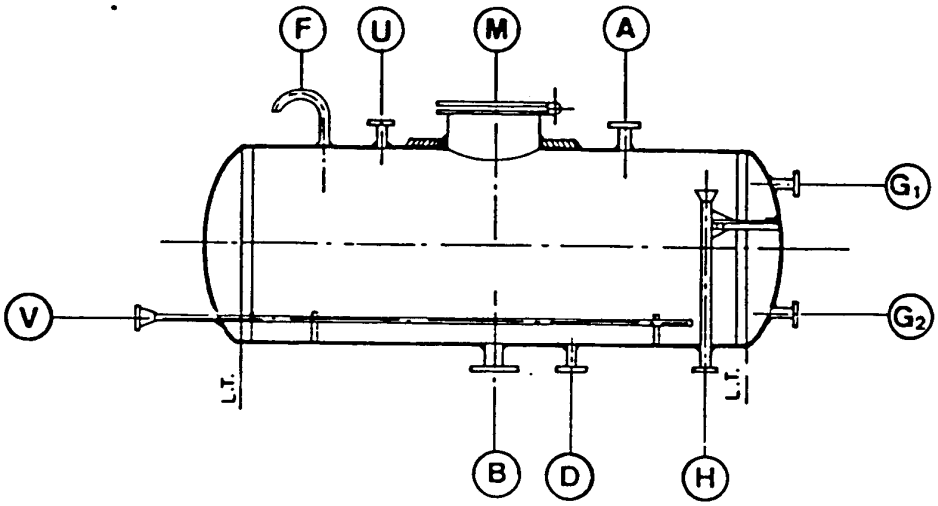
TEP/DD

PIEDS POUR PETITS APPAREILS
LEGS FOR SMALL APPARATUS

PS.TCS. 209

Date : février 1981

TOTAL TEP/DD	DESIGNATION DES TUBULURES NAME OF CONNECTIONS	PS.TCS. 210 Page 1 de 1 Date: février 1981



A	ENTRÉE PRODUIT PRINCIPAL	Main product inlet
B	SORTIE PRODUIT PRINCIPAL	Main product outlet
C	SORTIE CONDENSATS	Condensate outlet
D	DRAIN, VIDANGE	Drain
E	ENTRÉE SORTIE D'EAU	Water inlet, outlet
F	ÉVENT	Vent
G	NIVEAU A GLACE	Level gage
H	TROP PLEIN	Overflow
J	ENTRÉE PRODUIT AUXILAIRE	Auxiliary product inlet
K	SORTIE PRODUIT AUXILAIRE	Auxiliary product outlet
L	NIVEAU LI_LRC_LAH...	Level LI_LAH_LRC...
M	TROU D'HOMME	Manhole
N	TROU DE POING _ TUBULURE SUPPORT	Handhole_Nozzle support
P	PRISE DE PRESSION	Pressure connection
R	VERS OU RETOUR D'ÉCHANGEUR	To or from of exchanger
S	ÉCHAPPEMENT - SOUPAPE	Escape or safety valve
T	PRISE TEMPÉRATURE	Temperature connection
U	CONNEXION D'UTILITÉ_ECHANTILLON	Utility connection_Sample
V	ENTRÉE VAPEUR	Steam inlet
W	TUBULURE FUTURE	Spare connection

NOTA: LORSQUE L'ON SE TROUVERA EN PRÉSENCE DE PLUSIEURS TUBULURES DE MÊME REPÈRE ON LES DÉSIGNERA PAR EXEMPLE: $A_1, A_2 \ldots A_n$
When it happens in the presence of many connections of same mark it indicate them by exemple: $A_1, A_2 \ldots A_n$

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	EPAISSEURS MINIMALES DES TUBULURES SUR APPAREILS MINIMUM THICKNESS OF WALL NOZZLES WELDED ON VESSELS	PS.TCS. 211 Page 1 de 2 Date: Février 1981

EPAISSEURS MINIMALES DES PAROIS TUBULAIRES POUR TUBULURES SOUDEES SUR LES APPAREILS
MINIMAL WALL THICKNESS OF TUBES FOR NOZZLES WELDED ON VESSELS

SUIVANT ASME SECTION VIII § UG 45
ACCORDING TO ASME SECTION VIII § UG 45

TUBULURES REALISEES EN TUBES SANS SOUDURE — SEAMLESS TUBE NOZZLES

DN " N.D. "	EPAISSEUR THICKNESS STD.USAS B.36-10 mm SCH.	EPAISSEUR THICKNESS STD.USAS B.36-10 −12,5% mm	MARGES DE CORROSION ET SCHEDULES CORRESPONDANTS CORROSION ALLOWANCE AND CORRESPONDING SCHEDULES 1 mm	1/16" 1,6 mm	2 mm	1/8" 3,2 mm	4 mm	3/16" 4,8 mm	6 mm	1/4" 6,4 mm
3/4	2,87 40	2,51	3,51 160	4,11 160	4,51 160	5,71 XXH	6,51 XXH	7,31 XXH**	8,51 XXH*	8,91 XXH*
1	3,38 40	2,96	3,96 80	4,56 160	4,96 160	6,16 XXH	6,96 XXH	7,76 XXH	8,96 XXH**	9,36 XXH*
1 1/4	3,56 40	3,12	4,12 80	4,72 160	5,12 160	6,32 XXH	7,12 XXH	7,92 XXH*	9,12 XXH**	9,52 XXH**
1 1/2	3,68 40	3,22	4,22 80	4,82 160	5,22 160	6,42 XXH	7,22 XXH	8,02 XXH	9,22 XXH**	9,62 XXH**
2	3,91 40	3,42	4,42 80	5,02 160	5,42 160	6,62 160	7,42 160	8,22 XXH	9,42 XXH	9,82 XXH**
2 1/2	5,16 40	4,52	5,52 80	6,12 160	6,52 160	7,72 160	8,52 XXH	9,32 XXH	10,52 XXH	10,92 XXH
3	5,49 40	4,81	5,81 80	6,41 80	6,81 160	8,01 160	8,81 160	9,61 160	10,81 XXH	11,22 XXH
3 1/2	5,74 40	5,03	6,03 80	6,63 80	7,03 80	8,23 XXH	9,03 XXH	9,83 XXH	11,03 XXH	11,43 XXH
4	6,02 40	5,27	6,27 80	6,87 80	7,27 80	8,47 120	9,27 120	10,07 160	11,27 160	11,67 160
5	6,55 40	5,73	6,73 80	7,33 80	7,73 80	8,93 120	9,73 120	10,53 120	11,73 160	12,13 160
6	7,11 40	6,22	7,22 80	7,82 80	8,22 80	9,42 80	10,22 120	11,02 120	12,22 120	12,62 160
8	8,18 40	7,17	8,17 60	8,77 60	9,17 80	10,37 80	11,17 100	11,97 100	13,17 100	13,57 120
10	9,27 40	8,10	9,10 60	9,70 60	10,10 60	11,30 80	12,10 80	12,90 80	14,10 100	14,50 100
12	9,52 STD	8,32	9,32 XH	9,92 XH	10,32 XH	11,52 60	12,32 60	13,12 80	14,32 80	14,72 80
14	9,52 30	8,32	9,32 40	9,92 XH	10,32 60	11,52 60	12,32 60	13,12 60	14,32 80	14,72 80
16	9,52 30	8,32	9,32 40	9,92 40	10,32 40	11,52 60	12,32 60	13,12 60	14,32 60	14,72 80
18	9,52 STD	8,32	9,32 30	9,92 XH	10,32 XH	11,52 40	12,32 40	13,12 60	14,32 60	14,72 60
20	9,52 STD	8,32	9,32 30	9,92 30	10,32 30	11,52 40	12,32 40	13,12 40	14,32 60	14,72 60
24	9,52 STD	8,32	9,32 XH	9,92 XH	10,32 XH	11,52 30	12,32 30	13,12 40	14,32 40	14,72 40
30	9,52 STD	8,32	9,32 20	9,92 20	10,32 20	11,52 30	12,32 30	13,12 30	14,32 ** 30	14,72 ** 30

* EPAISSEUR NOMINALE NORMALISEE MAXIMALE INFERIEURE A L'EPAISSEUR MINIMALE ASME
MAXIMUM STANDARD NOMINAL THICKNESS LOWER THAN ASME MINIMUM THICKNESS
EPAISSEUR NOMINALE NORMALISEE MAXIMALE INFERIEURE A L'EPAISSEUR MINIMALE ASME POUR LES PAROIS DONT L'EPAISSEUR EST EGALE A LA COTE NOMINALE MOINS LA TOLERANCE DE 12,5 %.

** MAXIMUM STANDARD NOMINAL THICKNESS LOWER THAN ASME MINIMUM THICKNESS FOR WALLS OF WHICH THICKNESS IS EQUAL TO NOMINAL SIZE REDUCED OF 12,5 % ALLOWANCE.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	EPAISSEURS MINIMALES DES TUBULURES SUR APPAREILS MINIMUM THICKNESS OF WALL NOZZLES WELDED ON VESSELS	PS.TCS. 211 Page 2 de 2 Date : Février 1981

TUBULURES REALISEES EN TOLE ROULEE SOUDEE 18" ≤ DN ≤ 42"
WELDED STEEL PIPE NOZZLES 18" ≤ NC ≤ 42"

MARGES DE CORROSION CORROSION ALLOWANCE	//			1/16"		1/8"		3/16"		1/4"
	mm	0	1	1,5	2	3,2	4	4,8	6	6,4
EPAISSEURS MINIMUM, MINIMUM THICKNESS	mm	9	10	10,6	11	12,2	13	13,8	15	15,4
EPAISSEURS MINIMUM NOMINALES DES TOLES A METTRE EN OEUVRE MINIMUM NOMINAL THICKNESS OF STEEL PLATES PLATES TO BE USED	mm	9,5	10,5	11,5	11,5	13	13,5	14,5	15,5	16

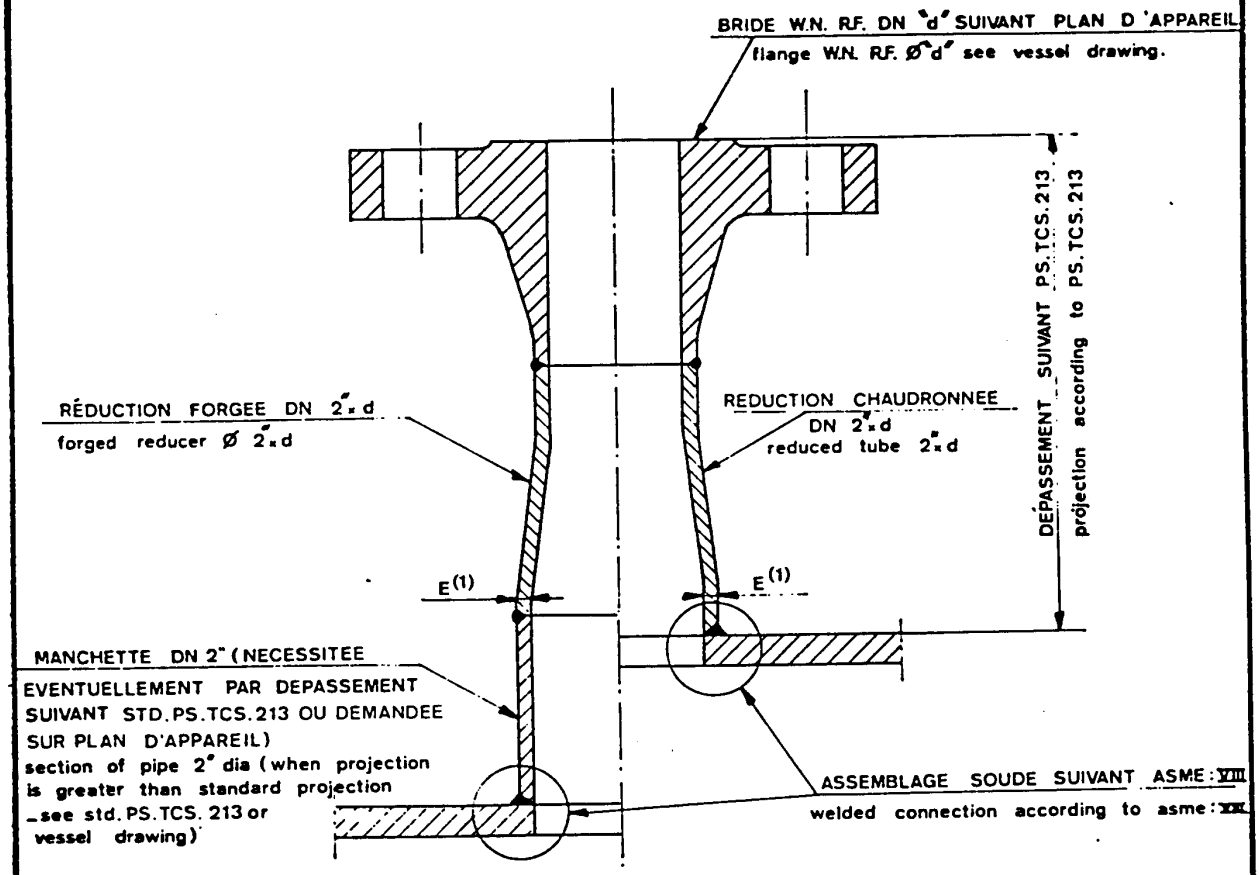
NOTA: L'EPAISSEUR RELEVEE SUR LES PAROIS TUBULAIRES, LORS DE LA RECEPTION DES APPAREILS, DEVRA AVOIR LA PLUS GRANDE DES DEUX VALEURS SUIVANTES :
- EPAISSEUR MINIMUM DONNEE PAR CE STANDARD
- EPAISSEUR CALCULEE SOUS L'EFFET DE LA PRESSION, COMPTE TENU DU COEFFICIENT DE JOINT ET DE LA MARGE DE CORROSION. TOUTEFOIS, L'EPAISSEUR POURRA ETRE LIMITEE A CELLE DE LA PAROI DU RECIPIENT SUR LAQUELLE EST SOUDEE LA TUBULURE, MARGE DE CORROSION COMPRISE.

NOTE: THE THICKNESS VALUE MEASURED ON PIPE WALL DURING THE ACCEPTANCE OF THE VESSEL SHALL BE THE GREATEST OF THE FOLLOWING VALUES :
- MINIMUM THICKNESS GIVEN BY THIS STANDARD
- THICKNESS CALCULATED UNDER PRESSURE EFFECT TAKING INTO ACCOUNT BOTH JOINT EFFICIENCY AND CORROSION ALLOWANCE.

HOWEVER, THE THICKNESS SHALL BE LIMITED TO THAT OF THE VESSEL WALL ON WHICH THE NOZZLE IS WELDED, CORROSION ALLOWANCE INCLUSIVE.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TUBULURE DN < 2" SUR APPAREILS NOZZLE DN < 2" ON VESSELS	PS.TCS. 212 Page 1 de 1 Date: février 1981



NOTE: (1) E EN ACCORD AVEC LE STD. PS.TCS. 211
(1) E according to std. PS.TCS. 211

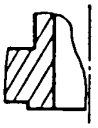
Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL · TEP/DD	DEPASSEMENT ET PERCAGE DES TUBULURES PROJECTION AND DRILLING OF NOZZLES	PS.TCS. 213 Page 1 de 1 Date: février 1981

F.P
F.F



F.S
R.F



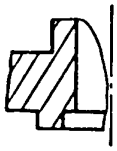
E.M.L
L.M.F
E.M.E
S.M.F



E.F.L
L.F.F
E.F.E
S.F.F



E.D.M.L
L.F.T
E.D.M.E
S.T.F



E.D.F.L
L.G.F
E.D.F.E
S.G.F



R.T.J

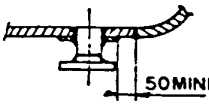


E.M.L : EMBOITEMENT MALE LARGE
L.M.F : large male face
E.M.E : EMBOITEMENT MALE ETROIT
S.M.F : small male face
E.F.L : EMBOITEMENT FEMELLE LARGE
L.F.F : large female face
E.F.E : EMBOITEMENT FEMELLE ETROIT
S.F.F : small female face
F.P : FACE PLATE
F.F : flat face
R.T.J : FACE POUR JOINT ANNULAIRE
R.T.J : ring type joint

E.D.M.L : EMBOITEMENT DOUBLE MALE LARGE
L.F.T : large tongue face
E.D.M.E : EMBOITEMENT DOUBLE MALE ETROIT
S.T.F : small tongue face
E.D.F.L : EMBOITEMENT DOUBLE FEMELLE LARGE
L.G.F : large groove face
E.D.F.E : EMBOITEMENT DOUBLE FEMELLE ETROIT
S.G.F : small groove face
F.S : FACE SURELEVEE
R.F : raised face

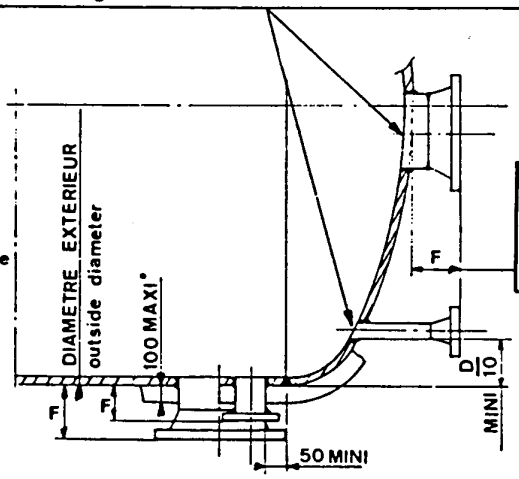
DANS LE CAS DE BRIDES A EMBOITEMENT, MONTER DE PREFERENCE LA PARTIE MALE SUR L'APPAREIL
where large male or tongued flanges are used, install flange male part on vessel preferencially

TUBULURES AVEC RENFORTS
nozzles with reinforcements

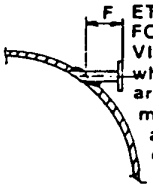


*LE DEPASSEMENT ' F ' DES TUBULURES EST VALABLE POUR UNE EPAISSEUR DE CALORIFUGE MAXIMUM DE 100 mm
*projection ' F ' is available for a 100mm maxi thickness of insulation

TUBULURES ARASEES SAUF INDICATION CONTRAIRE SUR LE PLAN DE CHAUDRONNERIE
flushed nozzles unless otherwise noted on vessel dwg

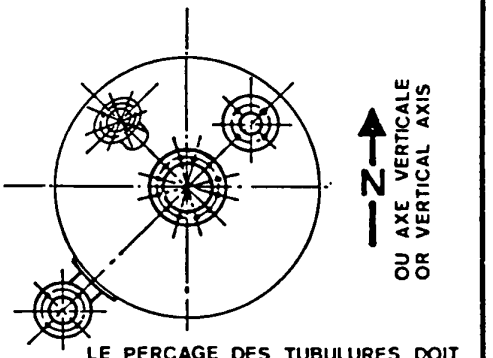


POUR LES TUBULURES NON RADIALES LA COTE ' F ' DOIT ETRE CALCULEE EN FONCTION DU O DE LA VIROLE
where non radial nozzles are used projecting value must be calculated in accordance with shell diameter



SI POSSIBLE ALIGNER LES FACES DE BRIDES DE TOUTES LES TUBULURES PAR RAPPORT A CELLE DE PLUS GRAND D.N
align whenever possible all nozzle external flange face in ratio to the flange having the largest N.D

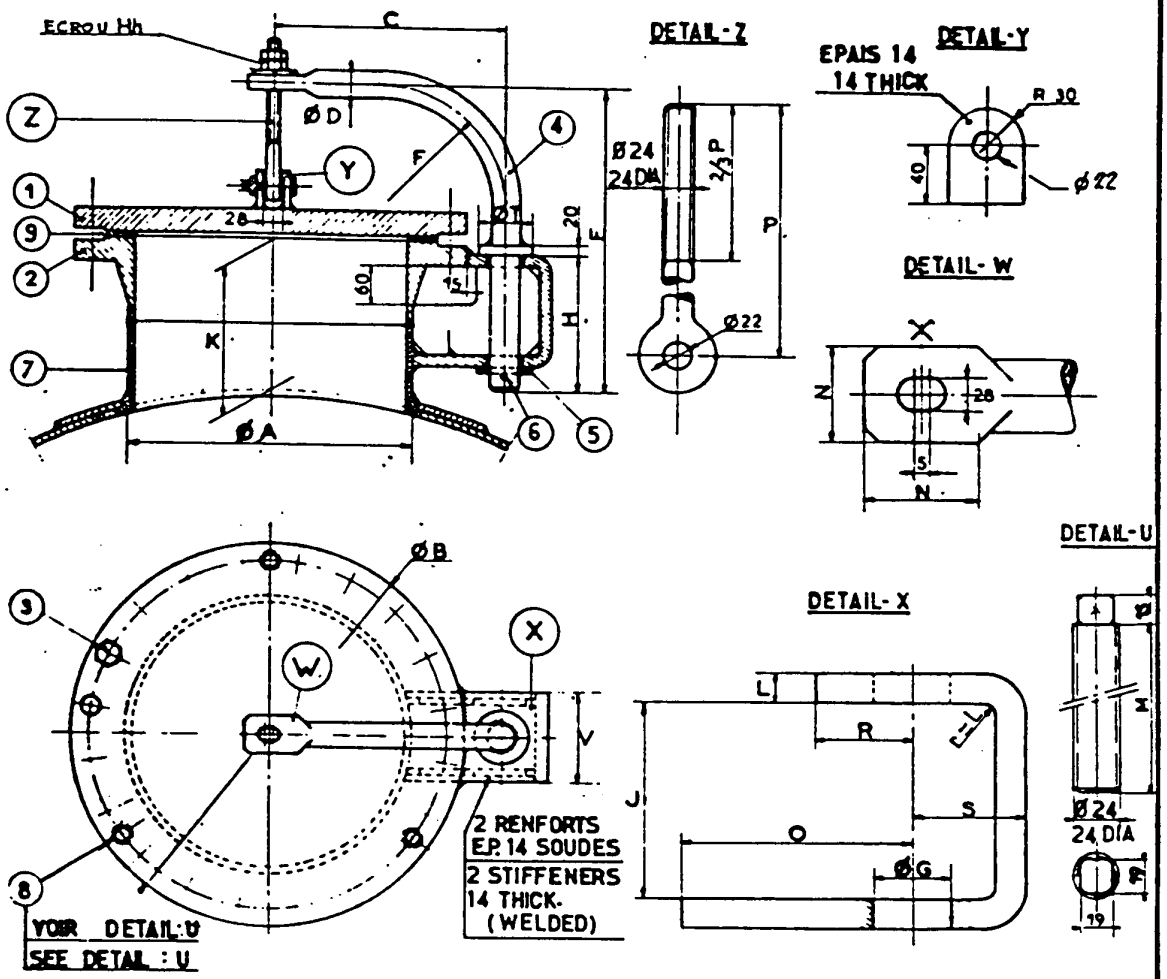
SERIE ✻ Rating ✻		150/300	600	900	1500	2500
P.N. bar eff.		10 à to 40	64			
D.N. mm		10 à to 40	64 à to 160	250		
D.N. DES TUBULURES Nozzles nominal size		DEPASSEMENT PAR RAPPORT AU O EXTERIEUR DE L APPAREIL F: Value projecting from vessel O.D.				
pouces / inches	mm					
de from 0 à to 4	de from 0 à to 100	200	200	200	200	300
de from 5 à to 10	de from 125 à to 250	200	250	300	350	550
de from 12 à to 20	de from 300 à to 500	250	300	350	450	600 pour / for 12"
> 20	> 500	300	350	450	550	300 mm
DRAINS SUR APPAREILS VERTICAUX Drains on vertical vessels		250	QUELS QUE SOIENT LE DN, LA SERIE ET LA PN For every size and rating			



LE PERCAGE DES TUBULURES DOIT ETRE EXECUTE:
-HORS AXE NORD-SUD POUR LES TUBULURES A AXE VERTICAL.
-HORS AXE VERTICAL POUR LES TUBULURE A AXE HORIZONTAL.
-HORS AXE RAYONNANT ET VERTICAL POUR LES TUBULURES A AXE INCLINE.
Nozzles drilling to be:
-Straddling the north south axis for vertical centerline nozzles.
-Straddling the vertical axis for horizontal centerline nozzles.
-Straddling the radiant and vertical axis for inclined nozzles.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TROU D'HOMME VERTICAL A POTENCE (BRIDE W.N.) VERTICAL MANHOLE WITH DAVIT (W.N. FLANGE)	PS.TCS. 214 Page 1 de 2 Date: février 1981



NOTA
NOTE
DANS LE CAS DE SIMPLE OU DOUBLE EMBOITEMENT LA COTE "K" EST COMPTEE A PARTIR DE LA PORTEE DE JOINT.
WHERE LARGE MALE OR TONGUED FLANGES ARE USED "K" TO BE MEASURED FROM GASKET SEAT
LE POIDS INDIQUE EST CELUI DU MATERIEL NOMENCLATURE Y COMPRIS UNE MANCHETTE EN TUBE SCH.20.
THE INDICATED WEIGHT IS THAT OF MATERIAL LISTED BELOW (INCL. NOZZLE SCH.20).

REPERE ITEM	DESIGNATION DESCRIPTION	MATIERE MATERIAL	QUANTITE QUANTITY	OBSERVATIONS OBSERVATIONS
9	JOINT GASKET	VOIR MATIERE DE L APPAREIL SEE VESSEL MATERIAL	1	
8	VIS DE DECOLLAGE RELEASE THREADED SCREW		3	
7	MANCHETTE SPOOL PIECE		1	
6	GOUPILLE COTTER PIN		1	
5	RONDELLE WASHER		1	
4	POTENCE DAVIT		1	
3	BOULON BOLT		SUIV. POSITION INDIQUEE TG POSITIONS	
2	BRIDE AVEC SUPPORT DETAIL: X FLANGE AND SUPPORT		1	
1	BRIDE PLEINE ET SON AXE DE SUSPENSION COMPLET BLIND FLANGE AND RELEVANT HANGING AXIS COMPLETE		1	

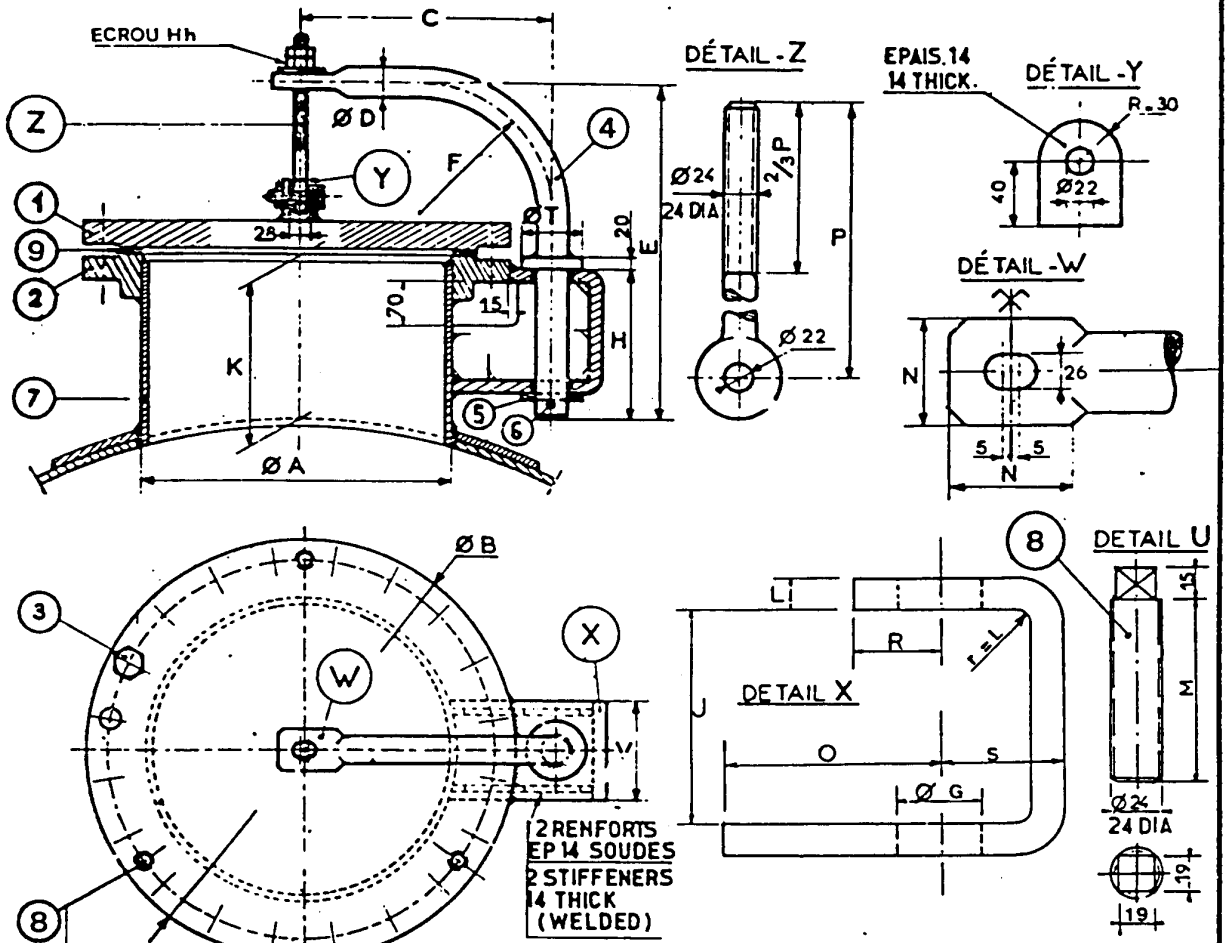
Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TROU D'HOMME VERTICAL A POTENCE (BRIDE S.O.) VERTICAL MANHOLE WITH DAVIT (S.O. FLANGE)	PS.TCS. 215 Page 2 de 2 Date: février 1981

D.N. N.D.	SERIE RATING	A	B	C	D	E	F	G	H	J	K	L	M	N	O	P	R	S	T	V	POIDS. WEIGHT.
18"	150 #	457	635	390	2"SCH. 80	540	300	62	220	150	300	20	80	80	159	300	70	100	100	140	200 kg
	300 #	457	711	430	2"SCH. 80	580	340	62	220	150	350	20	100	80	197	300	70	100	100	140	380 kg
	600 #	457	743	460	3"SCH. 160	740	340	92	280	200	450	20	130	100	228	350	85	120	130	180	630 kg
	900 #	457	787	480	3"SCH. 160	770	360	92	280	200	450	25	150	100	250	350	85	130	130	180	840 kg
20"	150 #	508	698	420	2"SCH. 80	550	300	62	220	150	300	20	85	80	166	300	70	100	100	140	250 kg
	300 #	508	775	460	3"SCH. 160	590	340	92	220	150	350	20	105	100	206	300	70	100	180	140	465 kg
	600 #	508	813	495	3"SCH160	800	400	92	280	200	450	25	130	100	241	450	85	130	130	180	760 kg
	900 #	508	857	520	3"SCH160	850	440	92	280	200	450	25	170	100	266	450	90	140	140	180	1030 kg
24"	150 #	610	813	480	2"SCH. 80	570	300	62	220	150	350	20	90	80	172	350	70	100	100	140	360 kg
	300 #	610	914	545	3"SCH. 160	650	410	92	220	150	350	20	110	100	237	350	85	120	130	180	700 kg
	600 #	610	940	560	3"SCH. 160	850	440	92	280	200	450	25	150	100	250	450	85	130	130	180	1125 kg
	900 #	610	1041	620	ROND ROD ø 95	940	500	98	280	200	500	25	190	130	310	450	95	140	140	180	1850 kg

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TROU D'HOMME VERTICAL A POTENCE (BRIDE S.O.) VERTICAL MANHOLE WITH DAVIT (S.O. FLANGE)	PS.TCS. 715 Page 1 de 2 Date : février 1981



NOTA
NOTE

DANS LE CAS DE SIMPLE OU DOUBLE EMBOITEMENT LA COTE K EST COMPTEE A PARTIR DE LA PORTEE DE JOINT.
WHERE LARGE MALE OR TONGUED FLANGES ARE USED K TO BE MEASURED FROM GASKET SEAT.
LE POIDS INDIQUE EST CELUI DU MATERIEL NOMENCLATURE Y COMPRIS UNE MANCHETTE EN TUBE SCH. 20.
THE WEIGHT INDICATED IS THAT OF MATERIAL LISTED BELOW (INCL. NOZZLE SCH. 20).

REPERE ITEM	DESIGNATION DESCRIPTION	MATIERE MATERIAL	QUANTITE QUANTITY	OBSERVATIONS REMARKS
	JOINT GASKET	VOIR MATIERE DE L'APPAREIL SEE VESSEL MATERIAL	1	
	VIS DE DECOLLAGE RELEASE THREADED SCHREW		3	
	MANCHETTE SPOOL PIECE		1	
	GOUPILLE COTTER PIN		1	
	RONDELLE WASHER		1	
	POTENCE DAVIT		1	
	BOULON BOLT		SUIVANT PRESSION ACCORDING TO PRESSURE	
	BRIDE AVEC SUPPORT DETAIL X FLANGE AND SUPPORT		1	
	BRIDE PLEINE ET SON AXE DE SUSPENSION COMPLET BLIND FLANGE AND RELEVANT HANGING AXIS COMPLETE		1	

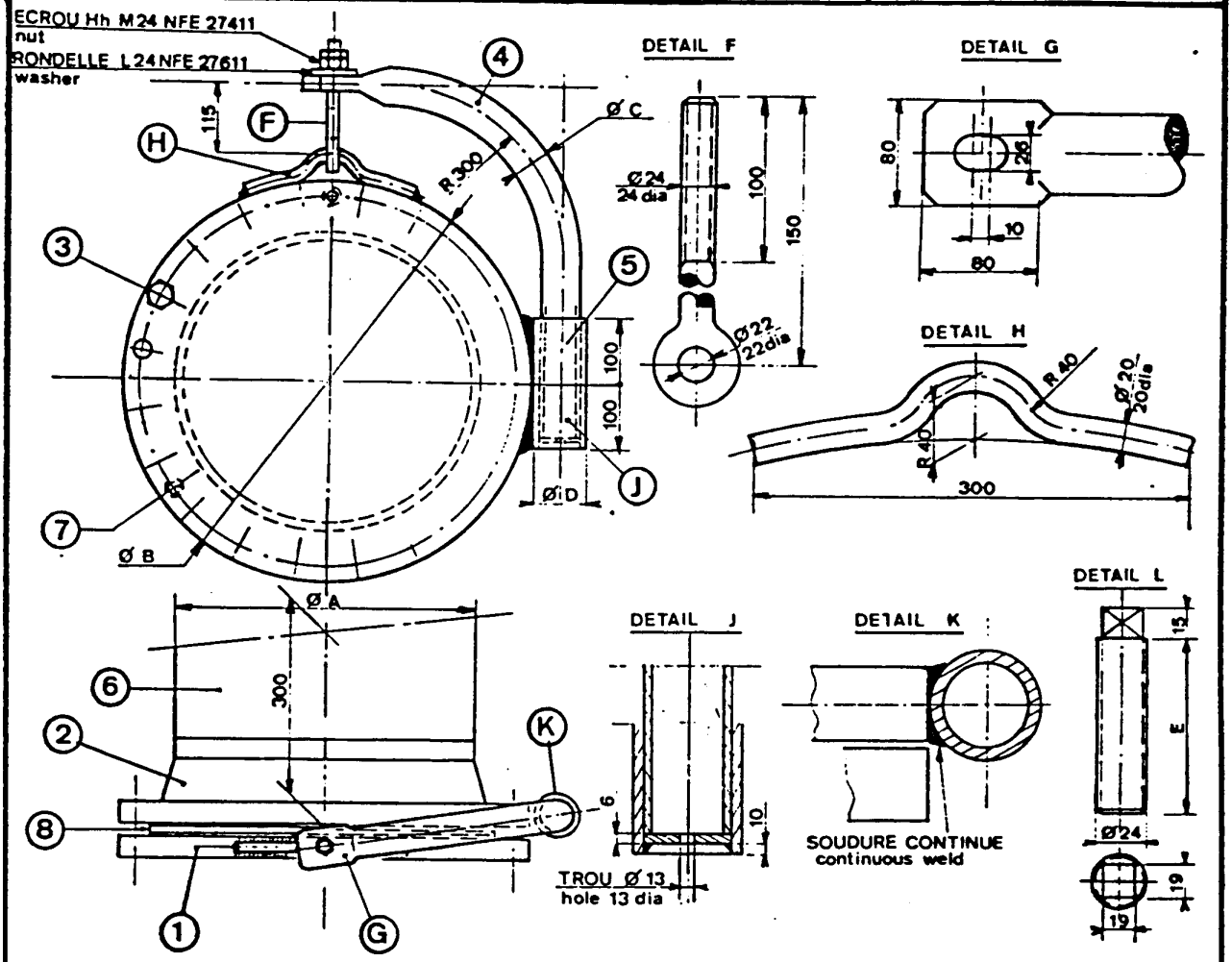
Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TROU D'HOMME VERTICAL A POTENCE (BRIDE S.O.) VERTICAL MANHOLE WITH DAVIT (S.O. FLANGE)	PS.TCS. 215 Page 2 de 2 Date: février 1981

D.N. N.D.	SERIE RATING	A	B	C	D	E	F	G	H	J	K	L	M	N	O	P	R	S	T	V	POIDS. WEIGHT.
18"	150 #	457	635	390	2"SCH. 80	540	300	62	220	150	300	20	80	80	159	300	70	100	100	140	200 kg
	300 #	457	711	430	2"SCH. 80	580	340	62	220	150	350	20	100	80	197	300	70	100	100	140	380 kg
	600 #	457	743	460	3"SCH. 160	740	340	92	280	200	450	20	130	100	228	350	85	120	130	180	630 kg
	900 #	457	787	480	3"SCH. 160	770	360	92	280	200	450	25	150	100	250	350	85	130	130	180	840 kg
20"	150 #	508	698	420	2"SCH. 80	550	300	62	220	150	300	20	85	80	166	300	70	100	100	140	250 kg
	300 #	508	775	460	3"SCH. 160	590	340	92	220	150	350	20	105	100	206	300	70	100	180	140	465 kg
	600 #	508	813	495	3"SCH160	800	400	92	280	200	450	25	130	100	241	450	85	130	130	180	760 kg
	900 #	508	857	520	3"SCH160	850	440	92	280	200	450	25	170	100	266	450	90	140	140	180	1030 kg
24"	150 #	610	813	480	2"SCH. 80	570	300	62	220	150	350	20	90	80	172	350	70	100	100	140	360 kg
	300 #	610	914	545	3"SCH. 160	650	410	92	220	150	350	20	110	100	237	350	85	120	130	180	700 kg
	600 #	610	940	560	3"SCH. 160	850	440	92	280	200	450	25	150	100	250	450	85	130	130	180	1125 kg
	900 #	610	1041	620	ROND ROD ø 95	940	500	98	280	200	500	25	190	130	310	450	95	140	140	180	1850 kg

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TROU D'HOMME HORIZONTAL A POTENCE (BRIDE W.N.) HORIZONTAL MANHOLE WITH DAVIT (W.N.FLANGE)	PS.TCS. 216 Page 1 de 1 Date: février 1981



NOTE
DANS LE CAS DE SIMPLE OU DOUBLE EMBOITEMENT LA COTE 300 EST COMPTEE A PARTIR DE LA PORTEE DE JOINT.
where large male or tongued flanges are used 300 to be mesured from gasket seat.
LE POIDS INDIQUE EST CELUI DU MATERIEL NOMENCLATURE Y COMPRIS UNE MANCHETTE EN TUBE SCH 20
the indicated weight is that of material listed below (incl. nozzle sch 20)

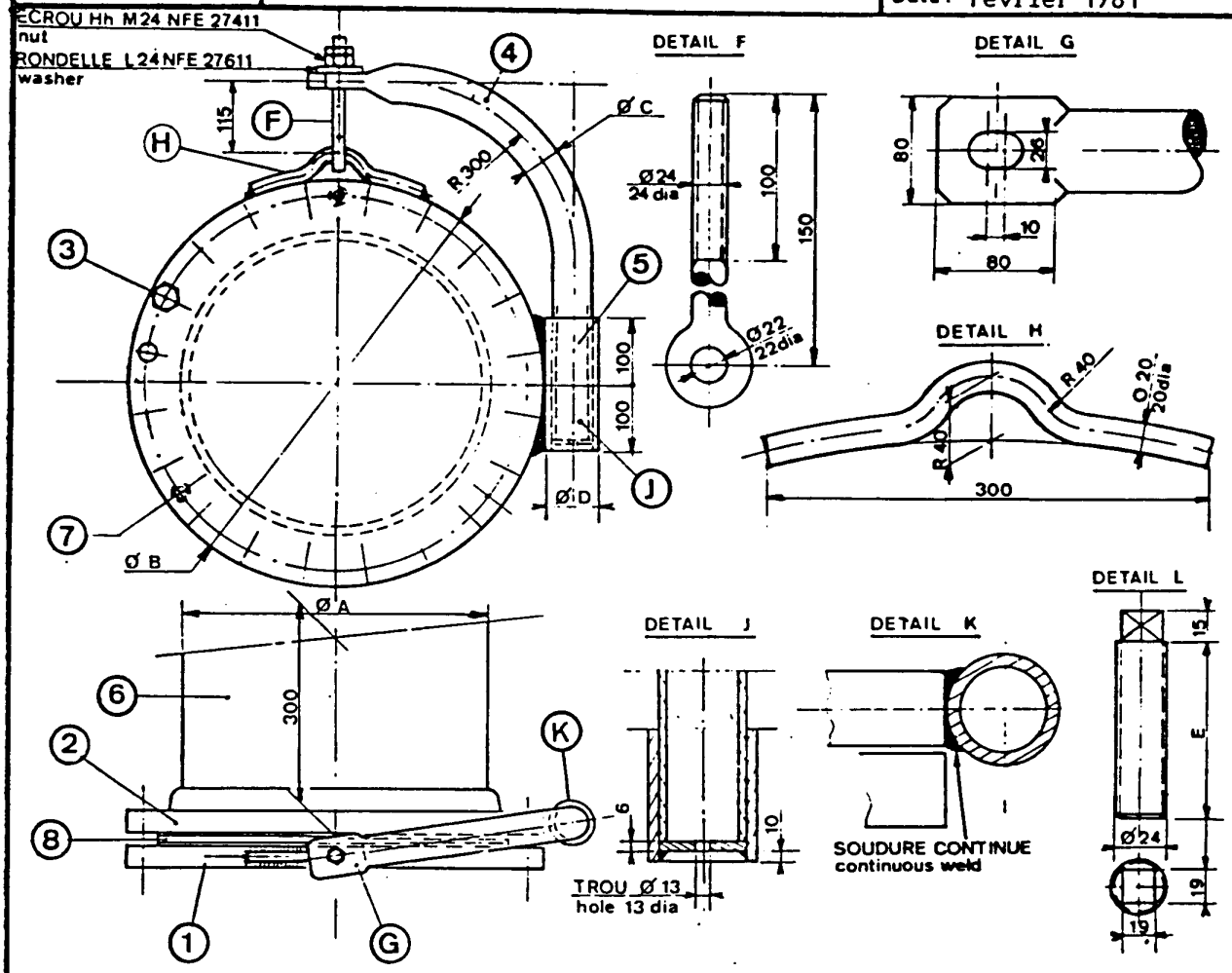
N°	DESIGNATION description	Nbr
8	JOINT gasket	1
7	VIS DE DECOLLAGE (DETAIL L) release threaded screw	3
6	MANCHETTE spool piece	1
5	FOURREAU sheath	1
4	POTENCE davit	1
3	BOULONNERIE bolting	*
2	BRIDE flange	1
1	BRIDE PLEINE ET SON AXE DE SUSPENSION COMPLET blind flange and complete relevant hanging axis	1

* SUIVANT PRESSION
according to pressure

DIA	PRESS	A	B	ØC SCH80	Ø D	E	POIDS weight
18"	150*	457	635	2"	2½" SCH40	80	215kg
20"		508	698	2"	2½" SCH40	85	270
24"		610	813	2"	2½" SCH40	90	375
18"	300*	457	711	2"	2½" SCH40	100	385
20"		508	775	2½"	3" SCH80	105	480
24"		610	915	3"	3½" SCH40	110	700
18"	400*	457	711	2½"	3" SCH80	105	460
20"		508	775	2½"	3" SCH80	110	575
24"		610	915	3"	3½" SCH40	115	840
18"	600*	457	743	3"	3½" SCH40	130	645
20"		508	813	3"	3½" SCH40	130	800
24"		610	940	3"	3½" SCH40	150	1110

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL · TEP/DD	TROU D'HOMME HORIZONTAL A POTENCE (BRIDE S.O.) HORIZONTAL MANHOLE WITH DAVIT (S.O.FLANGE)	PS.TCS. 217 Page 1 de 1 Date: février 1981



NOTE
DANS LE CAS DE SIMPLE OU DOUBLE EMBOITEMENT LA COTE 300 EST COMPTEE A PARTIR DE LA PORTEE DE JOINT.
where large male or tongued flanges are used 300 to be mesured from gasket seat.
LE POIDS INDIQUE EST CELUI DU MATERIEL NOMENCLATURE Y COMPRIS UNE MANCHETTE EN TUBE SCH 20
the indicated weight is that of material listed below (incl. nozzle sch 20)

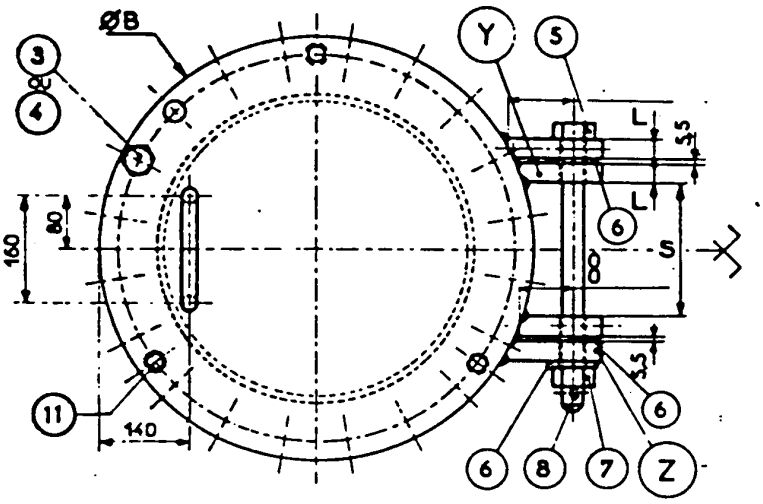
N°	DESIGNATION description	Nbr
8	JOINT gasket	1
7	VIS DE DECOLLAGE (DETAIL L) release threaded screw	3
6	MANCHETTE spool piece	1
5	FOURREAU sheath	1
4	POTENCE davit	1
3	BOULONNERIE bolting	•
2	BRIDE flange	1
1	BRIDE PLEINE ET SON AXE DE SUSPENSION COMPLET blind flange and complete relevant hanging axis	1

• SUIVANT PRESSION
according to pressure

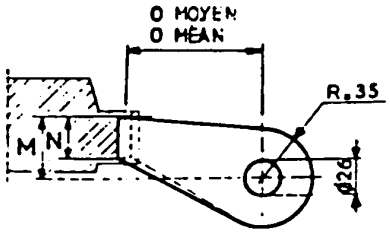
DIA	PRESS	A	B	OC SCH80	Ø D	E	POIDS weigh
18"	150*	457	635	2"	2½" SCH40	80	205
20"		508	698	2"	2½" SCH40	85	260
24"		610	813	2"	2½" SCH40	90	365
18"	300*	457	711	2"	2½" SCH40	100	390
20"		508	775	2½"	3" SCH80	105	460
24"		610	914	3"	3½" SCH40	110	695
18"	400*	457	711	2½"	3" SCH80	105	440
20"		508	775	2½"	3" SCH80	110	555
24"		610	914	3"	3½" SCH40	115	810
18"	600*	457	743	3"	3½" SCH40	130	595
20"		508	813	3"	3½" SCH40	130	750
24"		610	940	3"	3½" SCH40	150	1090

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

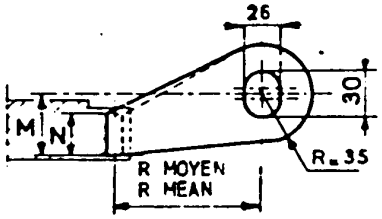
TOTAL TEP/DD	TROU D'HOMME A CHARNIERE (BRIDE W.N.) HINGED MANHOLE (WELDING NECK FLANGE)	PS.TCS. 218 Page 1 de 2 Date: février 1981

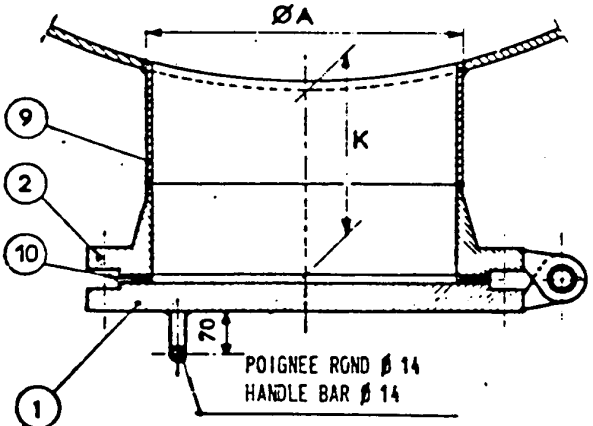


CHARNIERE SOUDEE SUR BRIDE "Y"
HINGE WELDED ON FLANGE "Y"



CHARNIERE SOUDEE SUR COUVERCLE "Z"
HINGE WELDED ON COVER "Z"





NOTA : SERRER LES BOULONS DU TROU D'HOMME AVANT DE SOUDER LES CHARNIERES SUR LE COUVERCLE POUR PERMETTRE LA COMPRESSION DU JOINT.
DANS LE CAS DE SIMPLE OU DOUBLE EMBOITEMENT LA COTE "K" EST COMPTEE A PARTIR DE LA PORTEE DE JOINT.

NOTE : TICHTEN THE MANHOLE COVER BOLTS BEFORE WELDING HINGES THE COVER TO PERMIT COMPRESSION OF THE GASKET WHERE LARGE MALE OR TONGUED FLANGES ARE USED, "K" TO BE MEASURED FROM GASKET SEAT.

LE POIDS INDIQUE EST CELUI DU MATERIEL NOMENCLATURE Y COMPRIS UNE MANCHETTE EN TUBE SCH. 20.
THE WEIGHTS INDICATED ARE THOSE LISTED IN THE BILL OF MATERIALS, INCLUDING A SCH. 20 TUBE COUPLING.

REPERE ITEM	DESIGNATION DESCRIPTION	MATIERE MATERIAL	QUANTITE QUANTITY	OBSERVATIONS REMARKS
11	VIS DE DECOLLAGE Ø d LONGUEUR L. LOOSENING SCREWS Ø d LENGHT L.	VOIR MATEIRE DE L'APPAREIL. SEE VESSEL MATERIAL.	3	ENTIEREMENT FILETEES. TOTALY THREADED.
10	JOINT. GASKET.		1	SUIVANT SPEC. SEE SPEC.
9	MANCHETTE Ø A. SPOOL PIECE Ø A.		1	EP. OU SCH. SUIVANT PL. TH. OR SCH. SEE DRAWING
8	GOUPILLE Ø 5. SPLIT PIN Ø 5.		1	SUIVANT PLAN OU SPEC. SEE DRAWING OR SPEC.
7	ECROU Ø 24. NUT Ø 24.		1	
6	RONDELLES. WASHERS.		3	
5	AXE CHARNIERE : VIS Ø 24 LONGUEUR H. HINGE BOLT : SCREW Ø 24 LENGHT H.		1	
4	TIGE FILETEE, ECROUS. THREAD STUD, NUTS.		SUIVANT DN & SERIE SEE ND AND RATING	
3	BOULON. BOLT.			
2	BRIDE S.O. AVEC CHARNIERE Y S.O. FLANGE WITH HINGE Y		1	
1	COUVERCLE, BRIDE PLEINE, POIGNEE, CHARNIERE Z COVER, BLIND FLANGE, HANDLE, HINGE Z		1	3 TROUS TARAUDES Ø d 3 TAPPED HOLES Ø d

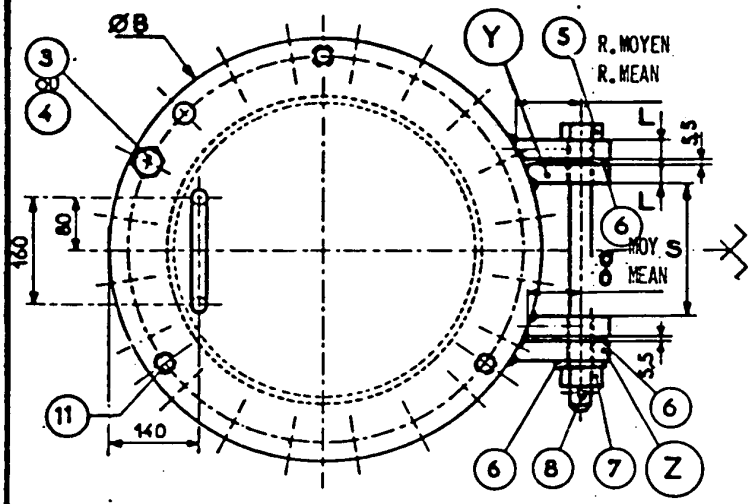
Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TROU D'HOMME A CHARNIERE (BRIDE W.N.) HINGED MANHOLE (W.N. FLANGE)	PS.TCS. 218 Page 2 de 2 Date: Février 1981

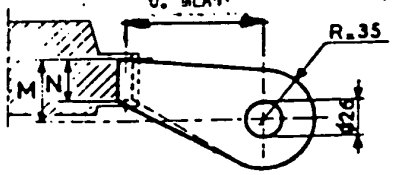
D.N. SERIE N.D. RATING		A	B				H		K	L	M	N	O		R	S	d	l	POIDS WEIGHT
18"	150 #	457	635				220		300	20	80	80	159		70	100	27	130	210 kg
	300 #	457	711				220		350	20	100	80	197		70	100	27	170	395 kg
	600 #																		kg
	900 #																		kg
20"	150 #	508	698				220		300	20	85	80	166		70	100	27	130	270 kg
	300 #	508	775				220		300	20	105	100	206		70	100	27	170	470 kg
	600 #																		kg
	900 #																		kg
24"	150 #	610	813				220		350	20	90	80	172		70	100	27	130	375 kg
	300 #	610	914				220		350	20	110	100	237		85	120	27	170	700 kg
	600 #																		kg
	900 #																		kg

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

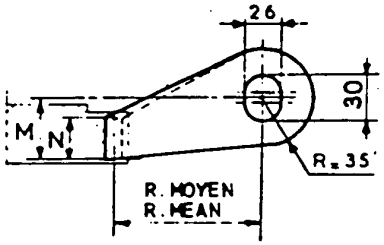
TOTAL TEP/DD	TROU D'HOMME A CHARNIERE (BRIDE S.O.) HINGED MANHOLE (SLIP-ON FLANGE)	PS.TCS. 219 Page 1 de 2 Date : Février 1981

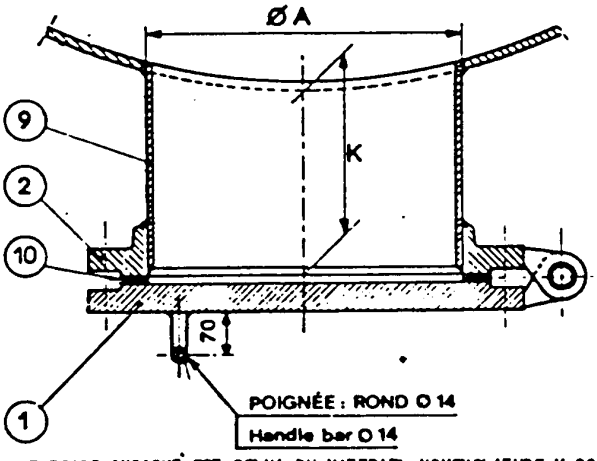


CHARNIERE SOUDEE SUR BRIDE "Y"
HINGE WELDED ON FLANGE "Y"



CHARNIERE SOUDEE SUR COUVERCLE "Z"
HINGE WELDED ON COVER "Z"





NOTA : SERRER LES BOULONS DU TROU D'HOMME AVANT DE SOUDER LES CHARNIERES SUR LE COUVERCLE POUR PERMETTRE LA COMPRESSION DU JOINT.
DANS LE CAS DE SIMPLE OU DOUBLE EMBOITEMENT, LA COTE "K" EST COMPTEE A PARTIR DE LA PORTEE DE JOINT.

NOTE : TIGHTEN THE MANHOLE BOLTS BEFORE WELDING HINGES THE COVER TO PERMIT COMPRESSION OF THE GASKET. WHERE LARGE MALE OR TONGUED FLANGES ARE USED, "K" TO BE MEASURED FROM GASKET SEAT.

LE POIDS INDIQUE EST CELUI DU MATERIEL NOMENCLATURE Y COMPRIS UNE MANCHETTE EN TUBE SCH. 20
THE WEIGHTS INDICATED ARE THOSE LISTED IN THE BILL OF MATERIALS, INCLUDING A SCH. 20 TUBE COUPLING

REPERE ITEM	DESIGNATION DESCRIPTION	MATIERE MATERIAL	QUANTITE QUANTITY	OBSERVATIONS. REMARKS.
11	VIS DE DECOLLAGE Ø d LONGUEUR L LOOSENING SCREWS Ø d LENGHT L	VOIR MATIERE DE L'APPAREIL SEE VESSEL MATERIAL	3	ENTIEREMENT FILETEES TOTALY THREADED.
10	JOINT GASKET		1	SUIVANT SPEC. SEE SPEC.
9	MANCHETTE Ø A SPOOL PIECE Ø A		1	EP. OU SCH. SUIVANT PL TH. OR SCH. SEE DRAWING
8	GOUPILLE Ø 5 SPLIT PIN Ø 5		1	SUIVANT PLAN OU SPEC.
7	ECROU Ø 24 NUT Ø 24		1	
6	RONDELLES WASHERS		3	
5	AXE CHARNIERE : VIS Ø 24 LONGUEUR "H" HINGE BOLT : SCREW Ø 24 LENGT "H"		1	
4	TIGE FILETEE, ECROUS THREADED STUD, NUTS		SUIVANT DN ET SERIE SEE DN AND RATING	SEE DRAWING OR SPEC.
3	BOULON BOLT			
2	BRIDE S.O. AVEC CHARNIERE S.O. FLANGE WITH HINGE "Y"		1	
1	COUVERCLE, BRIDE PLEINE, POIGNEE, CHARNIERE "Z" COVER, BLIND FLANGE, HANDLE, HINGE "Z"		1	3 TROUS TARAUDES Ø d 3 TAPPED HOLES Ø d

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

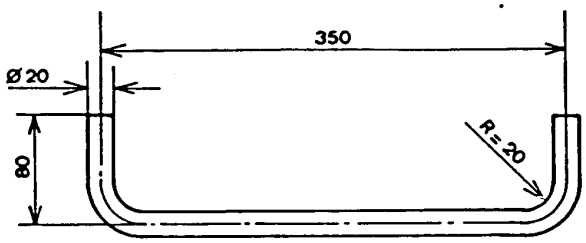
TOTAL TEP/DD	TROU D'HOMME A CHARNIERE (BRIDE S.O). HINGED MANHOLE (S.O. FLANGE)	PS.TCS. 219 Page 2 de 2 Date: février 1981

D.N. N.D.	SERIE. RATING	A	B	C	D	E	F	G	H	J	K	L	M	N	O	P	R	S	T	V	POIDS. WEIGHT.
18"	150 #	457	635						220		300	20	80	80	159		70	100			210 Kg
	300 #	457	711						220		350	20	100	80	197		70	100			395 Kg
	600 #																				
	900 #																				
20"	150 #	508	698						220		300	20	85	80	166		70	100			270 Kg
	300 #	508	775						220		300	20	105	100	206		70	100			470 Kg
	600 #																				
	900 #																				
24"	150 #	610	813						220		350	20	90	80	172		70	100			375 Kg
	300 #	610	914						220		350	20	110	100	237		85	120			700 Kg
	600 #																				
	900 #																				

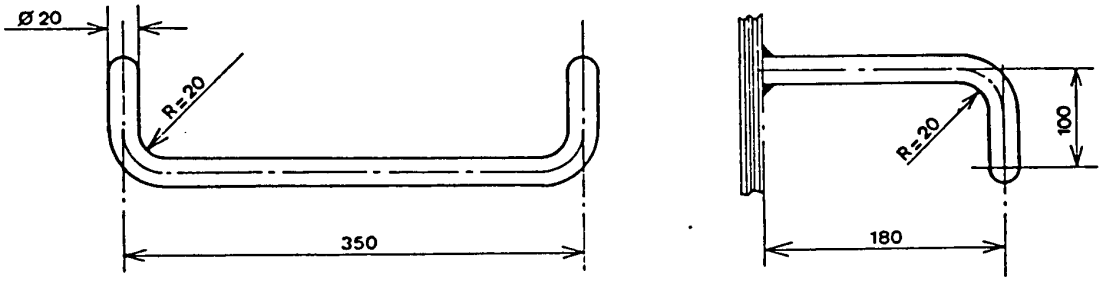
Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	POIGNEE ET ECHELONS POUR TROU D'HOMME HAND GRIP AND LADDER RUNG FOR MANHOLE	PS.TCS. 220 Page 1 de 1 Date : février 1981

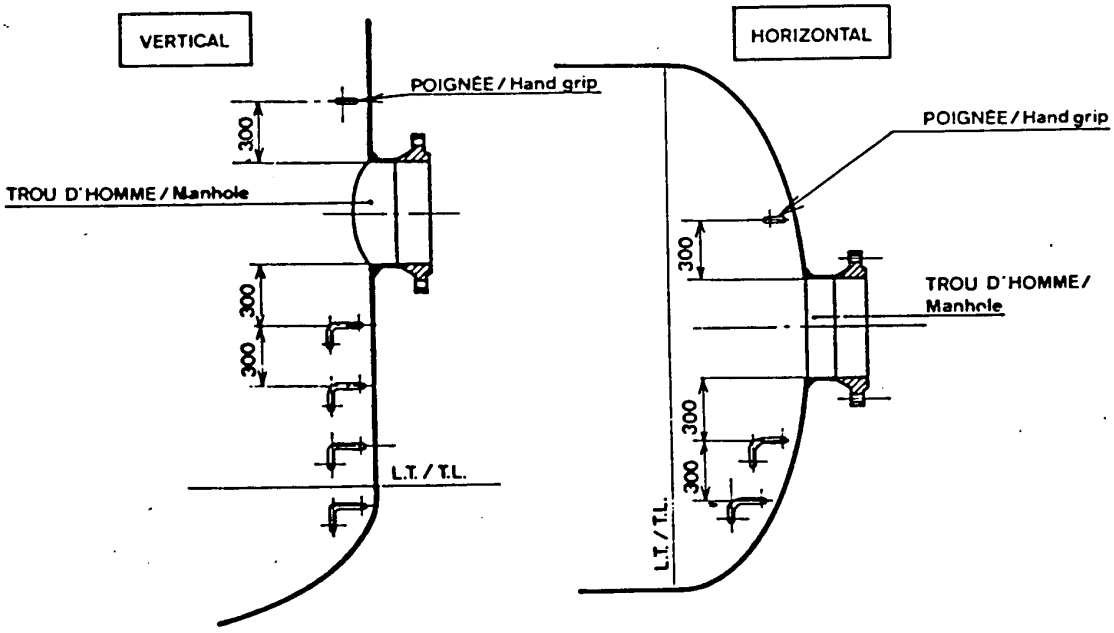
POIGNÉE / Hand grip



ÉCHELON / Ladder rung



EXEMPLE D'INSTALLATION / Installation example



NOTE 1 : MATIÈRE DE MÊME NUANCE QUE L'APPAREIL
Material is the same than vessel one

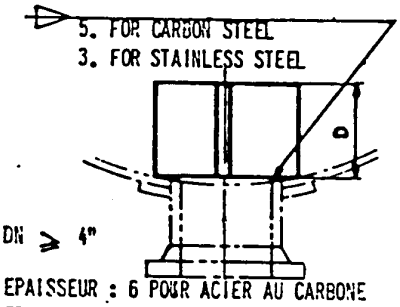
Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	ANTI-TOURBILLONS SUR SORTIE DE FLUIDE VORTEX BREAKERS ON FLUID OUTLET	PS.TCS. 221 Page 1 de 2 Date : février 1981

TYPE : 1

5. POUR ACIER AU CARBONE
3. POUR ACIER INOX OU MONEL

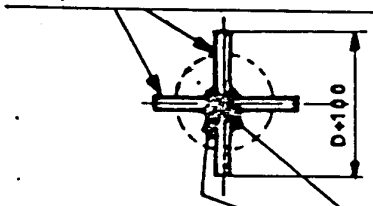
5. FOR CARBON STEEL
3. FOR STAINLESS STEEL



DN ≥ 4"

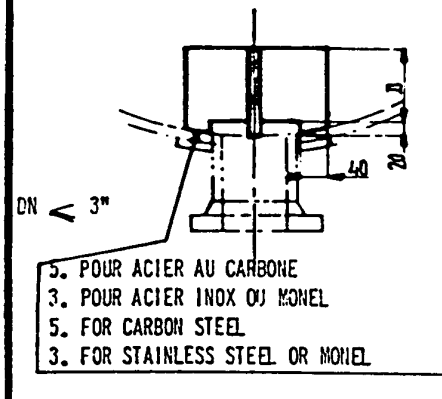
EPAISSEUR : 6 POUR ACIER AU CARBONE
EPAISSEUR : 3 POUR ACIER INOX OU MONEL

6. THICKNESS FOR CARBON STEEL
3. THICKNESS FOR STAINLESS STEEL



5. POUR ACIER AU CARBONE
3. POUR ACIER INOX OU MONEL

5. FOR CARBON STEEL
3. FOR STAINLESS OR MONEL



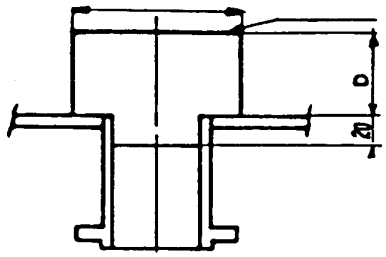
DN < 3"

5. POUR ACIER AU CARBONE
3. POUR ACIER INOX OU MONEL
5. FOR CARBON STEEL
3. FOR STAINLESS STEEL OR MONEL

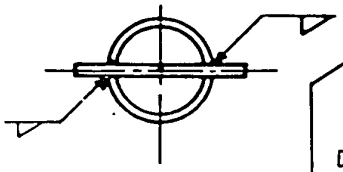
LES EPAISSEURS POUR ACIER AU CARBONE COMPRENNENT UNE SUREPAISSEUR DE CORROSION DE 1,5 MM.

TYPE : III

2 D EP. THK. 6

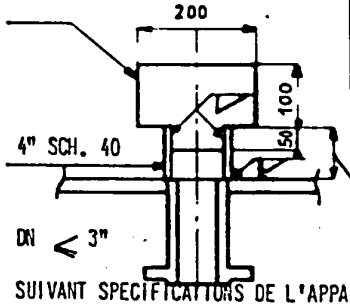


4" < DN ≤ 8"



TYPE : IV

EP. THK. 6

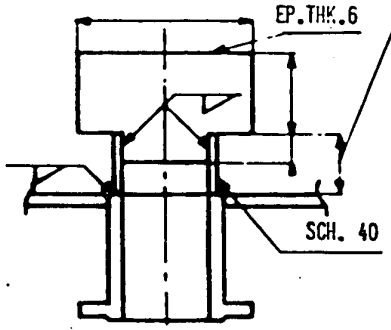


4" SCH. 40

DN < 3"

SUIVANT SPECIFICATIONS DE L'APPAREIL
AS PER VESSEL SPECIFICATIONS

EP.THK.6

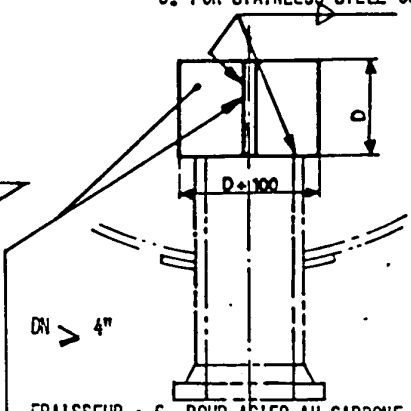


4" ≤ DN ≤ 8"

TYPE : II

(AVEC TUBE INTERIEUR)
(WITH INSIDE TUBE)

5. POUR ACIER AU CARBONE
3. POUR ACIER INOX OU MONEL
5. FOR CARBON STEEL
3. FOR STAINLESS STEEL OR MONEL

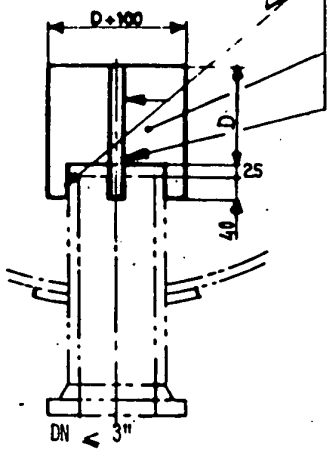


DN > 4"

EPAISSEUR : 6 POUR ACIER AU CARBONE
EPAISSEUR : 3 POUR ACIER INOX OU MONEL

6. THICKNESS FOR CARBON STEEL
3. THICKNESS FOR STAINLESS STEEL

5. POUR ACIER AU CARBONE
3. POUR ACIER INOX OU MONEL
5. FOR CARBON STEEL
3. FOR STAINLESS STEEL OR MONEL



DN < 3"

NOTA : 1 - TYPE ET MATERIAU SUIVANT PLAN DE L'APPAREIL
2 - "D" DIAMETRE NOMINAL DE LA TUBULURE DANS LES INDICATIONS DE COTES
3 - DISPOSITIONS ET DIMENSIONS VALABLES SI NON CONTREDITES PAR PLAN DE L'APPAREIL

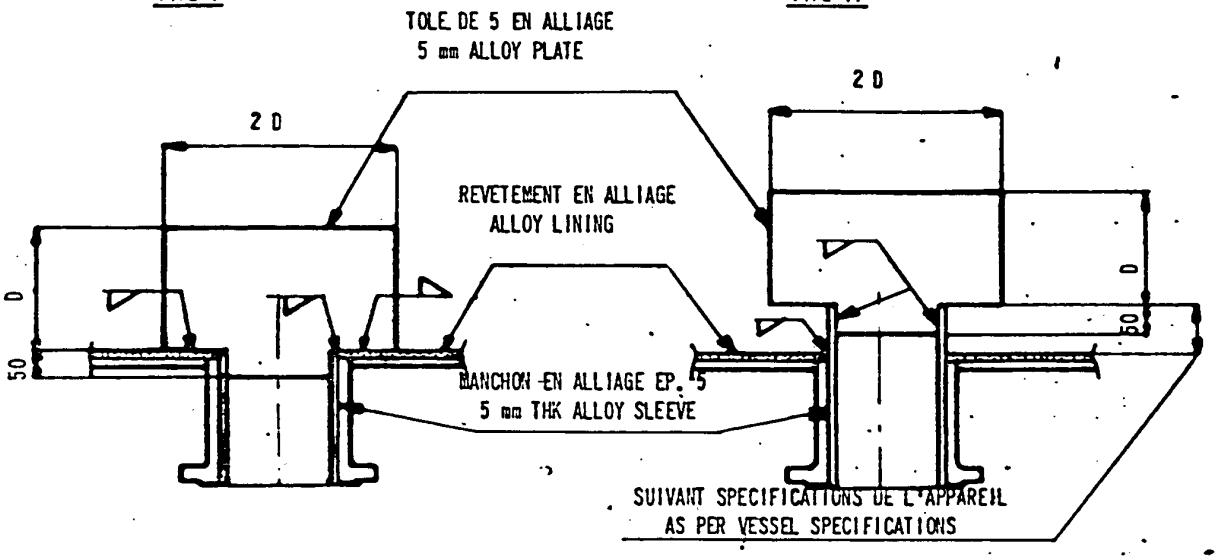
NOTE : 1 - TYPE AND MATERIAL ACCORDING TO VESSEL DRAWING
2 - WHERE SIZE IS INDICATED "D" = NOZZLE NOMINAL DIAMETER
3 - DETAILS AND DIMENSIONS SHOWN IN VESSEL DRAWING TAKE PRECEDENTE OVER THOSE SHOWN HEREON

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	ANTI-TOURBILLONS SUR SORTIE DE FLUIDE VORTEX BREAKERS ON FLUID OUTLET	PS.TCS. 221 Page 2 de 2 Date: Février 1981

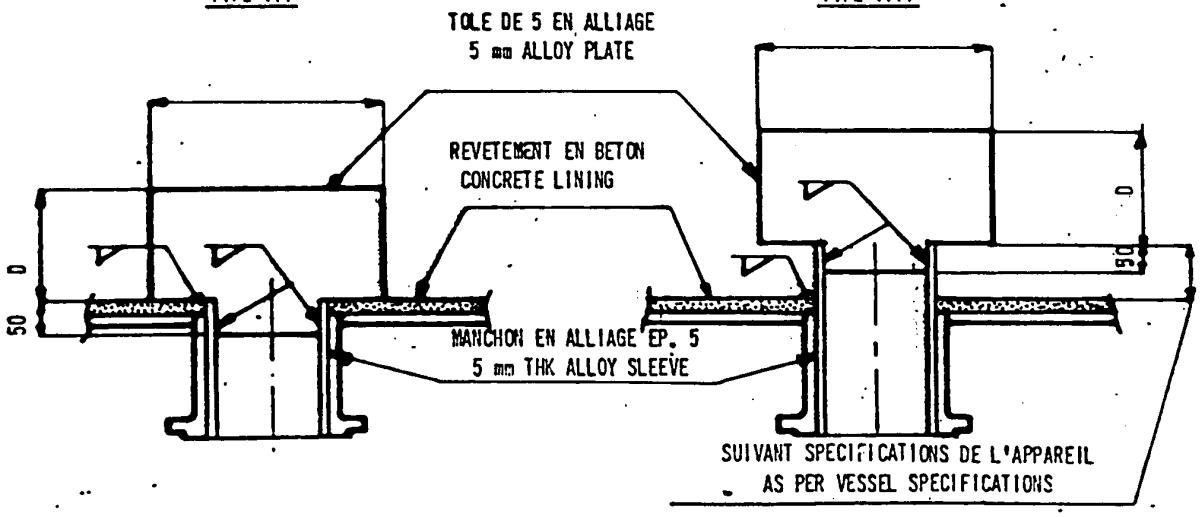
TYPE V

TYPE VI



TYPE VII

TYPE VIII

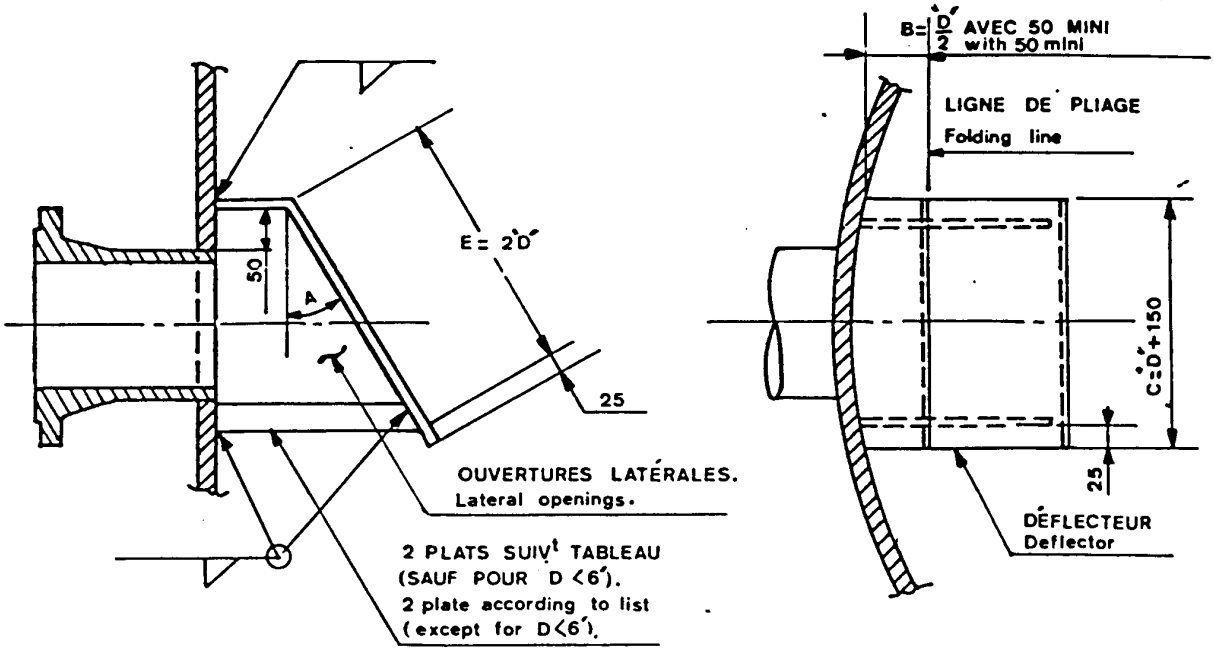


NOTA : 1 - TYPE ET MATERIAU SUIVANT PLAN DE L'APPAREIL.
2 - "D" DIAMETRE NOMINAL DE LA TUBULURE DANS LES INDICATIONS DE COTES.
3 - DISPOSITIONS ET DIMENSIONS VALABLES SI NON CONTREDITES PAR PLAN DE L'APPAREIL.

NOTE : 1 - TYPE AND MATERIAL ACCORDING TO VESSEL DRAWING.
2 - WHERE SIZE IS INDICATED, "D" = NOZZLE NOMINAL DIAMETER.
3 - DETAILS AND DIMENSIONS SHOWN IN VESSEL DRAWING TAKE PRECEDENCE OVER THOSE SHOWN HEREON.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL · TEP/DD	DEFLECTEUR SUR ENTREE DE VAPEUR DEFLECTOR ON STEAM INLET	PS.TCS. 222 Page 1 de 1 Date: février 1981



	ACIER AU CARBONE Carbon steel	ACIER INOX OU MONEL Stainless steel or monel
PLAT Plate	40 × 10	40 × 3
SOUDURE Welding	EPr 5 thk. 5	EPr 3 thk. 3
DEFLECTEUR Deflector	EPr 10 thk.10	EPr 5 thk. 5

NOTA : 1) LE MATERIEL SERA DEFINI SUR LE PLAN D'APPAREIL.
The material was defined on vessel drawing.

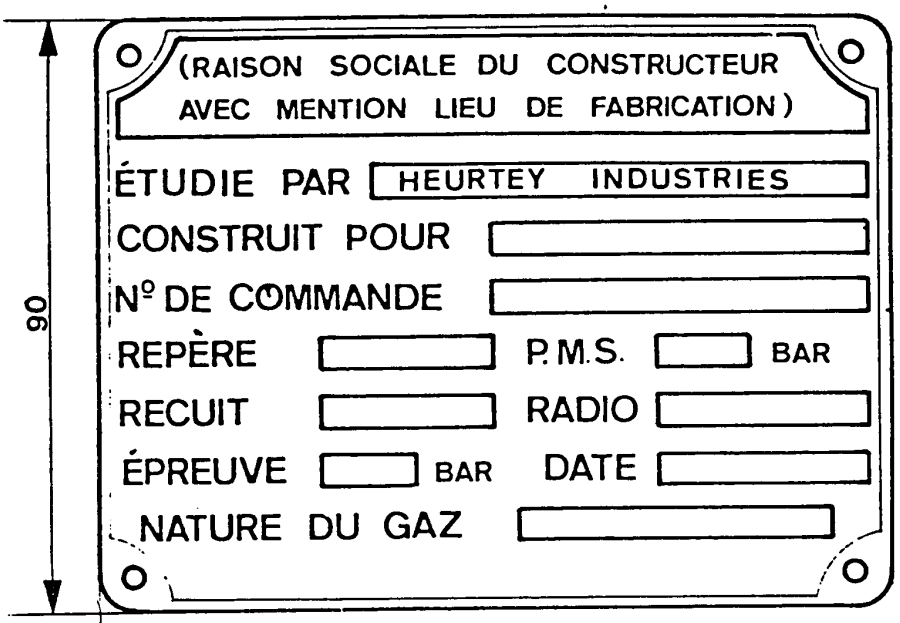
2) LE JET DE VAPEUR NE DOIT PAS ETRE DIRIGÉ SUR LA CUVETTE D'ÉTANCHÉITÉ DES PLATEAUX — "A" = 30° SAUF INDICATION CONTRAIRE.
CET ANGLE PEUT ETRE AJUSTÉ POUR EVITER DES REMOUS DANS LA CUVETTE D'ÉTANCHÉITÉ.
The steam jet do not ought to be towards on the tightness basin of the trays "A" = 30° except contrary indication.
This angle may be ajusted for avoid of eddies in of tightness basin.

3) D = DIAMETRE NOMINAL DE LA TUBULURE.
D = nominal diameter of the nozzle.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	PLAQUES SIGNALETIQUES POUR APPAREILS CHAUDRONNES NAMES PLATES FOR VESSELS	PS.TCS. 223 Page 1 de 1 Date: février 1981

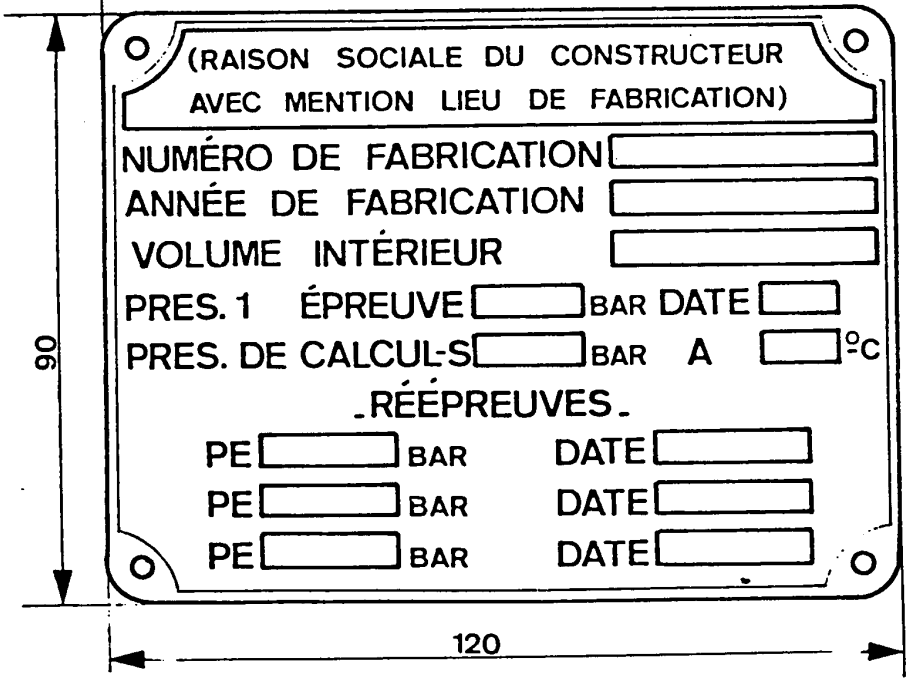
1-PLAQUE D'IDENTIFICATION (MARQUES DE SERVICE)



LA PLAQUE D'IDENTIFICATION SERA APPOSÉE SUR TOUS LES APPAREILS QUELLE QUE SOIT LEUR PRESSION.

MATIERE: ACIER INOXYDABLE 18/8 ÉPAISSEUR 2
COMPOSITION: CADRE, TEXTE ET BLOCS EN RELIEF, POLIS SUR FOND NOIR. LES INDICATIONS SUR LES BLOCS SERONT EN CREUX DE 0,2 mm MINIMUM.

2-PLAQUE MINÉRALOGIQUE (MARQUES D'IDENTITÉ)

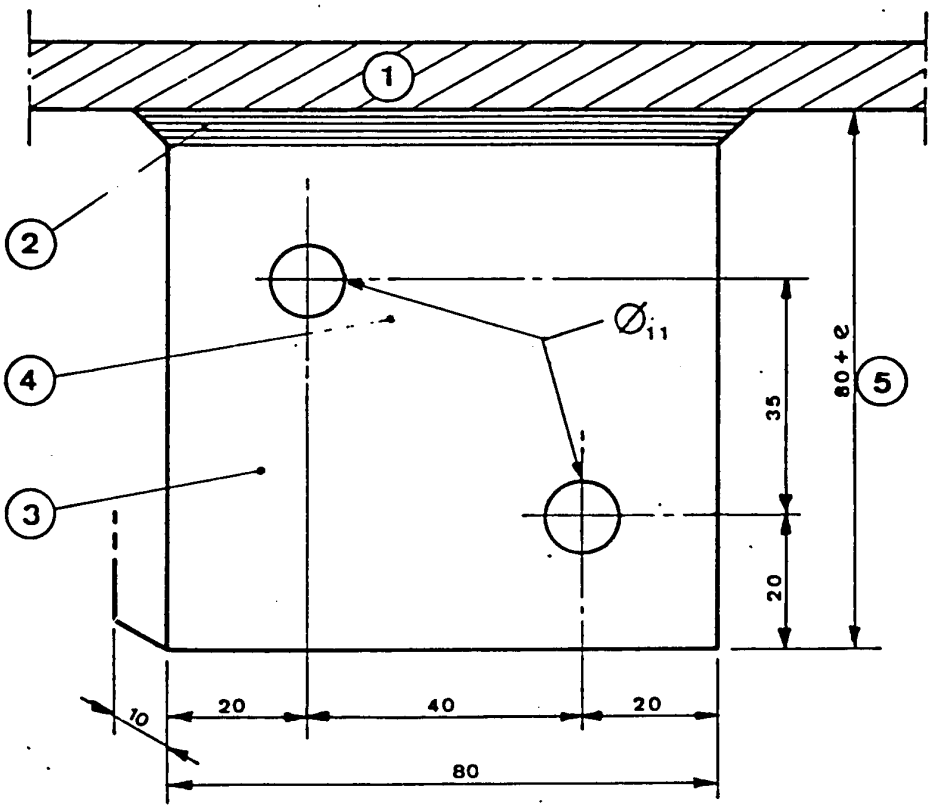


LA PLAQUE MINÉRALOGIQUE SERA APPOSÉE SUR TOUS LES APPAREILS REGLEMENTÉS PAR LE SERVICE DES MINES EN PLUS DE LA PLAQUE D'IDENTIFICATION.

3- UNE PLAQUE DE FIRME FACULTATIVE POURRA ETRE PROPOSÉE PAR LE CONTRUCTEUR SUIVANT SON MODÈLE STANDARD.

Rev.No.	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	PLAQUE DE MISE A LA TERRE EARTHING PLATE	PS.TCS. 224 Page 1 de 2 Date: février 1981



LEGENDE
Legend

-=-=-=-

1. APPAREIL
   Vessel

2. SOUDURE
   Welding

3. MATERIAU = CELUI DE L'APPAREIL
   Material: such as vessel material

4. SURFACES DE CONTACT ETAMEES
   Tinned contact surfaces

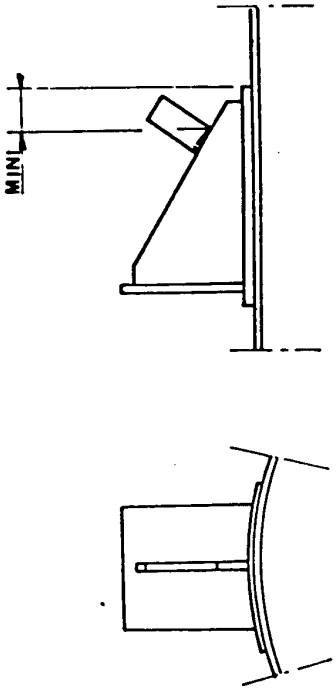
5. e = EPAISSEUR DU CALORIFUGE OU IGNIFUGE
   e = Fire proof material thickness.

Rev./Mod.	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	PLAQUE DE MISE A LA TERRE EARTHING PLATE	PS.TCS. 224 Page 2 de 2 Date: février 1981

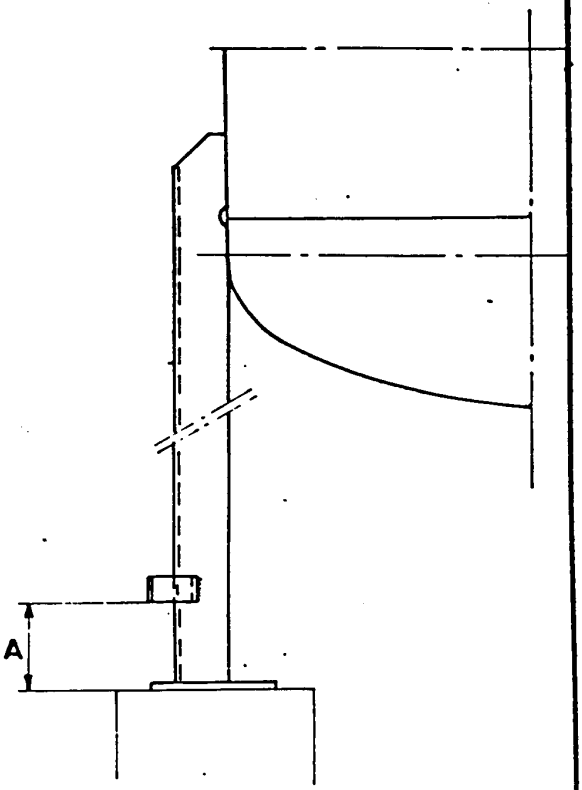
APPAREIL SUR CHAISES

vessel on chairs



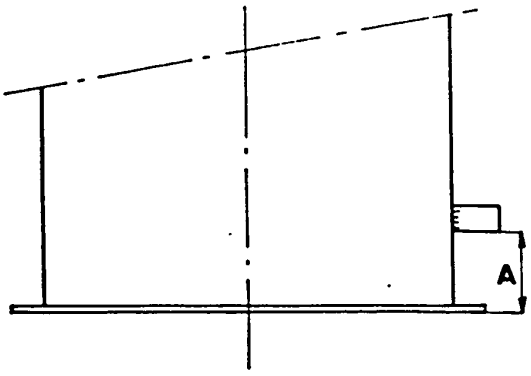
APPAREIL SUR PIEDS

vessel on legs

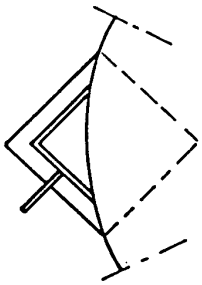


APPAREIL SUR JUPE

vessel on skirt



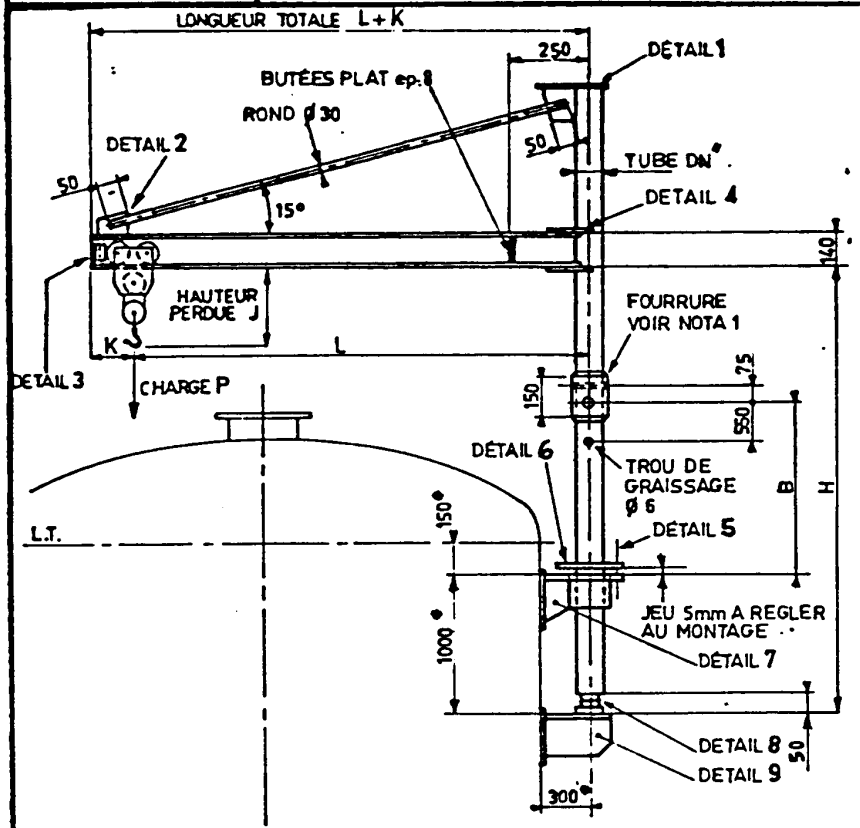
A = 300mm (sauf indication contraire)
except contrary

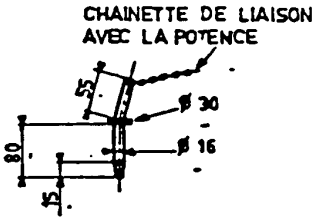


A = 300 mm (sauf indication contraire)
except contrary

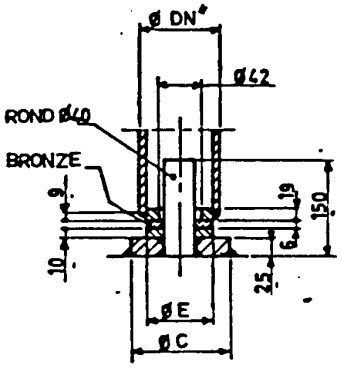
	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	POTENCES MONORAIL MONORAIL DAVIT	PS.TCS. 225 Page 1 de 5 Date: février 1981



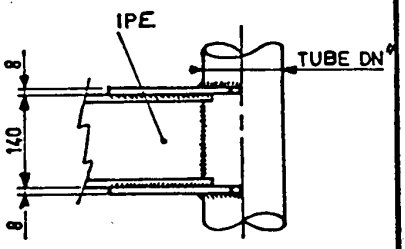


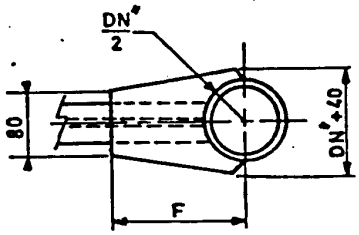
DOIGT
DETAIL : 5



GRAIN D'APPUI
DÉTAIL 8

FIXATION MONORAIL
DETAIL 4



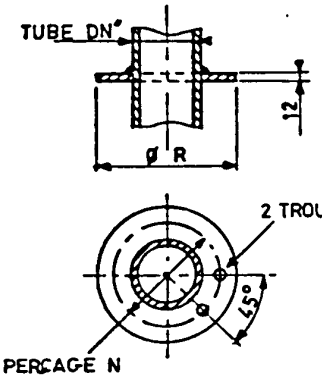


NOTA: 1) FOURRURE Ep:10 SOUDÉE AYANT 2 TROUS Ø 24 A 90° POUR BROCHE DE MANŒUVRE
2) LES COTES J ET K SONT POUR UN PALAN A MAIN NORMAL
3) LA COTE B SERA INDIQUÉE SUR PLAN D'APPAREIL
4) • COTES POUVANT ETRE MODIFIÉES SUR PLAN D'APPAREIL
5) LES SUPPORTS SERONT SOUDÉS SUR L'APPAREIL AVANT RECUIT EVENTUEL OU EPREUVE DES MINES
6) MATIÈRES VOIR F°2
7) SUR LA POTENCE LA VALEUR DE LA CHARGE POUVANT ETRE LEVEE DEVRA ETRE APPOSEE VISIBLEMENT PAR LE CONSTRUCTEUR.
8) LA FOURNITURE NE COMPREND NI LE CHARIOT PORTE-PALAN NI LE PALAN.

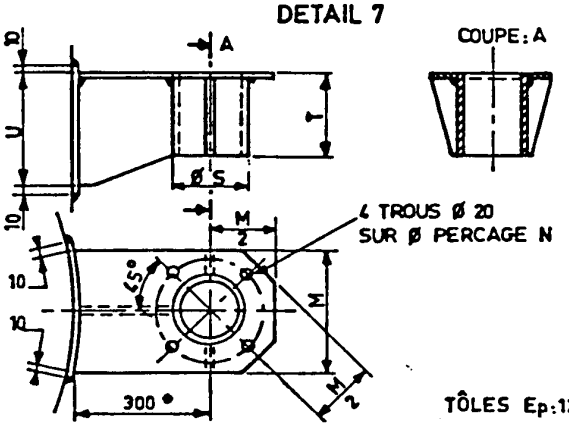
P kg	J	K
300	325	150
500	355	150
750	395	300
1000	435	300

Rect.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

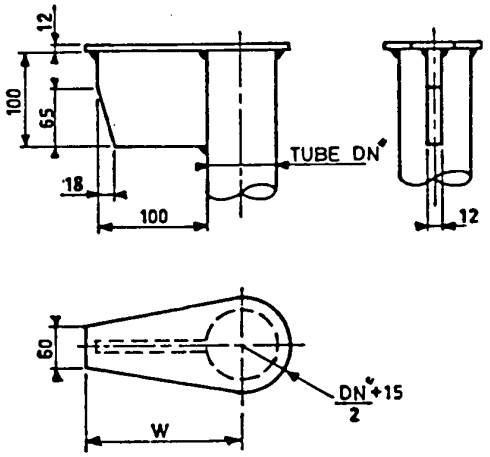
COLLERETTE D'ORIENTATION DE POTENCE
DETAIL 6



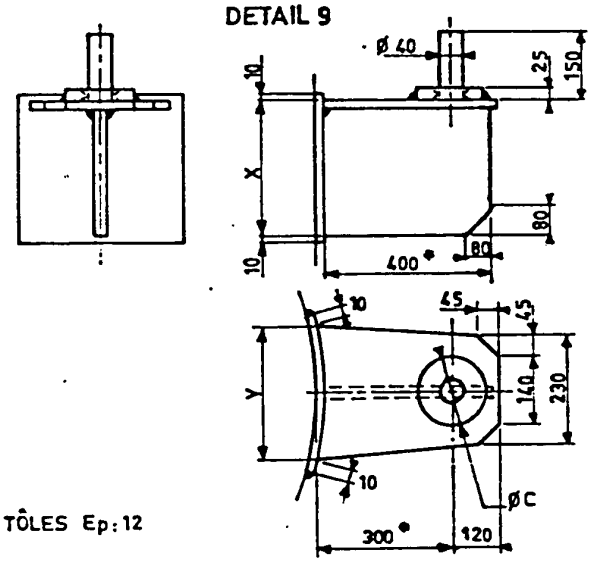
CONSOLE GUIDE DE POTENCE
DETAIL 7



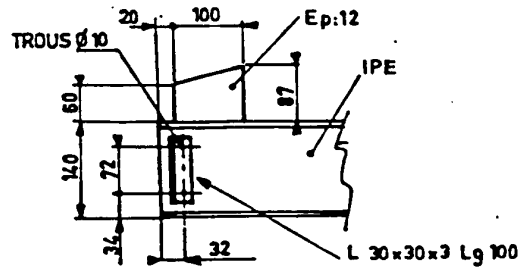
GOUSSET EN TETE DE POTENCE
DETAIL 1



CONSOLE SUPPORT DE POTENCE
DETAIL 9



GOUSSET ET BUTEES D'EXTREMITE DU MONORAIL
DETAILS 2,3



NOTA : 1) MATIÈRE=TUBE : ASTM A53 GRADE A OU B OU EQUIVALENT
=TÔLES: SUIVANT PLAN D'APPAREIL MAIS LA TÔLE DE RENFORT SERA DE MÊME MATIÈRE QUE L'APPAREIL
2) VALEUR DES COTES VARIABLES. VOIR FOLIO 3.
3) * COTES POUVANT ETRE MODIFIEES SUR PLAN D'APPAREIL.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

DN" / Sch.	6 / 40	6 / 80	8 / 40	8 / 80
C	175	175	220	220
E	154	146	202	193
F	200	200	230	230
M	300	300	400	400
N	260	260	350	350
R	300	300	400	400
S	194 x 10	194 x 10	245 x 10	245 x 10
T	200	200	250	250
U	300	300	350	350
W	200	200	225	225
X	300	300	350	350
Y	300	300	400	400

CHARGE 300kg

L / H	1000		1500		2000		2500		3000		3500		4000		4500	
	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch
2000	6	40	6	40	6	40	6	80	8	40	8	40	8	40	8	40
2500	6	40	6	40	6	80	8	40	8	40	8	40	8	40	8	80
3000	6	40	6	80	8	40	8	40	8	40	8	40	8	80	8	80
3500	6	40	6	80	8	40	8	40	8	40	8	80	8	80		
4000	6	40	8	40	8	40	8	40	8	80	8	80				
4500	6	80	8	40	8	40	8	80	8	80						
5000	6	80	8	40	8	40	8	80								
IPE	140		140		140		140		140		140		140		140	

CHARGE 500kg

L / H	1000		1500		2000		2500		3000		3500		4000		4500	
	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.
2000	6	40	6	80	8	40	8	40	8	40	8	80				
2500	6	40	8	40	8	40	8	40	8	80	8	80				
3000	6	80	8	40	8	40	8	80	8	80						
3500	6	80	8	40	8	80	8	80								
4000	8	40	8	40	8	80										
4500	8	40	8	80	8	80										
5000	8	40	8	80												
IPE	140		140		140		140		140		140					

Index	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

CHARGE 750 kg

L / H	1000		1500		2000		2500		3000	
	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.
2000	6	80	8	40	8	40	8	80	8	80
2500	8	40	8	40	8	80	8	80		
3000	8	40	8	80	8	80				
3500	8	40	8	80						
4000	8	40	8	80						
4500	8	80								
5000	8	80								
IPE	140		140		140		140		140	

CHARGE 1000 kg

L / H	1000		1500		2000		2500		3000	
	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.	DN"	Sch.
2000	8	40	8	40	8	80				
2500	8	40	8	80						
3000	8	40	8	80						
3500	8	80								
4000	8	80								
4500	8	80								
5000										
IPE	140		140		140					

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

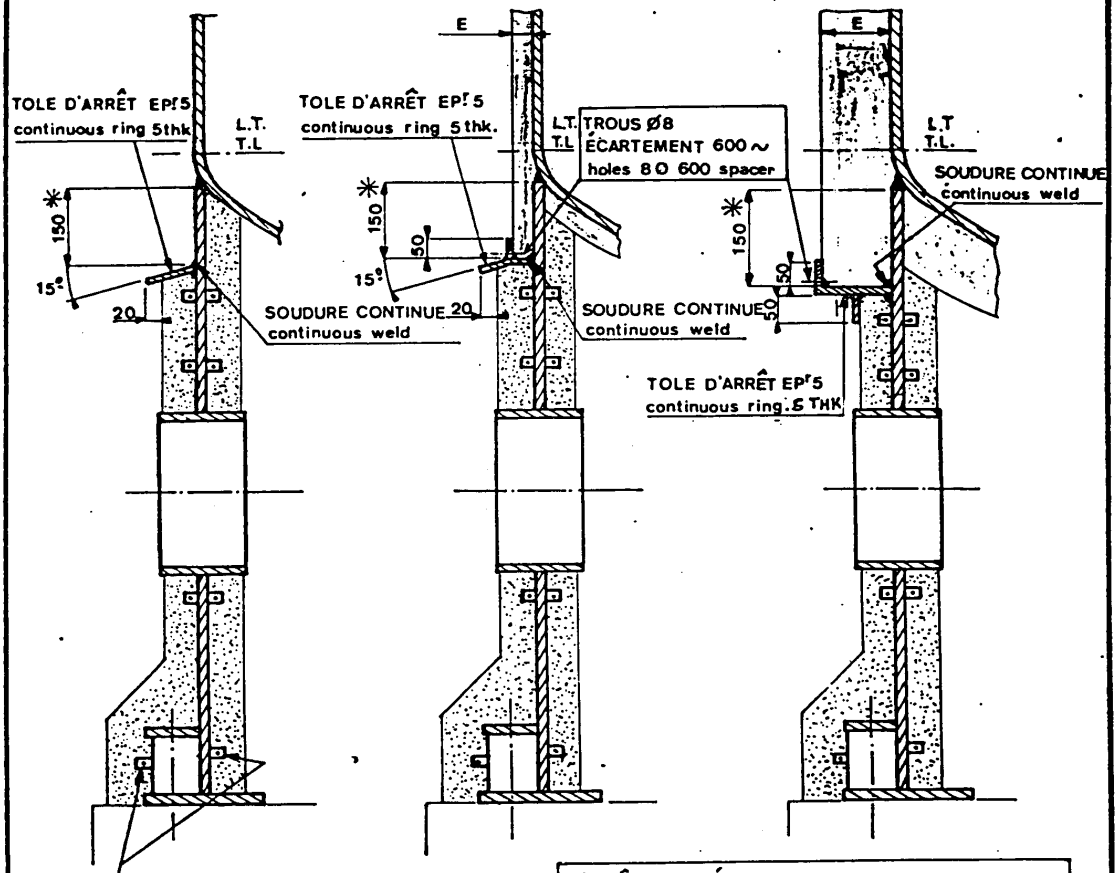
TOTAL TEP/DD	SUPPORTS D'IGNIFUGE FIRE-PROOFING SUPPORTS	PS.TCS. 229 Page 1 de 1 Date: février 1981

Ø DE LA JUPE Ø of the skirt	IGNIFUGEAGE INTÉRIEUR. inside fireproofing.	IGNIFUGEAGE EXTÉRIEUR. outside fireproofing.
⩽ 750	AUCUN none	BÉTON : 50 mm concrete "
⩾ 750	BÉTON : 50 mm concrete "	BÉTON : 50 mm concrete "

IGNIFUGEAGE DE JUPE DE COLONNE NON CALORIFUGÉE.
column skirt fireproofing without insulation.

IGNIFUGEAGE DE JUPE. COLONNE CALORIFUGÉE EP^r CALO ⩽ EP^r IGNIF.
column skirt fireproofing insulated insulation thk. ⩽ fireproofing thk.

IGNIFUGEAGE DE JUPE. COLONNE CALORIFUGÉE EP^r CALO. ⩾ EP^r IGNIF.
column skirt fireproofing insulated insulation thk. ⩾ fireproofing thk.



GOUJON RECTANGULAIRE 10mm x 3mm
SOUDÉ OU ÉCROU Ø 12 (CARRÉ).
10mm x 3mm rectangular welding stud
12Ø square nut welded.

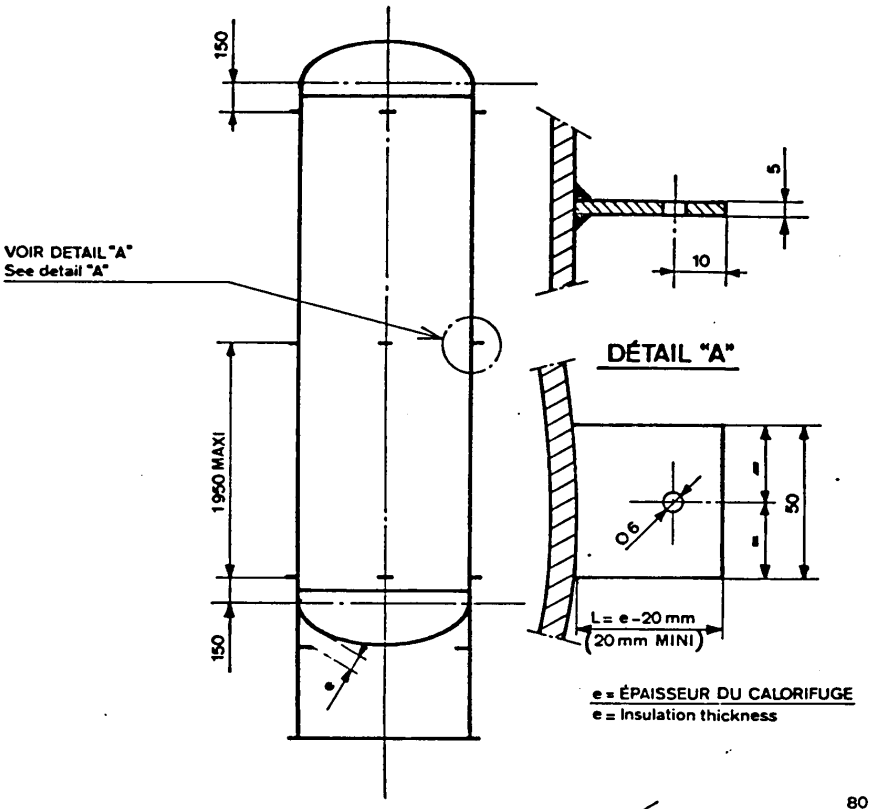
✳ CÔTE PORTÉE A 200 EN CHAUD. INOX
Dimensions increased to 200 in stainless vessel

ESPACEMENT : 300 mm DANS LES DEUX SENS POUR APPAREIL DE Ø ⩾ 900.
230 mm DANS LES DEUX SENS POUR APPAREIL DE Ø ⩽ 900.
space : 300 mm in both directions for skirt Ø ⩾ 900.
230 mm in both directions for skirt Ø ⩽ 900.

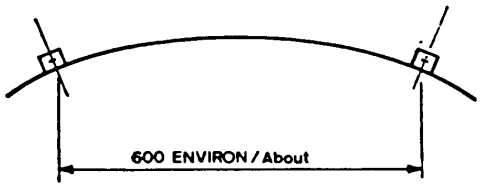
FOURNITURE ET POSE PAR CONSTRUCTEUR DE L'APPAREIL
supply and laying by the constructor.

Rev./No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

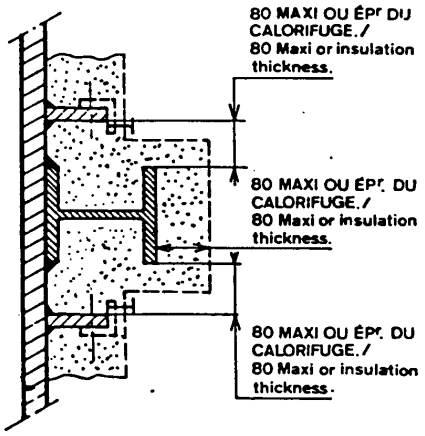
TOTAL TEP/DD	SUPPORTS CALORIFUGE POUR APPAREIL VERTICAL INSULATION SUPPORT FOR HOT SERVICE VERTICAL APPARATUS	PS.TCS. 230 Page 1 de 1 Date: février 1981



e = ÉPAISSEUR DU CALORIFUGE
e = Insulation thickness



CALORIFUGE POUR APPAREIL VERTICAL
Insulation for vertical vessel

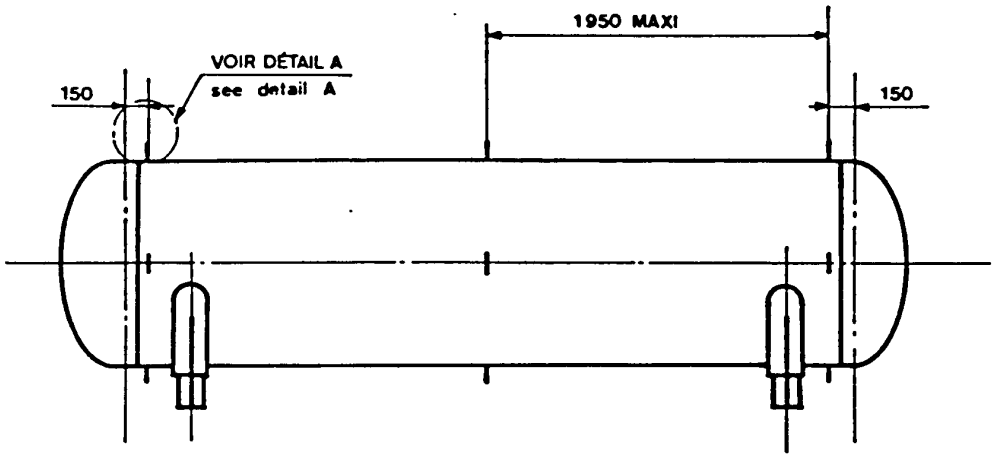


POUR APPAREIL VERTICAL AVEC ANNEAU RAIDISSEUR.
For vertical vessel with stiffener ring.

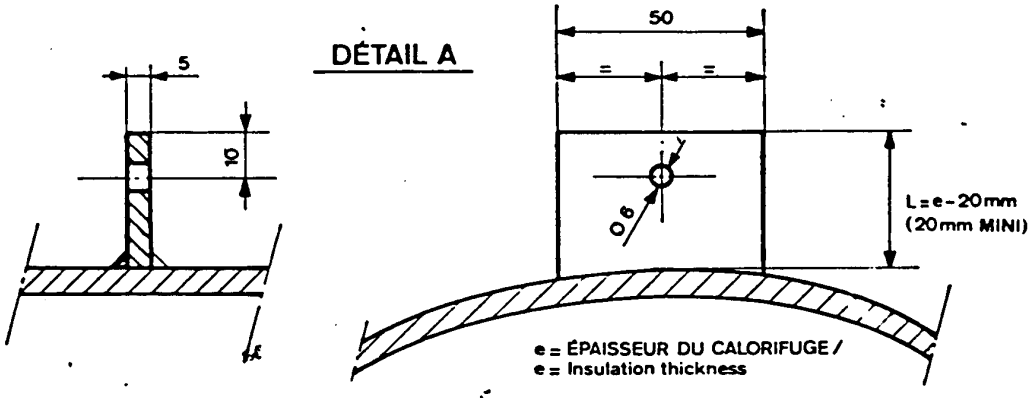
NOTE: MATIÈRE DE MÊME NUANCE QUE L'APPAREIL.
Material same as vessel material.

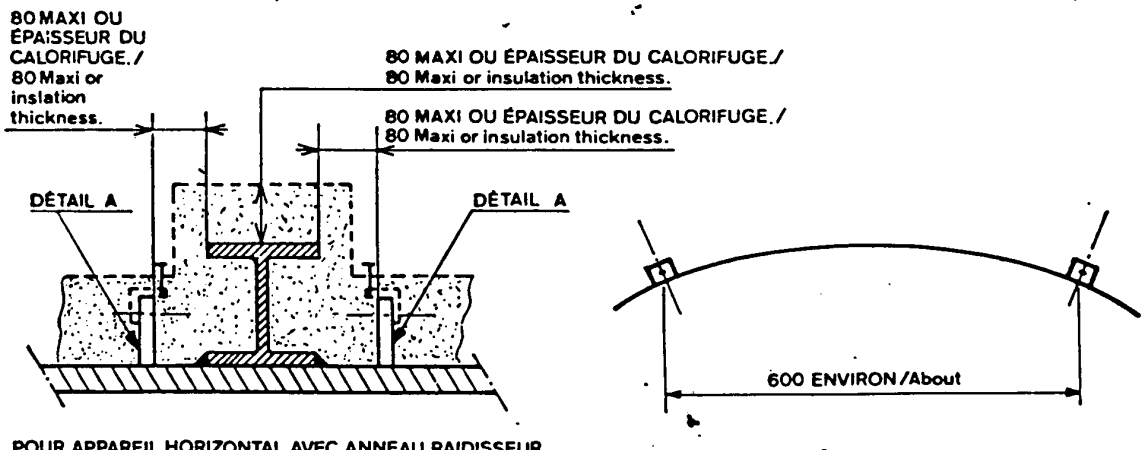
Repère	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	SUPPORTS CALORIFUGE POUR APPAREIL HORIZONTAL INSULATION SUPPORTS FOR HOT SERVICE HORIZONTAL APPARATUS	PS.TCS. 231 Page 1 de 1 Date: février 1981



DÉTAIL A





POUR APPAREIL HORIZONTAL AVEC ANNEAU RAIDISSEUR.
For horizontal vessel with stiffener ring.

NOTE: MATIÈRE DE MÊME NUANCE QUE L'APPAREIL.
Material same as vessel material.

CALORIFUGE POUR APPAREIL HORIZONTAL
Insulation for horizontal vessel

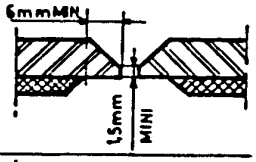
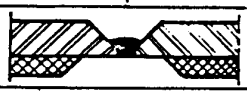
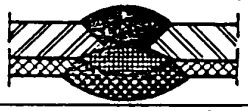
Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	SOUDAGE DES APPAREILS PLAQUES CLAD ALLOY VESSEL WELDING PROCEDURE	PS.TCS. 232 Page 1 de 5 Date: Février 1981

DETAILS DE SOUDURE. — WELDING DETAILS.

PREPARATION DES ASSEMBLAGES ET SUITE DES OPERATIONS DE SOUDURE

PREPARATION OF JOINT WELDED ASSY AND WELDING OPERATIONS

1 - CHANFREINER LA PLAQUE DE BASE EN REDUISANT LE CHAMP A 1,6 mm DE LARGE TAILLER LE REVETEMENT EN FORME DE TRANCHANT DEGAGEANT COMPLETEMENT LE METAL DE BASE SUR 6,5 mm.		1 - BEVEL BASE PLATE LEAVING A MINIMUM ROOT FACE OF 1,6 mm WIDE PROCEED TO SAME ON CLAD LEAVING PARENT METAL CLEAR OVER 6,5 mm IN WIDTH.
2 - SOUDER PAR POINTS LES PLAQUES DE BASE EN UTILISANT LES ELECTRODES ① NOTE 2 A.		2 - PROCEED TO SPOT WELDING OF BASE PLATES USING ELECTRODES ① NOTE 2 A.
3 - COMPLETER LA SOUDURE DES PLAQUES DE BASE EN UTILISANT LES ELECTRODES ① ET EN EVITANT LA PENETRATION DANS L'ALLIAGE DU REVETEMENT.		3 - PROCEED TO WELDING COMPLETION OF BASE PLATES USING ELECTRODES ① AVOIDING WELD PENETRATIONS INTO THE CLAD ALLOY.
4 - BURINER OU MEULER LE COTE REVETEMENT POUR ASSAINIR LA SOUDURE.		4 - CHIP OUT OR GRIND CLAD SIDE IN ORDER TO IMPROVE WELD.
5 - SOUDER LE REVETEMENT EN UTILISANT LES ELECTRODES EN ALLIAGE ② ET ③		5 - WELD CLAD USING ALLOY ② AND ③ ELECTRODES.

NOTES

1 - CES DETAILS NE CONSTITUENT PAS UNE METHODE INTANGIBLE DE SOUDURE. LE CONSTRUCTEUR DU RESERVOIR FOURNIRA DES INFORMATIONS COMPLEMENTAIRES CONCERNANT LA PREPARATION ET LA VALEUR DE SON PROCEDE EN ACCORD AVEC L'A.S.M.E. SECTION IX ET LE PRESSURE VESSEL CODE.

NOTES

1 - ABOVE DETAILS ARE NOT INTENDED TO PROVIDE A COMPLETE WELDING PROCEDURE. ADDITIONAL INFORMATIONS TO BE GIVEN BY THE VESSEL MANUFACTURER COVERING THE PREPARATION AND THE QUALIFICATION OF HIS PROCEDURE IN ACCORDANCE WITH A.S.M.E. SECTION IX AND PRESSURE VESSEL CODE.

METAL CONSTITUANT LES PLAQUES DE BASE BASE PLATE PARENT METAL	ELECTRODES
ACIER AU CARBONE CARBON STEEL	ASTM SFA 51
ACIER FAIBLEMENT ALLIE LOW ALLOY STEEL	* ASTM SFA 55

2 - CHOIX DES ELECTRODES COMME SUIT :

A - POUR LES PLAQUES DE BASE :

* QUAND ON UTILISE LES ELECTRODES SUIVANT ASTM SFA 55, CELLES-CI DOIVENT AVOIR UNE COMPOSITION CORRESPONDANT AUX PLAQUES DE BASE. POUR LES MATERIAUX NON CONCERNES PAR L'ASTM SFA 55, ETABLIR UNE SPECIFICATION PARTICULIERE.

2 - ELECTRODES TO BE SELECTED AS FOLLOWS :

A - FOR BASE PLATES :

* WHERE ASTM SFA 55 ARE USED THEIR COMPOSITION IS TO BE THAT OF THE BASE PLATE METAL. A SPECIAL SPECIFICATION IS TO BE DEVELOPPED FOR MATERIALS NOT COVERED BY ASTM SFA 55.

Rev./No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

DETAILS DE SOUDURE — WELDING DETAILS

PREPARATION DES ASSEMBLAGES ET SUITE DES OPERATIONS DE SOUDURE

PREPARATION OF JOINT WELDED ASSY AND WELDING OPERATIONS

NOTES

NOTES

B - POUR LE REVETEMENT :

B - FOR CLAD :

PLACAGE CLADDING ALLOY	ALLIAGE INTERMEDIAIRE ② INTERMEDIATE ALLOY ②		ALLIAGE ③ ALLOY ③	
AISI TYPE 405	* ASTM SFA 54	E 310	ASTM SFA 54	E 310 *
304 L AISI TYPE 321 347	ASTM SFA 54	E 309 CB	ASTM SFA 54	E 347
AISI TYPE 316 L	ASTM SFA 54	E 309 MO	ASTM SFA 54	E 316
MONEL	ASTM SFA 511	E NI CU 1	ASTM SFA 511	E NI CU 1

* MALGRE L'UTILISATION COURANTE DES ELECTRODES EN ACIER AUSTENITIQUE POUR SOUDER LES ACIERS DE REVETEMENT AU CHROME TYPE 405 ET AUTRES AFIN D'EVITER LA FRAGILISATION A CHAUD (PARTICULIERE A CHAQUE CAS MAIS GENERALEMENT $\geq$ T 444 °C) L'APPORT D'ACIER AUSTENITIQUE NE PEUT ETRE ADMIS A CAUSE DES ECARTS DE DILATATION. DANS CE CAS ON UTILISERA DES E 430, AVEC PRECHAUFFAGE ET POSTCHAUFFAGE. LES E 310 SONT PREFERABLES POUR T $<$ 444 °C (SAUF CONTRE INDICATIONS DUES AU CYCLE THERMIQUE).

* ALTHOUGH IT IS GENERAL PRACTICE TO WELD TYPE 405 AND OTHER STRAIGHT CHROMIUM CLAD STEELS WITH AUSTENITIC ELECTRODES TO AVOID POST HEAT BRITTELENESS (SPECIFIC FOR EACH CASE BUT GENERALY AT T $\geq$ 444 °C) AUSTENITIC WELD DEPOSITS CAN'T BE TOLERATED BECAUSE OF DIFFERENTIAL THERMAL EXPANSION. IN THIS CASE 430 SHOULD BE USED WITH APPROPRIATE PRE AND POST HEAT TREATMENT HOWEVER E 310 IS PREFERRED FOR T $<$ 444 °C. UNLESS EXTREME THERMAL CYCLING CONDITION EXISTS.

NOTES (SUITE)

3 - POUR LES FACES DE BRIDES, LE DEPOT DE METAL PAR SOUDURE SERA FAIT DANS UN LAMAGE USINE AU BORD INTERNE DE LA BRIDE. L'EPAISSEUR DE LA BRIDE TERMINEE SERA CONFORME AU STANDARD A.N.S.I.

4 - ON EVITERA LA DILUTION DE L'ALLIAGE D'APPORT LORS DE LA PASSE FINALE. GENERALEMENT 2 PASSES SONT NECESSAIRES POUR EVITER LA DILUTION SUR LA SURFACE EXPOSEE A LA COROSION, CE QUI EST PARTICULIEREMENT IMPORTANT DANS LE CAS DE REVETEMENTS MINCES. LE CONSTRUCTEUR DU RESERVOIR DEVRA S'ASSURER, PAR UN EXAMEN APPROPRIE, QUE LES PASSES DE SOUDURE FINALES REMPLISSENT CETTE CONDITION.

NOTES (CONTINUATION)

3 - FOR FLANGE FACINGS THE DEPOSIT SHALL BE MADE IN A MACHINED GROOVE FOR WELDING. FINISHED FLANGE THICKNESS SHALL BE A.N.S.I. STANDARD.

4 - DILUTION OF THE FINAL ALLOY WELD PASS SHOULD BE AVOIDED. GENERALLY 2 PASSES ARE REQUIRED TO AVOID DILUTION OF THE SURFACE EXPOSED TO CORROSION. THIS IS PARTICULARLY IMPORTANT WHEN THIN CLADDING IS USED. THE VESSEL MANUFACTURER SHALL ENSURE BY SUITABLE EXAMINATION THAT THE FINAL WELD PASS MEETS THE REQUIREMENT.

DETAILS DIVERS — MISCELLANEOUS DETAILS

SOUDURE DES BOSSAGES

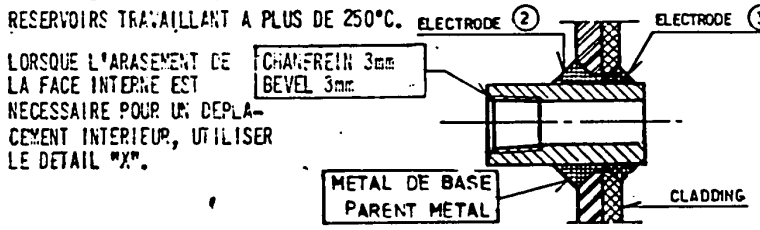
LES BOSSAGES SERONT PRIS DANS DES BARRES D'ACIER ALLIE DE MEME COMPOSITION QUE LE REVETEMENT DU RESERVOIR. LE FILETAGE SERA CONIQUE SUIVANT A.N.S.I. B 21.

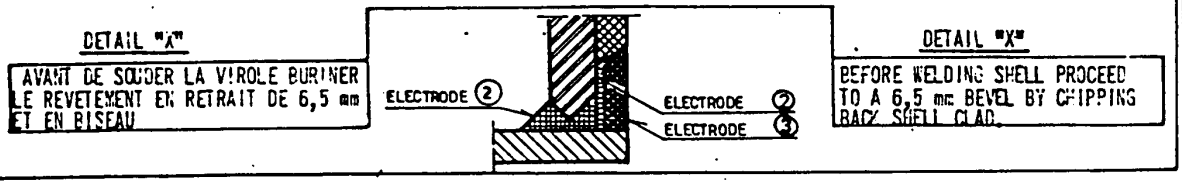
ON N'UTILISERA PAS CE MODE DE CONSTRUCTION POUR LES RESERVOIRS TRAVAILLANT A PLUS DE 250°C.

WELDING OF THREADED COUPLINGS

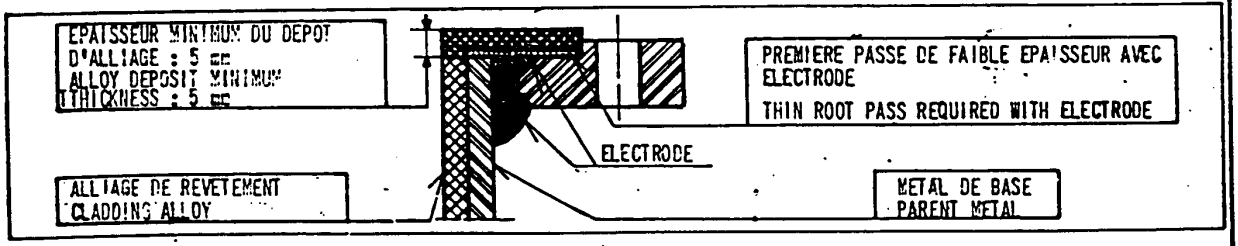
COUPLINGS SHALL BE MADE OF THE SAME ALLOY AS THE VESSEL CLADDING SAMPLED FROM ALLOY STEEL BARS. THREADS AS PER A.N.S.I. B 21.

THIS CONSTRUCTION MODE IS NOT TO BE USED FOR VESSELS OPERATING BEYOND 250°C.

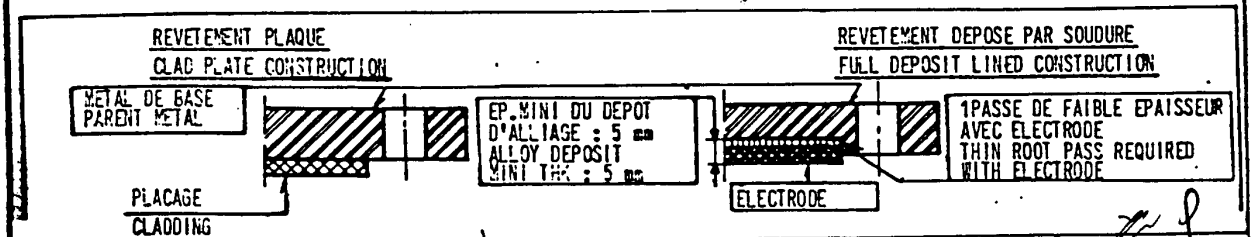




TUBULURE DE TROU D'HOMME — MANHOLE NOZZLE



COUVERCLE DE TROU D'HOMME ET BRIDE PLEINE — MANHOLE COVER AND BLIND FLANGE

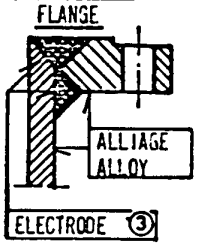


Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

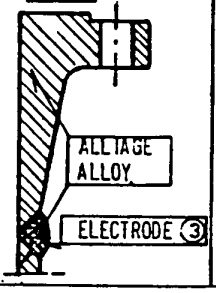
TUBULURE EN ALLIAGE MASSIF — SOLID ALLOY NOZZLE NECK

MONTAGE DES BRIDES — FLANGES CONNECTION

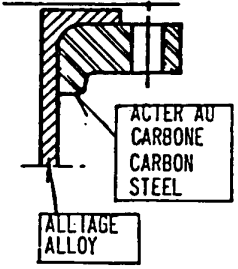
BRIDE PLATE EN ALLIAGE — ALLOY PLATE FLANGE



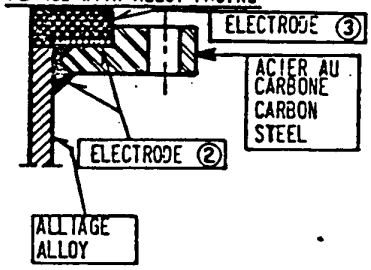
BRIDE A COLLERETTE EN ALLIAGE — ALLOY W.N. FLANGE



BRIDE TOURNANTE EN ACIER AU CARBONE SUR COLLET EN ALLIAGE — CARBON STEEL LAP JOINT FLANGE



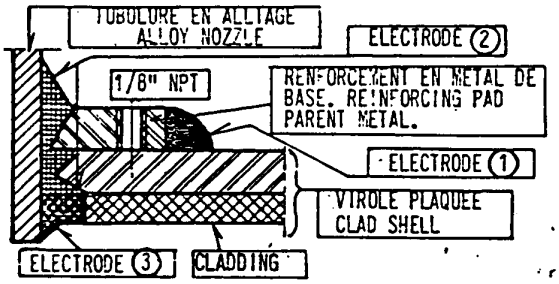
BRIDE PLATE OU S.O. EN ACIER AU CARBONE AVEC FACE EN ALLIAGE — CARBON STEEL FLANGE OR S.O. FLANGE WITH ALLOY FACING



PIQUAGE SUR VIROLE — SHELL CONNECTION

1 - NE PAS UTILISER DE TUBULURES EN ACIER INOXYDABLE AUSTENITIQUE MASSIF SUR DES VIROLES EN ACIER FERRITIQUE QUAND LA TEMPERATURE DEPASSE 250° C.

2 - ON N'UTILISERA PAS DE BRIDES PLATES EN ACIER AU CARBONE AVEC FACES REVETUES D'ALLIAGE QUAND L'ACIER AU CARBONE EST SUSCEPTIBLE D'ETRE ATTAQUE PAR L'HYDROGENE.

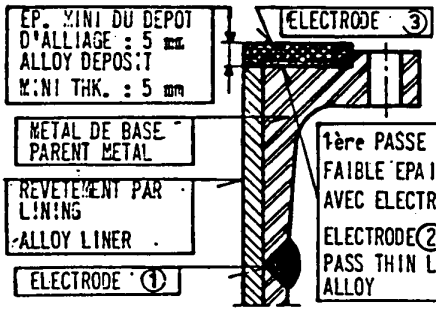


1 - WHERE THE TEMPERATURE EXCEEDS 250° C, SOLID AUSTENITIC STAINLESS NOZZLES FOR FERRITIC STEEL SHELLS SHALL NOT BE USED.

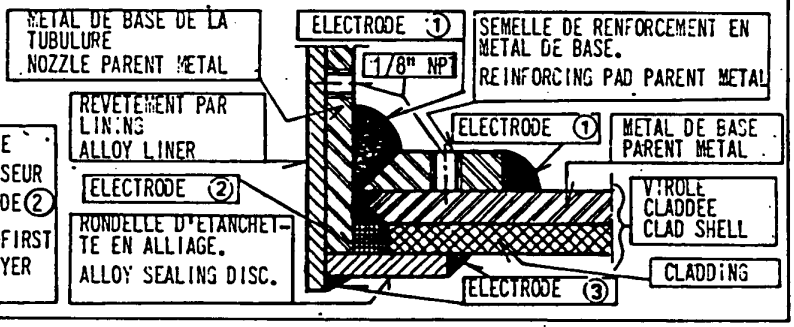
2 - WHERE CARBON STEEL IS SUBJECT TO HYDROGEN ATTACK CARBON STEEL PLATE FLANGES WITH ALLOY FACING SHALL BE USED UNDER CONDITIONS.

TUBULURE A REVETEMENT INTERIEUR PAR LINING — ALLOY LINED NOZZLE NECK

MONTAGE DES BRIDES - FLANGES CONNECTION



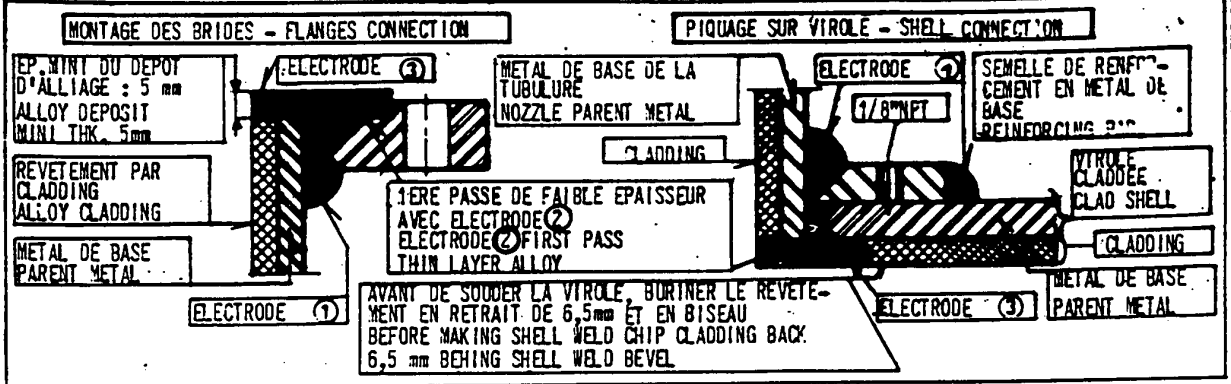
PIQUAGE SUR VIROLE - SHELL CONNECTION



Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	SOUDAGE DES APPAREILS PLAQUES CLAD ALLOY VESSEL WELDING PROCEDURE	PS.TCS. 232 Page 5 de 5 Date: Février 1981

TUBULURE A REVETEMENT INTERIEUR PAR CLADDING — ALLOY CLAD NOZZLE NECK



LEGENDE. — LEGEND.

METAL DE BASE	▨	PARENT METAL
REVETEMENT EN ALLIAGE MASSIF OU ALLIAGE	▨	SOLID ALLOY OR ALLOY LINER
PLACAGE	▩	ALLOY CLADDING
ELECTRODE DE SOUDURE EN METAL DE BASE ①	■	PARENT METAL WELDING ELECTRODE ①
ELECTRODE EN ALLIAGE INTERMEDIAIRE ②	■	INTERMEDIAIRE ALLOY WELDING ELECTRODE ②
ELECTRODE EN ALLIAGE ③	■	ALLOY WELDING ELECTRODE ③

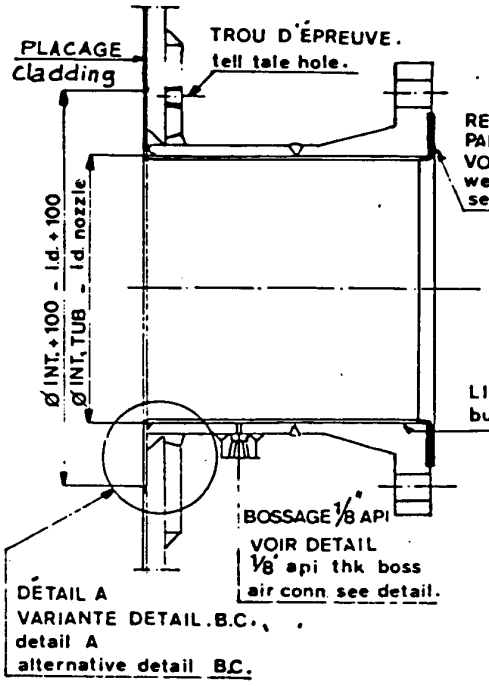
NOTES GENERALES. — DESIGN NOTES.

1 - LES BRIDES REPRESENTEES SONT A FACES SURELEVEES DE 1.6 mm. POUR LES AUTRES TYPES DE FACES DE BRIDES ON EFFECTUERA UN DEPOT D'ALLIAGE D'AU MOINS 3,5 mm. D'EPAISSEUR SOUS LA FACE DU JOINT.	1 - FLANGE FACINGS SHOWN ARE 1,6 mm. RAISED FACE WHERE OTHER FLANGE FACINGS ARE SPECIFIED AT LEAST 3,5 mm THICKNESS OF ALLOY DEPOSIT SHALL BE PROVIDED BENEATH THE GASKET FACE.
2 - LES PRISES D'APPAREIL SUR TUBULURES SERONT TARAUDES 1/8" NPT, GRAISSES ET LAISSES DEBOUCHEES.	2 - TEST HOLES IN NOZZLES SHALL BE TAPPED 1/8" NATIONAL PIPE THREAD GREASED AN LEFT UNPLUGGED.
3 - LES RENFORCEMENTS ET GARNITURES DES TUBULURES SUBIRONT UN ESSAI D'ETANCHEITE A L'AIR SOUS PRESSION. 2 BAR.	3 - REINFORCING PADS AND NOZZLE LINERS SHALL BE TESTED TIGHT WITH 2 BAR AIR.

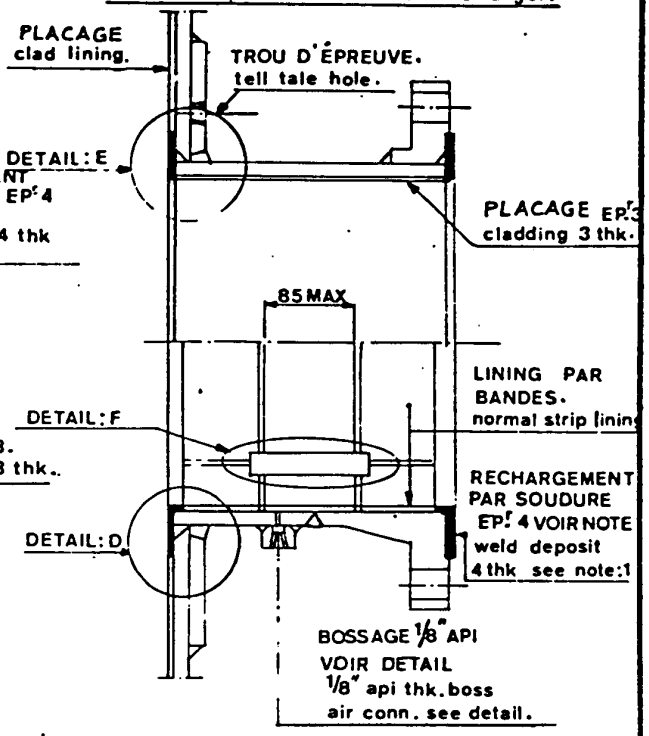
	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	REVETEMENT DES TUBULURES NOZZLES LINING	PS.TCS. 233 Page 1 de 2 Date: février 1981

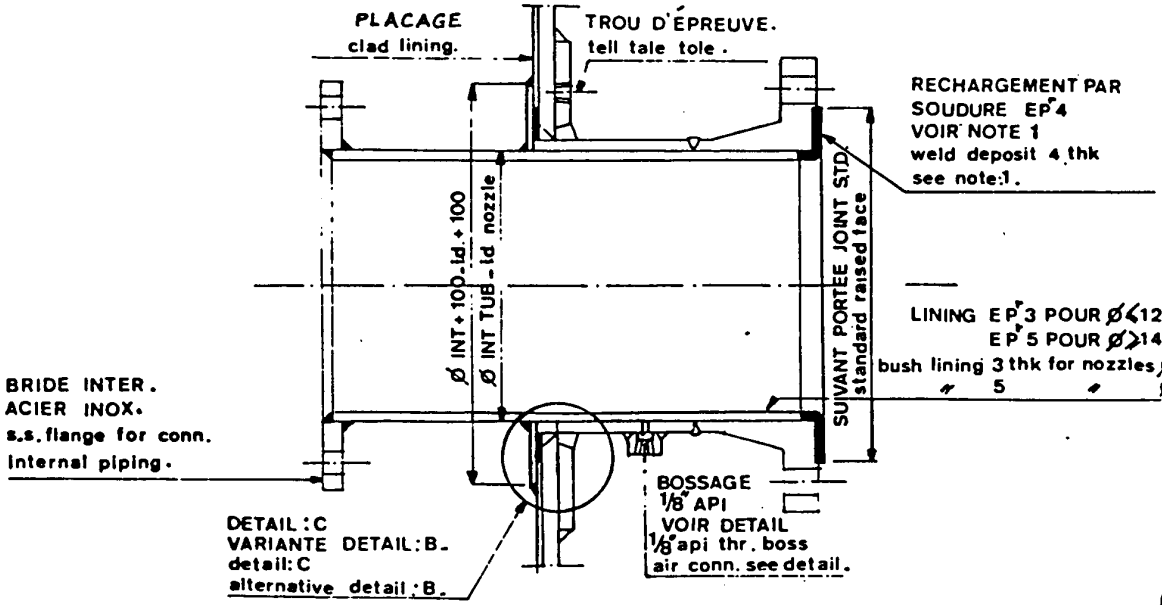
TUBULURES O ⩽ 12" AVEC LINING.
bush lined nozzle 12" and smaller.



TUBULURES O 14" ET TROU D'HOMME AVEC PLACAGE OU LINING PAR BANDES.
clad or strip lined nozzle 14" and larger.



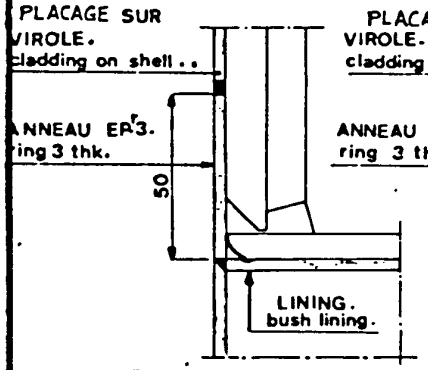
TUBULURES AVEC LINING PROLONGÉ A L'INTERIEUR ⩽ 12".
bush lined internal conneted nozzle ⩽ 12".



	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

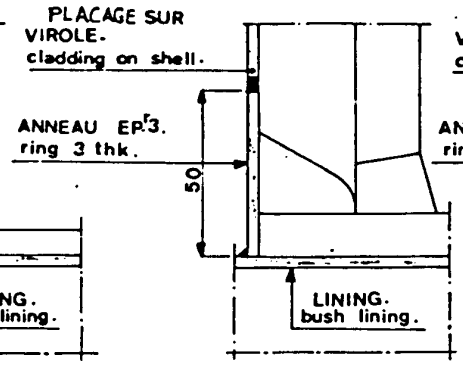
TOTAL TEP/DD	REVETEMENT DES TUBULURES NOZZLES LINING	PS.TCS. 233 Page 2 de 2 Date: février 1981

DETAIL : A

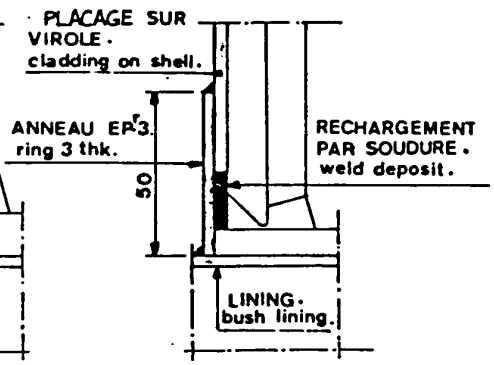


UTILISÉ POUR ECHANGEURS CHAUDS SUR LA VIROLE SEULEMENT
to be used for heat exchangers on shell side only.

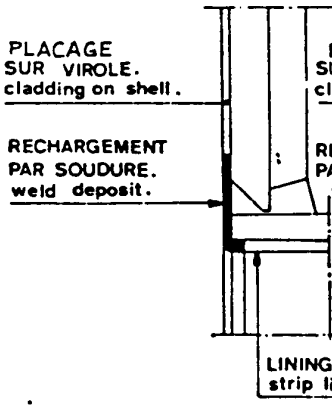
DETAIL : B



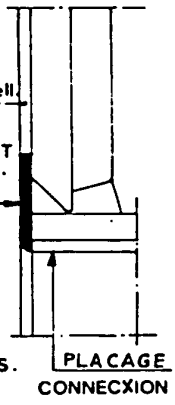
DETAIL : C



DETAIL : D

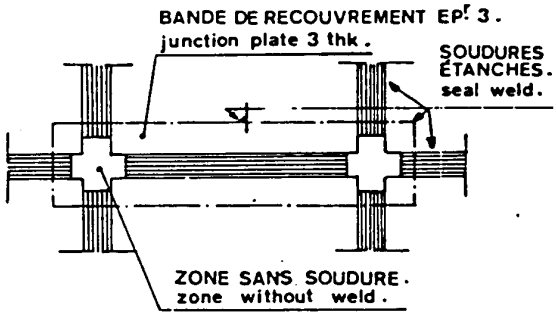


DETAIL : E

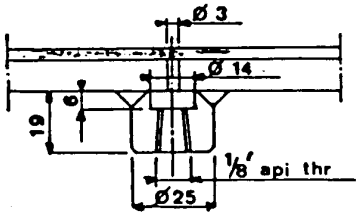


CONNECXION D'AIR, INUTILE POUR PLACAGE
air conn. not required for cladding.

DETAIL : F



BOSSAGE POUR ÉPREUVE.
air connection.

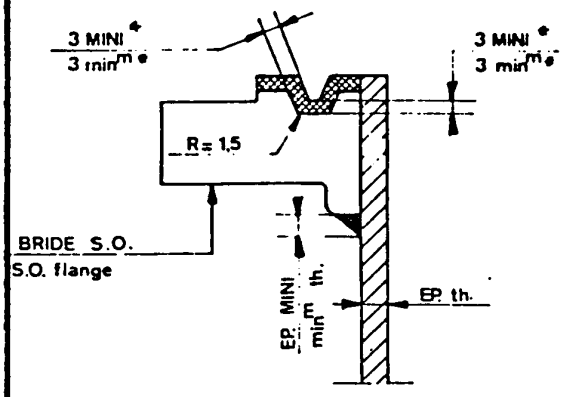


NOTES : 1) _ LE REVÊTEMENT DES TUBULURES EST DE MÊME NUANCE MÉTALLURGIQUE QUE CELLE DU PLACAGE DE LA VIROLE.
- nozzles lining : same metallurgic shade as shell cladding.

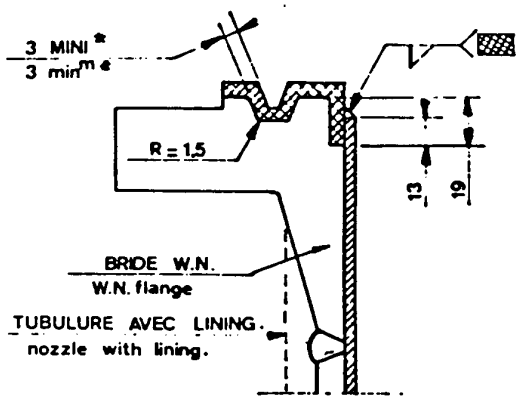
2) _ POUR LES REVÊTEMENTS EN ACIERS INOXYDABLES FERRITIQUES TYPE 11_13 Cr. LES DÉTAILS D ET E POURRONT EGALEMENT ÊTRE UTILISÉS POUR DES TUBULURES AVEC LINING.
- for ferritic stainless steel lining : 11_13 Cr type, the details D and E may be used for nozzles with bush lining.

Rev.No	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

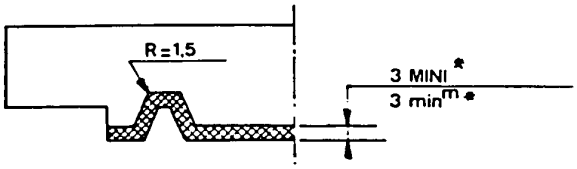
TOTAL · TEP/DD	RECHARGEMENT PORTEE DE JOINT RTJ LINED RING JOINT FACE	PS.TCS. 234 Page 1 de 1 Date: février 1981



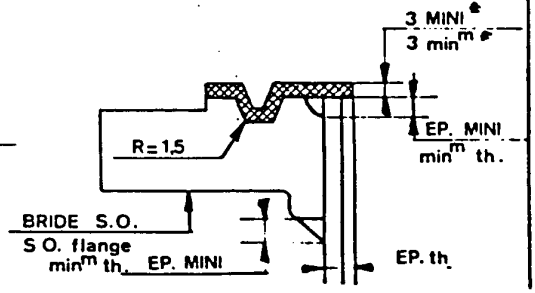
TUBULURES ACIER ALLIÉ $1\frac{1}{2}" \leq DN \leq 4"$
alloy steel necks $1\frac{1}{2}" \leq dn \leq 4"$



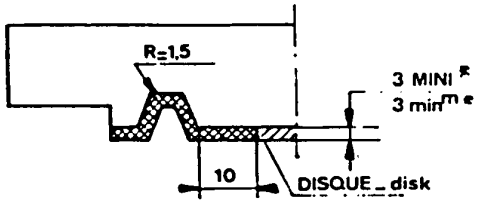
TUBULURES AVEC LINING $DN \geq 6"$
lined necks $dn \geq 6"$



BRIDES PLEINES $1\frac{1}{2}" \leq DN \leq 4"$
covers $1\frac{1}{2}" \leq dn \leq 4"$



TUBULURES AVEC CLADDING $DN \geq 6$
clad necks $dn \geq 6"$



DISQUE EN ACIER ALLIÉ - alloy steel disk

SOUDURE PAR POINTS OU PAR BOUCHON DE SOUDURE AU CENTRE POUR TUBULURES DN 10", 12" ET 14".
3 SOUDURES ÉQUIDISTANTES SUR Ø 150 POUR TUBULURES DN 16" ET 18", ET SUR Ø 250 POUR TUBULURES DN 20" ET 24".
spot or plug weld at the center for dn 10", 12" & 14" nozzles, 3 welds equidistant on 150 mm dia. circle for nozzles dn 16" & 18", and on 250 mm circle for nozzles dn 20" et 24".

BRIDES PLEINES $DN \geq 6"$ - cover $dn \geq 6"$

LEGENDE - legend

- ACIER AU CARBONE. carbon steel.
- SOUDURE ACIER AU CARBONE. carbon steel weld.
- ACIER ALLIÉ. alloy steel
- SOUDURE ACIER ALLIÉ alloy steel weld

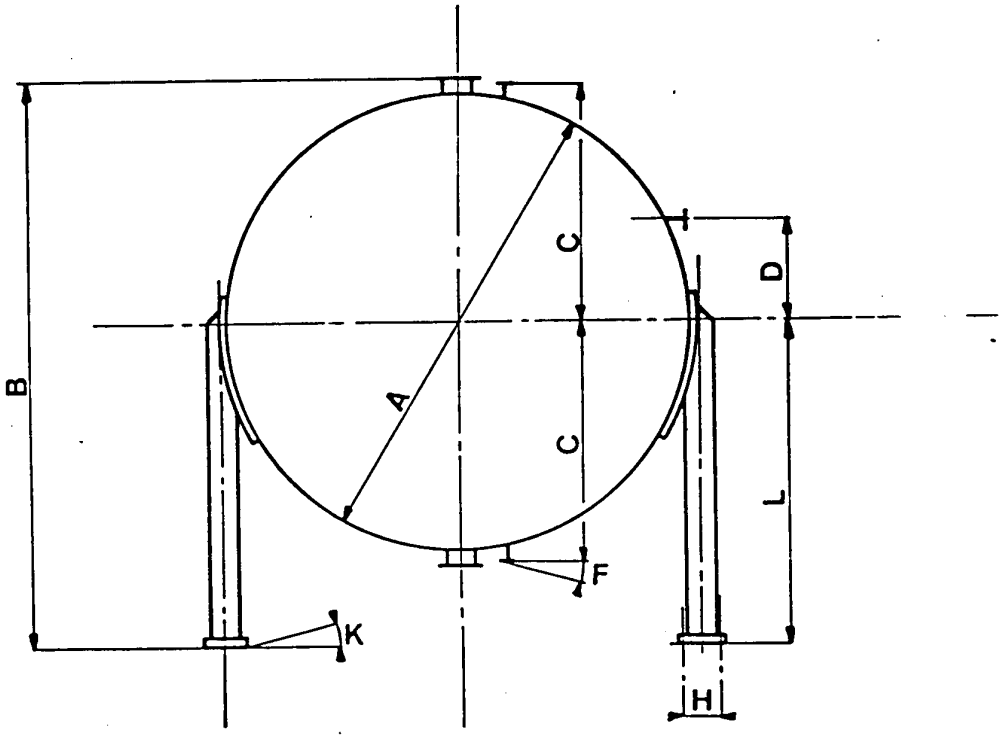
VOIR DESSIN DES APPAREILS POUR TYPE D'ACIER ALLIÉ ET NATURE DES ÉLECTRODES.
see vessel drawing for grade of alloy steel & type of electrodes.

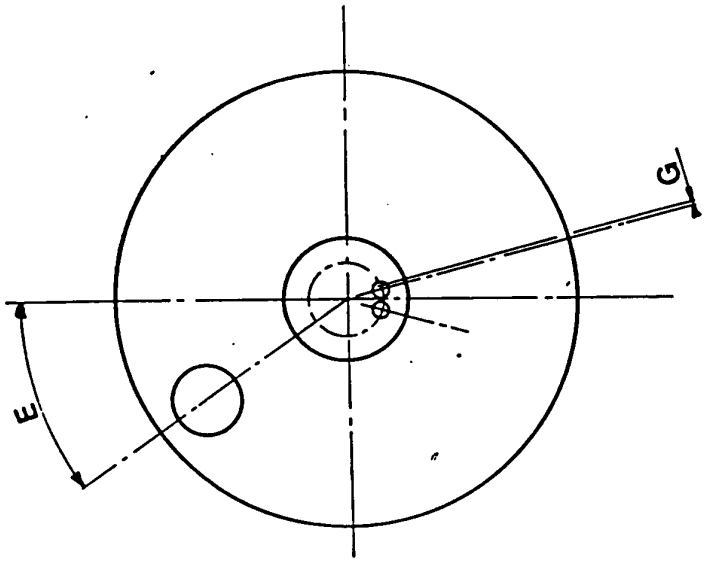
NOTE: VOIR ANSI B 16.5 POUR DIMENSIONS ET TOLÉRANCES DES FACES RING JOINT.
see Ansi B 16.5 for dimensions & tolerances of ring joint faces.

*ÉPAISSEUR UNIFORME DU METAL SAIN APRÈS USINAGE
*uniform thickness of sound metal after machining

Rev. No.	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	TOLERANCES DES RESERVOIRS SPHERIQUES SPHERICAL VESSELS TOLERANCES	PS.TCS 301 Page 1 de 2 Date: Février 1981





Rev.N°	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOLERANCES SUR LES DIMENSIONS EXTERIEURES
TOLERANCES ON OUTSIDE DIMENSIONS

A - ROTONDITE PERMISE PAR LE CODE
OUT OF ROUNDNESS AS PERMITTED BY CODE

B - TOLERANCE SUR LONGUEUR : $\pm$ 1,3 mm/m (MAX : 13 mm)
- TOLERANCE ON LENGTH : $\pm$ 1,3 mm/m (MAX : 13 mm)

C - TOLERANCE SUR LE DEPASSEMENT DES FACES DE BRIDES :
POUR LES TROUS D'HOMME : $\pm$ 6,5 mm
POUR LES TUBULURES : $\pm$ 5 mm
- FLANGE FACES PROJECTING FROM SHELL :
FOR MANHOLES : $\pm$ 6,5 mm
FOR NOZZLES : $\pm$ 5 mm

D - TOLERANCE SUR LES ELEVATIONS :
POUR LES TROUS D'HOMME : $\pm$ 13
POUR LES TUBULURES : $\pm$ 6,5 mm ENTRE LES AXES
DES CONNEXIONS DES PRISES D'INSTRUMENTS : $\pm$3 mm
- TOLERANCE ON ELEVATIONS :
FOR MANHOLES : $\pm$ 13 mm FOR NOZZLES : $\pm$ 6,5 mm
CLEARANCE BETWEEN INSTRUMENT BRANCH CONNECTIONS : $\pm$ 3mm

E - TOLERANCE SUR LA DEVIATION CIRCONFERENTIELLE ENTRE L'AXE REEL ET L'AXE THEORIQUE DES TUBULURES, TROUS D'HOMME, SUPPORT : $\pm$ 3 mm MESURES SUR LA VIROLE.
- CIRCUMFERENTIAL DEVIATION BETWEEN THE THEORICAL AXIS AND CENTERLINE OF NOZZLES, MANHOLES AND SUPPORTS SHALL NOT EXCEED $\pm$ 3 mm MEASURED ON SHELL.

F - ANGLE DU PLAN DE LA FACE D'UNE BRIDE OU D'UN SUPPORT PAR RAPPORT AU PLAN THEORIQUE : $\pm$ 1/2°
- ANGLE OF FLANGE FACE IN RATIO TO THE THEORICAL PLANE : $\pm$ 1/2°.

G - TOLERANCE SUR L'ORIENTATION DU PERCAGE DES TUBULURES : $\pm$ 1,5 mm MESURES SUR LE CERCLE DE PERCAGE.
- ORIENTATION OF NOZZLE BOLT HOLES : $\pm$ 1,5 mm MEASURED ON DRILLING CIRCLE.

K - TOLERANCE SUR PERCAGE DE L'EMBASE :
SUR DIAMETRE MESURES A 90° : $\pm$ 5 mm
ENTRE TROUS : $\pm$ 5 mm
- BASE RING DRILLING TOLERANCE :
MEASURED ON PERPENDICULAR DIAMETERS : $\pm$ 5 mm
MEASURED ON DRILLING CIRCLE BETWEEN HOLES : $\pm$ 5 mm

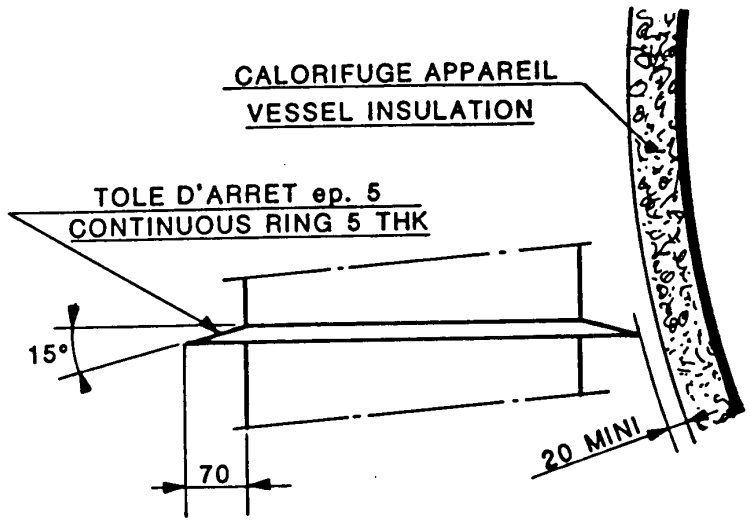
L - TOLERANCE SUR L'EQUERRAGE DE L'EMBASE :
- TOLERANCE ON BASE RING PERPENDICULARITY :

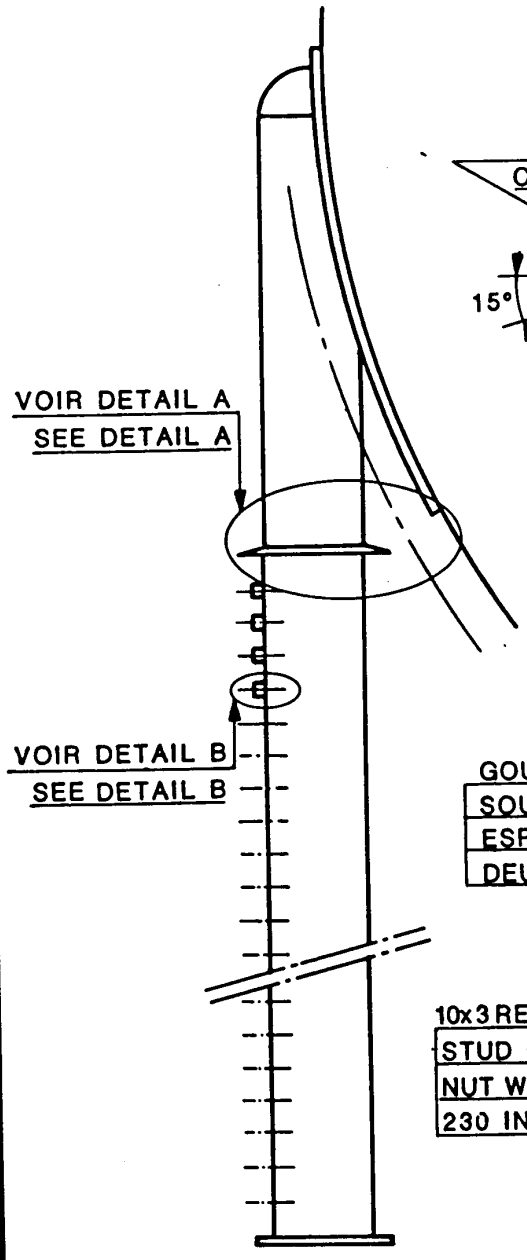
	K $\leq$ 1,2 m	: $\pm$ 3 mm
1,2 m	K $\leq$ 2,1 m	: $\pm$ 5 mm
	K $>$ 2,1 m	: $\pm$ 6,5 mm

Rev.N°	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

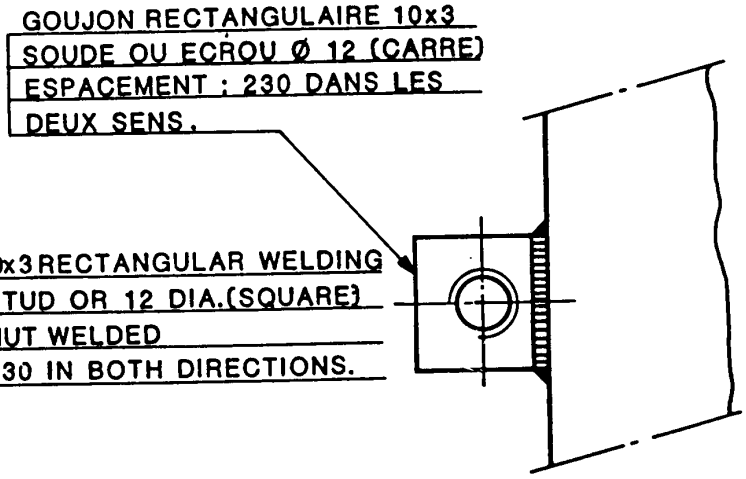
TOTAL TEP / DDP	SUPPORTS D'IGNIFUGE FIREPROOFING SUPPORTS	PS.TCS.302 Page 1 de 1 Date : Fevrier 1981

DETAIL " A "

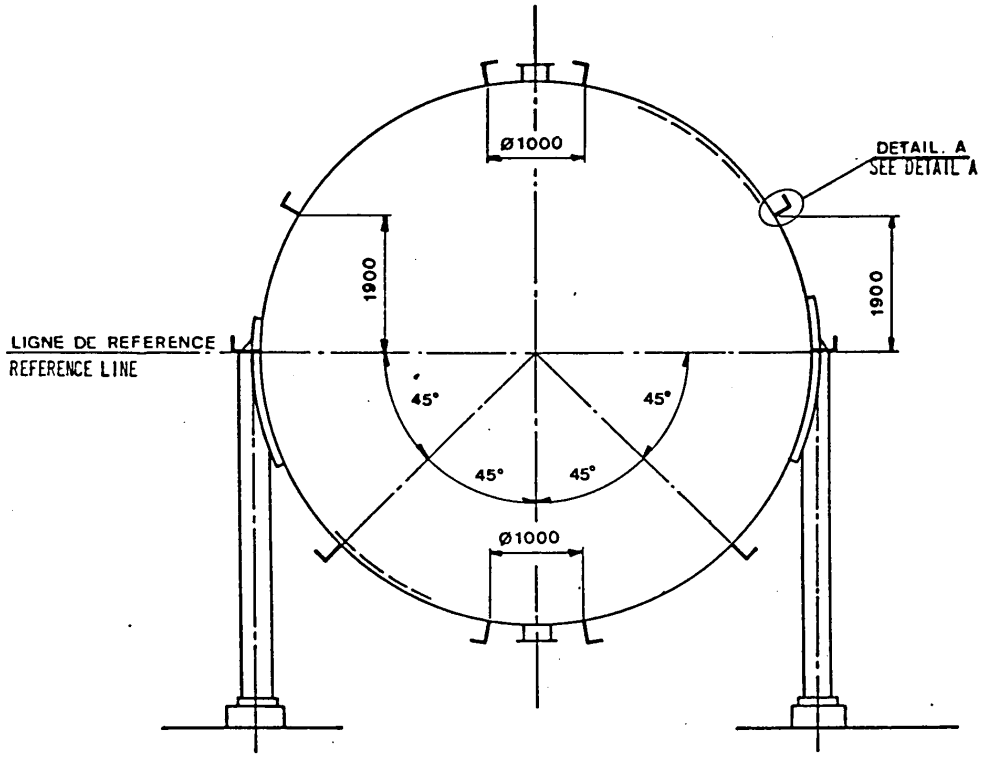




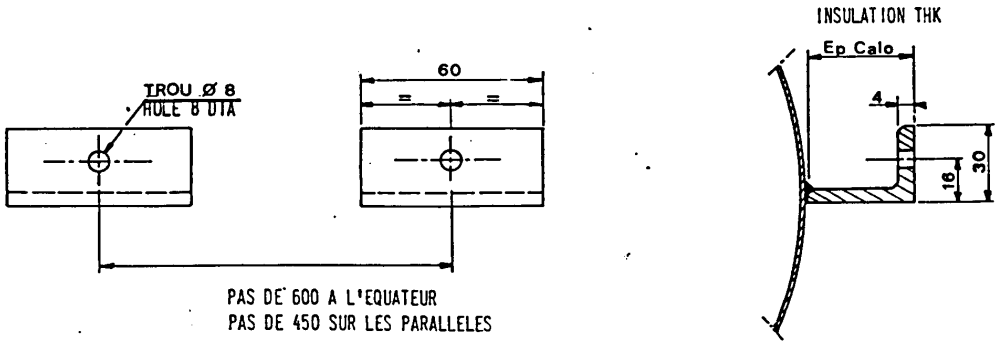
DETAIL " B "



TOTAL TEP/DD	SUPPORTS DE CALORIFUGE INSULATION SUPPORTS	PS.TCS 303 Page 1 de 1 Date: Février 1981



-DETAIL A-

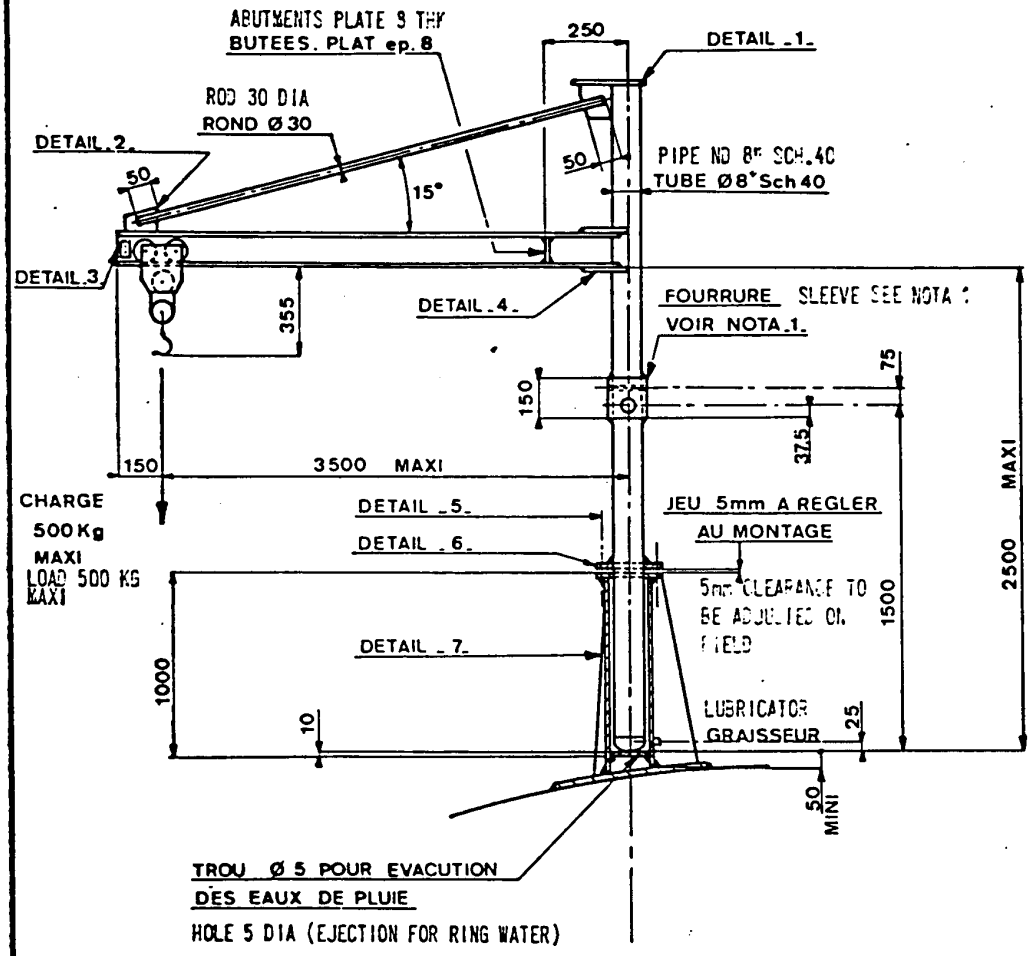


PAS DE 600 A L'EQUATEUR
PAS DE 450 SUR LES PARALLELES

600 SPACE AT EQUATOR
450 SPACE ON THE PARALLEL

Rev.N°	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	POTENCE MONORAIL ET GLISSIERE MONORAIL DAVIT AND SLIDE	PS.TCS 304 Page 1 de 5 Date: Février 1981



NOTA.

1) FOURRURE: Ep:10 SOUDEE AYANT 2 TROUS Ø 24mm A 90°
POUR BROCHE DE MANŒUVRE.
SLEEVE 10 THK WELDED WITH 2 HOLES 24 DIA AT 90° FOR HINGE PIN.

2) MATIERE: TUBE. ASTM A53 GRADE A OU B OU EQUIVALENT.
TÔLES. SUIVANT PLAN D'APPAREIL MAIS LA TOLE DE
RENFORT SERA DE MÊME MATIERE QUE L'APPAREIL.
MATERIAL – PIPE A 53 GRADE A OR B OR EQUAL PLATE ACCORDING TO VESSEL DRAWING.
BACK-UP PLATE – USE GRADE OF VESSEL.

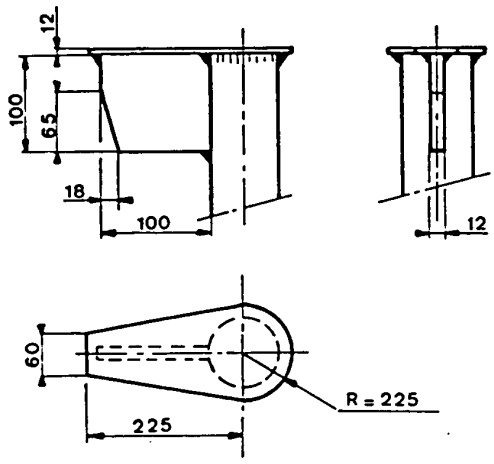
3) SUR LA POTENCE LA VALEUR DE LA CHARGE POUVANT ETRE
LEVEE DEVRA ETRE APPOSEE VISIBLEMENT PAR LE CONSTRUCTEUR.
ON THE DAVIT THE VALUE OF THE LOAD THAT CAN BE LIFTED WILL BE WISIBLY
INDICATED BY CONSTRUCTOR.

4) LA FOURNITURE NE COMPREND NI LE CHARIOT PORTE-PALAN
NI LE PALAN:
THE SUPPLY NEITHER INCLUDES THE TRAVELLING BLOCK NOR THE HOIST.

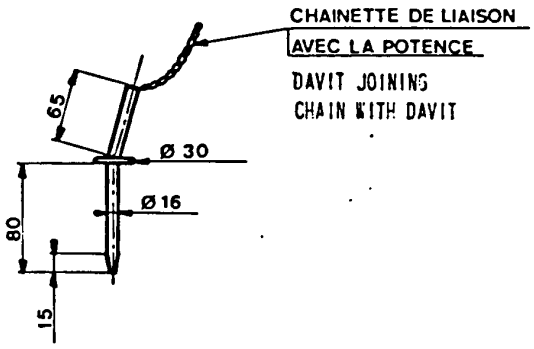
Rev.N°	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP/DD	POTENCE MONORAIL ET GLISSIERE MONORAIL DAVIT AND SLIDE	PS.TCS 304 Page 2 de 5 Date: Février 1981

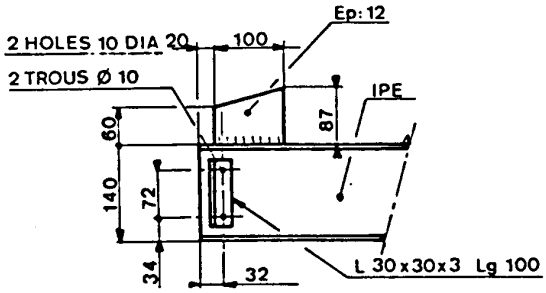
DETAIL 1



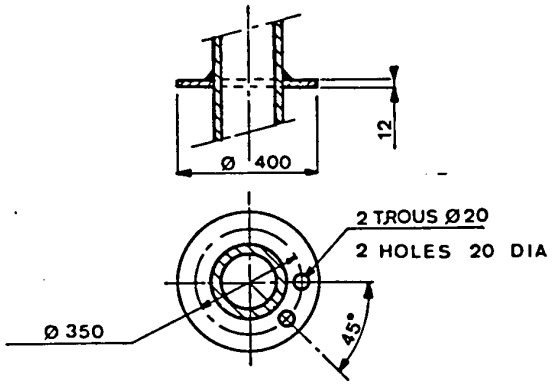
DETAIL 5



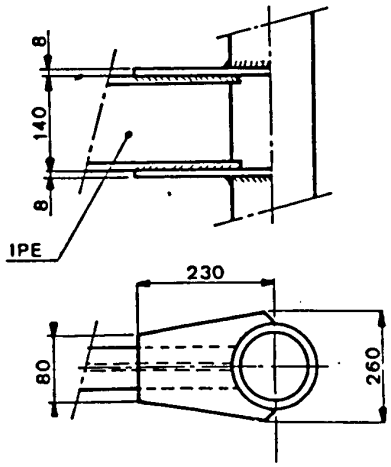
DETAIL 2 & 3



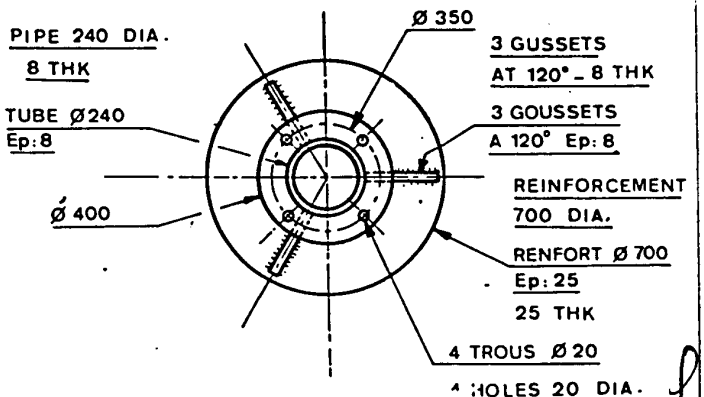
DETAIL 6



DETAIL 4

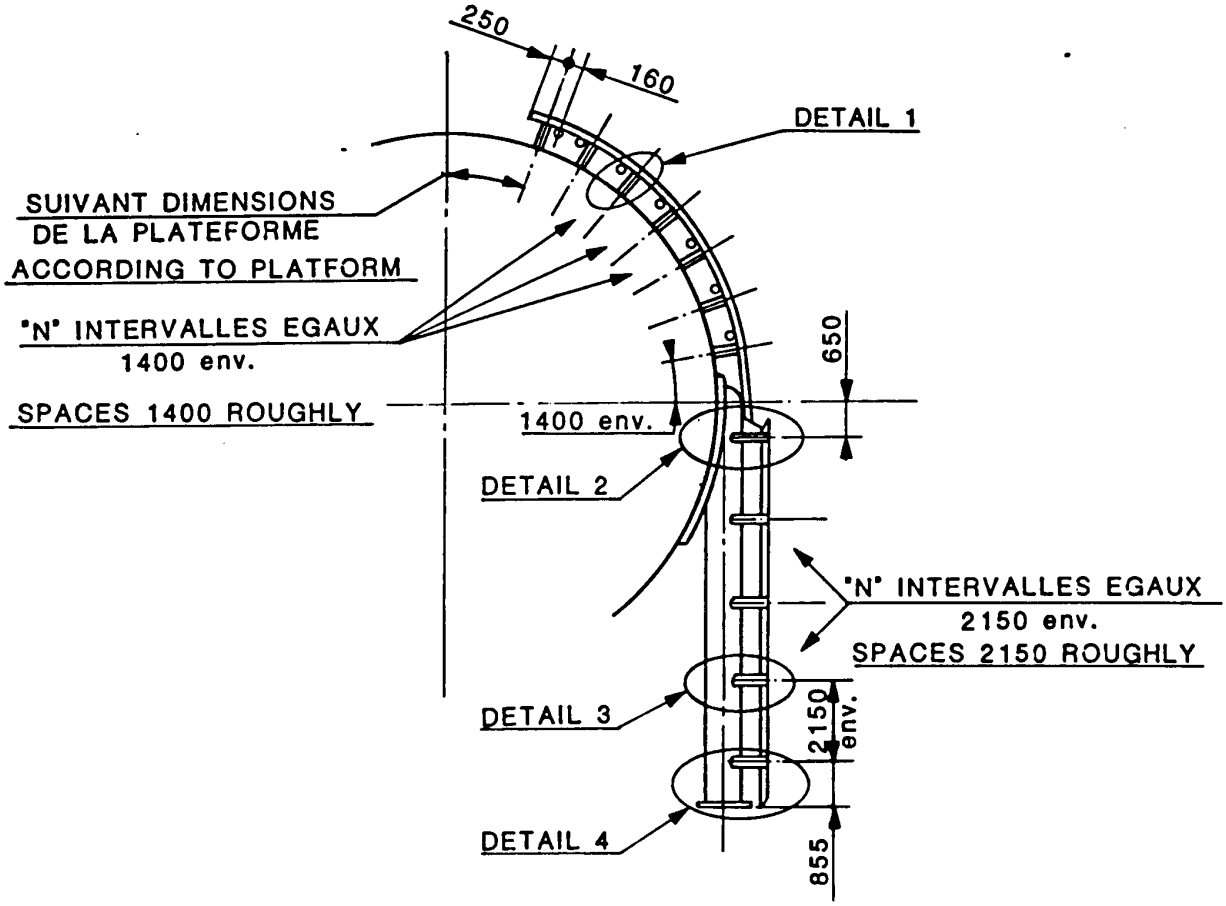


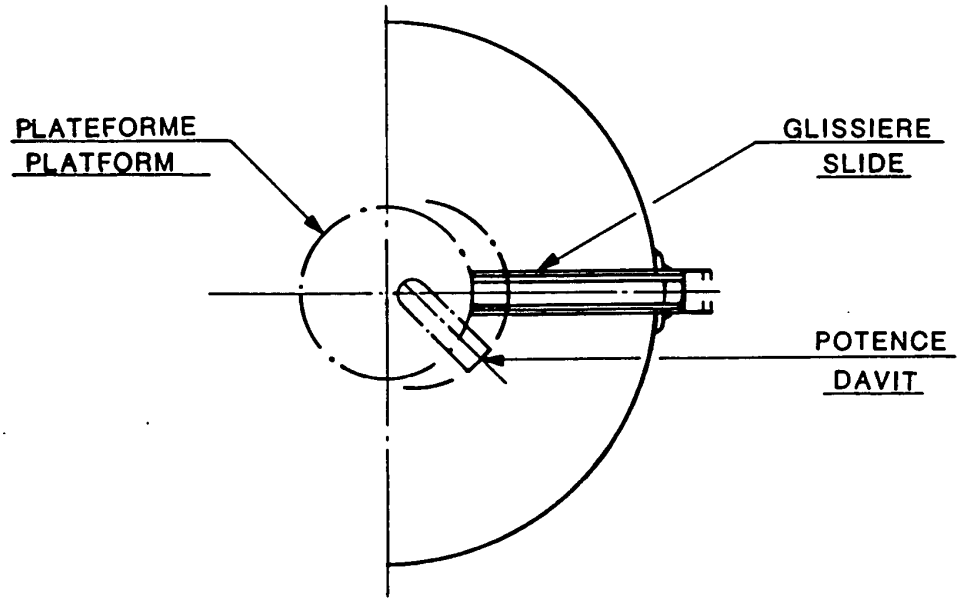
DETAIL 7



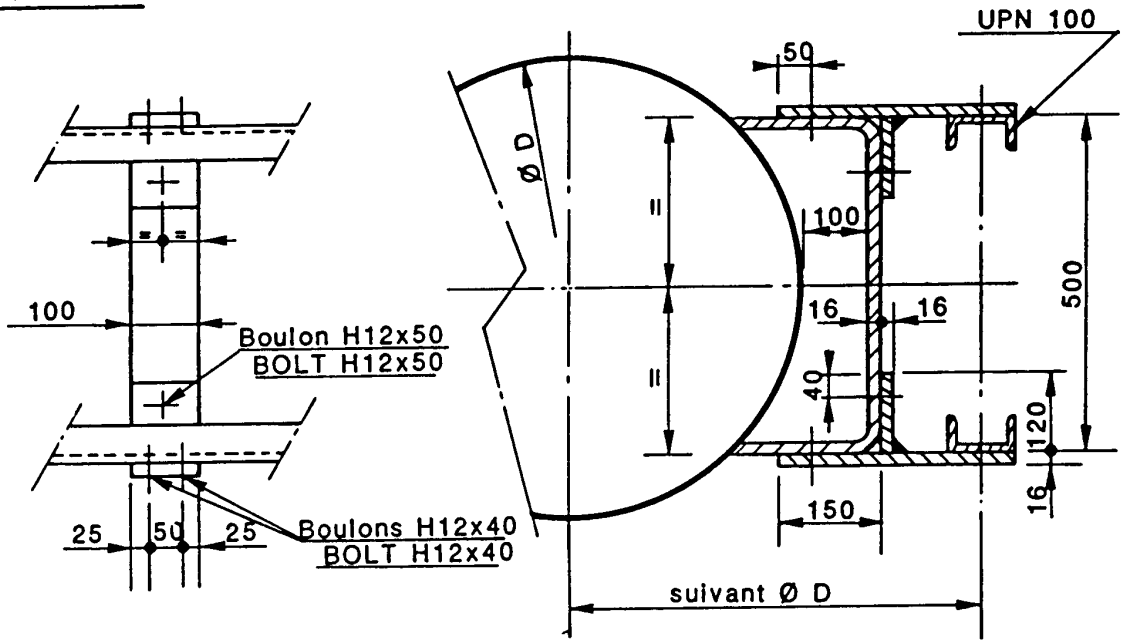
Rev.N°	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

TOTAL TEP / DDP	POTENCE MONORAIL ET GLISSIERE MONORAIL DAVIT AND SLIDE	PS.TCS.304 Page 3 de 5 Date : Avril 1981

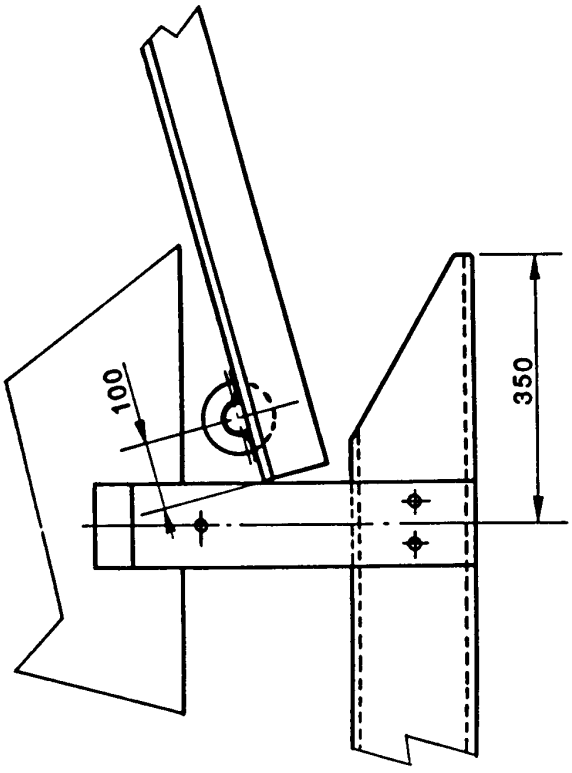




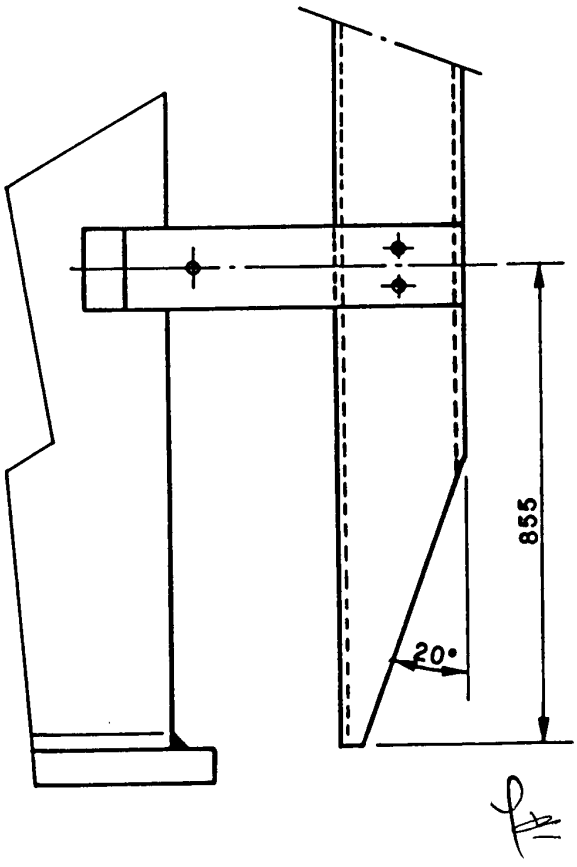
DETAIL 3



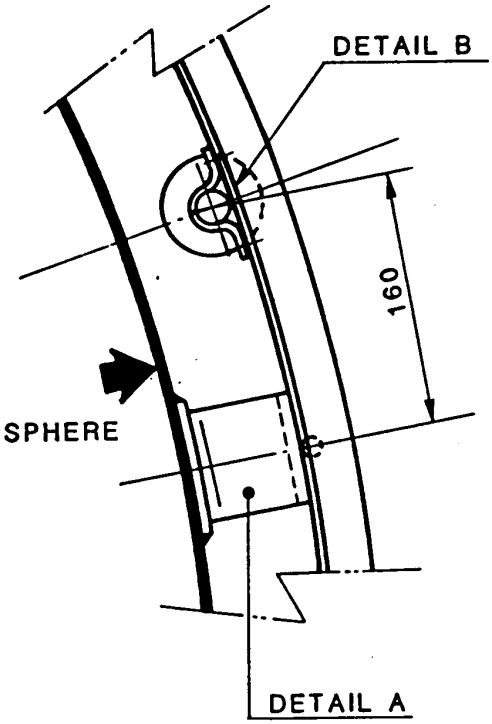
DETAIL 2



DETAIL 4

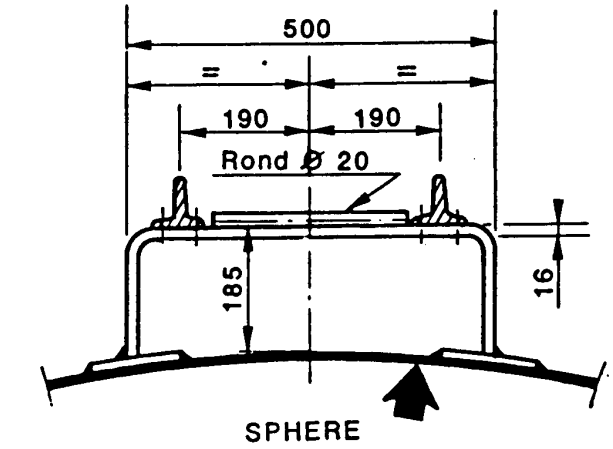


TOTAL TEP / DDP	POTENCE MONORAIL ET GLISSIERE MONORAIL DAVIT AND SLIDE	PS.TCS.304 Page 5 de 5 Date : Avril 1981

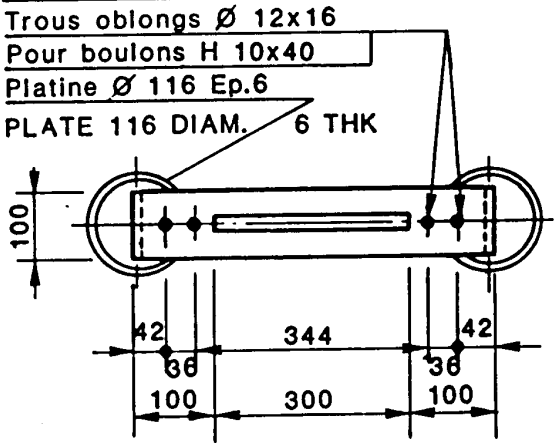


DETAIL 1

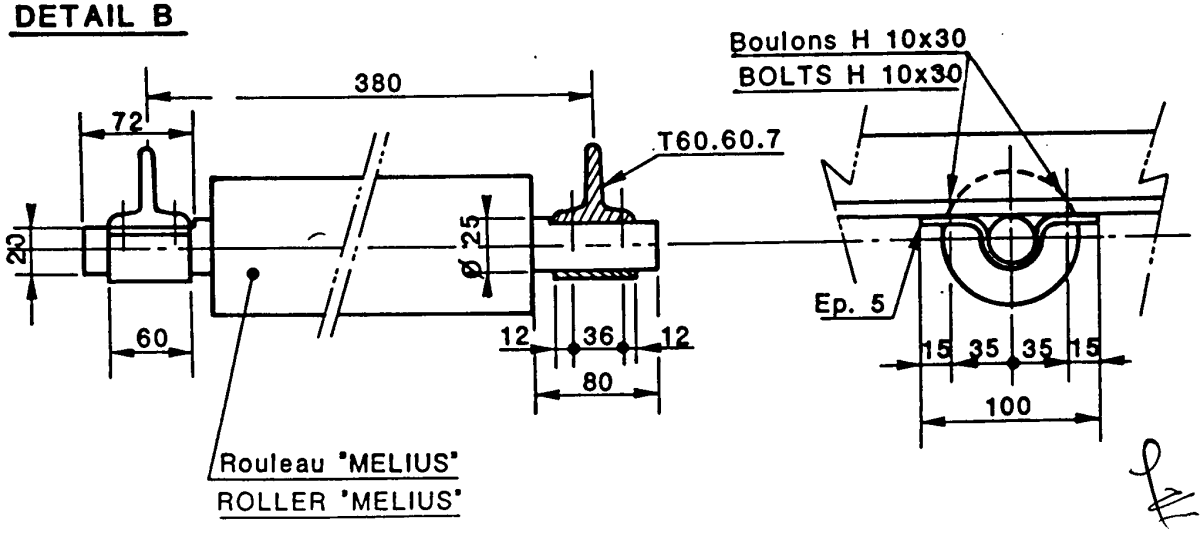
DETAIL A



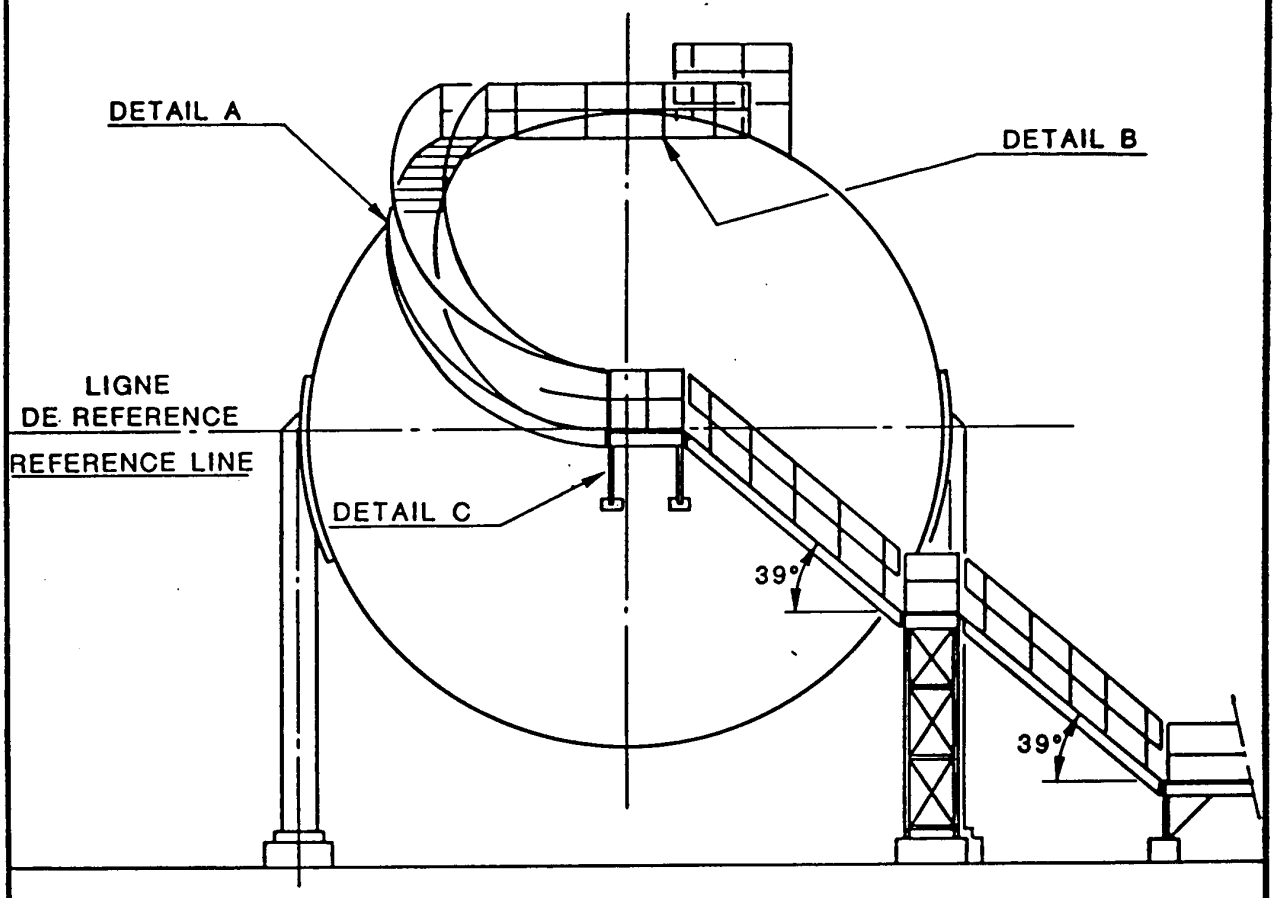
OVALIZED HOLES 12 DIA.x16
FOR BOLTS H 12x40
Trous oblongs Ø 12x16
Pour boulons H 10x40
Platine Ø 116 Ep.6
PLATE 116 DIAM. 6 THK

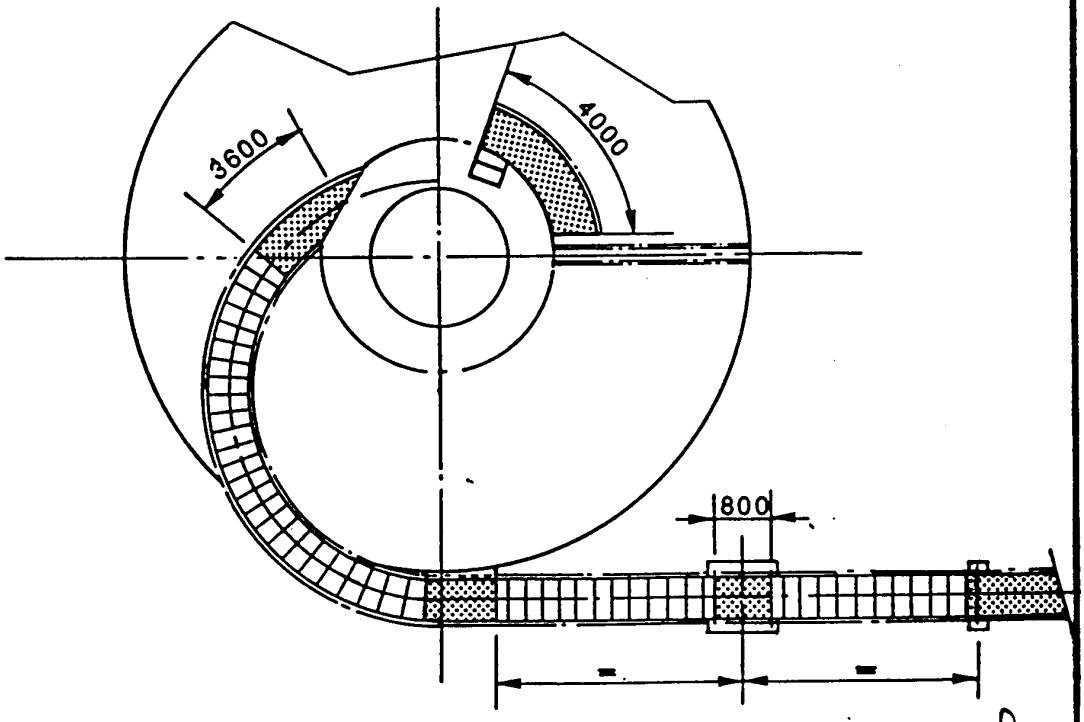


DETAIL B



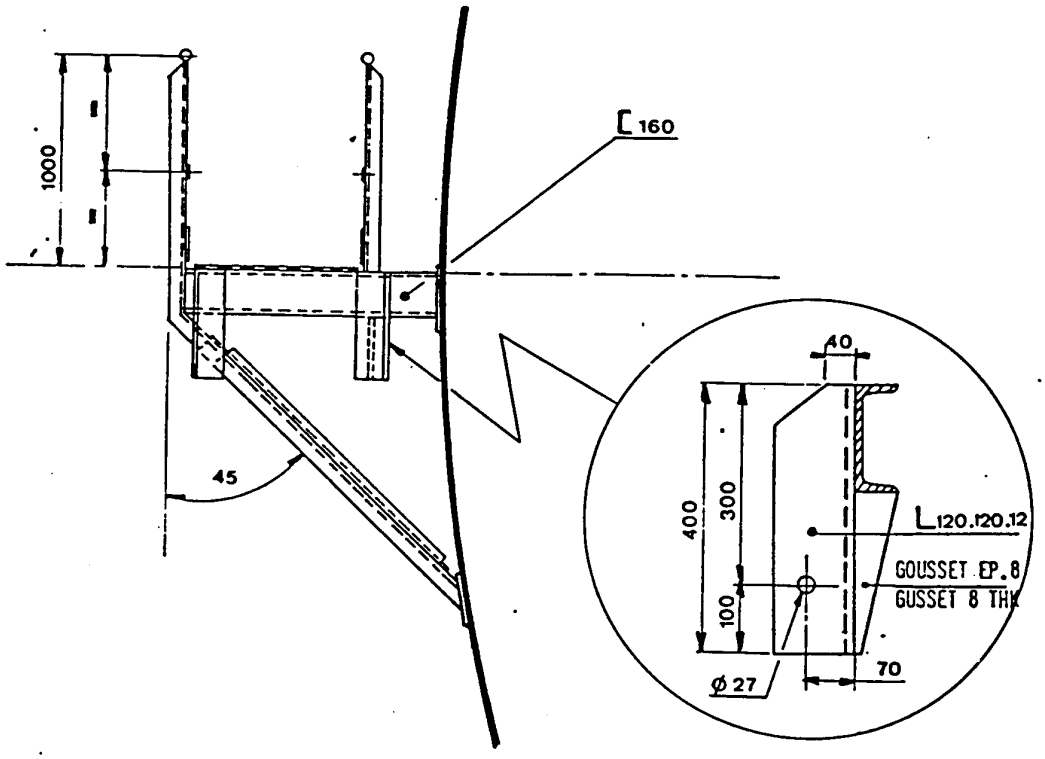
TOTAL TEP / DDP	PLATE-FORME ET ESCALIERS PLATFORM AND STAIRWAY	PS.TCS.305 Page 1 de 2 Date : Avril 1981



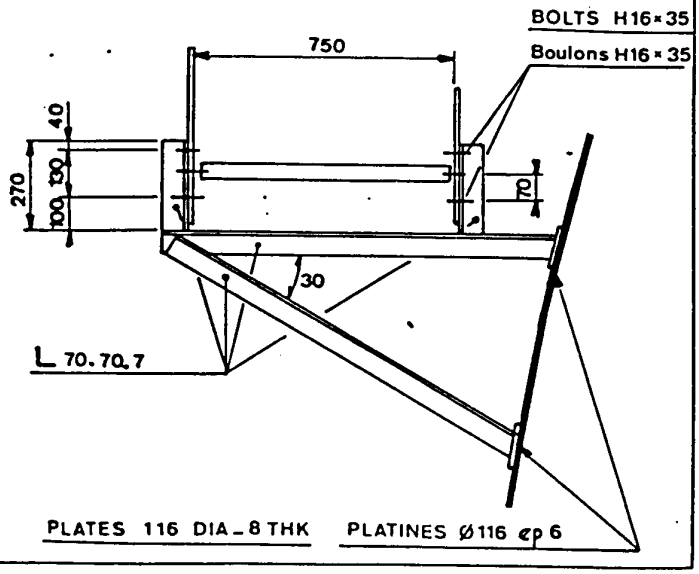


TOTAL TEP/DD	PLATE-FORME ET ESCALIER PLATFORM AND STAIRWAY	PS.TCS 305 Page 2 de 2 Date: Février 1981

DETAIL C

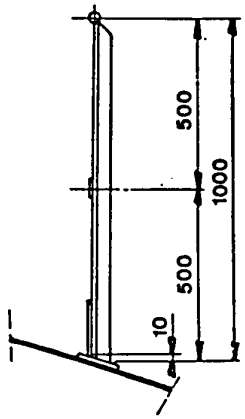


DETAIL A



PLATES 116 DIA - 8 THK PLATINES Ø116 ep 6

DETAIL B

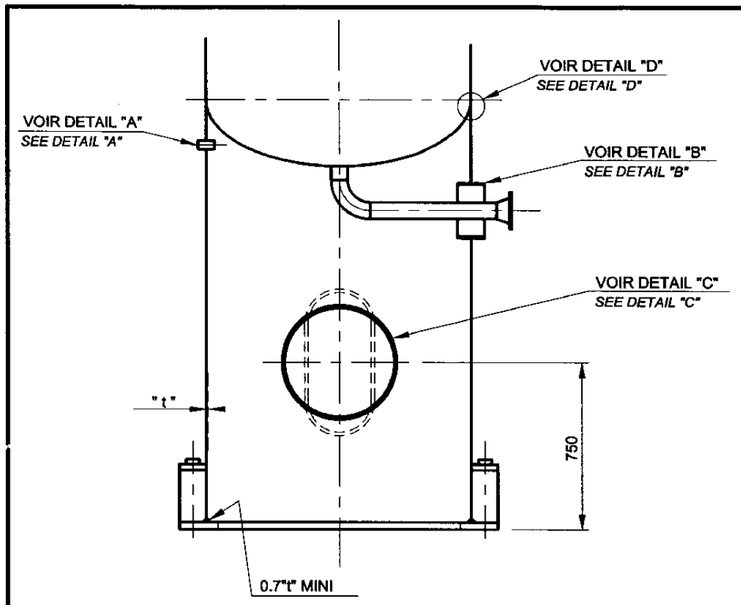


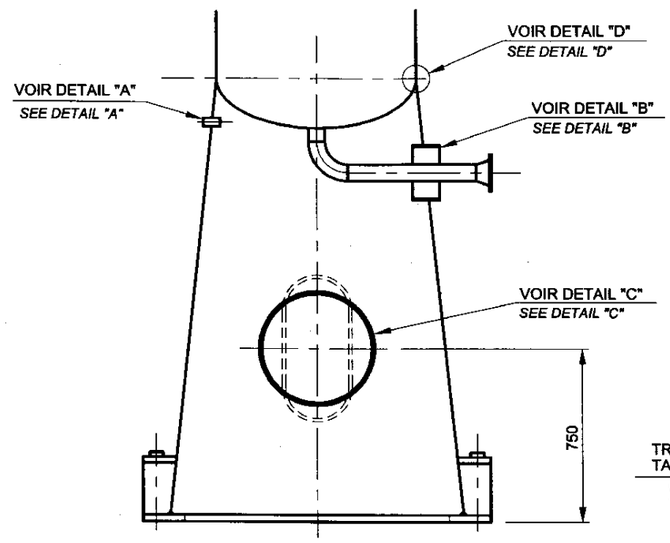
Rev.N°	DATE	OBJET DE LA REVISION	REDACTEUR	RESPONSABLE

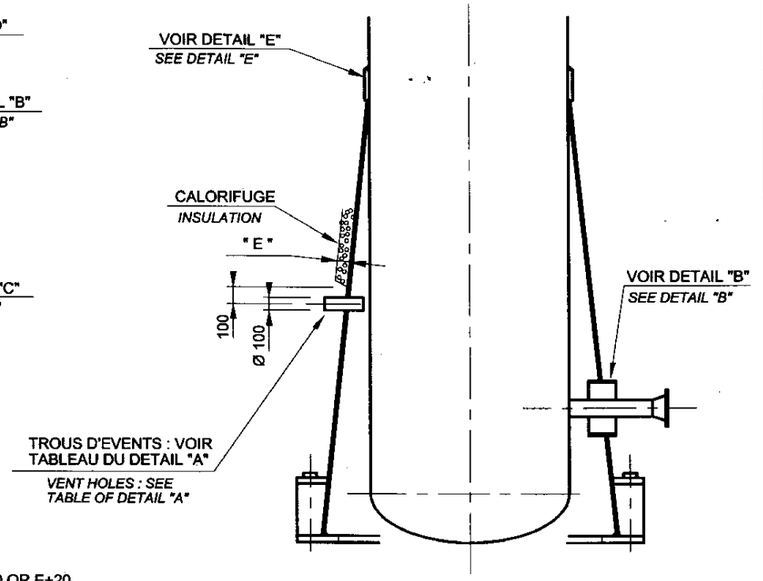
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Indice de revision du Standard general	0
Date: 20-12-2000	Visa: JP.B

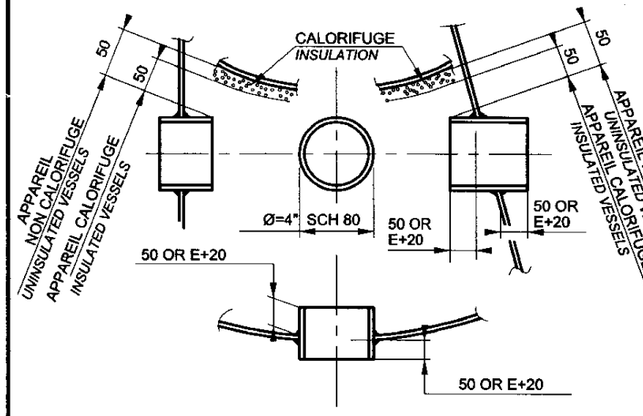
revision du Standard applique a l'activite	
Date:	Visa:





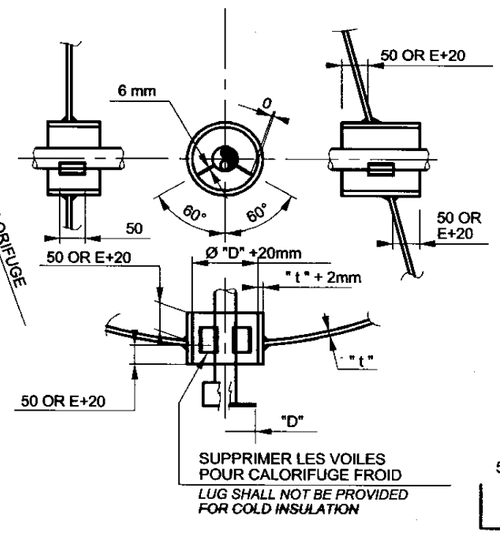


DETAIL "A"

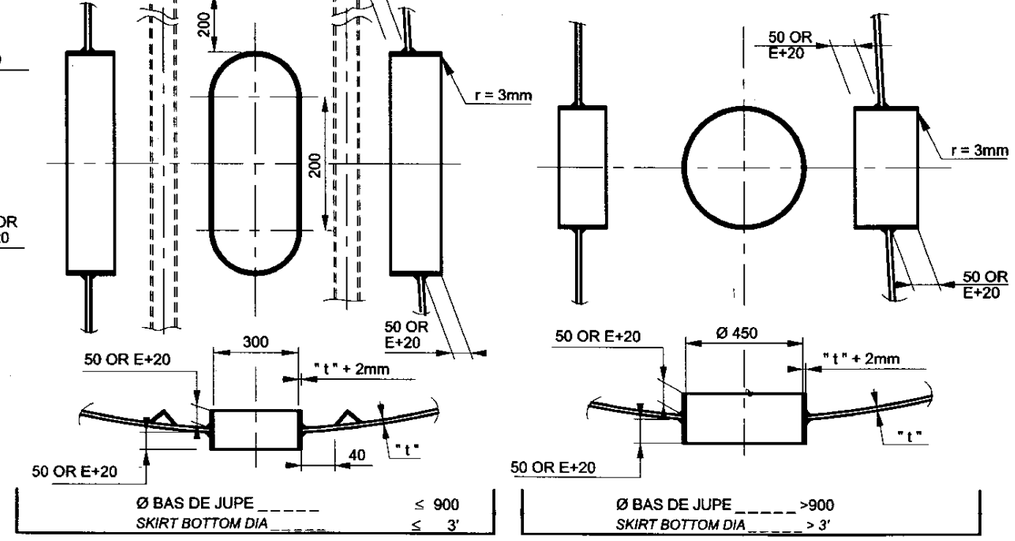


Ø VIROLE *SHELL DIA.*	Ø ≤ 1000	1000 < Ø ≤ 2000	Ø > 2000
NOMBRE D'EVENTS *NUMBER OF VENTS*	2	3	4

DETAIL "B"



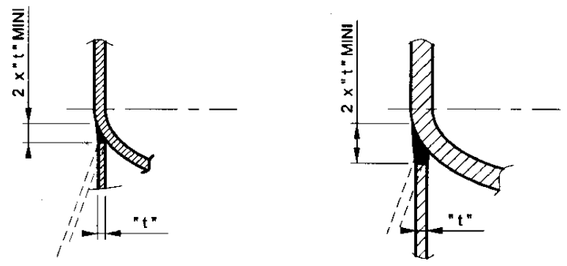
DETAIL "C"



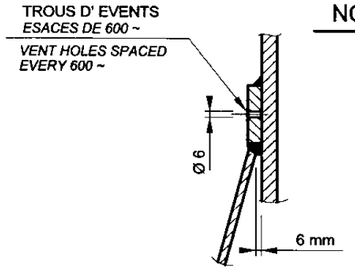
Ø BAS DE JUPE _____ ≤ 900
SKIRT BOTTOM DIA _____ ≤ 3'

Ø BAS DE JUPE _____ >900
SKIRT BOTTOM DIA _____ > 3'

DETAIL "D"



DETAIL "E"



Ø BAS DE JUPE *SKIRT BOTTOM DIA*	Ø ≤ 3000	Ø > 3000
TROUS D'ACCES *ACCESS HOLES*	1	2

NOTE:

E=EPAISSEUR CALORIFUGE OU IGNIFUGE
E = THICKNESS FIRE PROOF OR INSULATION

TECHNIP

STANDARD DE CONSTRUCTION
CONSTRUCTION STANDARD

APPAREILS A PRESSION
PRESSURE VESSELS

JUPES-TROUS D'ACCES-EVENTS

SKIRTS-ACCESS HOLES-VENTS

PARS OIL AND GAS Co.

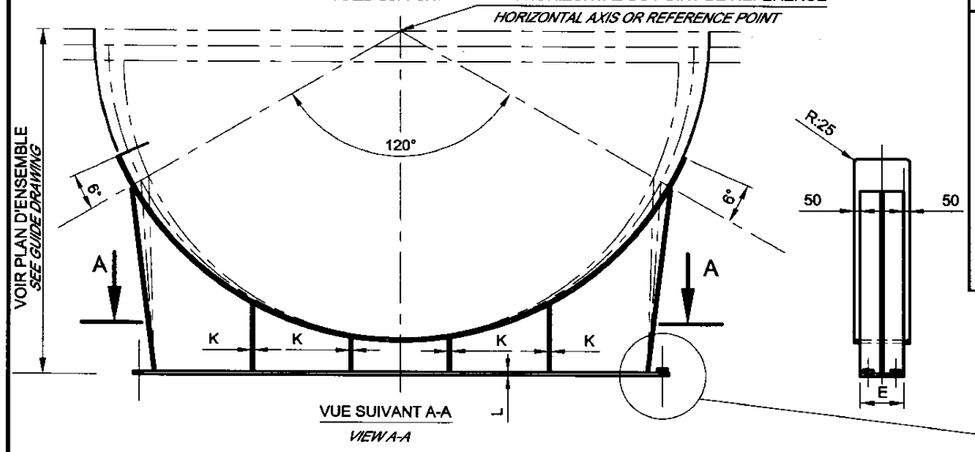
PROJECT N°	UNIT	DOC. TYPE	DISCIP.	SUBJ.	SERIAL N°	REV.
6340S	000	STC			000400301	0

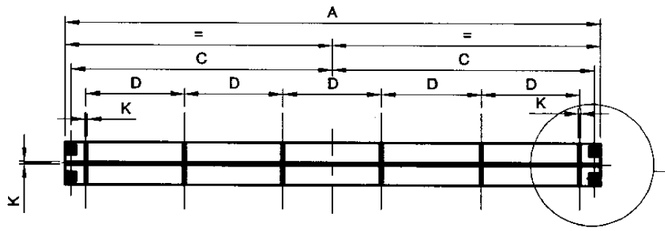
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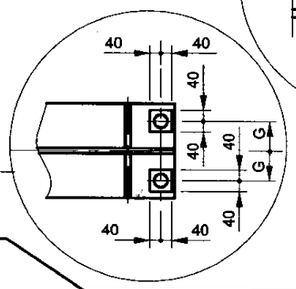
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Date: 20-12-2000	Visa: JP.B

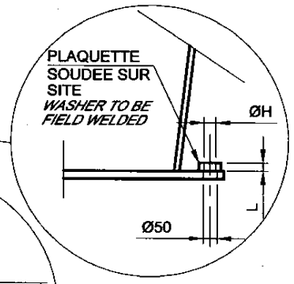
revision du Standard applique a l'activite	
Date:	Visa:

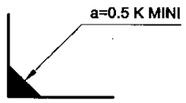
SUPPORT FIXE
FIXED SUPPORT





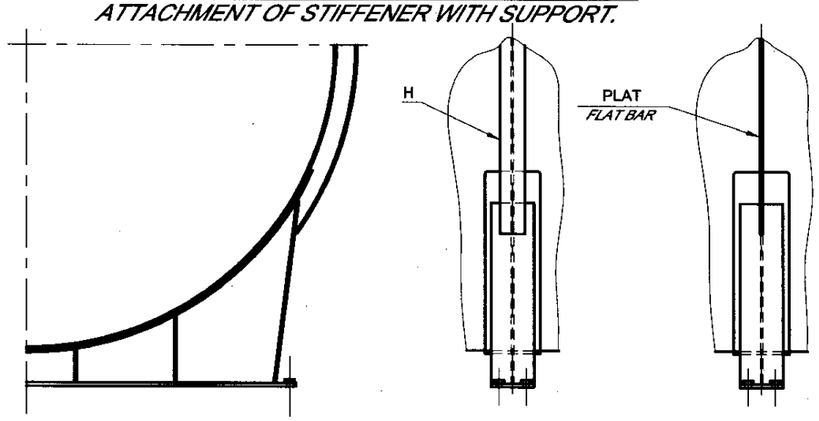




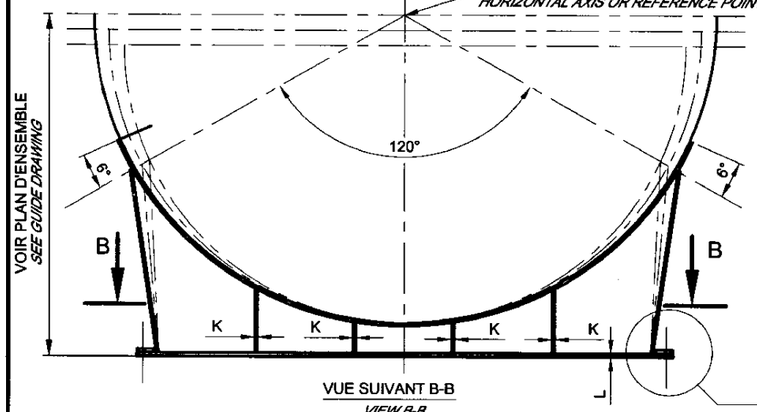


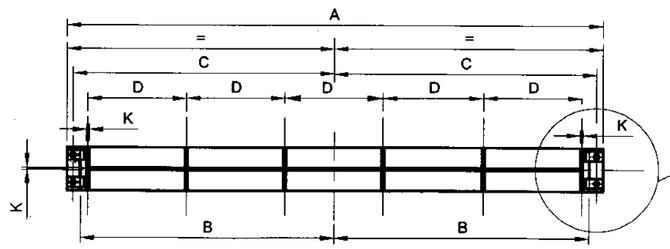
	Ø APPAREIL *VESSEL Ø*	A	B	C	D	E	F	G	H	Ø BOULON *BOLT DIAM.*	K	L	M	P	POIDS (KG) *WEIGHT (KG)*
TYPE V	3800≤Ø<4000	3385	1602.5	1642.5	625	300	30	100	33	27	16	20	60	50	665
	4000≤Ø<4500	3850	1825	1875	710	350	35	110	39	33	20	25	70	50	1095
	4500≤Ø<5000	4340	2060	2120	800	350	40	110	42	36	20	30	90	50	1295

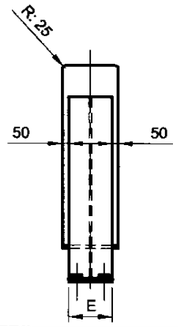
ACCOSTAGE D'UNE FRETTE AVEC UN SUPPORT.
ATTACHMENT OF STIFFENER WITH SUPPORT.

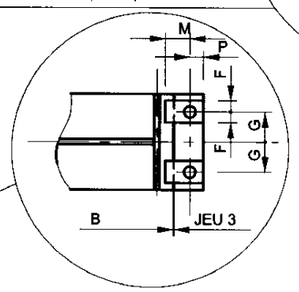


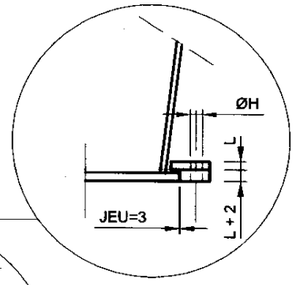
SUPPORT MOBILE
SLIDING SUPPORT

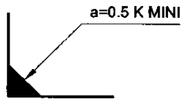












NOTES :

-LA FOURRURE AURA LA MEME EPAISSEUR QUE LA VIROLE
BUSHING SHALL HAVE SAME THICKNESS AS SHELL

-TOUTES LES SOUDURES DES SUPPORTS SERONT CONTINUES DE MEME QUE LA LIAISON FOURRURE -APPAREIL
ALL SUPPORT WELDS AND BUSHING -VESSEL WELDING SHALL BE CONTINUOUS WELDS

-PREVOIR SUR LA FOURRURE 2 TROUS Ø6 POUR EVENT
BUSHING SHALL BE PROVIDED WITH 2 HOLES DIA. 6 FOR VENT

-LE POIDS EST DONNE POUR 1 SUPPORT
WEIGHT IS GIVEN FOR 1 SUPPORT

-LA MATIERE DE LA FOURRURE SERA DE MEME NUANCE QUE L'APPAREIL SAUF INDICATION CONTRAIRE PORTEE SUR LES PLANS
BUSHING MATERIAL SHALL BE SAME GRADE AS VESSEL UNLESS OTHERWISE SPECIFIED

-TOUTE LES DIMENSIONS SONT EN MM
ALL DIMENSIONS IN MM

TECHNIP

STANDARD DE CONSTRUCTION
CONSTRUCTION STANDARD

APPAREILS A PRESSION
PRESSURE VESSELS

PARS OIL AND GAS Co.

BERCEAUX POUR APPAREILS HORIZONTAUX 3800≤Ø<5000
SUPPORT SADDLES FOR HORIZONTAL VESSELS 3800≤Ø<5000

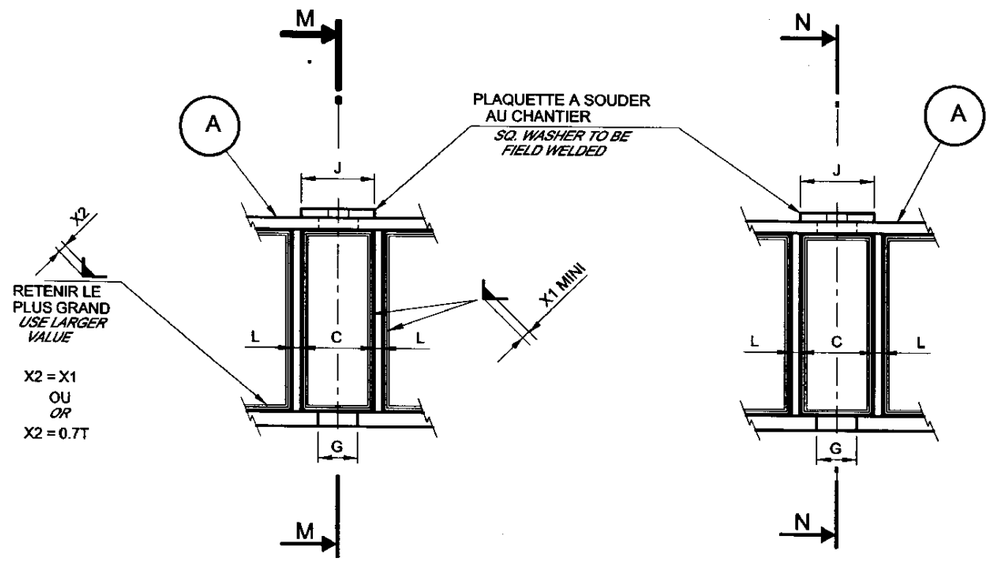
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6340S 000	STC	000400303-1				0

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JUPE DROITE / *CYLINDRICAL SKIRT*



JUPE CONIQUE / *CONICAL SKIRT*

NOTE.3

DIA BOULONS D'ANCRAGE (2)	24*	27*	30	33*	36*	39	42*	45	48*	52*	56	60*	64	68	72*
NEAREST ANCHOR BOLT SIZE	*1"*	*1.1/8"*	-	*1.1/4"*	*1.3/8"*	*1.1/2"*	*1.5/8"*	*1.3/4"*	*1.7/8"*	*2"*	*2.1/4"*	*2.3/8"*	*2.1/2"*	*2.3/4"*	*2.7/8"*
SECTION RESISTANTE CM^2	3.53	4.59	5.61	6.94	8.17	9.76	11.21	13.06	14.73	17.58	-	23.62	26.76	-	34.6
TRANSVERSAL WORKING AREA SQ.IN	*0.551*	*0.728*	-	*0.929*	*1.155*	*1.405*	*1.680*	*1.980*	*2.3*	*2.65*	*3.42*	*3.85*	*4.29*	*5.26*	*5.78*
A	ANNEAU SUPERIEUR CONTINU / *CONTINUOUS UPPER RING*														
B	160	160	160	160	180	180	180	180	180	210	210	210	210	230	230
C	100	100	100	100	120	120	120	120	120	135	135	135	135	155	155
D	215	215	215	215	235	250	250	250	250	290	290	290	290	310	310
E MINI (1)	20	20	20	20	20	25	25	25	30	30	30	35	35	40	45
F MINI (1)	16	16	20	20	25	25	30	30	30	35	35	35	40	40	45
H	28	31	34	37	40	43	46	49	52	56	60	64	68	72	76
G	54	57	60	63	66	69	72	75	78	87	91	95	99	103	107
J	100	100	100	100	120	120	120	120	120	140	140	140	140	160	160
K	90	90	90	90	100	100	100	100	100	110	110	110	110	120	120
L MINI (1)	12	12	12	12	14	14	14	14	14	16	16	16	16	16	16
SOUDURE "X1" *WELD*	6	6	6	6	6	6	6	6	6	8	8	8	8	8	8
R	250		250		300		300		300		350		350		400

NOTES:

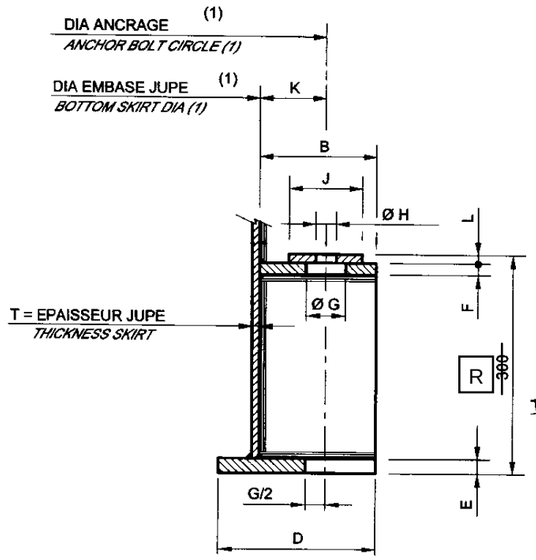
(1) -VOIR PLAN D'ENSEMBLE
-SEE ASSEMBLY DRAWING

(2) -BOULONS MARQUES * SONT RECOMMANDES POUR STANDARDISATION GENIE CIVIL
*-ANCHOR BOLTS MARKED * ARE RECOMMENDED FOR STANDARDIZATION OF "CIVIL WORK"*

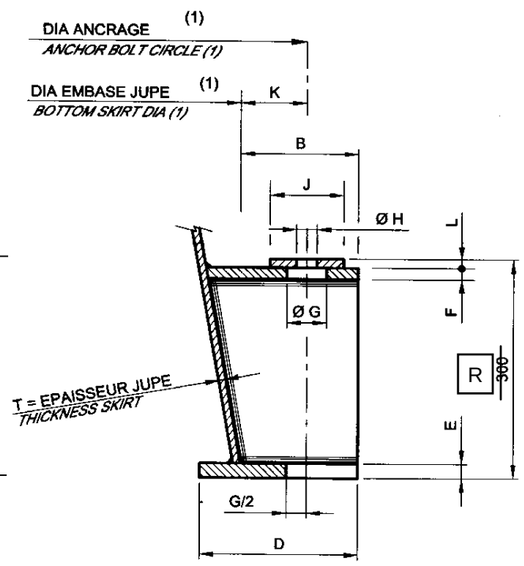
(3) ALL MARKED UP (/) SIZES ARE NOT APPLICABLE

(4) IF BOLT SIZE IS MORE THAN M72, ANCHOR CHAIRS STANDARD SHALL BE PROVIDED ACCORDING TO 6340S-L-C015

SECTION -MM-



SECTION -NN-



TECHNIP

STANDARD DE CONSTRUCTION / *CONSTRUCTION STANDARD*

APPAREILS A PRESSION / *PRESSURE VESSELS*

CHAISES D'ANCRAGE

ANCHOR CHAIRS

PARS OIL AND GAS Co.

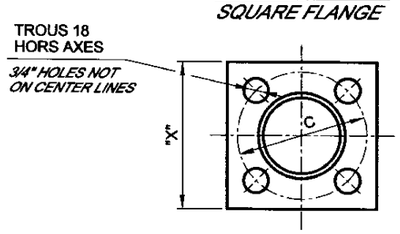
PROJECT N°	UNIT	DOC. TYPE	DISCIP.	SUBJ.	SERIAL N°	REV.
6340S	000	STC			000400304	0

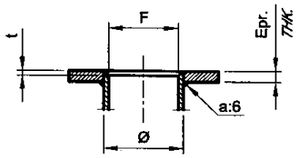
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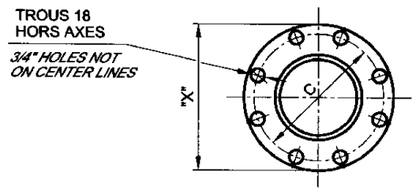
revision du Standard appliqué a l'activite	
Date:	Visa:

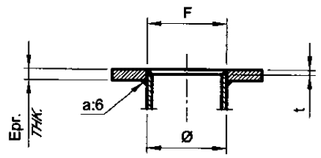
BRIDE CARREE
SQUARE FLANGE





BRIDE RONDE
ROUND FLANGE





		Ø INTERIEUR BRIDE *INSIDE DIAMETER* "F"	Ø PERCAGE *BOLT CIRCLE* "C"	"X"	BOULONS *BOLTS* Nb	BOULONS *BOLTS* Ø	EPR. BRIDE * *FLANGE THICKNESS* * A.C *C.S*	EPR. BRIDE * *FLANGE THICKNESS* * INOX MONEL *S.S. OR MONEL*	A.C *C.S* *t*	INOX OU MONEL *S.S OR MONEL* *t*
BRIDE CARREE *SQUARE FLANGE*	1"1/2	50 mm	100	110	4	16	16	10	6	4
	2"	62 mm	110	120	"	"	"	"	"	"
	3"	91 mm	140	140	"	"	"	"	"	"
	4"	116 mm	170	170	"	"	"	"	"	"
	6"	170 mm	230	210	"	"	"	"	"	"
	8"	223 mm	270	260	"	"	"	"	"	"
BRIDE RONDE *ROUND FLANGE*	10"	277 mm	340	390	8	"	20	12	"	"
	12"	328 mm	390	440	"	"	"	"	"	"
	14"	359 mm	420	470	"	"	"	"	"	"
	16"	410 mm	470	520	"	"	22	16	10	6
	18"	462 mm	520	570	12	"	"	"	"	"
	20"	512 mm	570	620	"	"	"	"	"	"
	24"	614 mm	670	720	"	"	"	"	"	"

NOTA:
NOTE

- DANS LE CAS D'UTILISATION DE JOINTS LA FOURNITURE DE CES DERNIERS SERONT A LA CHARGE DU CONSTRUCTEUR DE L'APPAREIL.
  IF GASKETS ARE USED THEY SHALL BE SUPPLIED BY THE VESSEL MANUFACTURER.
- * EPAISSEUR DE CORROSION PRISE EN COMPTE 3MM
- * *CORROSION THICKNESS TAKING INTO ACCOUNT 3MM*

TECHNIP

STANDARD DE CONSTRUCTION
CONSTRUCTION STANDARD

APPAREILS A PRESSION
PRESSURE VESSELS

BRIDES DE TUYAUTERIES INTER. DEMONTABLES

FLANGES FOR REMOVABLE INTERNAL PIPING

PARS OIL AND GAS Co.

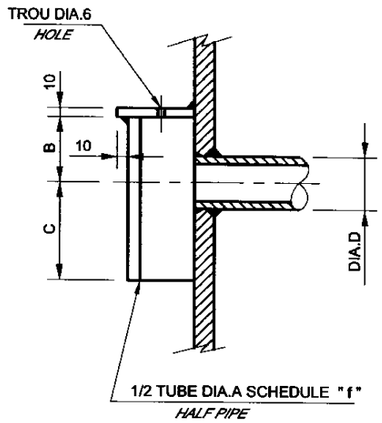
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6340S	000	STC			000400401	0

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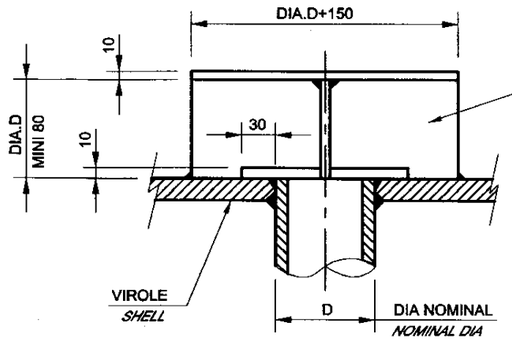
DEFLECTEUR POUR TUBULURES DE NIVEAU
VORTEX BREAKER FOR LEVEL NOZZLES

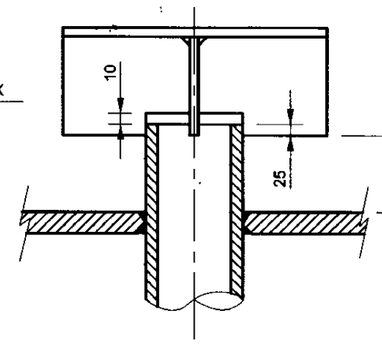


DIA.D	DIA.A	B	C	SCHEDULE "f" EPAISSEUR DE CORROSION / CORROSION ALLOWANCE 1.5mm	3mm	6mm
≤ 2"	4"	60	90	XS	XXS	XXS
3"	8"	80	90	XS	XS	XXS
4"	10"	100	100	XS	XS	XXS

ANTI-TOURBILLON
VORTEX BREAKER

TYPE 1 *

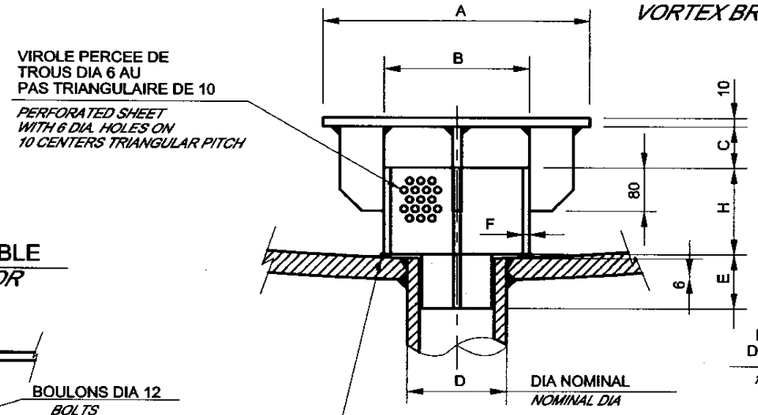


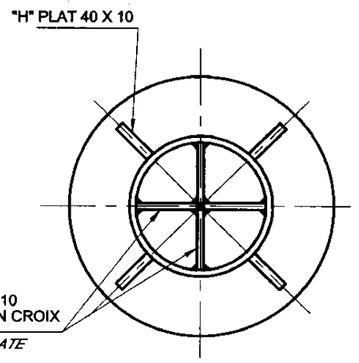


VOIR PLAN D'ENSEMBLE
REFER TO GENERAL DWG.

ANTI-TOURBILLON
VORTEX BREAKER

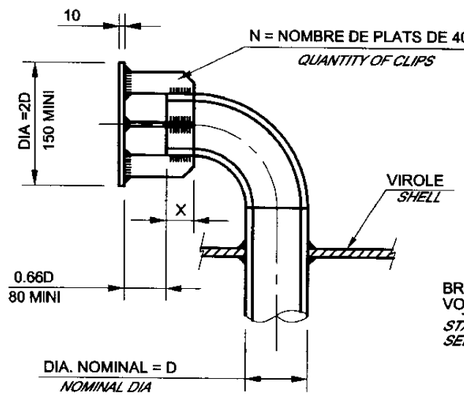
TYPE 2 *



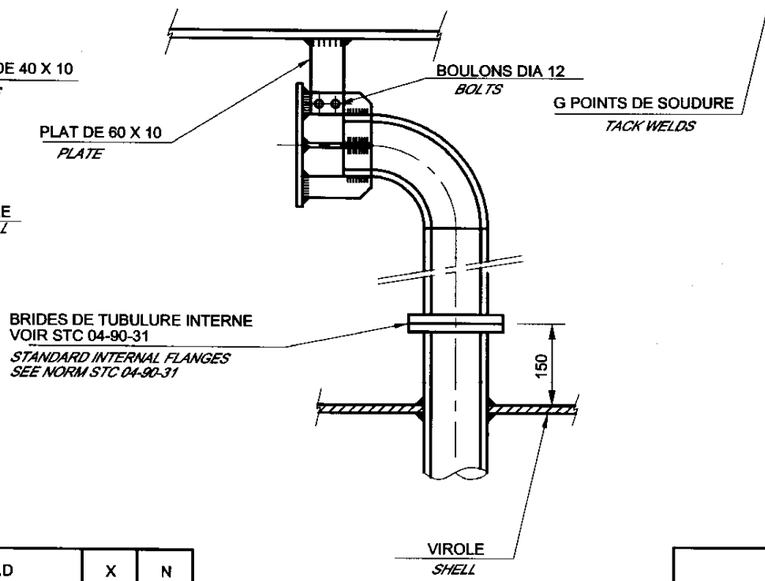


DEFLECTEUR SUR TUBULURE INTERNE
DEFLECTOR FOR INTERNAL PIPE

DEFLECTEUR FIXE
FIXED DEFLECTOR



DEFLECTEUR DEMONTABLE
REMOVABLE DEFLECTOR



DIA.D	X	N
1" 1/2 ≤ Ø ≤ 3"	30	3
4" ≤ Ø ≤ 8"	50	3
Ø ≥ 10"	50	4

	DIA NOMINAL TUBULURE / NOMINAL DIA. OF NOZZLE — TYPE 2									
D	≤ 2"	3"	4"	6"	8"	10"	12"	14"	16"	18"
A	210	230	300	350	500	550	650	700	800	900
B	100	130	150	200	300	350	400	450	500	600
C	50	50	80	80	100	100	130	130	150	150
H	100	130	150	150	150	200	250	280	350	350
E	100	100	100	100	130	130	130	150	150	150
F	10	10	10	10	10	10	10	10	10	10
G	3	4	4	6	8	10	10	12	12	12
H	2	2	3	3	3	3	4	4	4	4

★ POUR PIECES INTERIEURES EN INOX OU MONEL PARTOUT TOLE EPAISSEUR 4MM
FOR INTERNAL PARTS MADE OF S.S. OR MONEL USE 3/16" THICK PLATE EVERYWHERE
- EPAISSEUR DE CORROSION PRISE EN COMPTE 3MM
CORROSION THICKNESS TAKING INTO ACCOUNT 3MM

TECHNIP

STANDARD DE CONSTRUCTION
CONSTRUCTION STANDARD

APPAREILS A PRESSION
PRESSURE VESSELS

PARS OIL AND GAS Co.

AGENCEMENT INTERIEUR DES APPAREILS
INTERNAL ARRANGEMENT OF VESSELS

PROJECT N°	UNIT	DOC. TYPE	DISCIP.	SUBJ.	SERIAL N°	REV.
6340S	000	STC			000400402	0

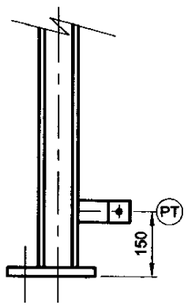
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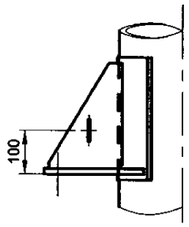
revision du Standard appliquee a l'activite	
Date:	Visa:

ELEVATION DES PRISES DE TERRE
ELEVATION EARTH CONNECTION

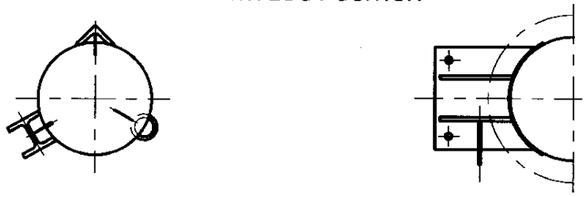
PIEDS
LEGS



CHAISES
BRACKETS

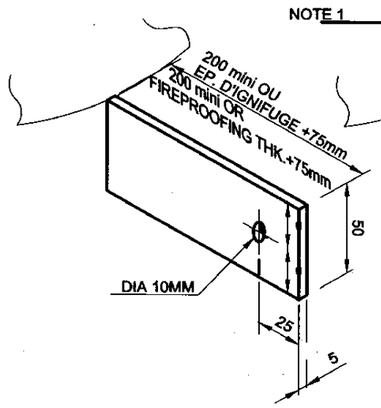


POSITION PRISE DE TERRE
EARTH LUG POSITION

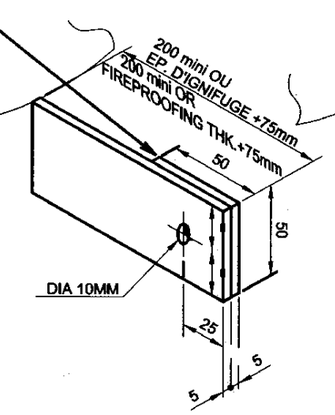


REVISED BY 6340S-L-E020 EXCEPT ELEVATION

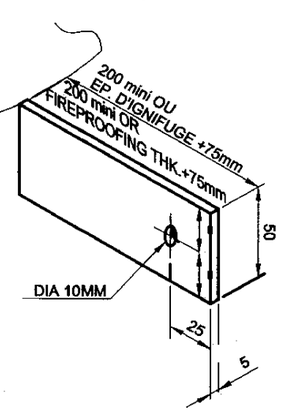
"PT" INOX
STAINLESS STEEL



"PT" NORMALE
CARBONE STEEL



"PT" ALUMINIUM
ALUMINIUM

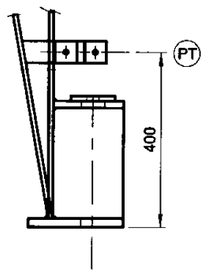


NOTES:

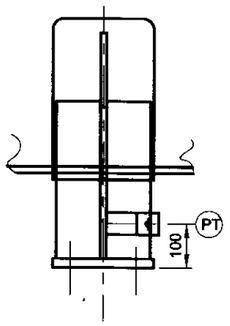
(1)- PLAQUETTE DE CUIVRE BRASEE FACE EXTERIEURE ETAMEE
COPPER PLATE BRAZED EXTERNAL FACE TINNED

(2)- ORIENTATIONS VOIR PLAN D'ENSEMBLE
ORIENTATIONS SEE DRAWING

JUPES
SKIRTS



BERCEAUX
SADDLES



TECHNIP

STANDARD DE CONSTRUCTION
CONSTRUCTION STANDARD

APPAREILS A PRESSION
PRESSURE VESSELS

PARS OIL AND GAS Co.

PRISE DE TERRE
EARTH CONNECTION

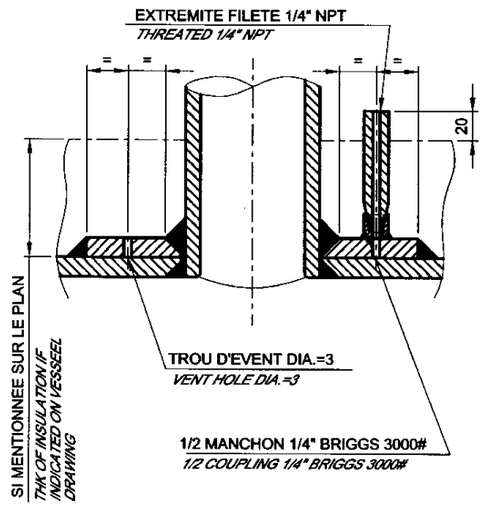
PROJECT N°	UNIT	DOC. TYPE	DISCIP.	SUBJ.	SERIAL N°	REV.
6340S	000	STC			000400505	~~0~~

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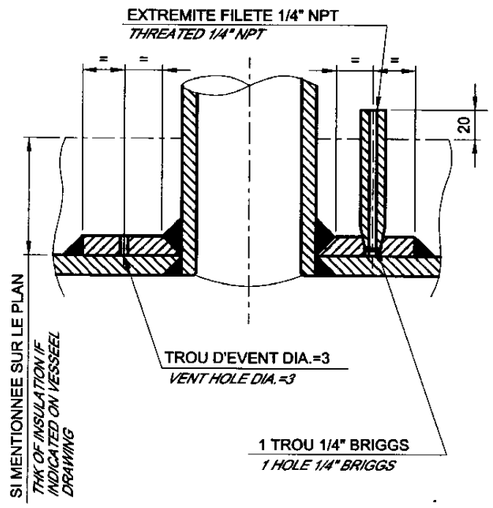
Indice de revision du Standard general	0
Date: 20-12-2000	Visa: JP.B

revision du Standard applique a l'activite	
Date:	Visa:

EPAISSEUR ≤ 16
EPAISSEUR ≤ 16



EPAISSEUR > 16
THICKNESS > 16



	DIAMETRE NOMINAL EN POUCE *NOMINAL DIAMETER IN INCHES*										
	3	4	6	8	10	12	14	16	18	20	≥ 24
NOMBRE D'EVENTS *NOMBER OF VENTS*	1	1	1	1	1	1	1	1	2	2	2
LARGEUR DU RENFORT STANDARD (MM) *STANDARD REINFORCEMENT WIDTH*	60	70	70	100	130	160	170	180	210	230	280

NOTA
NOTE

- LES TROUS PERCES SUR LE RENFORT SERONT EQUIDISTANTS
  HOLES DRILLED IN REINFORCEMENT SHALL BE EQUIDISTANT
- LES TROUS D'EVENT Ø3 SERONT BOUCHES PAR SOUDURE POUR L'ESSAI
  FOR THE TEST VENT HOLES Ø3 WILL BE OBTURED BY WELDING
- LE TROU 1/4" EST TOUJOURS PLACE A LA PARTIE BASSE DU RENFORT
  HOLE 1/4" IS ALWAYS SITUATED AT LOWER PART OF REINFORCEMENT
- APRES ESSAI LE TROU DE 1/4" SERA MUNI D'UN TUBE FILETE DEPASSANT DU CALORIFUGE.
  AFTER TEST 1/4" HOLE SHALL BE FILLED WITH THREADED TUBE EXTENDING BEYOND INSULATION
- LE TUBE FILETE SERA PREVU UNIQUEMENT SUR LES APP. CALORIFUGES.
  THREATED TUBE SHALL ONLY BE PROVIDED ON INSULATED VESSEL.
- SAUF INDICATION CONTRAIRE, TOUTES LES TUBULURES PROCESS SERONT EQUIPEES DE RENFORTS.
  UNLESS OTHERWISE SPECIFIED, ALL PROCESS NOZZLES ARE PROVIDED WITH REINFORCING PADS.

TECHNIP

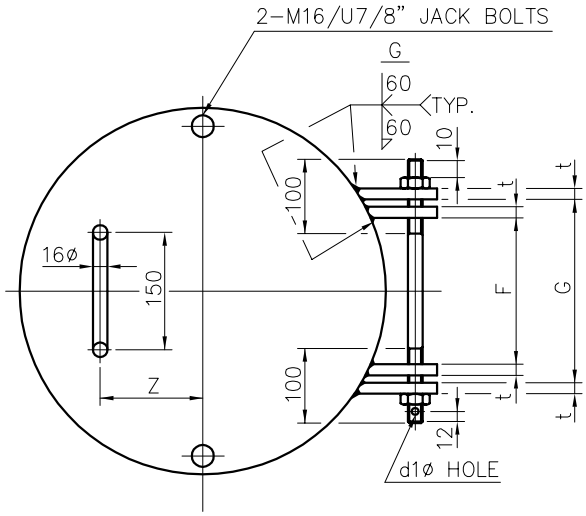
STANDARD DE CONSTRUCTION
CONSTRUCTION STANDARD

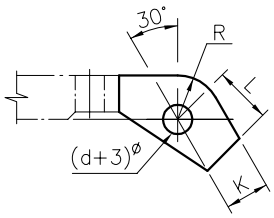
APPAREILS A PRESSION
PRESSURE VESSELS

DETAILS RENFORTS DE TUBULURES
NOZZLE REINFORCEMENTS DETAILS

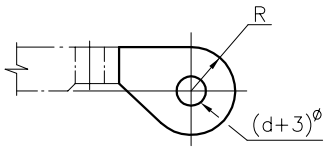
PARS OIL AND GAS Co.

PROJECT N°	UNIT	DOC. TYPE	DISCIP.	SUBJ.	SERIAL N°	REV.
6340S 000		STC			000400905	0

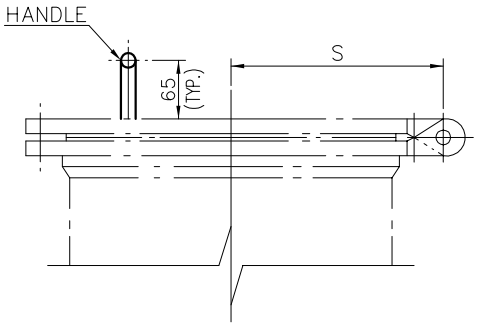


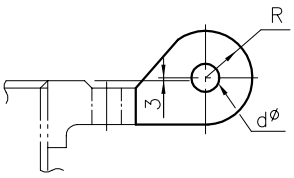


COVER HINGE
(FOR TOP M.H.)



COVER HINGE
(FOR SIDE & BOTT. M.H.)





MANHOLE HINGE

RAT'G	SIZE A (mm)	SIZE B (inch)	S	F	G	t	d	R	K	L	d1	SPLIT PIN DIA.xLENG.	HINGE BOLT SIZE U	HINGE BOLT SIZE M	HINGE BOLT LENG.	Z
150#	400	16	330	250	297	22	28	37	38	70	5.5	5 x 40	U1	M24	417	180
	450	18	356	265	312	22	↓	40	41	74	5.5	5 x 40	↓	↓	432	200
	500	20	380	300	347	22	↓	43	45	76	5.5	5 x 40	↓	↓	457	220
	600	24	440	355	402	22	↓	48	50	86	5.5	5 x 40	↓	↓	522	270
300#	400	16	360	270	317	22	↓	57	58	77	5.5	5 x 40	↓	↓	437	180
	450	18	390	300	347	22	↓	60	62	80	5.5	5 x 40	↓	↓	467	200
	500	20	420	340	387	22	28	64	55	84	5.5	5 x 40	U1	M24	507	220
	600	24	495	410	463	25	35	70	72	100	6.5	6 x 60	U1 1/4	M30	605	270
600#	400	16	380	295	342	22	28	75	78	82	5.5	5 x 40	U1	M24	462	200
	450	18	405	320	373	25	28	83	84	85	5.5	5 x 40	U1	M24	497	220
	500	20	440	355	408	25	35	89	90	88	6.5	6 x 60	U1 1/4	M30	550	240
	600	24	505	420	479	28	35	102	103	100	6.5	6 x 60	U1 1/4	M30	625	280

NOTES

1. MATERIALS TO BE JIS SS400 ASTM A36, A283 Gr.C
2. JACK BOLTS TO BE TYPE 304SS UNLESS SPECIFIED ON ENG'G DWG.
3. FILLET WELDS TO BE ALL AROUND AND SIZE OF WELD TO BE 6mm AND OVER UNLESS OTHERWISE NOTED.
4. HINGE LUGS NOT TO MEET BOLTS IN MANHOLE.
5. DIMENSIONS TO BE GIVEN IN mm.
6. DIRECTION OF HINGE (LEFT, RIGHT OR ORIENTATION) TO BE FOLLOWED TO ENG'G DWG.

LG Engineering & Construction Corp.

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
◇0	'01.11.03	FOR SP 9&10 PROJECT	SRK	HCS	KSL	

MANHOLE COVER HINGE (ANSI)

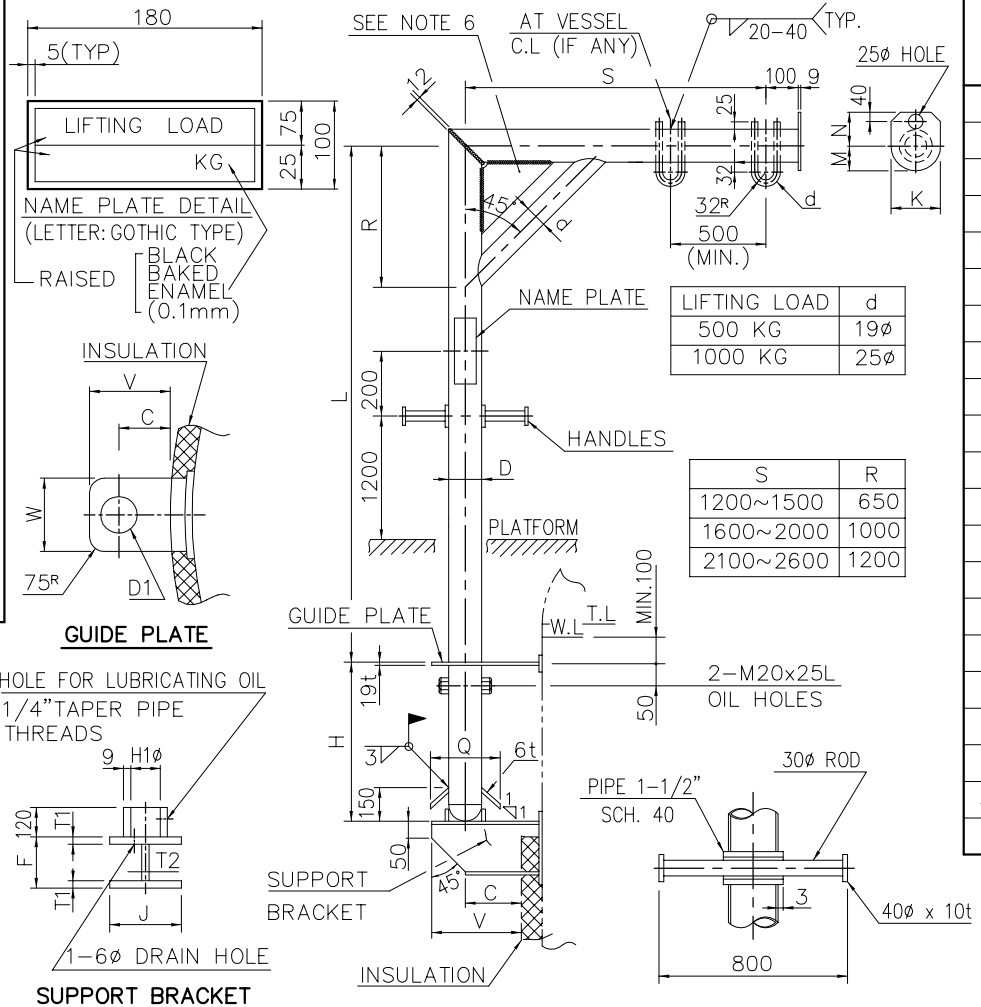
C | STD. NO. 6340S-L-C008

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
◇						
◇1	'03.12.21	AS PER COMPANY'S COMMENTS	SRK	HCS	KSL	
◇0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

LG Engineering & Construction Corp.

C STD. NO. 6340S-L-C011

PIPE DAVIT



LIFTING LOAD	d
500 KG	19ø
1000 KG	25ø

S	R
1200~1500	650
1600~2000	1000
2100~2600	1200

GUIDE PLATE

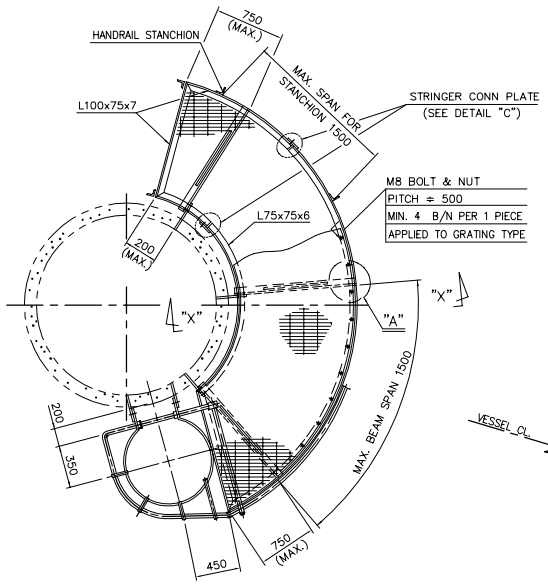
SUPPORT BRACKET

S	LIFTING LOAD 500 KG					LIFTING LOAD 1000 KG					L (MAX.)	H
	TYPE NO.	D			R	TYPE NO.	D			R		
		MM	IN	SCH.			MM	IN	SCH.			
600	1	100	4	80	500	–	–	–	–	–	2300	750
700	2	100	4	80	500	–	–	–	–	–	2300	750
800	3	150	6	80	750	103	150	6	80	750	2500	750
900	4	↑	↑	↑	↑	104	150	6	80	750	↑	750
1000	5					105	150	6	80	750		900
1100	6					106	200	8	80	1000		900
1200	7					107	200	8	80	↑	2500	900
1300	8			↓		108	↑	↑	↑		2800	1100
1400	9			80		109			↓		2800	1100
1500	10			80		110			80		2800	1100
1600	11			↑		111			80		3000	1250
1700	12					112			↑		↑	↑
1800	13					113						
1900	14					114						
2000	15	↓	↓	↓	↓	115	↓	↓	↓	↓	3000	1250
2100	16	150	6	80	750	116	200	8	80	1000	3300	1400
2200	17	200	8	80	1000	–	–	–	–	–	↑	1400
2300	18	↑	↑	↑	↑	–	–	–	–	–		1550
2400	19					–	–	–	–	–		↑
2500	20					–	–	–	–	–		
2600	21	200	8	80	1000	–	–	–	–	–	3300	1550

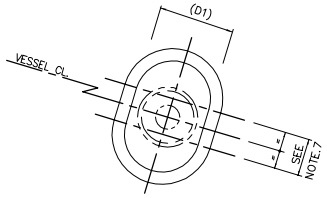
D IN	d	W	V	C	D1	F	H1	J	T1	T2	K	M	N	Q	D3	D4 MM	D4 IN
4	2"SCH.80	250	305	175	120	180	120	150	16	9	140	70	140	180	55	25	SCH.80
6	3"SCH.80	300	350	200	172	200	175	205	16	9	190	95	160	230	70	40	SCH.80
8	4"SCH.80	400	425	225	222	250	225	255	19	12	240	120	185	280	70	40	SCH.80

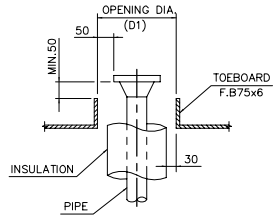
NOTES

1. ALL DIMENSIONS ARE IN mm.
2. FOR TYPE NO. AND MATERIALS OF PADS SEE ENG.G DWG.
3. MATERIALS EXCEPT PADS TO BE AS FOLLOWS
   JIS : STPG 370/SS 400
   ASTM : A53/A283 Gr.C
4. FILLET WELDS TO BE ALL AROUND AND SIZE OF WELDS TO BE 6MM AND OVER.
5. WHEN "S AND/OR L" IS OVER THAN THIS STANDARD, THE STRENGTH CALCULATION OF THE DAVIT SHALL BE CARRIED OUT BY VENDOR.
6. WHEN "S" IS LESS THAN 1100 mm, 10t PLATE SHALL ONLY BE USED.

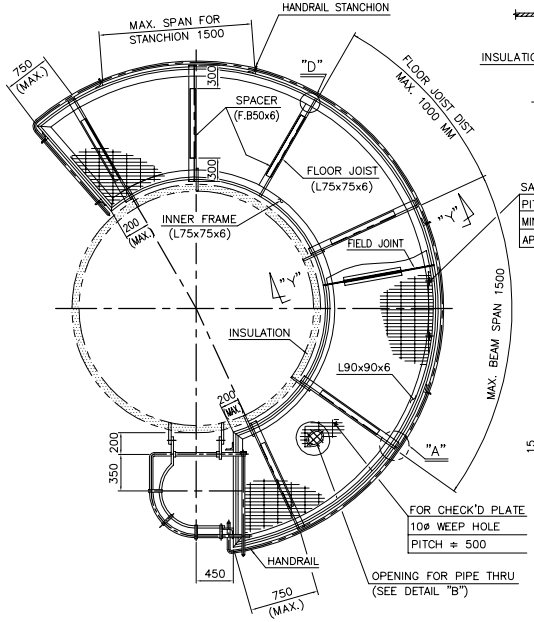


TYPE "I"
(UPTO 1500 WIDTH)

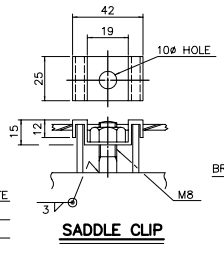




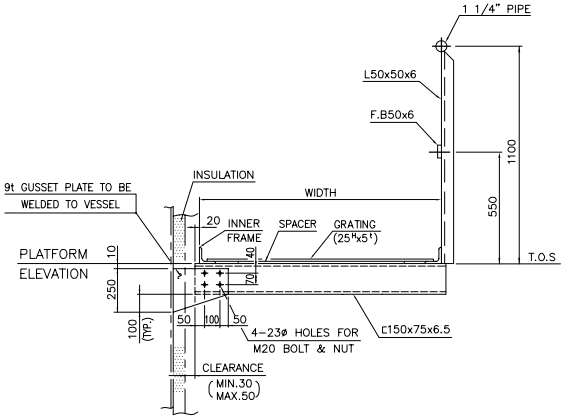
DETAIL "B"



TYPE "II"
(OVER 1501 WIDTH)

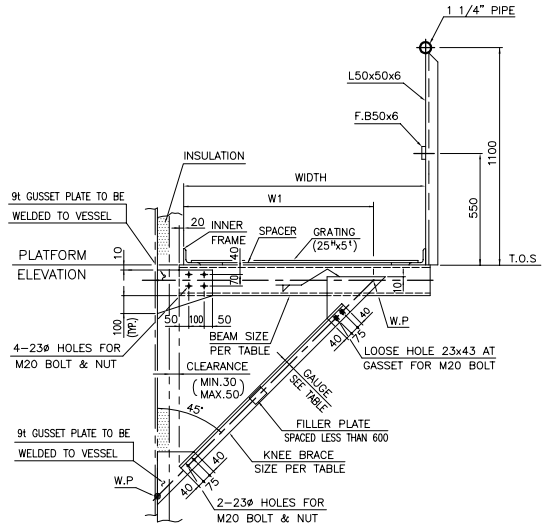


SADDLE CLIP



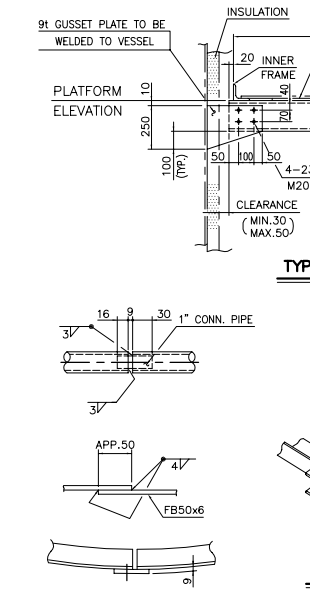
TYPE I (CANTILEVER TYPE)
(MAX. WIDTH :1500)

SECTION "X"-"X"

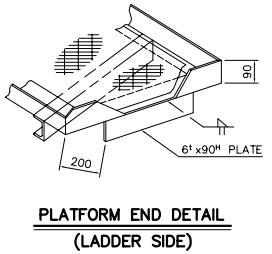


TYPE II (KNEE BRACE TYPE)
(OVER 1501 WIDTH)

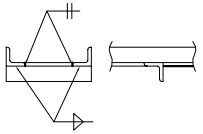
SECTION "Y"-"Y"



DETAIL "C"



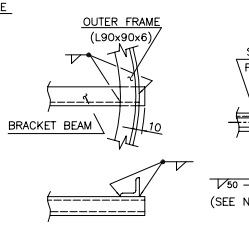
PLATFORM END DETAIL
(LADDER SIDE)

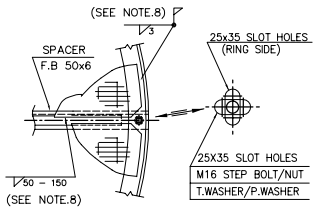


DETAIL "D"

WIDTH (MM) / TYPE	UPTO 1500 I	1501~2100 II	2101~2700 II	2701~3300 II
LENGTH W1(MM)	-	WIDTH-500	WIDTH-700	WIDTH-1000
BEAM SIZE	C150x75x6.5	C150x75x6.5	C200x80x7.5	C250x90x9
KNEE BRACE	-	2-L65x65x6	2-L75x75x6	2-L75x75x6
GAUGE LENGTH	-	35	40	40

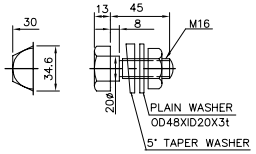
MEMBER TABLE





WELDED CONNECTION **BOLTED CONNECTION**

DETAIL "A"



STEP BOLT & NUT DETAIL

NOTE

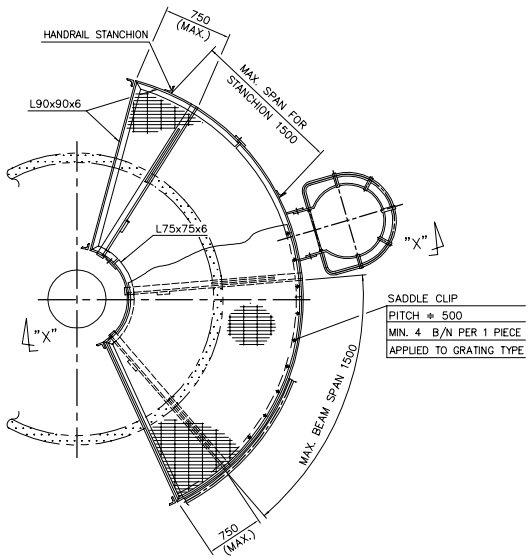
1. ALL DIMENSIONS ARE IN mm.
2. MATERIALS USED FOR PLATFORMS & LADDER TO BE AS FOLLOWS.
   -BOLT & NUT -19°C & OVER ; SS400,A307 Gr.B
   -20°~-100°C ; A320 Gr.L7+A194 Gr.4 OR SUS304 , A320 Gr.L8
   -101°C & LOWER SUS304,A320 Gr.L8
   -STRUCTURAL STEEL ; SS41,SGP,A36
   -FOR MATERIALS DIRECTLY WELDED TO THE SHELL ; SEE ENG'G DWG.
3. FLOOR TYPE ; GRATING
4. TOS ; TOP OF STRUCTURE , THIS IS THE SPECIFIED LEVEL ON ENG'G DWG UNLESS OTHERWISE NOTED.
5. FLOOR FIXING METHOD.
   -GRATING ; CLAMPING
6. GRATING SHALL BE HOT DIP GALV.
7. FOR PIPE SIZE 4" AND UNDER: 80
   FOR PIPE SIZE 6" ~12" : 120
   FOR PIPE SIZE 12" OVER : 150
8. THESE WELDING DETAILS ARE REQUIRED FOR CHECK'D PLATE FLOOR.

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
◇						
◇						
◇	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

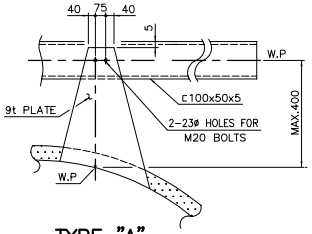
LG Engineering & Construction Corp.

PLATFORM FOR VERTICAL VESSEL

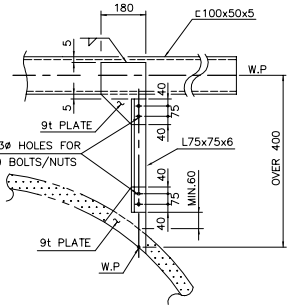
C | STD. NO. 6340S-L-C013-1/3



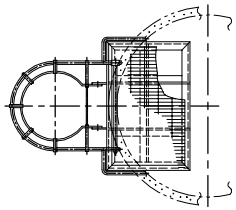
TYPE "I"
(UPTO 1500 WIDTH)

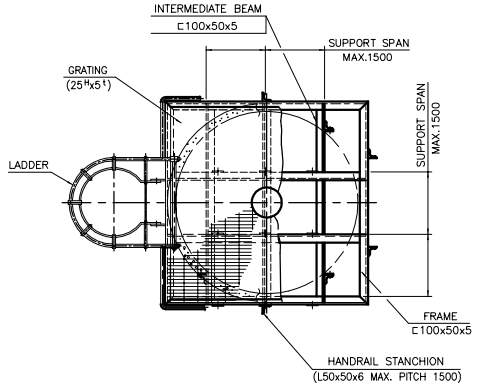


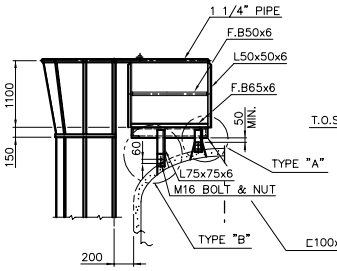
TYPE "A"

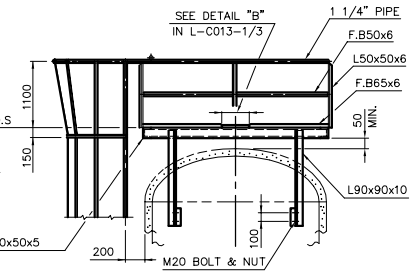


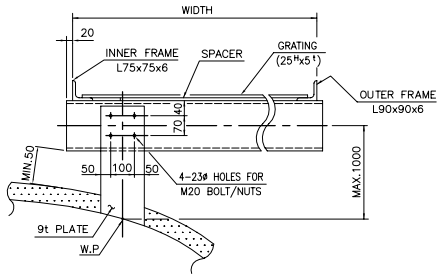
TYPE "B"



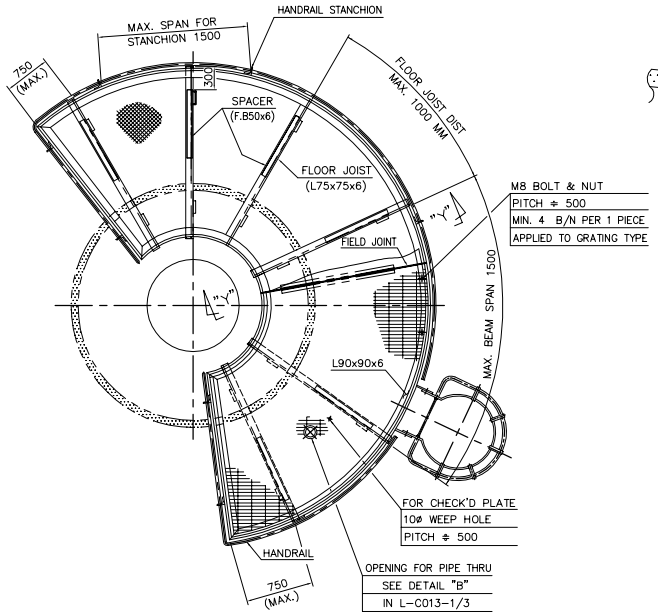




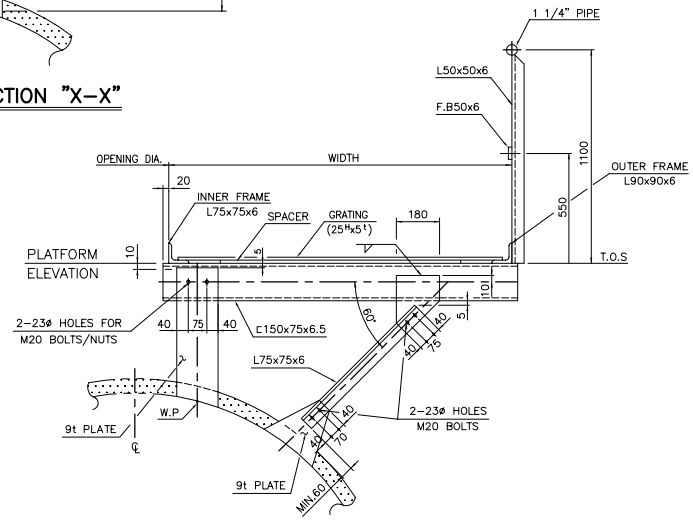




SECTION "X-X"



TYPE "II"
(OVER 1501 WIDTH)



SECTION "Y-Y"

NOTE

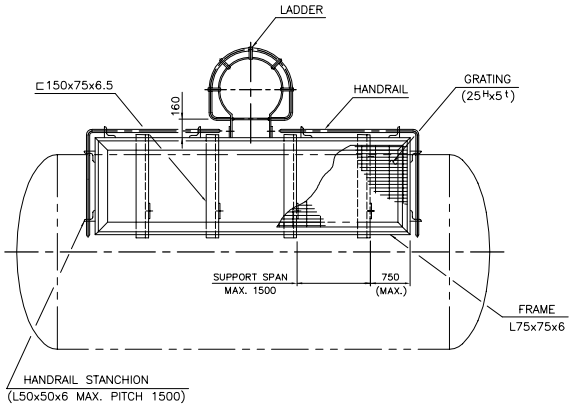
1. SEE NOTE IN L-C013-1/3
2. DETAILS TO BE SPECIFIED IN L-C013-1/3 SHALL BE APPLIED.
3. IF FLOOR IS MADE FROM CHECK'D PLATE, REFER TO WELDING DETAIL ON THE L-C013-1/3.

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
◇						
◇						
◇0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

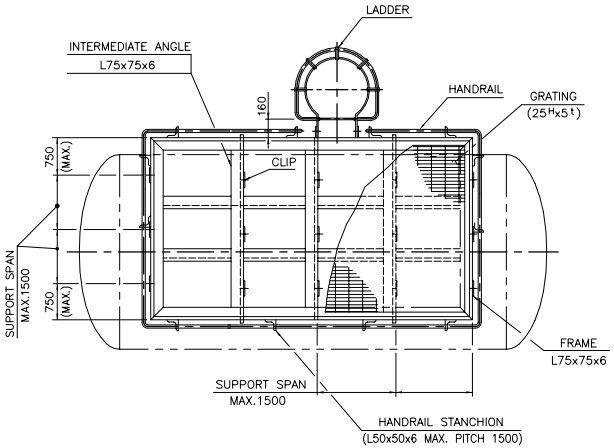
LG Engineering & Construction Corp.

TOP PLATFORM FOR VERTICAL VESSEL

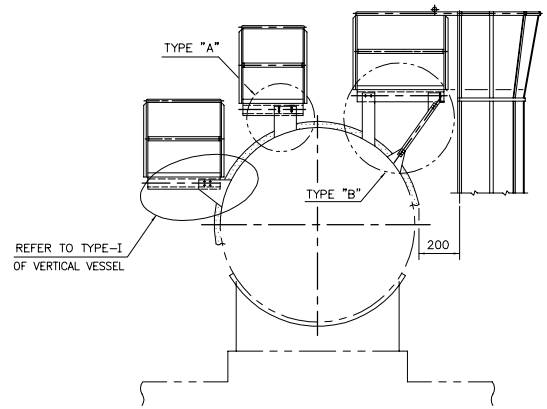
C | STD. NO. 6340S-L-C013-2/3

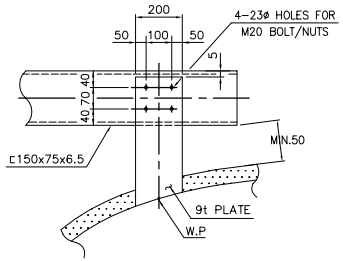


UP TO 1500 WIDTH
(FOR TYPE "A")

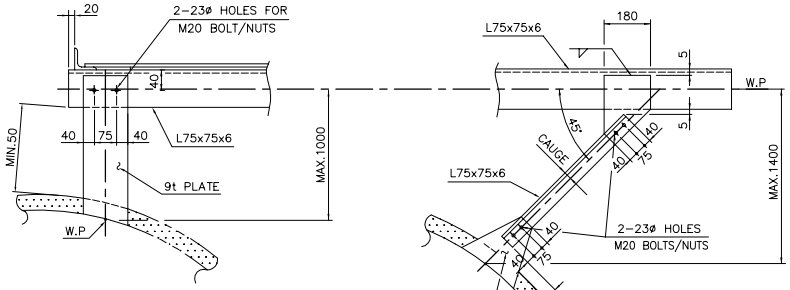


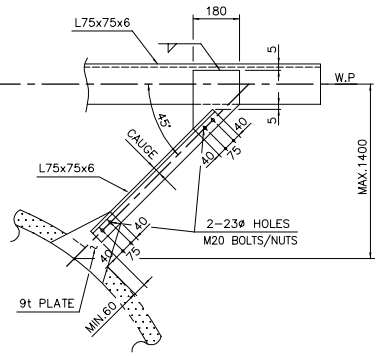
OVER 1501 WIDTH
(FOR TYPE "B")





TYPE "A"

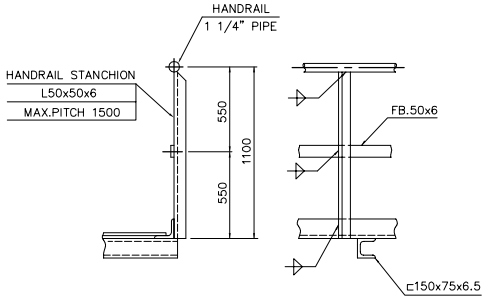




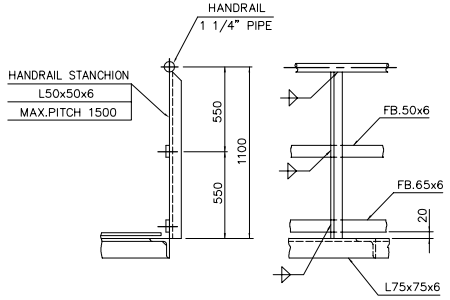
TYPE "B"

NOTE

1. SEE NOTE IN L-C013-1/3
2. WHEN THE PLATFORM ELEVATION IS LOWER THAN 4M, SAFETY CAGE IS NOT REQUIRED.
3. DETAILS TO BE SPECIFIED IN L-C013-1/3 SHALL BE APPLIED.
4. IF FLOOR IS MADE FROM CHECK'D PLATE, REFER TO WELDING DETAIL ON THE L-C013-1/3.



FOR TYPE "A"



FOR TYPE "B"

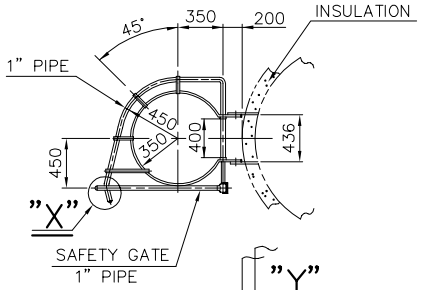
DETAILS OF HANDRAIL
FOR HORIZONTAL VESSEL

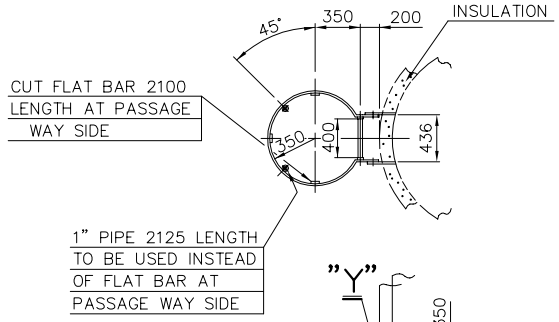
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

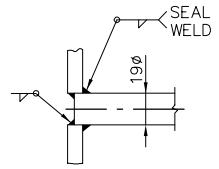
LG Engineering & Construction Corp.

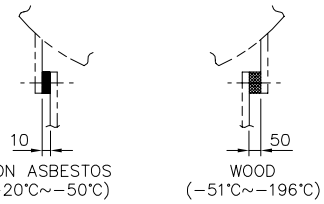
PLATFORM
FOR HORIZONTAL VESSEL

C | STD. NO. 6340S-L-C013-3/3



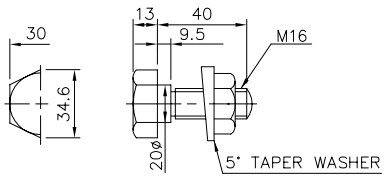






STEP BAR WELD DETAIL

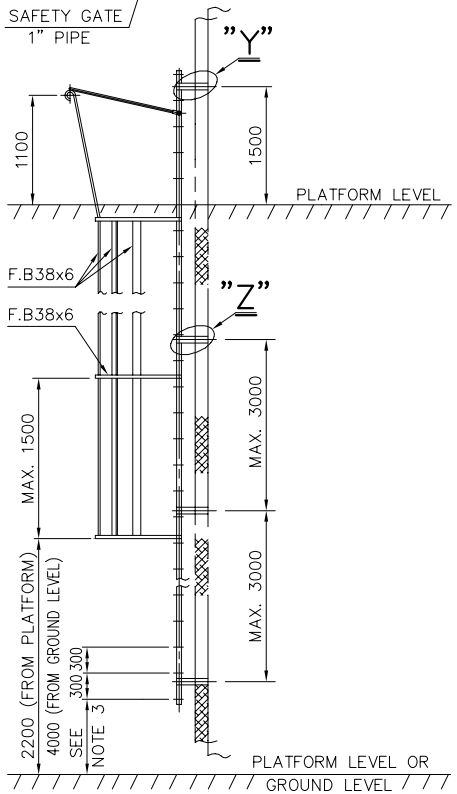
COLD TYPE



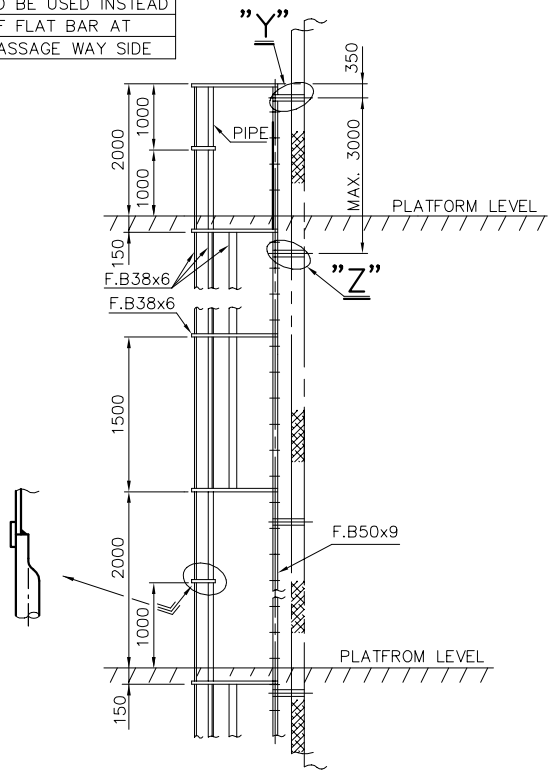
STEP BOLT & NUT DETAIL

NOTE

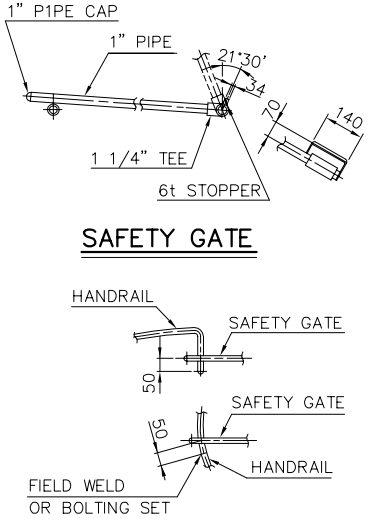
1. ALL DIMENSIONS ARE IN mm.
2. MATERIALS USED FOR PLATFORMS & LADDERS TO BE AS FOLLOWS.
   −BOLT & NUT −19°C & OVER SS400,A307 Gr.B
   −GALVANIZED FOR C/S AND LOW ALLOY
   −20~100°C A320 L7 + A194 Gr.4 OR SUS 304,A320 L8
   −101°C AND LOWER SUS 304,A320 L8.
   −STRUCTURAL STEEL ; SS400,SGP,A36
   −FOR MATERIALS DIRECTLY WELDED TO THE SHELL SEE ENG'G DWG.
3. IN CASE OF PLATFORM: 100MM
   IN CASE OF GROUND LEVEL: EXTEND TO GROUND, MECHANICAL ENGINEER TO CONFIRM ITS SELECTED DISTANCE AND SUITABILITY WITH FOUNDATION.



TYPE "A"

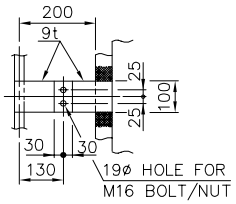


TYPE "B"

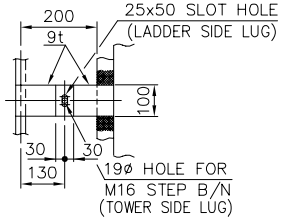


SAFETY GATE

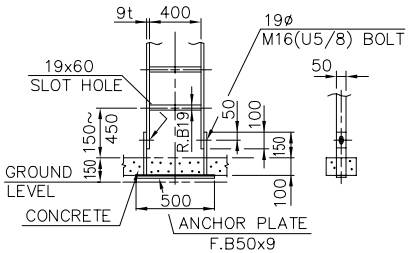
DETAIL "X"



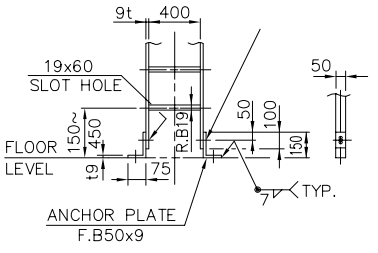
DETAIL "Y"



DETAIL "Z"



IN CASE OF FIXED ON THE GRONUD



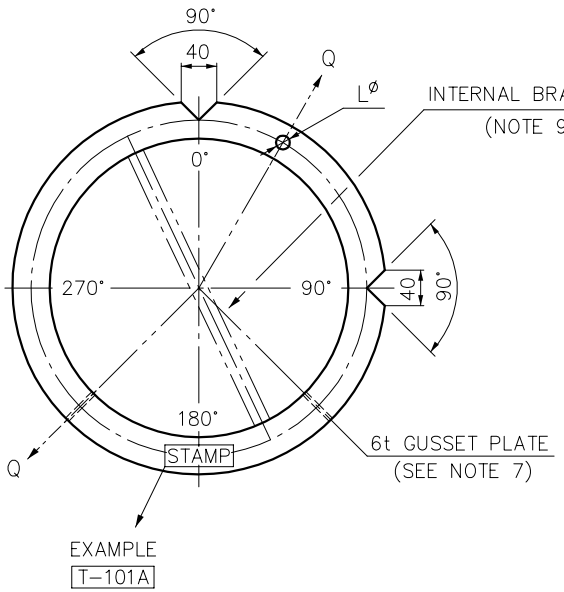
IN CASE OF FIXED ON THE STRUCTURE

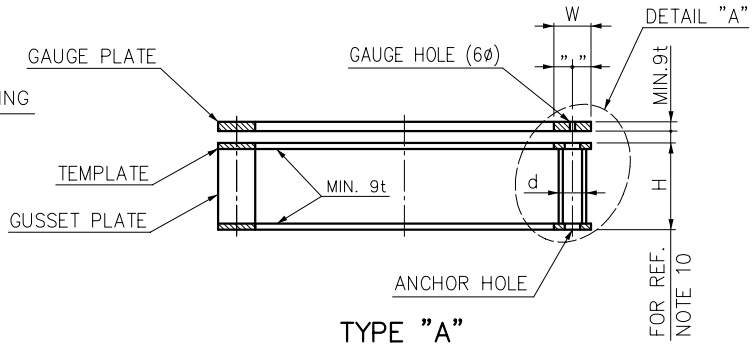
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
◇						
◇						
◇	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

LG Engineering & Construction Corp.

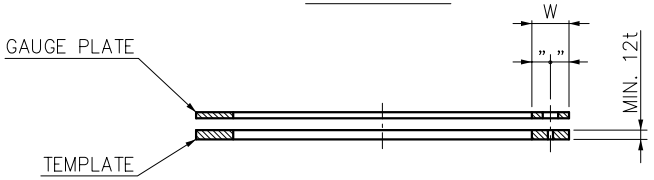
LADDER

C | STD. NO. 6340S−L−C014



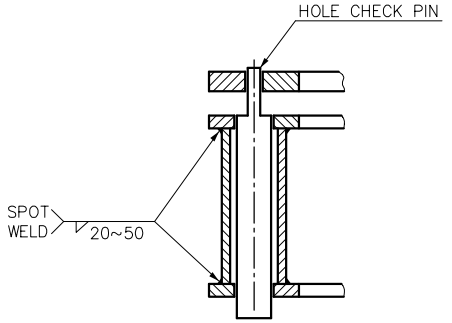


TYPE "A"

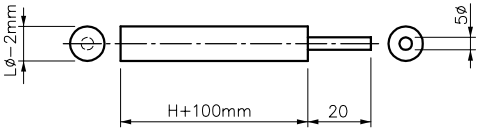


TYPE "B"

SECTION Q-Q



DETAIL "A"



HOLE CHECK PIN
(APPLIED TYPE "A")

BOLT SIZE	M16	M20	M24	M30	M36	M42	M48	M56	M64	M72	M80	M90	M100
H	–	–	262	264	314	316	316	369	369	422	475	475	475
W	100	100	100	100	110	120	130	150	160	170	180	200	220
L	19	23	26	33	39	45	52	62	70	78	86	96	107
d	–	–	1½"	1½"	2"	2"	2½"	3"	3"	4"	4"	6"	6"

BOLT SIZE	U 5/8	U 3/4	U1	U1 1/4	U1 1/2	U1 3/4	U2	U2 1/4	U2 1/2	U2 3/4	U3	U3 1/2	U4
H	–	–	262	264	314	316	316	369	369	422	475	475	475
W	100	100	100	100	110	120	130	150	160	170	180	200	220
L	19	22	28	35	41	48	55	62	69	75	82	96	107
d	–	–	1½"	1½"	2"	2"	2½"	3"	3"	4"	4"	6"	6"

NOTES

1. ALL DIMENSIONS ARE ALL IN MM.
2. TYPE "A" TO BE USED FOR CHAIR TYPE.
   TYPE "B" TO BE USED FOR SINGLE RING BEAM TYPE.
3. BOLT HOLES IN TEMPLATE AND BASE PLATE TO BE DRILLED USING THE SAME GAUGE PLATE.
4. THE REINFORCEMENT OF TEMPLATE SHALL BE DESIGNED NOT TO DEFORM DURING TRANSPORTATION.
5. ORIENTATION MARK 0°, 90° AND ITEM NUMBER TO BE MARKED WITH WHITE PAINT ON THE UPPER FACE OF TEMPLATE.
6. ITEM NUMBER TO BE STAMPED AT 180° ON THE UPPER FACE OF TEMPLATE.
7. IN CASE THE BOLT PITCH IS 600mm AND OVER, GUSSET PLATES TO BE REQUIRED.
8. MATERIAL TO BE AS FOLLOWS.
   PLATE : JIS SS400 / ASTM A283 Gr.C
   PIPE : JIS SGP / ASTM A53 Gr.A OR B
9. TEMPLATE, B.C.D 3000 AND OVER, MAY BE SEGMENT TYPE WITH INTERNAL ADEQUATE BRACING.
   IN THIS CASE, THE ASSEMBLED TEMPLATE TO BE KEPT THE ORIGINAL DIMENSIONS.
10. DIMENSION "H" TO BE COINCIDENT WITH THE HEIGHT OF ACTUAL ANCHOR CHAIR PLUS WASHER ON THE FABRICATION DWG.
11. TEMPLATE SHALL BE APPLIED TO THE SKIRT TYPE EQUIPMENT UNLESS SPECIFIED ON ENG'G DWG.
12. RUST PREVENTED PAINT FOR TEMPLATE TO BE PROVIDED AT DELIVERY.

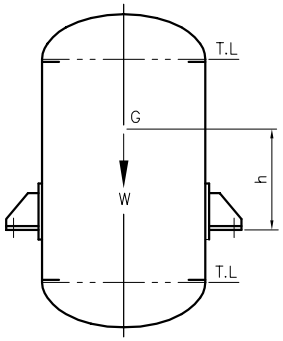
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
◇						
◇						
0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

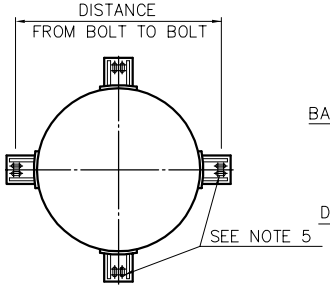
LG Engineering & Construction Corp.

TEMPLATE

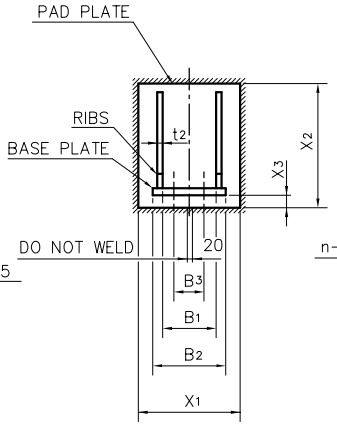
D | STD. NO. 6340S-L-C016

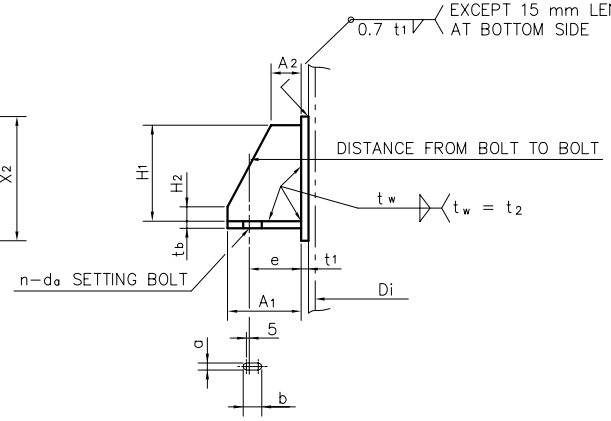
VESSEL I.D Di	NO. OF LUG	d_a	n	a	b	e	A_1	A_2	t_b	t_1	t_2	B_1	B_2	B_3	H_1	H_2	X_1	X_2	X_3	MAXIMUM ALLOWABLE OPERATION WEIGHT (KG)	MAXIMUM ALLOWABLE MOMENT ARMS DUE TO EARTHQUAKE	MAXIMUM LENGTH OF VESSEL
550 ~ 790	4	M16	1	19	38	150 ~ 350	e+50	50	12	6	6	160	200	–	270	50	250	340	30	4450	2500	5500
800 ~ 1040	4	M20	1	23	46	150 ~ 350		50	12	6	9	180	230	–	300	50	280	370	30	10300	2500	7500
1050 ~ 1540	4	M24	1	27	54	170 ~ 370		70	12	6	9	200	250	–	350	70	300	420	30	22150	2500	9000
1550 ~ 2040	4	M24	2	27	54	200 ~ 400		70	16	9	12	250	310	130	400	70	360	470	30	72950	2500	12500
2050 ~ 2540	4	M30	2	35	70	220 ~ 420		100	16	9	12	340	400	180	500	100	480	600	40	118800	2500	16000
2550 ~ 3000	4	M36	2	41	82	250 ~ 450		100	19	9	12	420	480	220	600	100	560	700	40	176700	2500	19000





4–SUPPORT LUGS

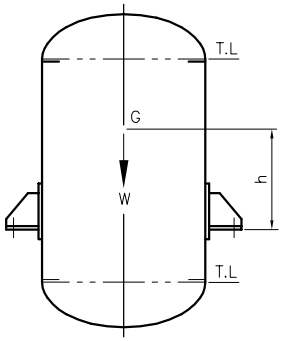


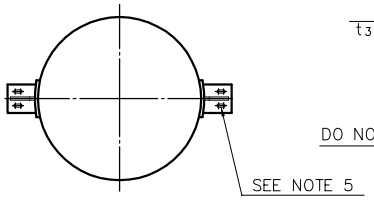


	ALLOWABLE LOAD W (Kgf)	MAX. SHELL O.D Do	MAX. CENTER OF GRAVITY HIGHT h	BOLT POSITION A3	LUG SUPPORT H1	H2	B1	B2	A1	t2	t3	C	K	SET BOLT B3	BOLT SIZE M	BOLT SIZE UNC	d M	d UNC	MIN. SHELL THICK
1	1030	UP TO 250	3000	110 ~ 160	280	220	170	140	A3+40	12	9	30	20	80	20	3/4	23	22	3.0
2	2410	260 ~ 540	3000	120 ~ 170	330	260	240	200	A3+50	16	12	50	20	120	20	3/4	23	22	3.0

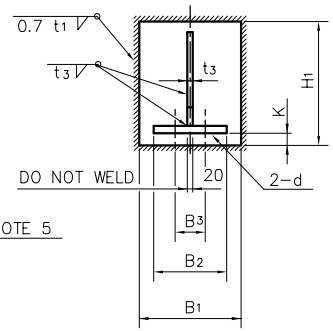
REMARKS ;

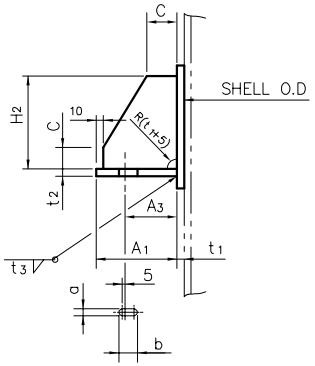
1. ALL DIMENSIONS ARE IN MM.
2. BOLT SIZE AND B.C.D. SEE ENG'G DWG.
3. IN CASE DIMENSION "e" EXCEEDS STANDARD DIMENSION SPECIFIED ON ABOVE TABLE VENDOR SHALL CHECK AND FINALIZE ALL DIMENSIONS.
4. MATERIALS SHALL BE AS FOLLOW;
   LUG – JIS SS400, ASTM A36 OR EQUIVALENT
   PAD PLATE – SEE ENG'G DWG.
5. SLOT HOLE DIRECTION WILL BE CONFIRMED BY PIPING ENGINERING GROUP OF PURCHASER.





2–SUPPORT LUGS





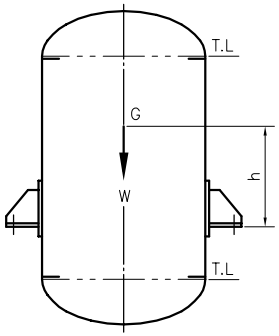
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

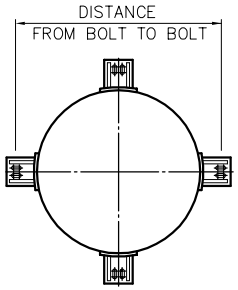
LG Engineering & Construction Corp.

SUPPORT LUGS FOR VERTICAL VESSEL (LIGHT DUTY)

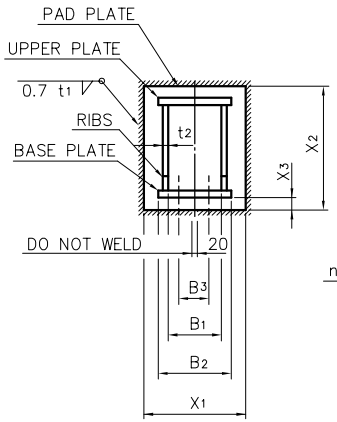
C | STD. NO. 6340S–L–C019

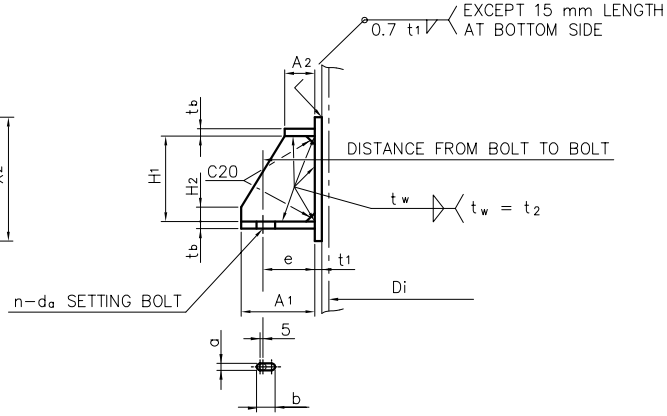
VESSEL I.D Di	NO. OF LUG	d_a	n	a	b	e	A_1	A_2	t_b	t_1	t_2	B1	B2	B3	H_1	H_2	X_1	X_2	X_3	MAXIMUM ALLOWABLE OPERATION WEIGHT (KG)	MAXIMUM ALLOWABLE MOMENT ARMS DUE TO EARTHQUAKE	MAXIMUM LENGTH OF VESSEL
550 ~ 790	4	M16	2	19	38	150 ~ 350	e+50	50	12	6	6	160	200	80	270	50	250	350	25	8950	2500	8000
800 ~ 1040	4	M20	2	23	46	150 ~ 350		50	16	6	9	180	230	90	300	50	280	380	25	20650	2500	11000
1050 ~ 1540	4	M24	2	27	54	170 ~ 370		70	16	6	9	200	250	100	350	70	300	430	25	44300	2500	13000
1550 ~ 2040	4	M30	2	35	70	200 ~ 400		70	19	12	12	260	320	140	400	70	370	490	25	109400	2800	16000
2050 ~ 2540	4	M36	2	41	82	220 ~ 420		100	19	12	12	340	400	180	500	100	480	620	40	178200	3500	19000
2550 ~ 3000	4	M36	2	41	82	250 ~ 450		100	22	12	12	420	480	220	600	100	560	720	40	214000	4000	19000





4–SUPPORT LUGS

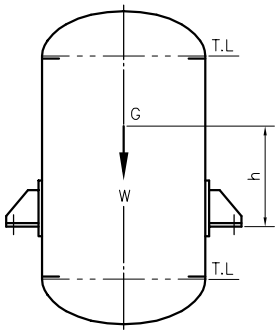


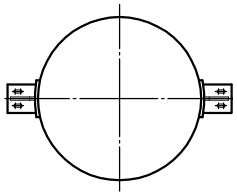


	ALLOWABLE LOAD W (Kgf)	MAX. SHELL O.D Do	MAX. CENTER OF GRAVITY HIGHT h	BOLT POSITION A3	LUG SUPPORT									SET BOLT					MIN. SHELL THICK
					H_1	H_2	B_1	B_2	A_1	t_2	t_3	C	K	B_3	BOLT SIZE		d		
															M	UNC	M	UNC	
1	1030	UP TO 250	3000	110 ~ 160	280	220	170	140	A3+40	12	9	30	20	80	20	3/4	23	22	3.0
2	2410	260 ~ 540	3000	120 ~ 170	330	260	240	200	A3+50	16	12	50	20	120	20	3/4	23	22	3.0

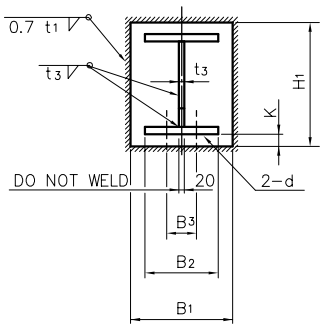
REMARKS ;

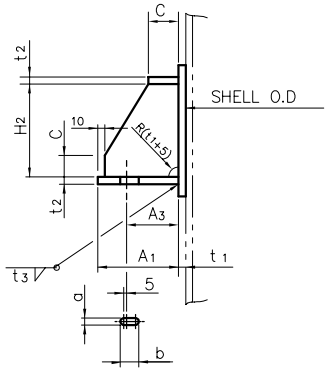
1. ALL DIMENSIONS ARE IN MM.
2. BOLT SIZE AND B.C.D. SEE ENG'G DWG.
3. IN CASE DIMENSION "e" EXCEEDS STANDARD DIMENSION SPECIFIED ON ABOVE TABLE, VENDOR SHALL CHECK AND FINALIZE ALL DIMENSIONS.
4. MATERIALS SHALL BE AS FOLLOW;
   LUG – JIS SS400, ASTM A36 OR EQUIVALENT
   PAD PLATE – SEE ENG'G DWG.





2–SUPPORT LUGS



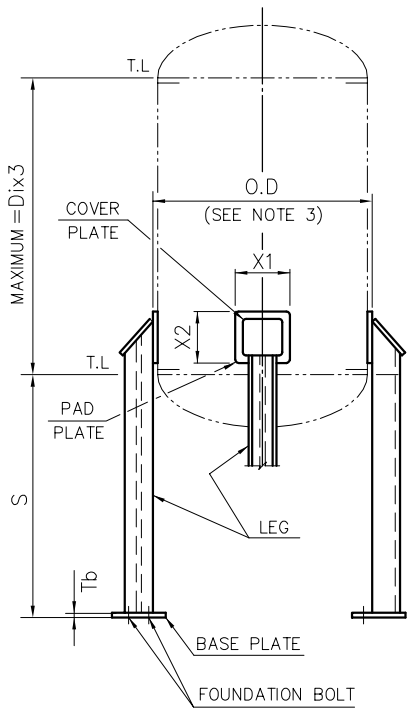


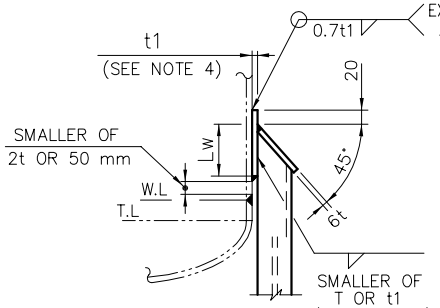
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

LG Engineering & Construction Corp.

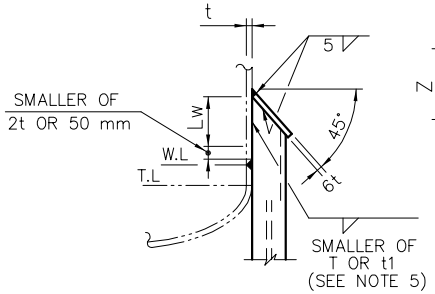
SUPPORT LUGS FOR VERTICAL VESSEL (HEAVY DUTY)

C | STD. NO. 6340S–L–C020



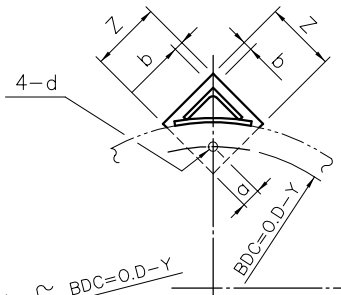


WITH PAD PLATE

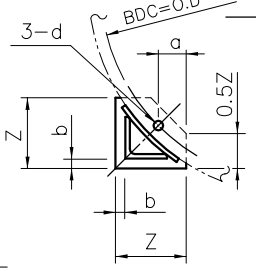


WITHOUT PAD PLATE

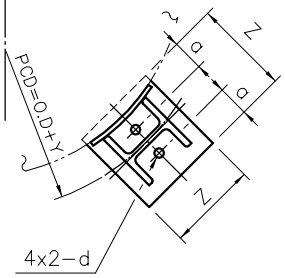
"A" TYPE



"B" TYPE



"C" TYPE



CONTINUOUS WELD AS FAR AS POSSIBLE

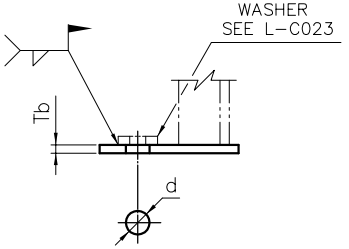


TABLE-2 HOLE SIZE

BOLT SIZE	BASE PLATE
	d
M16	23
M20	27
M24	32
M30	38
M326	44

NOTES

1. ALL DIMENSIONS ARE IN MM.
2. LEG SIZE, BOLT SIZE AND BCD (PCD), SEE ENG'G DWG.
3. OUTSIDE DIAMETER "O.D" IS DEFINED AS SHELL OUTSIDE DIAMETER OR PAD PLATE OUTSIDE DIAMETER WHEN PROVIDED.
4. THICKNESS OF PAD PLATE "t1" SHALL BE EQUAL TO SHELL PLATE THICKNESS. BUT NEED NOT BE THICKER THAN 12 MM.
5. SIZE OF FILLET WELD TO ATTACH LEG IS 12 MM FOR TYPE "A", "B" AND 15 MM FOR TYPE "C".
6. DIMENSIONS WITH ASTERISK MARK IN COLUMN "A" AND "Y" ON TABLE-1 IS FOR REFERENCE. VENDOR SHALL BE CHECK AND FINALIZE THESE DIMENSIONS.
7. WHEN APPLICABLE INSIDE DIAMETER OF SHELL IS NOT TABLE-1. THE DIMENSIONS FOR THE LARGER SHELL DIAMETER SHALL BE APPLIED.
8. HOLE DIAMETER "d" IS FOR THE ANCHOR BOX TYPE FOUNDATION. THOSE FOR THE PRE-INSTALLED ANCHOR BOLT FOUNDATION IS REFERRED TO TABLE-2.
9. PAD PLATES SHALL BE PROVIDED FOR ALL LEGS EXCEPT THAT SHELL MATERIAL IS THE SAME GRADE AS THE LEGS.
10. IN CASE VESSELS ARE SUBJECT TO REACTION LOADS FROM PIPING OR LOADS DUE TO VIBRATION FROM AGITATORS, PUMP, COMPRESSORS, ETC. VENDOR SHALL CHECK VIBRATION STABILITY OF VESSELS AND TAKE COUNTERMEASURES. IF NECESSARY.
11. FOR VESSEL WITH INSIDE DIAMETER LESS THAN 400 MM, RING TYPE BASE PLATE SHALL BE APPLIED AS FOLLOWS:

LEG SIZE	D1	D2
L50x50x6	"Di"+100	"Di"-160
L65x65x6	"Di"+120	"Di"-160

WHERE: "D1" MEANS INSIDE DIA. OF BASE PLATE AND "D2" MEANS OUTSIDE DIA. OF BASE PLATE

12. MATERIAL SHALL BE AS FOLLOWS:
    LEG - ASTM A36 OR EQUIVALENT
    COVER & BASE PLATE - ASTM A36 OR EQUIVALENT
    PAD PLATE - SEE ENG'G DWG.

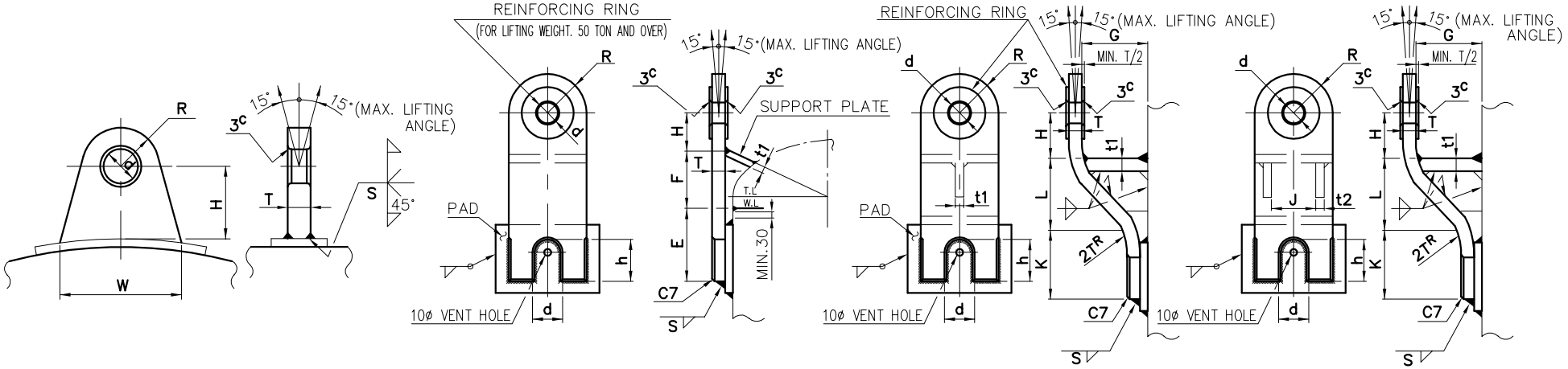
TYPE	SHELL INSIDE DIAMETER	LEG			BASE PLATE					FOUNDATION BOLT			PAD PLATE SIZE		WELD LENGTH OF LEG	MAX. OPER. WEIGHT (KG)	
		LEG SIZE	Q'TY	ST'D HEIGHT	SIZE			Q'TY TH'K	HOLE DIA.							CASE OF STANDARD HEIGHT	CASE OF STANDARD HEIGHT+500MM
	Di			Sa	Z	a	b	tb	d	SIZE	Q'TY	Y	X1	X2	Lw	(S=Sa)	(S=Sa+500)
A	300 ~ 350	L-65x65x6	3	600	SEE REMARK 11			12	19	M16	3	-	145	170	140	960	600
	400 ~ 550	L-75x75x9	3	700	125	* 45	25	14	23	M20	3	* 51	165	185	155	1660	1090
	600 ~ 750	L-90x90x10	3	800	140	* 55	25	16	23	M20	3	* 43	190	210	180	2380	1640
B	700 ~ 850	L-90x90x10	4	900	140	* 55	25	16	23	M20	4	* 44	190	210	180	2980	2140
	900 ~ 1050	L-100x100x13	4	1000	150	* 60	25	20	27	M24	4	* 41	210	230	200	4390	3250
	1100 ~ 1350	L-130x130x15	4	1200	190	* 75	25	20	35	M30	4	* 43	255	300	270	7100	5500
	1400 ~ 1650	L-150x150x15	4	1400	210	* 90	30	20	35	M30	4	* 44	285	350	320	8220	6580
C	1000 ~ 1350	H-150x150x7x10	4	1200	210	40	-	20	23	M20	8	* 143	210	360	330	7630	6070
	1400 ~ 1750	H-175x175x7.5x11	4	1400	235	50	-	22	23	M20	8	* 168	240	415	385	10050	8240
	1800 ~ 2150	H-200x200x8x12	4	1600	260	60	-	25	27	M24	8	* 192	265	470	440	12740	10690
	2200 ~ 2550	H-250x250x9x14	4	1800	315	60	-	25	35	M30	8	* 240	320	580	650	19890	17080
	2600 ~ 2850	H-300x300x10x15	4	1900	365	70	-	25	35	M30	8	* 287	370	690	660	27860	24250
	2900 ~ 3000	H-350x350x10x16	4	2000	420	80	-	0	41	M36	8	* 333	425	800	770	36000	31680

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.07.29	FOR SP 9&10 PROJECT	LJG	CHT	WJM	

LG Engineering & Construction Corp.

SUPPORT LEGS FOR VERTICAL VESSEL

C | STD. NO. 6340S-L-C021



TYPE "A" TYPE "B" TYPE "C" TYPE "D"

TYPE	LIFTING MAX. WEIGHT(TON)	T	H	d	W	R	S	NO. OF LUG
A	1	16	40	25	90	30	7	2
	2	16	50	32	120	40	7	
	3	19	60	40	150	50	9	
	4	22	70	45	170	55	10	
	5	22	80	50	200	65	10	

NOTE

1. ALL DIMENSIONS ARE IN mm.
2. MATERIAL – CARBON STELL.
3. THIS DWG. IS BASED ON 2 POINT SHACKLING.
4. FOR TAILING LUG. SEE ENG'G DWG.
5. THICKNESSES INDICATED ON THIS TABLE ARE MINIMUM AND MUST BE VERIFIED BY VENDOR.

TYPE		LIFTING MAX. WEIGHT (TON)	T	t1	t2	E	H	d	R	h	K	L	G	J	S	C & D WEIGHT (Kg)	F (2:1 ELLIP.) SHELL DIAMETER 500	1000	1500	2000	2500	3000	3500	4000	4500	5000	NO. OF LUG
B	C	10	22	16	–	150	70	50	65	70	100	220	150	–	10	14.5	80	140	190	230	270	300	330	360	390	420	2
		20	25	16	–	180	90	70	100	90	130	280	200	–	15	31.8	100	170	230	280	330	370	410	440	470	500	
		35	30	19	–	220	110	90	130	110	160	320	230	–	15	37.6	100	180	240	290	340	390	430	470	510	550	
		50	36	22	–	260	130	110	160	140	210	380	260	–	20	99.8	100	180	250	300	350	400	440	480	520	560	
		75	45	25	–	300	150	130	190	160	240	420	280	–	20	161											
	D	100	55	30	25	340	170	140	210	180	260	460	300	250	25	246.7											
		150	70	36	25	400	200	170	250	210	310	550	350	300	30	429.2	100	180	250	300	350	400	450	500	550	600	
		200	80	45	30	450	220	190	280	240	360	640	400	340	35	640.5											
		250	100	50	30	500	230	210	290	280	420	680	400	350	45	851.9											

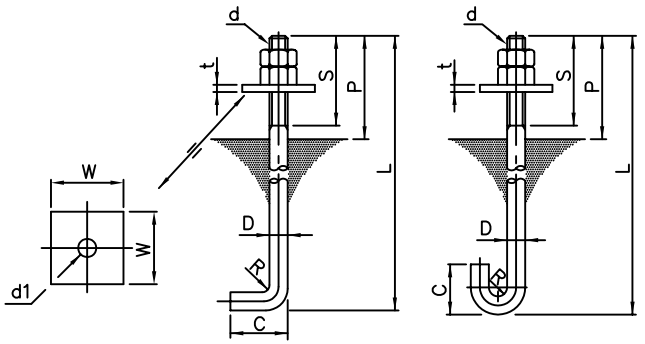
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

LG Engineering & Construction Corp.

LIFTING LUGS

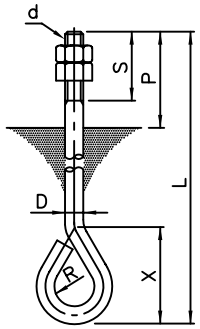
C | STD. NO. 6340S-L-C022

<APPLICATION>
1. TANK,HORIZONTAL DRUM & H/EX.
2. LEG-SUPPORTED VESSEL.
3. BASE TYPE I (SEE STD NO;L-C015)



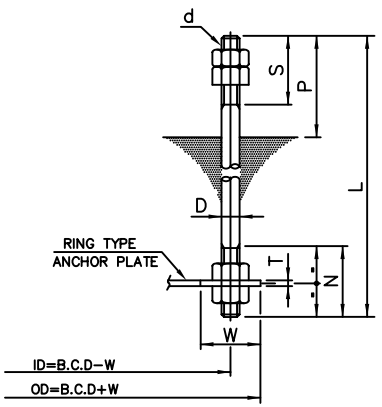
TYPE I

<APPLICATION>
1. VESSEL HEIGHT INCLUDING SKIRT SKIRT IS LESS THAN 30 METER & BOLT DIA. ≤ M48
2. TANKS DESIGNED BY API 650 or 620
3. BASE TYPE II (SEE STD NO:L-C015)

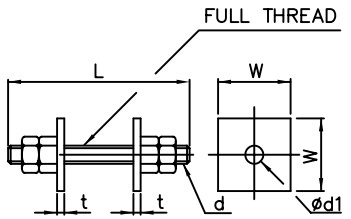


TYPE II

<APPLICATION>
1. VESSEL HEIGHT INCLUDING SKIRT IS 30 METER AND OVER.
2. BLOT DIA. IS LARGER THAN M48.
3. BASE TYPE II (SEE STD NO:L-C015)



TYPE III



SETTING BOLTS

BOLT SIZE	WASHER		
	W	t	d1
M16	40	9	18
M20	45	12	22
M24	60	16	26
M30	70	16	33
M36	80	19	39
M42	90	19	45
M48	110	19	52
M56	130	22	60
M64	150	22	68

BOLT SIZE d	BOLT DATA			TYPE I									TYPE II					TYPE III					
	D	ROOT AREA (CM^2)	M.A.L (kg)	L "L"	L "J"	S	P	C	R	d1	W	t	L	S	P	R	X	L	S	P	N	W	T
M 10	10	0.52	780	260		35	40	40	30	11	50	9											
M 12	12	0.76	1140	360		45	50	50	36	14	50	9											
M 16	16	1.50	2250	450		50	55	65	48	18	60	9											
M 20	20	2.35	3520	550		60	65	80	60	22	60	9											
M 24	24	3.38	5070		700	90	95	115	72	26	70	16	920	90	320	50	200						
M 30	30	5.40	8100		800	110	115	140	90	33	70	16	1090	120	340	60	200						
M 36	36	7.88	11800		950	130	140	160	108	39	80	19	1300	140	400	70	240	1300	140	400	110	150	16
M 42	42	10.83	16200		1100	150	160	200	126	45	90	19	1460	160	410	90	320	1460	160	410	130	175	19
M 48	48	14.24	21300		1200	170	180	225	144	52	100	19	1630	170	430	100	360	1630	170	430	150	200	19
M 56	56	19.67	29500										1890	190	490	120	400	1890	190	490	160	240	25
M 64	64	25.97	38900										2110	220	510	140	410	2110	220	510	190	260	25
M 72	72	33.70	50500															2375	260	570	220	280	28
M 80	80	42	63600															2640	290	640	240	300	32
M 90	90	54.77	82100															2910	310	660	270	350	32
M 100	100	68.67	103000															3180	330	680	300	380	36

NOTES

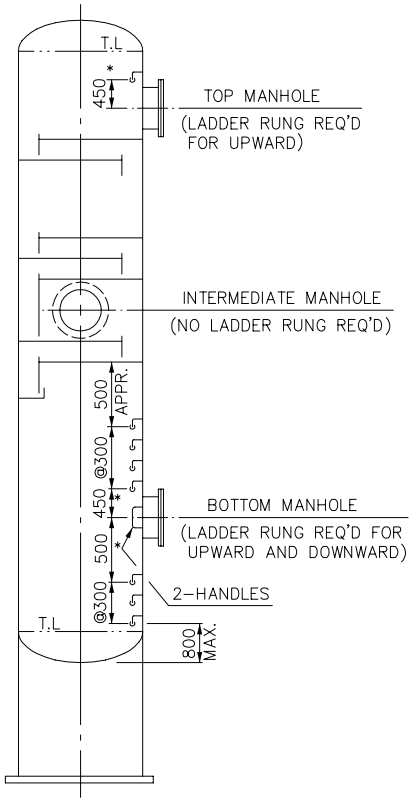
1. ALL DIMENSIONS ARE IN MM.
2. MATERIAL TO BE ASTM A307B OR EQUIVALENT.
3. FOR TYPE II & TYPE III TEMPLATE TO BE REQUIRED. (SEE STD. NO. L-C016)
5. WHEN WOODEN PILLOWS ARE USED, DON'T SHORTEN BURIED LENGTH BUT ADD THE HEIGHT OF WOODEN PILLOW. (SEE STD NO. "L-C044")
6. IN CASE THAN THE ACTUAL ANCHOR CHAIR HEIGHT IS LARGER THAN THAT OF STD. NO. L-C015, ADD THE AUGMENTED HEIGHT TO P & L.
7. THREAD STANDARD ; SEE STD. NO. "L-C025"
8. FOR SETTING BOLT, FOLLOWING NOTES TO BE APPLIED.
   ~. FOR BOLT SIZE, LENGTH "L" AND MATERIAL, SEE ENG'G DWG.

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
◇						
1	'04.07.15	AS PER API CODE & SPEC.	LDH	SSP	SCK	
0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL	

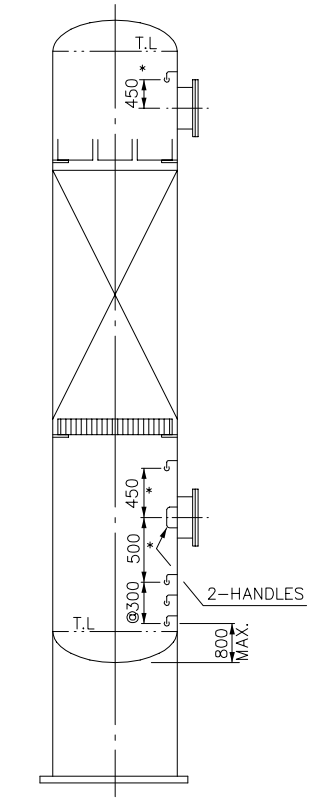
LG Engineering & Construction Corp.

FOUNDATION BOLTS (METRIC)

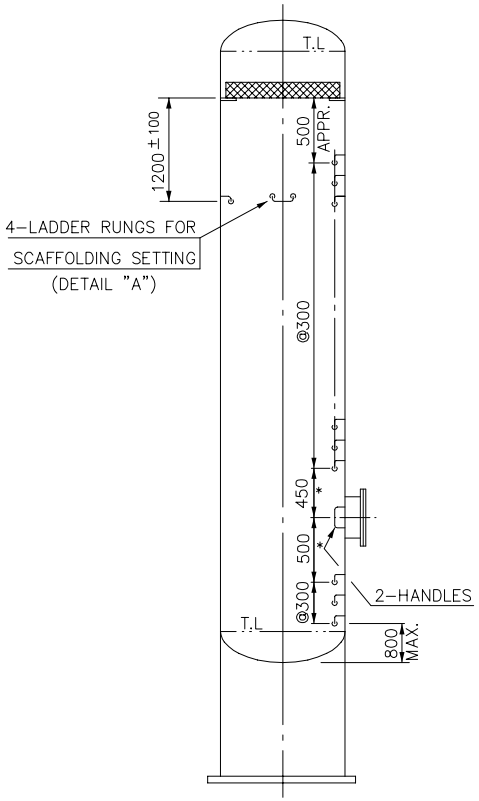
C | STD. NO. 6340S-L-C023



TRAY TOWER



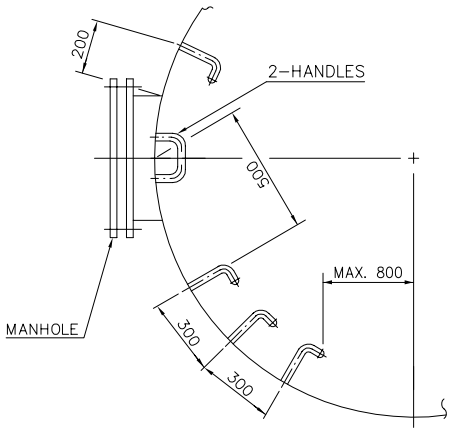
PACKED TOWER



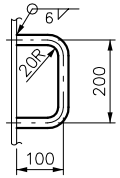
VERTICAL VESSEL W/ DEMISTER

NOTES

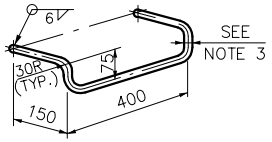
1. ALL DIMENSIONS ARE IN MM.
2. FOR MATERIAL SEE ENG'G DWG.
3. DIA. OF ROD TO BE 19$^{\phi}$ FOR CARBON STEEL AND 16$^{\phi}$ FOR STAINLESS STEEL.
4. IN GENERAL , LADDER RUNGS AS MARKED WITH * SHALL BE PROVIDED OTHER RUNGS ARE NOT NECESSARY, WHEN
   - DISTANCE BETWEEN MANHOLE CENTER LINE AND DOWNWARD HEAD OR INTERNAL DOES NOT EXCEED 800MM.
   - THERE IS NO INTERNAL UPWARD OF MANHOLE.
5. ORIENTATION OF LADDER RUNG SHALL BE THE SAME AS MANHOLE.
6. D < 1500 ; TEMPORARY LADDER MAY BE PREPARED BY CLIENT.



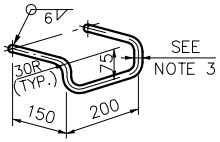
HORIZONTAL VESSEL



HANDLE
(SEE NOTE 3)



BENDED TYPE



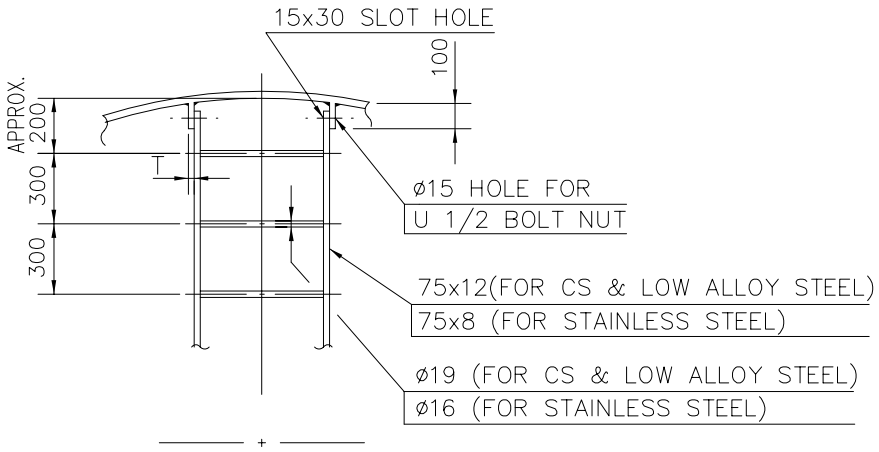
DETAIL "A"

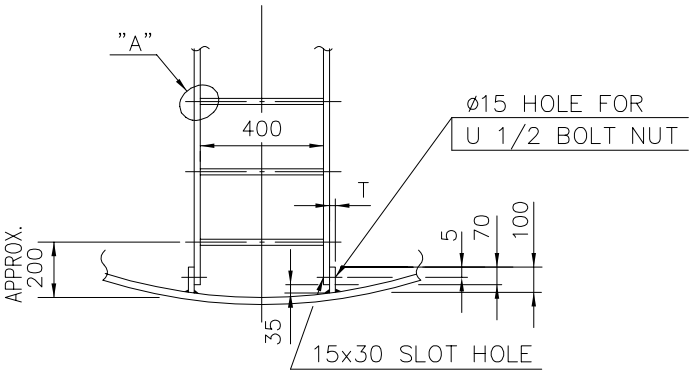
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.12.01	FOR SP 9&10 PROJECT	SRK	HCS	KSL	

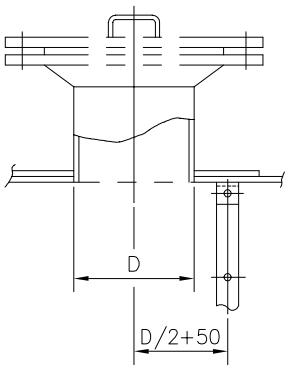
LG Engineering & Construction Corp.

INTERNAL LADDER RUNG

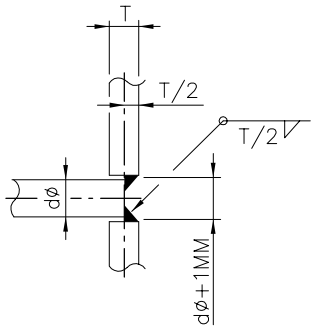
C | STD. NO. 6340S – L – C027







CORROSION ALLOWANCE	MIN. THICKNESS : T	
	C.S & LOW ALLOY STEEL	STAINLESS STEEL
0~1.6	9	8
3.2	12	12
4.8	16	–
6.4	20	–



DETAIL "A"

NOTES

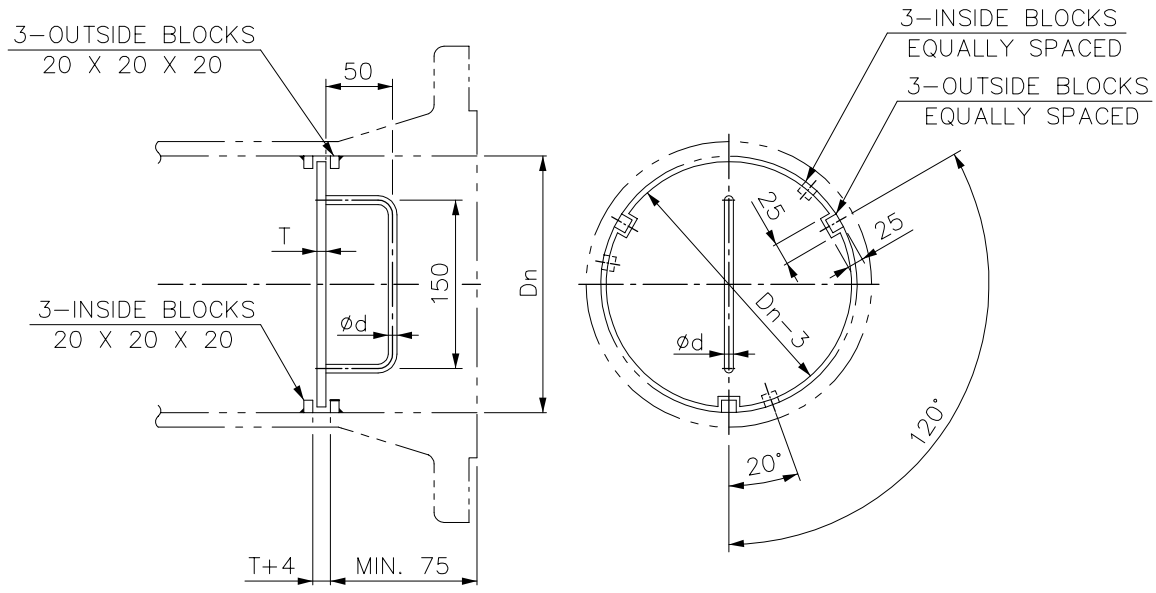
1. ALL DIMENSIONS ARE IN MM.
2. FOR MATERIAL SEE ENG'G DWG.
3. IN GENERAL , LADDER RUNGS AS MARKED WITH * SHALL BE PROVIDED
   OTHER RUNGS ARE NOT NECESSARY, WHEN
   – DISTANCE BETWEEN MANHOLE CENTER LINE AND
     DOWNWARD HEAD OR INTERNAL DOES NOT EXCEED 800MM.
   – THERE IS NO INTERNAL UPWARD OF MANHOLE.
4. D < 1500 ; TEMPORARY LADDER MAY BE PREPARED BY CLIENT.

LG Engineering & Construction Corp.

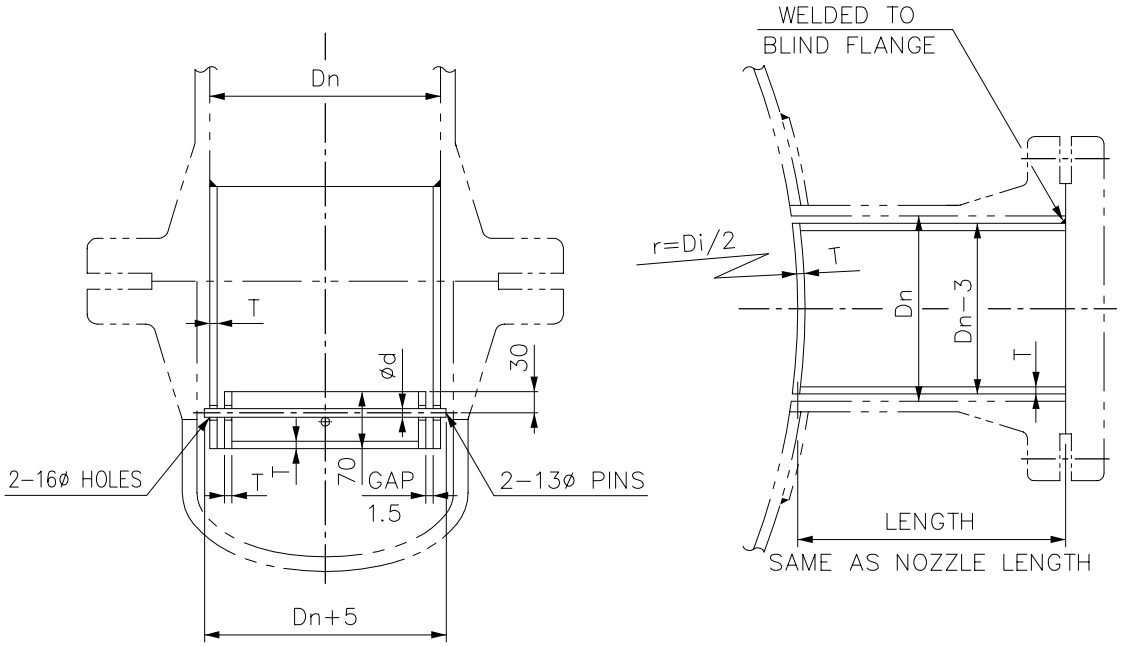
INTERNAL LADDER FOR HORIZONTAL VESSEL

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.12.01	FOR SP 9&10 PROJECT	SRK	HCS	KSL	

C | STD. NO. 6340S – L – C028



TYPE "A"



TYPE "B" TYPE "C"

NOTE

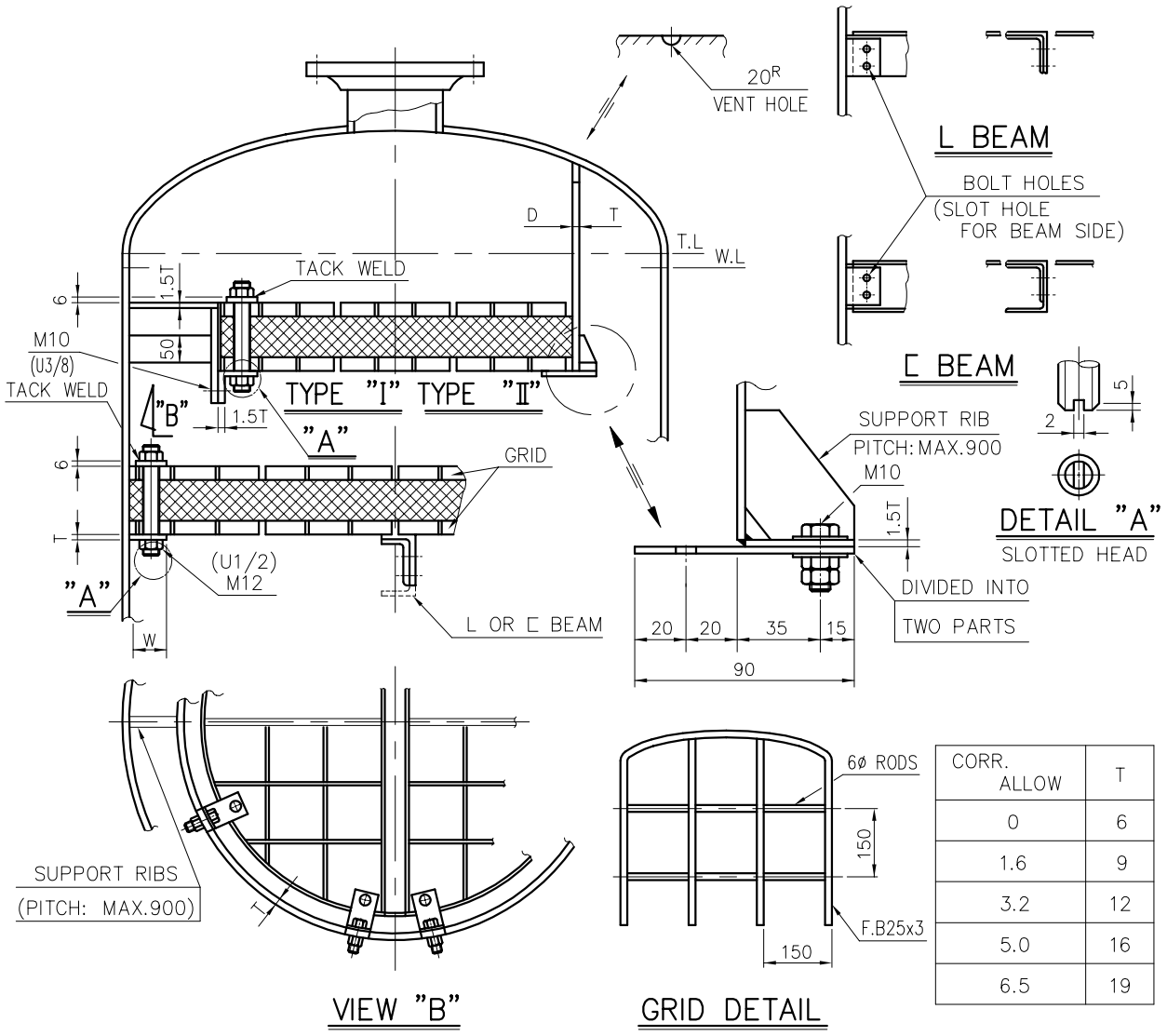
1. ALL DIMENSIONS ARE IN MM.
2. Di MEANS VESSEL I.D., Dn MEANS NOZZLE I.D.
3. FOR MATERIAL AND TYPE SEE ENG'G DWG.
4. T (MIN. THICK.) & d
   1) C.S. & LOW ALLOY STEEL --- T : 6 MM, d : 16 MM.
   2) STAINLESS STEEL --- T : 4 MM, d : 12 MM.

◇0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL		
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD	

LG Engineering & Construction Corp.

RETAINING PLATE

C STD. NO. 6340S-L-C040



VIEW "B"

GRID DETAIL

CORR. ALLOW	T
0	6
1.6	9
3.2	12
5.0	16
6.5	19

VESSEL I.D.	W	SUPPORT BEAM					
		NO.	C.A=0	C.A=1.6	C.A=3.2	C.A=5.0	C.A=6.5
– 1200	38	0	–	–	–	–	–
1200 – 1800	50	1	L75x50x6	L75x75x6	L75x75x9	L75x75x12	L75x75x12
1800 – 2400	65	1	L100x50x6	L100x100x7	L100x100x10	L100x100x10	L100x100x13
2400 – 3300	65	2	L125x50x6	L125x75x7	L150x75x10	L125x75x13	L125x75x13
3300 – 4000	75	3	L150x65x6	⊏125x65x6	⊏150x75x6.5	⊏150x75x9	⊏150x75x9

NOTES

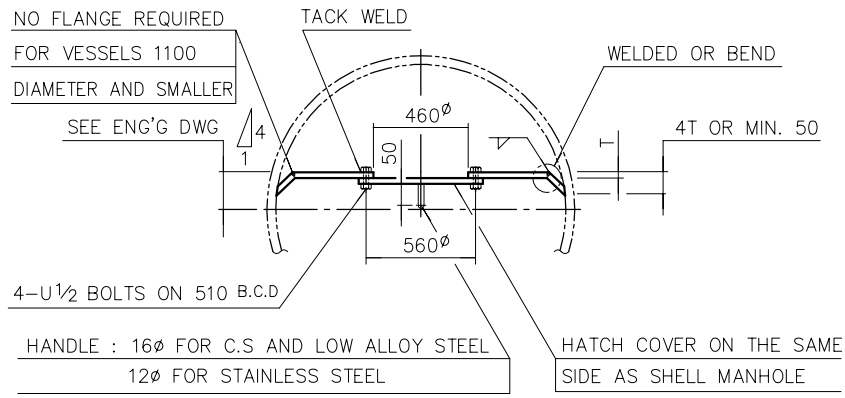
1. ALL DIMENSIONS ARE IN MM.
2. FOR DEMISTER TYPE AND MATERIALS, SEE ENG'G DWG.
   SUPPORT BEAM TO BE ARRANGED AS CROSS DIRECTION OF DEMISTER SEGMENTS.
3. EACH SEGMENTS OF DEMISTER TO BE SIZED TO PASS THROUGH NEARST MANHOLE IN VESSEL.

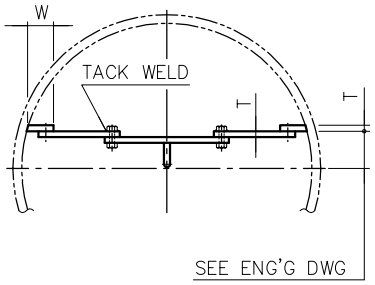
LG Engineering & Construction Corp.

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.12.01	FOR SP 9&10 PROJECT	SRK	HCS	KSL	

DEMISTER AND SUPPORT

C | STD. NO. 6340S – L – C041

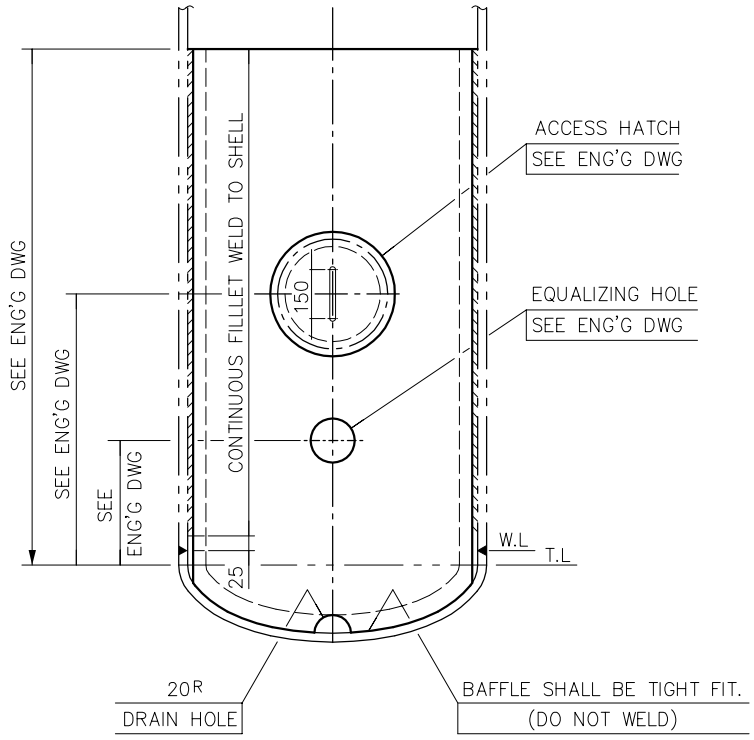




TYPE–I

TYPE–II

DIMENSION ARE SAME AS TYPE–I



VESSEL I.D	W
~ 900	38
901 ~ 1800	50
1801 ~ 3300	65
3301 ~ 5800	80
5800 ~	100

C.A	MIN. THICKNESS : T	
	C.S & LOW ALLOY STEEL	STAINLESS STEEL
0	6	4
1.6	9	7
3.2	12	10
4.8	16	–
6.4	20	–

NOTE

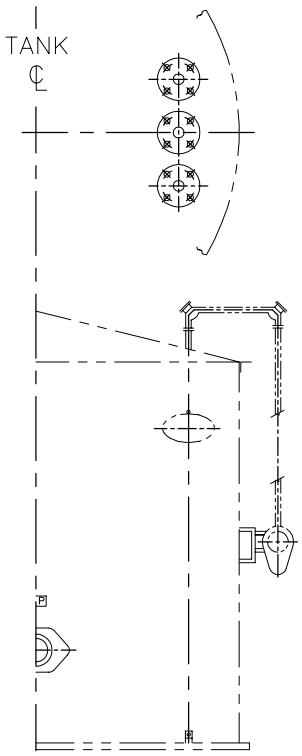
1. ALL DIMENSIONS ARE IN MM.
2. FOR MATERIAL AND TYPE SEE ENG'G DEG.
3. MINIMUM THICKNESS OF HATCH COVER
   C.S & LOW ALLOY STEEL : 6 MM
   STAINLESS STEEL : 4 MM
4. BAFFLE SHALL BE SUITABLY REINFORCED.
5. ACCESS HATCH SHALL BE REQUIRED ONLY WHEN EQUALIZING HOLE IS 400 I.D AND UNDER.
6. FOR TYPE I & II, SEE ENG'G DWG.

LG Engineering & Construction Corp.

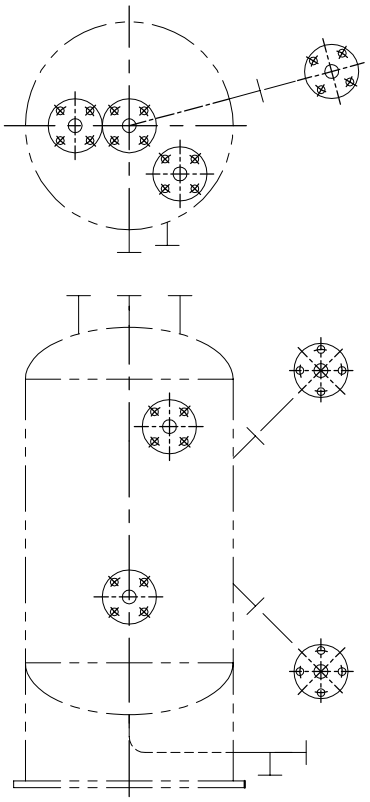
PREFERENTIAL REBOILER BAFFLE

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.12.01	FOR SP 9&10 PROJECT	SRK	HCS	KSL	

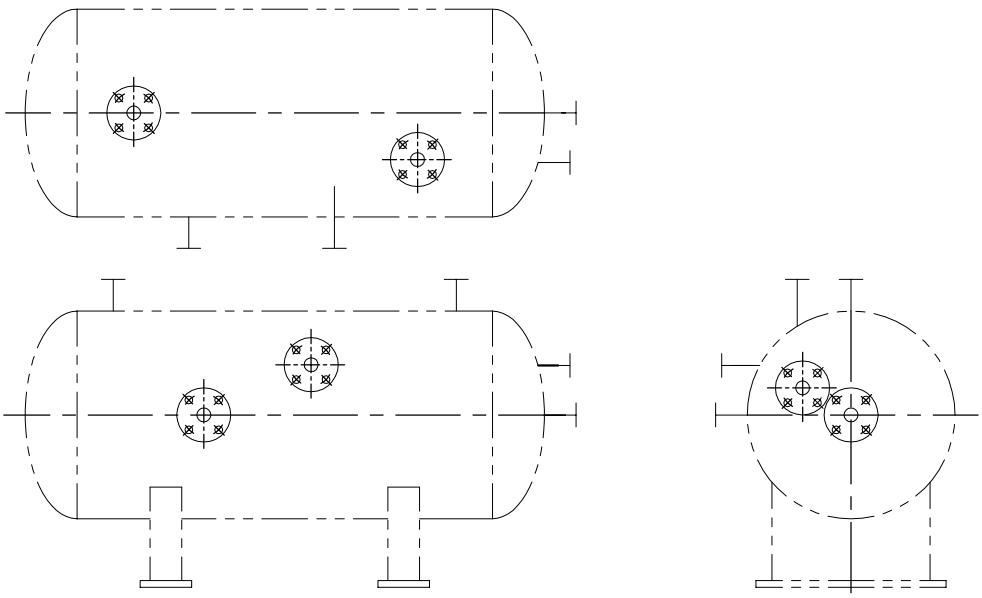
C | STD. NO. 6340S – L – C043



(a) FLOATING TYPE
LEVEL GAGE NOZZLE



(b) VERTICAL VESSEL

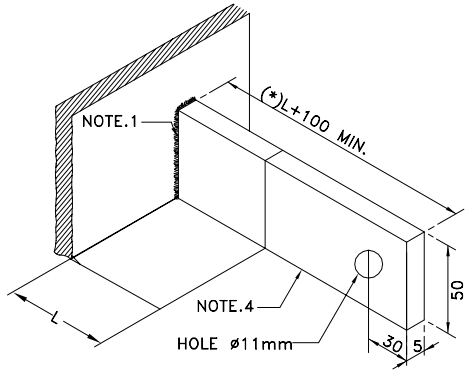


(C) HORIZONTAL VESSEL

	LG Engineering & Construction Corp.						
◇							
◇0	'03.07.29	FOR SP 9&10 PROJECT	LDH	HCS	KSL		
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD	

BOLT HOLE ORIENTATION
FOR FLANGE

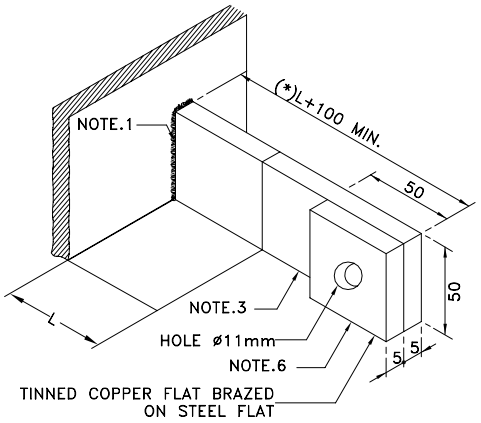
C | STD. NO. 6340S-L-C046



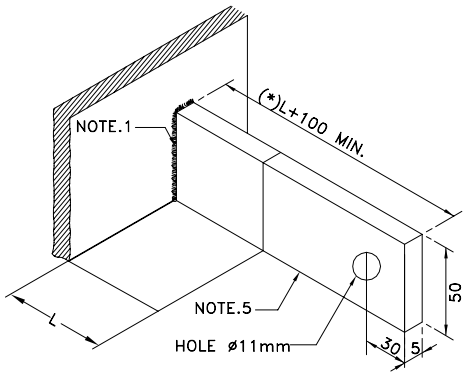
STAINLESS STEEL OR CARBON STEEL

NOTES

1. WELDED ALL AROUND TO METALLIC STRUCTURE OR EQUIPMENT
2. FOR SITE WELDING : STEEL BAR TO BE WELDED TO DOUBLER PLATE IN WORKSHOP. DOUBLER PLATE THEN TO BE WELDED TO STRUCTURE.
3. CARBON STEEL FLAT BAR 50x(100+L)x5THK
4. STAINLESS STEEL FLAT BAR 50x(100+L)x5THK
5. ALUMINIUM STEEL FLAT BAR 50x(100+L)x5THK
6. TINNED COPPER BAR 50x50x5THK
7. L : THICKNESS OF INSULATION OR FIRE PROOF COATING



CARBON STEEL



ALUMINIUM STEEL

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
0	'03.08.08	FOR SP 9&10 PROJECT	DHL	HCS	KSL	

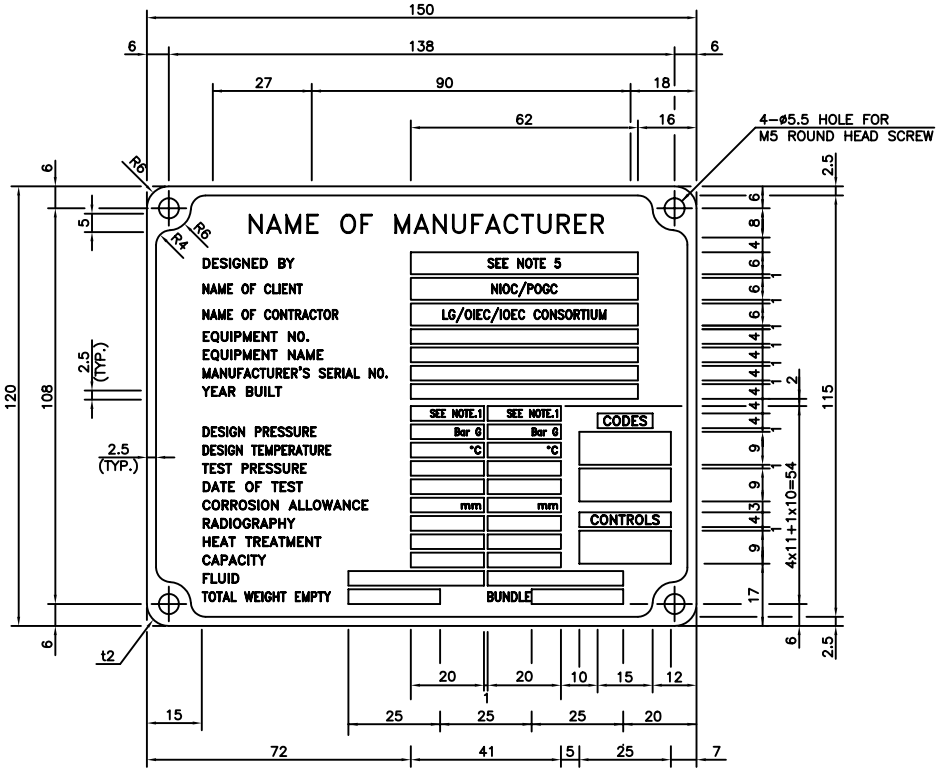
LG Engineering & Construction Corp.

EARTH CONNECTION

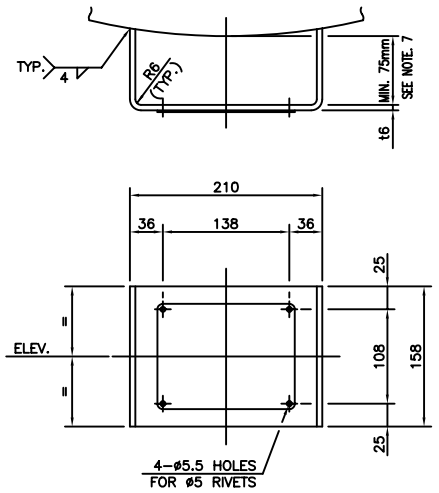
E | STD. NO. 6340S– L – E020

NOTES

1. FOR HEAT EXCHANGER WRITE DOWN: SHELL – CHANNEL
   FOR DOUBLE WALLED VESSEL WRITE DOWN : VESSEL – JACKET
   FOR VESSEL WITH HEATING COIL WRITE DOWN: VESSEL–COIL
   FOR VESSEL WITH TUBE COIL WRITE DOWN: VESSEL–COIL
   FOR VESSEL WITH DOUBLE COMPARTMENT WRITE DOWN ITEM NUMBER OF EACH COMPARTMENT
   ALL LETTERS, BLOCKS, AS WELL AS EDGES, SHALL HAVE RAISED POLISHED FACE – RELIEF 0.5MM APPROX.
2. BLACK BACKGROUND
3. DESIGN OF THIS PLATE IS INDICATIVE ONLY–MANUFACTURER MAY SUBSTITUTE HIS OWN EXISTING PLATE PROVIDED THAT IT INCLUDES ALL DATA SHOWN IN THIS STANDARD
4. ASME MARK ON MANUFACTURER NAME PLATE IS NOT ALLOWED FOR VESSELS NOT PROVIDED WITH OFFICIAL ASME STAMP
5. NAME OF MANUFACTURER
6. MATERIAL : STAINLESS STEEL (THICKNESS: 2mm)
   4–RIVETS COPPER 5 DIA HEAD 'C'
7. MIN 20mm CLEARANCE FROM INSULATION SURFACE



MANUFACTURER'S NAME PLATE



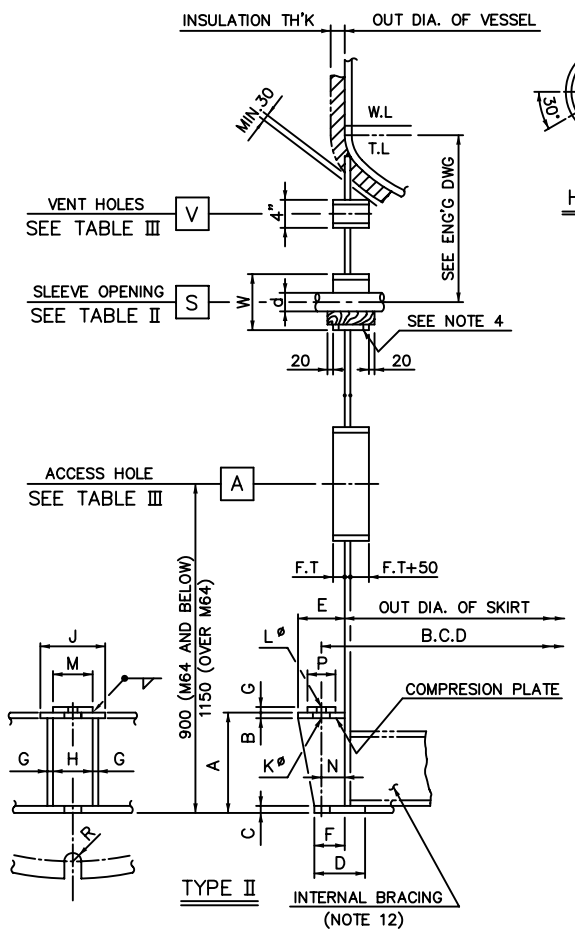
NAME PLATE BRACKET

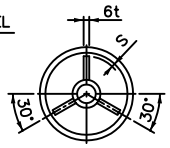
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
1	'04.11.29	FOR SP 9&10 PROJECT	DHL	HCS	KSL	
0	'03.09.22	FOR SP 9&10 PROJECT	DHL	HCS	KSL	

LG Engineering & Construction Corp.

MANUFACTURER NAME PLATE

E | STD. NO. 6340S– L – E021

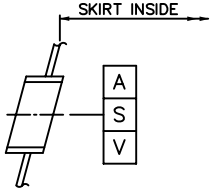




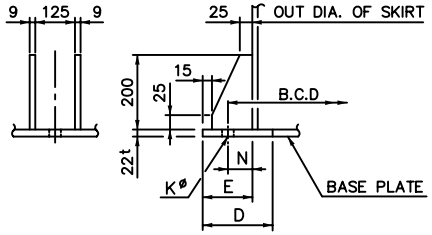
HOT TYPE



COLD TYPE



FLARED SKIRT



TYPE I

TABLE I. BASE DETAIL

BOLT SIZE		A	(*) B	(*) C	(*) D	E	F	G	H	J	K	L	M	(*) N	P	R	MIN. BOLT PITCH	BOLT SIZE
U (inch)	M (mm)																	M (mm)
5/8	16	–	–	22	–	–	–		–	–	19	–	–	50	–	–	–	16
3/4	20	–	–	22	–	–	–		–	–	23	–	–	50	–	–	–	20
1	24	250	16	22	125	100	70	12	80	140	40	26	110	55	70	25	160	24
1 1/4	30	250	16	25	130	110	75	14	90	150	50	33	120	60	75	30	170	30
1 1/2	36	300	19	25	135	120	80	14	90	160	60	39	120	65	85	30	180	36
1 3/4	42	300	22	28	145	140	100	16	105	180	70	45	140	75	100	35	190	42
2	48	300	25	32	160	150	110	16	115	190	80	52	150	80	110	40	200	48
2 1/4	56	350	28	32	170	160	115	19	130	210	90	60	170	85	115	45	210	56
2 1/2	64	350	32	36	180	170	125	19	140	220	100	68	180	90	125	50	220	64
2 3/4	72	400	36	40	190	190	140	22	145	240	100	78	190	95	140	50	240	72
3	80	450	40	45	220	200	145	25	170	270	120	86	220	105	145	60	260	80
3 1/2	90	450	45	50	240	220	165	25	180	280	130	96	230	115	165	65	300	90
4	100	450	50	50	260	240	185	25	190	290	140	107	240	125	180	70	340	100

TABLE Ⅱ. SLEEVE OPENING

PIPE SIZE d (inch)	W	PIPE SIZE d (inch)	W	PIPE SIZE d (inch)	W	PIPE SIZE d (inch)	W
3/4	4	3	8	12	500	24	850
1	6	4	10	14	550	26	850
1 1/2	6	6	12	16	600	28	850
2	8	8	350	18	650	30	900
2 1/2	8	10	450	20	700	32	950

TABLE Ⅲ. ACCESS OPENING

VESSEL DIAMETER	VENT NO. REQ'D		ACCESS HOLE	
	HOT	COLD	NO. REQ'D	INDIA.
d< 1500	2	2	1	450
1500 ≦ d< 3000	4	2	2	450
3000 ≦ d< 3500	4	4	2	450
3500 ≦ d< 4500	4	4	2	500
4500 ≦ d	6	4	2	500

NOTE

1. ALL DIMENSIONS ARE IN mm.
2. FOR OPENING SIZE AND ORIENTATION, SEE ENG'G DWG.
3. TH'K AND MATERIALS OF REINFORCING RING (Ⓥ,Ⓢ,Ⓐ) TO BE AS FOLLOWS.
   – NOMINAL SIZE 12" & SMALLER
   JIS SGP, ASTM A120 HEAVY EXTRA.
   – NOMINAL SIZE OVER 12"
   JIS SS400 12 mm, ASTM A283 Gr.C 12 mm.
   BUT, RING TH'K MAY NOT EXCEED SKIRT TH'K.
4. WOOD USED FOR SKIRT OPENING TO BE TIGHTED WITH 2 SETS SCREWS AND APITON.
5. CLEARANCE "S" BETWEEN SLEEVE AND RIB TO BE AS FOLLOWS.
   $S \geqq \alpha \cdot \Delta\theta + 1$ mm
   WHERE α : COEFFICIENTS OF THERMAL EXPANSION
   $\Delta\theta$: TEMP. DIFFEERENCE (DESIGN TEMP.–AMB TEMP.)
6. FOR MATERIALS OF SKIRT, B.C.D, TYPE AND OUTSIDE DIA. OF VESSEL, SEE ENG'G DWG.
7. USE THE DIMENSIONS ON TABLES, UNLESS SPECIFIED ON ENG'G DWG.
8. IN CASE THAT PITCH OF FOUNDATION BOLT IS 450 mm AND LESS, COMPRESSION PLATE TO BE CONTINUOUS.
9. SIZE OF FILLET WELDS TO BE SAME AS THE SKIRT TH'K (MIN. 6 mm)
10. OUTSIDE DIA. OF SKIRT TO BE SAME AS OUTSIDE DIA. OF VESSEL, UNLESS OTHERWISE SPECIFIED ON ENG'G DWG.
11. F.T ; FIRE PROOFING THICKNESS.
12. TO PREVENT ANY DEFORMATION DURING ERECTION, SKIRT INTERNAL BRACING TO BE PROVIDED, IF REQUIRED.
13. VENDOR SHALL CHECK ASTERISKED (*) DIMENSIONS ON TABLE–I INCREASE THEM.

FOR REFERENCE

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	APPD
1	'00.12.06	GENERAL REVISION	RJH			
0	'96.2.29	ORIGINAL ISSUE	LJG	CHT	WJM	

LG Engineering & Construction Corp.

TYPICAL SKIRT & BASE DETAILS FOR VERTICAL VESSELS

C | STD. NO. 6340S – L – C015